



**CENTRE FOR
DISTANCE AND
ONLINE EDUCATION**

BUSINESS ANALYTICS

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**Centre for Distance and Online Education
NMIMS Deemed-to-be-University**

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Unit 1: Introduction to Business Analytics: Understanding Business Analytics, Benefits and Challenges in Business Analytics, Types of Business Analytics, Descriptive Analytics, Diagnostic Analytics, Predictive Analytics, Prescriptive Analytics, Nature and Scope of Business Analytics, Stages of a Business Analytics Project, Identifying Information Gaps.

Unit 2: Data Preprocessing and Visualisation: Using Data in Analytics, Data Collection Methods, Types of Data, Data Cleansing: Common Issues with Data, Missing Data, Identification of Outliers, Treating Missing Values, Imputation Methods, Data Visualisation, Column and Bar Charts, Line Charts, Pie Charts, Scatter Charts.

Unit 3: Introduction to Probability Theory: Principles of Counting, Permutations, Combinations, Discrete Probability Distributions, Probability Mass Function (PMF), Types of Discrete Probability Distributions, Continuous Probability Distributions, Probability Density Function (PDF), Types of Continuous Probability Distributions, Difference Between PDF and PMF.

Unit 4: Statistical Inference: Hypothesis Testing, p-values and High p-values, One-Sample Hypothesis Tests, Two-Sample Hypothesis Tests, Types of Errors, Chi-Square Test, Chi-Square Test for Independence, Chi-Square Goodness of Fit Test.

Unit 5: Introduction to Data Mining: Data Mining, Data Mining Architecture, Scope of Data Mining, Ethical Considerations and Challenges in Data Mining, Data Mining Techniques, Supervised Learning, Unsupervised Learning, Unsupervised Learning Techniques, Association Rule Mining, Clustering.

Unit 6: Predictive Modelling in Business Analytics: Understanding the Predictive Modelling Concept, Importance of Predictive Modelling, Advanced Models in Business Analytics, Integration of Technology, Framing Predictive Models, Types of Predictive Models, Trends in Predictive Analytics, Best Practices and Common Pitfalls When Implementing Models.

Unit 7: Simple Linear Regression: Concept of Simple Linear Regression, Simple Linear Regression Model, Estimated Regression Equation, Least Squares Method, Assessing the Fit of Simple Linear Regression Model, Sum of Squares Method, The Co-efficient of Determination, Model Assumptions.

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Unit 8: Multiple Linear Regression: Understanding the Concept of Multiple Regression, Application of Multiple Regression in Various Industries, Forward, Backward and Stepwise Regression, Forward Regression, Backward Regression, Stepwise Regression, Assessing the Fit of Multiple Regression Model, Validation of Multiple Regression Models, Coefficient of Multiple Determination (R-Square), Residual Analysis in Multiple Regression, Challenges in Multiple Regression.

Unit 9: Logistic Regression: Concept of Logistic Regression Model, Estimation of Parameter in Logistic Regression, Interpretation of Logistic Regression Parameters, Accuracy of Logistics Model, Accuracy Paradox, Sensitivity, Specificity and Precision, Receiver Operating Characteristic (ROC) Curve, Area Under the Curve (AUC) and Lorenz Curve.

Unit 10: Forecasting and Time-Series: Concept of Forecasting, Techniques for Forecasting, Understanding the Theory of Forecasting Techniques, Concept of Time-Series Analysis, Time-Series Pattern, Understanding the Techniques Used in Time-Series Analysis, Application of Time-Series Principles.

Unit 11: Decision Theory: Concept of Decision Theory, Components and Application of Decision Theory, Branches of Decision Theory, Payoff Tables in Decision-Making, Decision-Making without Probabilities, Optimistic Approach, Minimax Approach, Conservative Approach, Decision-Making With Probabilities, Expected Monetary Value, Decision Tree.

Unit 12: Decision Trees: Concept of Decision Trees, Elements of Decision Trees, Working of Decision Trees, Types of Decision Trees, Advantages and Risks Associated with Decision Trees, Chi-Square Automatic Interaction Detection (CHAID), CHAID Tree Development, Bonferroni Correction, Classification and Regression Tree, Gini Impurity Index, Entropy, Ensemble Method, Random Forest.

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INTRODUCTION TO BUSINESS ANALYTICS

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**Case Objective**

The objective of this caselet is to illustrate the comprehensive impact of UnitedHealth's implementation of business analytics in transforming its customer engagement strategies and enhancing operational efficiency.

INTRODUCTORY CASELET**UNITEDHEALTH GROUP'S PREDICTIVE ANALYTICS IN HEALTHCARE**

UnitedHealth Group utilised predictive analytics to improve patient care within its network significantly. The healthcare provider identified patients at risk of developing chronic diseases or those likely to experience rehospitalisation by analysing extensive datasets that include patient medical histories, treatment outcomes, and lifestyle choices. This proactive approach allows for early intervention through customised care plans, which enhances patient outcomes and optimises resource allocation within the healthcare system.

UnitedHealth's analytics capabilities extended to managing healthcare costs and improving service delivery. They managed the hospital staffing and resource needed by leveraging data to predict patient admission rates and peak times for different treatments. Furthermore, predictive analytics aided in developing new health services and programs that target the specific requirements of their patient population, leading to more efficient healthcare delivery and reduced operational costs. This strategic use of data ensures that patients receive the right care at the right time, enhancing overall patient satisfaction and loyalty which boosted UnitedHealth's revenue.

(Source: <https://digitaldefynd.com/IQ/business-analytics-case-studies/>)

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Grasp the fundamentals of business analytics
- Understand various types of business analytics
- Discuss the nature and scope of business analytics
- Recognise the stages of a business analytics project for analysis

1.1 INTRODUCTION

In the era of globalisation and aggressive competition, data has become an essential part of any organisation. With the rise of several business intelligence tools, business professionals can easily access data for making data-driven decisions. Data-driven organisations consider data an important asset as it helps them to make the right decision at the right time. However, in order to stay competitive, it is not enough for organisations to merely make accurate decisions; they must also be able to anticipate future trends. In such cases, predictive insights allow organisations to mitigate risks, analyse market trends, and address growth challenges proactively.

Business analytics is the scientific process of transforming data into insights to make better decisions. It is used for data-driven or fact-based decision-making, which is often seen as more impartial than other alternatives. Business analytics utilises descriptive analytics, diagnostic analytics, predictive analytics, and prescriptive analytics for performing analysis on business data and support decision-making processes.

In this chapter, you will learn about business analytics, explore different types of business analytics and understand the scope and nature of business analytics along with the various stages of business analytics project.

1.2 UNDERSTANDING BUSINESS ANALYTICS

Individuals and business managers face increasing complexity when making decisions with important economic implications. They must deal with uncertain data and a large amount of information. In today's business environment, analytics has become essential as it integrates data, information technology, statistical analysis, and quantitative methods to provide more informed and effective insights for decision-making. Analytics helps in fact-based decision-making by transforming data into actionable strategies, benefiting areas like Customer Relationship Management (CRM) and supply chain optimisation. Multi National Companies (MNCs) like Harrah's entertainment, Amazon, and United Parcel Service (UPS)

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demonstrate analytics' significance in strategic decision-making, showing a strong correlation between analytics use and organisational performance. Understanding analytics becomes crucial for effective management in contemporary business environments amid the overwhelming abundance of data.

Business analytics is the methodological exploration of business data using statistical tools with an aim to drive decision-making. The applications of business analytics can be found in various industries, such as:

- ❑ **Retail:** Retailers leverage analytics for forecasting, inventory management and personalised marketing. Through customer segmentation and predictive analytics, retailers optimise stock levels and create personalised shopping experiences, fostering customer loyalty.
- ❑ **Healthcare:** Analytics is essential in patient care management, resource allocation and disease prevention. Predictive analytics helps hospitals in anticipating patient admission rates, optimising resource allocation and identifying potential disease outbreaks, ultimately improving patient outcomes.
- ❑ **Finance:** Financial institutions employ analytics for fraud detection, credit risk assessment and investment portfolio optimisation. By analysing a large amount of financial datasets, financial organisations can make data-driven decisions, minimise risks and enhance their overall financial performance.
- ❑ **Manufacturing:** Manufacturers use analytics to optimise production processes, enhance supply chain management and improve product quality. Predictive maintenance analytics, for allows manufacturers to anticipate equipment failures, reducing downtime and operational costs.
- ❑ **Marketing:** Marketing analytics is employed for targeted campaigns, customer behaviour analysis and market segmentation. By understanding customer preferences and behaviours, businesses can tailor their marketing strategies, resulting in more effective and impactful campaigns.

The effectiveness of business analytics is supported by a combination of advanced tools and technologies that facilitate data processing, analysis and interpretation. These tools and technologies are:

- ❑ **Data visualisation tools:** Tools such as Tableau, Power BI, Qlik and Excel enable organisations to create visually compelling and interactive dashboards. These tools facilitate the communication of complex data insights in an accessible manner, aiding decision-makers in understanding trends and patterns.
- ❑ **Statistical analysis systems:** Tools such as SAS, R and SPSS provide powerful statistical analysis capabilities, allowing

businesses to derive meaningful insights from data. These tools are essential for advanced statistical modelling, hypothesis testing and data interpretation.

- ❑ **Machine Learning Platforms:** Platforms such as TensorFlow, Amazon SageMaker, Microsoft Azure Machine Learning, and other machine learning platforms empower organisations to implement sophisticated algorithms for predictive analytics and pattern recognition. Machine learning is instrumental in uncovering hidden patterns and trends within datasets.
- ❑ **Big data technologies:** Technologies such as Apache Hadoop, Spark and SPSS enable businesses to handle large-scale data processing. In an era of Big Data, these tools enable organisations to process and analyse massive datasets efficiently, extracting valuable insights from the wealth of available information.

With the large amount of data generated today, organisations must adopt well-defined strategies to leverage insights effectively, ensuring they stay competitive and make informed decisions in an increasingly data-driven environment. The strategic significance of business analytics are:

- ❑ **Identifying trends:** Business analytics enables organisations to scan large datasets to identify and capitalise on emerging trends. By recognising patterns in customer behaviour, market dynamics and industry trends, businesses can position themselves proactively and increase their operation.
- ❑ **Risk mitigation:** Analytics can help organisations assess potential risks and develop strategies to mitigate them through potential analytics. Whether it is forecasting financial uncertainties, anticipating supply chain disruptions, or identifying operational vulnerabilities, analytics empowers businesses to make informed decisions that safeguard against potential pitfalls.
- ❑ **Capitalising on opportunities:** Analytics acts as a catalyst for innovation by helping organisations uncover unexplored opportunities. Whether identifying new customer segments, optimising pricing strategies or innovating products based on market demands, businesses can strategically position themselves to capitalise on emerging opportunities.
- ❑ **Performance optimisation:** Business analytics provides a continuous feedback loop for organisations to monitor and optimise performance. By analysing operational data, businesses can streamline processes, identify bottlenecks and enhance overall efficiency, thereby ensuring sustained success.

After being introduced to business analytics, it is necessary to know the benefits and challenges of business analytics in real-life problems which will be provided in the next section.

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1.2.1 BENEFITS AND CHALLENGES IN BUSINESS ANALYTICS

Exploring the benefits and challenges of business analytics is important to grasp its potential impact. Recognising the advantages helps organisations use analytics effectively, while acknowledging challenges ensures they are ready to overcome obstacles and make the most of analytics solutions in their operations. The benefits of business analytics are as follows:

- ❑ **Informed decision-making:** Business analytics enables businesses to make well-informed choices by deriving valuable insights from data. This approach guides organisations toward strategies based on intelligence and analysis, rather than relying solely on intuition.
- ❑ **Increased efficiency:** Business analytics enhances operational efficiency and resource utilisation by optimising the processes and resources. The careful application of data-driven insights leads to improved performance across various organisational functions.
- ❑ **Competitive advantage:** Business analytics helps businesses in gaining a competitive edge by harnessing the analytics process to uncover market trends, customer behaviours and strategic opportunities to stay ahead in dynamic industries.
- ❑ **Customer understanding:** Business analytics helps in enabling businesses to comprehend customer preferences and behaviour, customising products and services for personalised engagement and satisfaction.
- ❑ **Risk management:** Business analytics plays a pivotal role in identifying potential risks, offering a proactive approach to risk management by analysing patterns and predicting potential challenges before they impact the business.
- ❑ **Strategic insights:** Business analytics uncovers the insights for strategic decision-making and operational excellence from historical data.

While business analytics offers significant benefits, it is important to address the challenges it presents. The key challenges are outlined of business analytics are as follows:

- ❑ **Data quality and integration:** Ensuring the accuracy and seamless integration of diverse data sources pose challenges, impacting the reliability and effectiveness of analytics insights.
- ❑ **Privacy and security concerns:** Managing sensitive data requires robust privacy measures and security protocols to safeguard information from unauthorised access and potential breaches.

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- ❑ **Skill shortages:** The demand for skilled analytics professionals outstrips the available talent pool, posing a challenge for organisations aiming to build proficient analytics teams.
- ❑ **Cost of implementation:** Implementing robust analytics systems may involve substantial upfront costs, creating a financial hurdle for organisations, especially operating on a small scale.
- ❑ **Resistance to change:** Overcoming resistance within organisations to adopt an analytics-driven decision-making culture can be challenging, requiring effective change management strategies.

Having explored the benefits and challenges of business analytics, it is essential to study its various types. Each type serves a specific purpose in analysing data and driving business decisions, as outlined in the next section.



SELF ASSESSMENT QUESTIONS

1. Business analytics enhances operational efficiency by applying intuition-based strategies. (True/False)
2. What role do machine learning platforms play in analytics?
 - a. Predictive analytics and pattern recognition
 - b. Data visualisation
 - c. Statistical analysis
 - d. Inventory management
3. Business analytics can navigate data abundance, predict future trends and deliver measurable outcomes, contributing to revenue growth. (True/False)



ACTIVITY

Engage in a group discussion exploring the transformative journey of business analytics. Discuss its evolution, benefits, challenges and real-world applications. Share insights on how analytics shapes decision-making in various industries.

1.3 TYPES OF BUSINESS ANALYTICS

Business analytics serves as a tool for decision-making by using data to identify patterns, forecast trends, and support strategic decision-making choices. It is divided into different types to address various aspects of decision-making and problem-solving, providing a thorough approach to data analysis.

Each type of business analytics serves a distinct purpose, targeting specific questions and objectives within an organisation. This categorisation helps businesses to implement customised analytical

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techniques at various stages of their operations. By using these categories, organisations can enhance decision-making, improve efficiency, and align their strategies with data-driven insights to promote innovation.

The different types of business analytics namely, descriptive analytics, diagnostic analytics, predictive analytics and prescriptive analytics are shown in Figure 1.1:

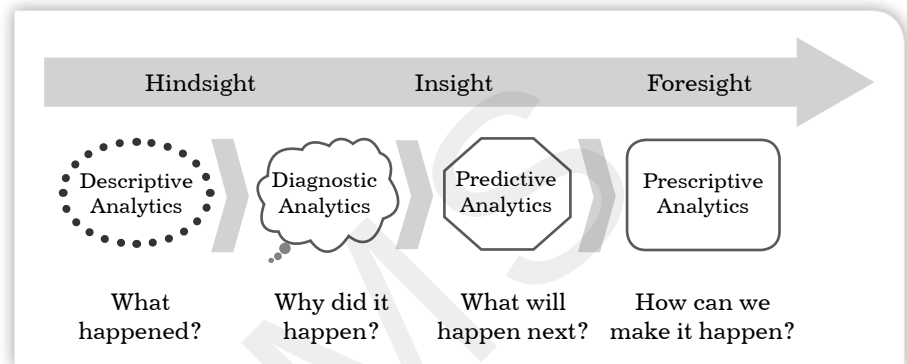


Figure 1.1: Types of Business Analytics

1.3.1 DESCRIPTIVE ANALYTICS

Descriptive analytics is the foundational layer of business analytics, focusing on the interpretation and summarisation of historical data to provide a clear understanding of past events and trends.

Descriptive analytics does not predict the future or tell us what to do next. Instead, it sets the stage for making decisions by giving us a clear picture of what has already happened. It is like understanding the past to better plan for the future.

The key features of descriptive analytics are as follows:

- ❑ **Looking back in time:** Descriptive analytics deals with information about things that have already occurred. It's like looking at a timeline of events in business.
- ❑ **Summarising facts:** Imagine you have a big storybook of data. Descriptive analytics help us summarise this information. It turns complex details into simple, easy-to-understand facts.
- ❑ **Understanding performance:** Businesses have different parts, like a puzzle. Descriptive analytics help us see how each piece of the puzzle has performed in the past. For example, how many products were sold last month?
- ❑ **Creating a story:** It is a bit like creating a story with data. By looking at the past, we can build a narrative that explains what happened, making it easier for everyone to understand.

1.3.2 DIAGNOSTIC ANALYTICS

Building upon descriptive analytics, diagnostic analytics takes us beyond the surface, probing deeper to uncover the reasons behind past events. It is like a detective in the world of business data, investigating to understand not just what happened but why it happened.

In essence, while descriptive analytics provides the groundwork, diagnostic analytics adds layers of depth by unravelling the mysteries behind business events, empowering organisations to not only comprehend their history but also obtain valuable insights for future decisions. The key features of diagnostic analytics are as follows:

- ❑ **Uncovering causes:** While descriptive analytics shows us what occurred, diagnostic analytics digs into the causes behind those events. It is like figuring out why a particular event happened in the business timeline.
- ❑ **Identifying patterns:** Diagnostic analytics goes beyond summarising facts; it identifies patterns in the data. For example, recognising trends in business performance and understanding why certain trends emerged.
- ❑ **Diagnosing problems:** Just as a detective solves a mystery, diagnostic analytics helps in diagnosing problems in business performance. For instance, if sales drop, diagnostic analytics seeks to reveal the specific factors contributing to the decline.
- ❑ **Enhancing understanding:** By exploring the underlying reasons behind data trends, diagnostic analytics improves your overall understanding. It is similar to interpreting the nuances within a business narrative, thereby enriching the story and providing deeper insights.

1.3.3 PREDICTIVE ANALYTICS

Predictive analytics builds on diagnostic analytics by allowing organisations to forecast future outcomes. This approach acts like a strategic advisor in business data by identifying potential scenarios and facilitating proactive decision-making. The working of predictive analytics is shown in Figure 1.2:

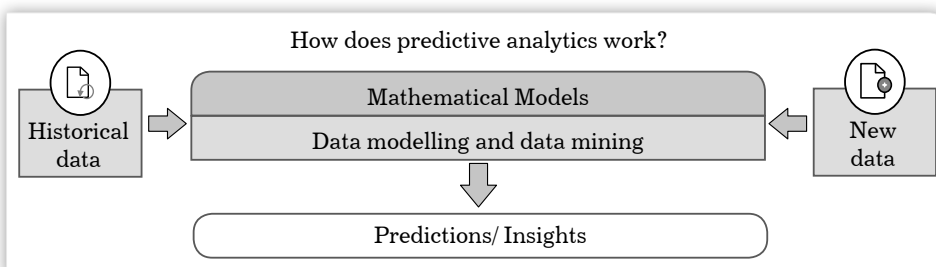


Figure 1.2: Working of Predictive Analytics

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Diagnostic analytics explains the reasons behind past events, while predictive analytics focuses on future possibilities. This approach provides organisations with tools to anticipate, plan and strategise effectively. Predictive analytics acts as a guide, helping businesses navigate uncertain situations and maintain a competitive edge in a constantly changing environment.

The key features of predictive analytics are as follows:

- ❑ **Anticipating trends:** While diagnostic analytics explains past trends, predictive analytics forecasts future patterns. It is like having a crystal ball that foresees where the puzzle pieces of business performance might align in the future.
- ❑ **Forecasting outcomes:** Predictive analytics is forward-focused, projecting potential outcomes based on historical data and identified patterns. Imagine it as a compass, providing direction for future business endeavours.
- ❑ **Identifying opportunities:** Just as a strategic advisor identifies opportunities for growth, predictive analytics highlights areas where businesses can capitalise. It is like having a guidebook that points out the promising avenues ahead.
- ❑ **Risk mitigation:** Predictive analytics aids in identifying potential risks on the horizon. It is similar to having a radar system that alerts businesses to potential challenges, allowing them to proactively navigate and mitigate risks.

1.3.4 PRESCRIPTIVE ANALYTICS

Prescriptive analytics develops from the insights gained through diagnostic and predictive analytics to provide strategic guidance in business decision-making. It not only predicts future outcomes but also recommends specific actions, helping to make proactive decisions based on different scenarios. The working of a prescriptive model is shown in Figure 1.3:

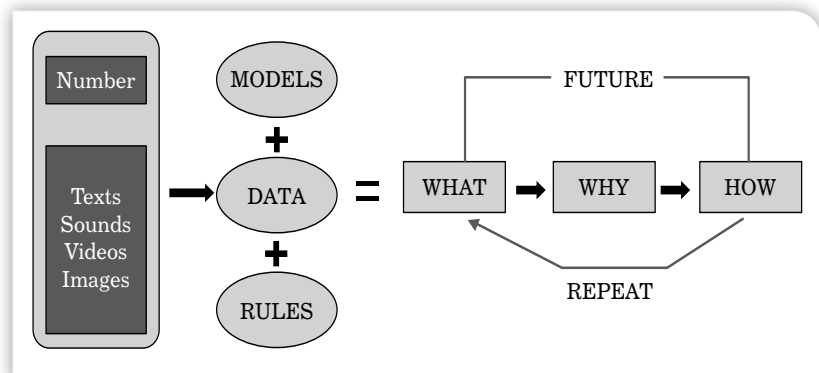


Figure 1.3: Prescriptive Analytics Model

Prescriptive analytics provides organisations with essential tools to anticipate future developments, plan effectively and navigate strategically. This type of analytics acts as a guide, assisting businesses in exploring new opportunities and maintaining a competitive edge in a constantly changing environment.

The key features of prescriptive analytics are as follows:

- ❑ **Forecasting future trends:** Unlike diagnostic analytics, which examines past trends, prescriptive analytics focuses on anticipating future patterns. It serves as a valuable tool for businesses, providing insights into how various factors may align to influence performance.
- ❑ **Predicting outcomes:** Prescriptive analytics aims to project potential outcomes based on historical data and established patterns. It acts as a guide in assisting businesses by making informed decisions for their future initiatives.
- ❑ **Identifying growth opportunities:** Similar to a strategic advisor, prescriptive analytics highlights areas where businesses can seize growth opportunities. It functions as a comprehensive resource, directing organisations towards promising avenues for development.
- ❑ **Mitigating risks:** This type of analytics plays a crucial role in identifying potential risks that may arise. It operates like a proactive alert system, helping businesses recognise challenges in advance and enabling them to implement effective risk management strategies.

Business analytics includes various types, such as descriptive, diagnostic, predictive, and prescriptive analytics, each offering unique insights. Understanding the nature and scope of business analytics which we will study next, helps businesses make informed, data-driven decisions for improved performance.



SELF ASSESSMENT QUESTIONS

4. Predictive analytics anticipates future outcomes by analysing historical data and patterns. (True/False)
5. The role of business analytics is to uncover actionable insights through _____ analysis, transforming data into strategic decisions.
6. Diagnostic analytics goes beyond summarising facts and identifying patterns in the _____.
7. Diagnostic analytics aims to anticipate future trends and guide proactive decision-making. (True/False)

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ACTIVITY

Explore various types of analytics by analysing real-world cases. Discuss how each type affects decisions and solves business challenges effectively.

1.4 NATURE AND SCOPE OF BUSINESS ANALYTICS

Business analytics is like a smart tool for businesses because it uses data (like numbers and information) to help companies understand things better and make smart choices. It looks at patterns in data, predicts what might happen next and uses techniques to get important information from all that data.

Imagine you have a big box of different candies, and you want to know which one everyone likes the most. Business analytics is like sorting through the candies, figuring out patterns and predicting which candy will be the favourite. It is not just about looking at the past (like saying, “Yesterday, everyone liked chocolate”). It is also about guessing what might happen in the future (like saying, “Maybe tomorrow, more people will like gummy bears”).

The nature and scope of business analytics encompass a wide range of activities that use data to drive decision-making across diverse functional areas within an organisation. It is a dynamic field that continues to evolve with advancements in technology and the availability of increasingly large and complex datasets.

The following points show the nature of business analytics:

- ❑ **Data-driven decision-making:** Business analytics relies on data to support decision-making processes. It transforms raw data into actionable insights that facilitate strategic and operational decisions.
- ❑ **Statistical analysis and modelling:** Business analytics employs statistical methods and models to analyse past data, identify trends and make predictions about future outcomes. This includes regression analysis, clustering and machine learning algorithms.
- ❑ **Technological integration:** Business analytics leverages advanced technologies, including data visualisation tools, machine learning algorithms and Big Data technologies, to process and analyse large volumes of data efficiently.
- ❑ **Cross-functional application:** It is not limited to a specific business function. Business analytics can be applied across various departments, such as finance, marketing, operations and human resources, to enhance overall organisational performance.
- ❑ **Continuous improvement:** Business analytics is an iterative process. Organisations continually refine their analytical

models and strategies based on new data and changing business conditions.

The scope of business analytics are as follows:

- ❑ **Customer experience:** Ensuring a positive customer experience is vital for efficient business operations. Business analytics enables companies to comprehend the types of customers frequenting their establishments, understand their purchasing habits and adapt services to satisfy and build loyalty. This customisation is possible through insights gained from business analytics.
- ❑ **Inventory management:** Stream lining supply chains and reducing costs by understanding order frequency, demand for specific products and optimising supply chain operations. Business analytics allows organisations to scale their offerings sustainably, making strategic planning more effective.
- ❑ **Sales and marketing:** Designing targeted campaigns and identifying cross-sell and up-sell opportunities by analysing customer responses to product offers and marketing initiatives. Analysing demographics, average income and purchasing reasons helps predict patterns in consumer behaviour, allowing businesses to focus product messages and launch timing according to customer needs.
- ❑ **Hiring and recruitment:** Seeking collaboration with HR officials skilled in business intelligence and data analytics. This ensures that HR specialists can use data to bring on board skilled and qualified workers, contributing to the company's success. This approach also reduces hiring and training expenses.
- ❑ **Finance:** Leveraging Big Data and business analytics enables businesses to manage finances more skillfully. Insights into marketing spending and a comprehensive understanding of all financial transactions empower businesses to make informed decisions, effectively managing their resources. According to McKinsey & Company, a globally integrated analytics strategy for marketing spending could potentially unlock up to US \$ 200 billion.

As organisations start exploring business analytics, it is crucial to navigate the ethical considerations surrounding the collection, processing, and use of data. The ethical considerations in business analytics are as follows:

- ❑ **Privacy concerns:** Maintaining the privacy of individuals is paramount. Businesses must adhere to data protection regulations, obtain informed consent and implement robust privacy policies to ensure the responsible handling of personal information.
- ❑ **Transparency:** Building trust through transparency in data practices. Organisations should communicate openly about their

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data collection and usage policies, providing stakeholders with a clear understanding of how their data is being utilised.

- ❑ **Bias and fairness:** Acknowledging and addressing biases in data and algorithms is crucial. Organisations must actively work to identify and rectify biases to ensure fair and equitable outcomes in decision-making processes, avoiding unintended consequences.
- ❑ **Security measures:** Safeguarding data from unauthorised access and cyber threats is non-negotiable. Implementing stringent security measures, including encryption and access controls, protects sensitive information and mitigates the risk of data breaches.
- ❑ **Social impact:** Considering the broader social impact of their organisation analytics initiatives. This involves assessing how data-driven decisions may influence society and actively working to minimise any negative consequences. Responsible and ethical analytics practices contribute to the overall wellbeing of communities and societies at large.

Next, we will learn about the stages of a business analytics project.



SELF ASSESSMENT QUESTIONS

8. One application of business analytics is in designing targeted campaigns and identifying cross-sell and up-sell opportunities in _____.
9. Business analytics is confined to specific business functions and cannot be applied across different departments. (True/False)
10. Business analytics relies on data to support decision-making processes and continuously refines analytical models based on new data. (True/False)



ACTIVITY

Explore business analytics applications in pharmaceuticals, inventory, sales, hiring and finance. Discuss real-world impacts on decision-making and efficiency, fostering understanding of analytics in diverse industries.

1.5 STAGES OF A BUSINESS ANALYTICS PROJECT

The stages of a business analytics project are like different steps you take when you want to use data to understand things and make smart choices in a company. It is a way to tackle tricky business problems by using special methods with data and getting valuable insights that help in making better decisions. These stages guide the process,

making sure everything is organised and leading to useful outcomes for the organisation. Figure 1.4 shows the strategic building blocks of business analytics:

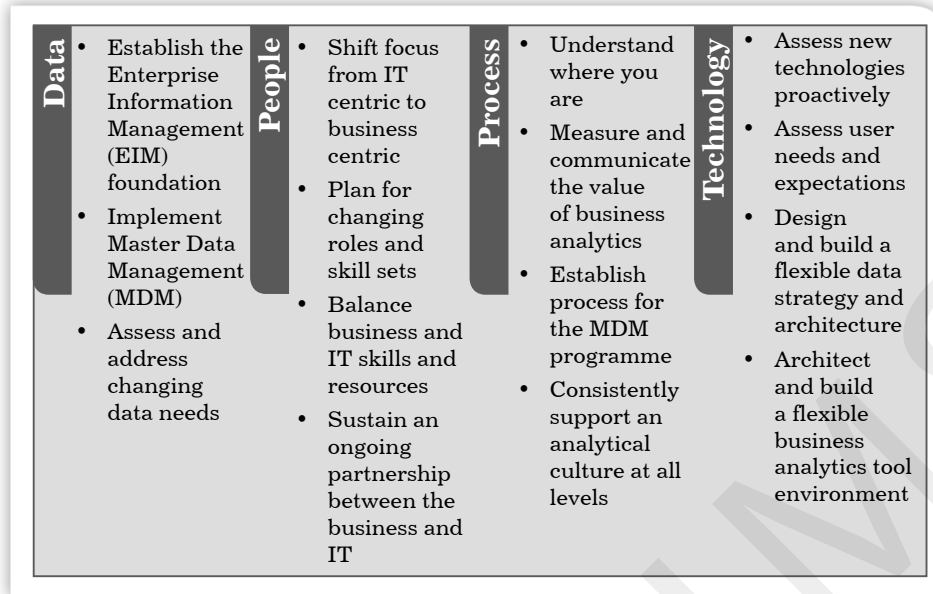


Figure 1.4: Strategic Building Blocks of Business Analytics

The stages of a business analytics project are as follows:

1. **Identifying the business problem:** Every business faces challenges, and this stage is about recognising and defining those challenges. For instance, a restaurant might identify a drop in customer satisfaction and want to figure out why.
2. **Collecting data:** To understand a problem, data is gathered. This could include customer feedback, sales numbers, employee schedules—basically, any information that might offer insights into the situation.
3. **Analysing data:** Now, it is time to dig into the data. This involves using tools and techniques to study the information thoroughly. Imagine it like detectives examining evidence to understand the details of a case.
4. **Predicting outcomes:** With a clear understanding of the data, the next step is predicting what might happen in the future. Using past patterns, they can estimate how things will unfold. If customers typically order more during the holidays, they might predict a similar trend next year.
5. **Finding the best solution:** Now that they have predictions, it is about brainstorming solutions. For our restaurant example, they might consider improving service, changing the menu or running promotions—exploring different ways to address the drop in customer satisfaction.

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6. **Making a decision:** After considering various solutions, a decision is made. They choose the strategy they believe will be most effective in solving the problem. It is like picking the best course of action based on the insights gained from data analysis.

1.5.1 IDENTIFYING INFORMATION GAPS

Identifying information gaps in business analytics is about pinpointing areas where there is a shortage or lack of necessary data, which is crucial for making well-informed decisions. For business analytics, having thorough and accurate data is like having a superpower—it makes the analysis and decision-making process super effective.

When there are information gaps, it suggests that some parts of the business or its processes might not be completely clear or are not well represented in the data that is available. The process of spotting these information gaps involves carefully looking at the data sources you already have, understanding exactly what you need for your analysis and figuring out where there are gaps or things missing. This process is super important because the success of business analytics really depends on having top-notch, complete data to analyse. It is like having all the puzzle pieces to see the full picture of what is happening in the business.

For example, imagine a retail store using data to figure out what to stock. Identifying information gaps would be like noticing they only get monthly sales data, not daily. This gap means they might miss quick changes in what customers want each day. So, they decide to fix it by getting daily sales data. Now, they can adjust their stock every day, making sure they always have what customers are looking for, and the shop runs more smoothly. It is like making sure you have all the right ingredients for your favourite recipe!



SELF ASSESSMENT QUESTIONS

11. The stages of a business analytics project involve using data to tackle business problems and gain valuable insights for better decision-making. (True/False)
12. The evolution of business analytics has seen a shift driven by automation and the proliferation of _____.
13. Identifying information gaps in business analytics is about pinpointing areas where there is a shortage or _____, which is crucial for making well-informed decisions.



ACTIVITY

Collaborate in a simulated business analytics project. Identify a problem, collect data, analyse, predict outcomes and decide solutions collaboratively.

1.6 SUMMARY

- ❑ Business analytics, using diverse tools, relies on data for decisions, necessitating adaptation to technological advances and increasing data complexity in organisations.
- ❑ Business analytics offers benefits like informed decision-making and competitive advantage, but faces challenges such as data quality, privacy concerns and organisational resistance to change.
- ❑ Business analytics spans from describing historical data for informed decisions to diagnosing past events, predicting future trends and prescribing actions for optimised outcomes in strategic decision-making.
- ❑ Business analytics scope covers customer experience, inventory management, sales, marketing, hiring, recruitment and finance.
- ❑ Stages of a business analytics project involve identifying challenges, gathering data, analysing with tools, forecasting trends, brainstorming solutions and making decisions based on data insights.
- ❑ Identifying information gaps in business analytics requires recognising areas lacking essential data, examining existing sources to understand and fill those gaps meticulously.



KEY WORDS

- ❑ **Automation:** It employs technology for tasks without human intervention, hence streamlining processes.
- ❑ **Big data:** It refers to vast and complex datasets that exceed the capabilities of traditional data processing methods.
- ❑ **Cloud computing:** It is a technology that enables the storage, access and processing of data and applications over the Internet.
- ❑ **Data sourcing:** It refers to the process of collecting and acquiring data from different internal and external sources to support analysis.
- ❑ **Metrics:** They are quantifiable measures that are used to assess, track and evaluate performance in various aspects of business.
- ❑ **Simple linear regression:** It analyses the relationship between two variables, predicting the dependent variable based on the independent variable using a linear equation.

N O T E S

1.7 MULTIPLE CHOICE QUESTIONS

1. Which analytics type delves into predicting potential outcomes based on historical data?
 - a. Descriptive analytics
 - b. Diagnostic analytics
 - c. Predictive analytics
 - d. Prescriptive analytics
2. In the context of business analytics, what does identifying information gaps involve?
 - a. Predicting future trends
 - b. Recognising areas with insufficient data
 - c. Analysing past data patterns
 - d. Designing targeted campaigns
3. Which stage of a business analytics project involves brainstorming solutions?
 - a. Find the best solution
 - b. Analyse data
 - c. Predict outcomes
 - d. Business problem
4. In business analytics, what does “continuous improvement” mean?
 - a. A one-time analytical process
 - b. Exclusive to the finance department
 - c. Stagnation in analytical strategies
 - d. Iterative refinement based on new data
5. Which attribute corresponds to descriptive analytics?
 - a. Envisaging prospective trends
 - b. Anticipating contingent outcomes
 - c. Discerning latent growth opportunities
 - d. Reflecting on historical data
6. What does diagnostic analytics aim to uncover?
 - a. Future predictions
 - b. Opportunities for growth
 - c. Causes behind past events
 - d. Mitigation of potential risks

7. What does descriptive analytics focus on?
 - a. Anticipating trends
 - b. Understanding past events and trends
 - c. Forecasting future outcomes
 - d. Recommending optimal actions
8. What does HR analytics assist businesses in reducing?
 - a. Employee turnover
 - b. Customer complaints
 - c. Marketing expenses
 - d. Product development costs
9. What does the process of identifying information gaps in business analytics primarily involve?
 - a. Creating new data
 - b. Ignoring existing data
 - c. Carefully examining existing data
 - d. Focusing solely on complete data
10. What is the initial stage in a business analytics project?
 - a. Predicting outcomes
 - b. Finding the best solution
 - c. Analysing data
 - d. Identifying the business problem

1.8 DESCRIPTIVE QUESTIONS

1. How does business analytics empower organisations, influencing strategic decisions, optimising resources and enhancing competitiveness in dynamic markets?
2. What are the benefits and challenges of business analytics?
3. What sets prescriptive analytics apart from predictive analytics, and how does it elevate decision-making processes?
4. Describe the essential stages of a thorough business analytics project, highlighting challenges and critical decision points from inception to deployment.
5. What is the nature of business analytics, and where can it be applied?

N O T E S

1.9

HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. A retail company wants to enhance customer satisfaction. What aspect of business analytics would be most beneficial for understanding customer preferences?
 - a. Predictive analytics
 - b. Descriptive analytics
 - c. Diagnostic analytics
 - d. Prescriptive analytics
2. Within the integration of business analytics, which subtle challenge underscores the need for a sophisticated approach to ethical considerations and responsible data usage, ensuring clarity and transparency?
 - a. Harmonising diverse data sources for unified analytics
 - b. Implementing advanced encryption protocols for secure data privacy
 - c. Balancing transparency and privacy in interpretable algorithmic decision-making
 - d. Optimising data processing speed through interpretable parallel computing solutions
3. Suppose you are a business analyst tasked with explaining the key features of diagnostic analytics to a non-technical audience. Use relatable examples to elucidate how diagnostic analytics goes beyond summarising facts.
4. Suppose you are a consultant helping a local cafe enhance its operations. Describe how business analytics can be applied across different stages, starting from identifying the business problem to making decisions, to optimise the cafe's overall performance.

1.10

ANSWERS AND HINTS**ANSWERS FOR SELF ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Understanding Business Analytics	1.	False
	2.	a. Predictive analytics and pattern recognition
	3.	True
Types of Business Analytics	4.	True
	5.	data-driven
	6.	historical data

N O T E S

Topic	Q. No.	Answer
Nature and Scope of Business Analytics	7.	False
	8.	privacy
	9.	False
	10.	True
Stages of a Business Analytics Project	11.	True
	12.	integration
	13.	lack of necessary data

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	c. Predictive analytics
2.	b. Recognising areas with insufficient data
3.	a. Find the best solution
4.	d. Iterative refinement based on new data
5.	d. Reflecting on historical data
6.	c. Causes behind past events
7.	b. Understanding past events and trends
8.	a. Employee turnover
9.	c. Carefully examining existing data
10.	d. Identifying the business problem

HINTS FOR DESCRIPTIVE QUESTIONS

1. Business analytics empowers organisations through strategic decision-making, resource optimisation and improved competitiveness in dynamic markets. Refer to Section 1.2
2. Business analytics yields benefits and encounters challenges crucial to organisational success. Refer to Section 1.2
3. Prescriptive analytics stands out by not only predicting future outcomes but also recommending optimal actions, elevating decision-making processes with actionable insights. Refer to Section 1.3
4. Business analytics projects unfold in a sequential process with distinct stages. Refer to Section 1.5
5. Business analytics, with its distinctive nature and expansive scope, is a transformative force across organisational landscapes. Refer to Section 1.4

N O T E S

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	b. Descriptive analytics
2.	c. Balancing transparency and privacy in interpretable algorithmic decision-making
3.	Diagnostic analytics, like a detective, uncovers the reasons behind events. For example, in retail, if sales drop, diagnostic analytics helps identify specific factors contributing to the decline, such as changes in consumer behaviour or market trends. This goes beyond summarising sales data and provides actionable insights for business improvement.
4.	Beginning with identifying the business problem, analytics can unveil challenges like low customer satisfaction. Collecting data involves gathering sales, customer and feedback information. Analysing data reveals patterns, aiding in targeted improvements. Predicting outcomes anticipates future trends, such as popular menu items. Finding the best solution includes strategies like improving service. Making a decision finalises the chosen approach, ensuring an enhanced cafe experience based on comprehensive analytics.

1.11 SUGGESTED READINGS & REFERENCES**SUGGESTED READINGS**

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CASE STUDY

N O T E S

BOOSTING SALES AND CUSTOMER RETENTION WITH DATA ANALYTICS

BACKGROUND

TechGears, a mid-sized electronics retailer, faced stagnating sales and declining customer retention. Despite a wide product range, the company struggled to adapt to customer preferences and market trends. To address this, TechGears turned to business analytics to gain deeper insights into its operations, focusing on descriptive analytics, diagnostic analytics, predictive analytics and prescriptive analytics.

CHALLENGE

The following were the data challenges faced by the company:

- ❑ The company had vast amounts of historical sales data but lacked a coherent understanding of what had happened in previous quarters.
- ❑ While TechGears knew there had been drops in customer satisfaction and sales in certain regions, they struggled to find the reason behind it. Key issues such as product returns, customer complaints and low repeat purchases needed investigation.
- ❑ Management wanted to forecast future demand, identify trends and predict customer behaviour (e.g., what products customers might buy next, or how likely a customer was to stop purchasing from TechGears).
- ❑ Finally, TechGears needed concrete recommendations on how to address customer dissatisfaction and improve retention. They also sought advice on optimal marketing strategies and inventory management to boost sales.

SOLUTION

- ❑ **Descriptive analytics:** The analytics team began by using descriptive analytics to examine historical data, focusing on key metrics such as sales by region, product category and customer segment over the past two years. Data visualisation tools (e.g., dashboards) were employed to summarise customer purchases, revenue trends and product popularity.

Outcome: The team discovered that sales had been consistently declining in certain regions and that a specific group of products had not been selling well compared to others. It also showed customer engagement had dropped sharply in the last two quarters.



Case Objective

The purpose of this case study is to understand how TechGears utilised business analytics techniques to improve its product quality, customer retention and inventory management.

CASE STUDY

- ❑ **Diagnostic analytics:** Using diagnostic analytics, the team probed deeper to understand the reasons behind the sales decline. They analysed customer complaints, product return rates and the timing of the sales drop. This analysis showed that a surge in product returns coincided with negative feedback related to the quality of certain electronics.

Outcome: The analysis revealed that poor-quality products, along with inadequate customer support, were the primary drivers behind declining sales and customer dissatisfaction. This insight provided TechGears with actionable knowledge about which products and processes needed improvement.

- ❑ **Predictive analytics:** Moving forward, the company applied predictive analytics to forecast future trends. By analysing historical purchasing behaviour and customer demographics, the team developed models to predict future demand for certain product categories and customer attrition rates.

Outcome: Predictive models forecasted that if no intervention occurred, customer attrition would increase by 10 per cent over the next six months. Additionally, the analysis showed that demand for certain high-end electronics would rise during the holiday season, providing TechGears with insights into stocking decisions.

- ❑ **Prescriptive analytics:** Finally, prescriptive analytics was used to recommend actionable steps based on the findings. The system suggested offering personalised promotions to high-value customers to reduce attrition, optimising stock levels of high-demand products and improving the quality control process for underperforming items.

Outcome: The team recommended personalised marketing campaigns targeting loyal customers with special discounts on high-end electronics. In addition, they proposed stricter quality checks on the problematic product categories and a retraining program for customer service agents to enhance the customer experience.

RESULTS

By applying data analytics in the business problem, TechGears:

- ❑ **Improved customer retention:** Personalised promotions led to a 12 per cent increase in customer retention over three months.
- ❑ **Optimised inventory:** Stocking decisions based on predictive insights reduced stockouts during the holiday season, leading to a 20 per cent increase in sales of high-demand products.

CASE STUDY

N O T E S

- ❑ **Enhanced product quality:** Quality control measures helped reduce product returns by 25 per cent, improving overall customer satisfaction.

CONCLUSION

TechGears' experience highlights the transformative power of data analytics in addressing sales and customer retention challenges. By employing descriptive, diagnostic, predictive and prescriptive analytics, TechGears turned their stagnant business into a more efficient and customer-centric operation. The insights gained allowed the company to make informed decisions, optimise inventory, improve customer satisfaction and design effective marketing strategies. As more companies embrace data-driven decision-making, they can unlock significant improvements in performance and competitive advantage.



QUESTIONS

1. How did TechGears use descriptive analytics to identify sales trends and product performance?

(Hint: Descriptive analytics helped TechGears visualise historical sales data by region, product and customer segments, revealing patterns such as declining sales in certain regions and underperforming products.)

2. What prescriptive analytics strategies were recommended to address customer retention and sales growth?

(Hint: Personalised promotions, improved quality control and optimised inventory based on prescriptive analytics led to increased customer retention and sales, alongside better customer experiences.)

2

C H A P T E R

DATA PREPROCESSING AND VISUALISATION

CONTENTS

2.1	Introduction
2.2	Using Data in Analytics
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	Activity
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2.4	Data Cleansing: Common Issues with Data
2.4.1	Missing Data
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2.12	Suggested Readings & References

N O T E S

**Case Objective**

The objective of this caselet is to improve the layout of a retail store using data-based insights and visualisation to increase foot traffic, sales and customer satisfaction.

INTRODUCTORY CASELET**IMPROVING RETAIL STORE DESIGN USING DATA VISUALISATION**

Giba, a well-known clothing store experienced lower sales in its retail store, leading to an internal review that indicated the store layout could be responsible. Without the know-how, they looked for methods based on data. They worked with a data analytics company to collect various data, including point-of-sale records showing customer purchases, heatmaps generated by cameras showing traffic levels, and customer surveys measuring satisfaction.

By making use of analytics dashboards and heatmaps, the retailer obtained the highlighted busy 'hot' regions and identified neglected 'cold' spots, informing decisions on product placement and staffing strategies and examining customer mobility patterns identified congestion points and navigation challenges, improving the flow of the store. Moreover, these visual representations deciphered customer actions, revealing their browsing patterns and trends in decision-making, which in turn informed personalised merchandising and marketing approaches.

The results were significant which shows a 15 per cent increase in the number of people coming in, a 10 per cent rise in the rates of people making a purchase, improved customer satisfaction ratings, and reduced operational costs. The significant impact of data visualisation on understanding customer behaviour and improving store layouts is highlighted by this achievement, leading to concrete business improvements.



LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Describe the significance of data in analytics
- Evaluate how different data types are utilised in analytics
- Identify the common issues in data and how to rectify them
- Apply imputation methods to real-world datasets
- Employ visualisations to effectively communicate findings from data

2.1 INTRODUCTION

In the previous chapter, you have learned about business analytics, types of business analytics, nature and scope of business analytics and the stages of a business analytics project.

Data preprocessing and visualisation are some of the most important steps in business analytics. Preprocessing involves cleaning, transforming, and organising raw data ensuring higher quality and consistency using techniques like imputation which replaces missing values with estimates and helps data for accurate insights. Data visualisation, on the other hand, is the graphical representation of data through charts, graphs and dashboards. It allows businesses to quickly identify trends, patterns, and anomalies, which aids in informed decision-making. Together, preprocessing and visualisation help businesses derive actionable insights from vast amounts of data, improving efficiency and strategy development.

In this chapter, you will be introduced to the significance of data in analytics. Then you will learn about types of data, common issues with data, and imputation methods. Further, you will learn about data visualisation by using Excel.

2.2 USING DATA IN ANALYTICS

Data is collected to gain insights into business problems. Both profit and non-profit organisations heavily depend on reliable data to achieve their organisational goals, such as strategic planning, performance assessment, improving operations, and benchmarking against industry standards or competitors.

Some examples of how data is used in business include:

- ❑ **Annual reports:** Data is systematically presented alongside visualisations such as charts and graphs to provide stakeholders with a clear representation of the company's financial performance, including profitability, revenue and market share.

N O T E S

- ❑ **Audits and financial oversight:** Accountants audit financial records to ensure that balance sheet figures are precise and reflect the actual data, maintaining accuracy in reporting and regulatory compliance.
- ❑ **Financial analysis:** Financial analysts compile and analyse datasets to estimate a company's performance based on key indicators such as profitability, revenue growth, Return on Investment (ROI), Economic Value Added (EVA) and shareholder value.
- ❑ **Business analysis:** Business analysts process data to develop clear visualisations to convey complicated data in a way that is easy to understand. Business analysts are required to develop and maintain dashboards using data for tracking key business metrics.
- ❑ **Artificial Intelligence (AI):** AI-engineers are required to analyse and interpret large amounts of data to train AI models and to improve their accuracy, efficiency and scalability.

To use data for business, we need to first collect the data from various sources. Next, we will study different data collection sources and their considerations while collecting data.

2.2.1 DATA COLLECTION METHODS

Data collection involves obtaining and evaluating information from various sources to address research challenges, provide answers, assess results and forecast trends. It plays a crucial role in research, analysis, and decision-making across sectors such as social sciences, business and healthcare. Maintaining high accuracy in data collection is essential for making business decisions and ensuring research validity. Analysts or researchers must determine the types of data, their sources, and appropriate methods before starting the data collection process. Data collection for research or analysis involves two main methods:

- ❑ **Primary data collection:** This method refers to gathering original data directly from sources, allowing it to be customised to meet specific objectives. The key primary data collection sources include:
 - ◆ **Surveys and questionnaires:** Surveys and questionnaires are used to collect data through face-to-face interviews, phone calls, mail or online platforms.
 - ◆ **Observations:** Observation methods involve observing and recording behaviours or events in their natural setting without interfering.
 - ◆ **Experiments:** Experiment methods involve researchers who manipulate variables to observe their effects, providing insights into cause-and-effect relationships.

- ❑ **Secondary data collection:** This method involves using existing data that was originally collected for other purposes. The key secondary data collection sources include:
 - ◆ **Published sources:** Published sources include books, academic journals, magazines and newspapers.
 - ◆ **Online databases:** Online platforms that provide access to a variety of secondary data, such as research articles, statistics, and economic data.
 - ◆ **Publicly available data:** Information shared on public websites, platforms, or social media by individuals or organisations.
 - ◆ **Previous research studies:** Data from past research, which can provide valuable insights or a foundation for new research.

The collection of data involves numerous ethical considerations that are meant to achieve fairness, accuracy, and regard for the rights and privacy of individuals. They include the following considerations:

- ❑ **Privacy and consent:** The collection of information should respect the privacy of people, and explicit consent should be sought if it is appropriately needed. Only such information as absolutely required should be gathered, and participants should be informed of how their information will be used, stored and shared. That data which happens to be sensitive requires high levels of consent and security measures.
- ❑ **Transparency:** Organisations need to show their processes of data collection and cleaning clearly. Contributors need to know what information is being gathered and why. Transparency is also important in explaining how data will be processed so that the findings, which may affect a decision or might be a perception of the public, do not get misinterpreted.
- ❑ **Bias and fairness:** Depending upon the process used to collect the data, it may inherit biases in society or throughout history. For instance, a biased sampling that could lead to unfair or discriminatory outcomes might have been done if a certain group of people was underrepresented. Cleaning data for the elimination of bias is very challenging but necessary to make results fair and equitable.
- ❑ **Accuracy and integrity:** Ethical data cleaning seeks accuracy without alteration to ensure the cleanliness of data. Over-cleanliness might eliminate the very valid data points one aims to keep, thereby producing biased results. Data integrity and an audit trail are, therefore, extremely important to make sure that all the data remain a good representation of reality.

N O T E S

- ❑ **Security and confidentiality:** The data collected, especially containing personal identifiers, should be kept away from unauthorised access and breaches. More often, anonymised or pseudonymised data is used to ensure the security of a person's identity.
- ❑ **Purpose limitation:** Data must be collected for specific, legitimate purposes. Use of data for purposes other than those for which it is collected and particularly for activities that may cause harm to individuals or groups is unethical. Regular audits of data use help to support this principle.
- ❑ **Accountability:** Organisations must be accountable to themselves about their data practices. It is part of the ethical responsibility that there are avenues provided for participants to seek information about their data, correct errors in the data, or revoke consent.

Having explored the various methods of data collection, it is now essential to understand the different types of data that are commonly utilised in analytics which we will learn next.



SELF ASSESSMENT QUESTIONS

1. What is the primary purpose of a financial analyst in collecting and analysing diverse datasets within a business?
 - a. Assessing customer demographics
 - b. Evaluating operational compliance standards
 - c. Ascertaining a business's contribution to shareholders
 - d. Forecasting population dynamics
2. In secondary data collection, what source would be considered a 'published' source?
 - a. Government records
 - b. Online databases
 - c. Books and academic journals
 - d. Focus groups
3. Data imputation involves manipulating variables to observe their impact on the outcome. (True/False)



ACTIVITY

Explore the impact of data on decision-making in various industries, analysing how it shapes strategies and operations. Use examples to illustrate the practical influence of data-driven approaches.

2.3 TYPES OF DATA

Data comes in various forms, each with its own specific characteristics and purposes. When we refer to types of data, we are essentially discussing the different categories or classifications that information can belong to. Generally, data can be classified into several types based on its nature, source or intended use. Figure 2.1 illustrates the common types of data used in the business industry:

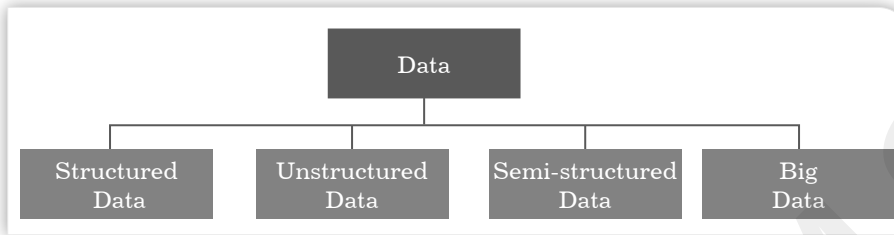


Figure 2.1: Common Types of Data Used in Business Analytics

The descriptions of common types of data used in the business industry are as follows:

- ❑ **Structured data:** This is a type of data that is prepared and easily handled using traditional data management tools such as spreadsheets, databases or tables. Structured data normally consists of numbers, percentages and other numerical values. Because of its organised nature, structured data is comparatively easy to analyse using statistical methods such as regression analysis or correlation analysis. Figure 2.2 presents car details sourced from the CarDekho website, offering a visualisation of structured data:

	A	B	C	D	E	F	G	H	I
	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	
1									
2	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner	
3	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner	
4	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner	
5	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner	
6	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner	
7	Maruti Alto LX BSIII	2007	140000	125000	Petrol	Individual	Manual	First Owner	
8	Hyundai Xcent 1.2 Kappa S	2016	550000	25000	Petrol	Individual	Manual	First Owner	
9	Tata Indigo Grand Petrol	2014	240000	60000	Petrol	Individual	Manual	Second Owner	
10	Hyundai Creta 1.6 VTVT S	2015	850000	25000	Petrol	Individual	Manual	First Owner	
11	Maruti Celerio Green VXI	2017	365000	78000	CNG	Individual	Manual	First Owner	
12	Chevrolet Sail 1.2 Base	2015	260000	35000	Petrol	Individual	Manual	First Owner	
13	Tata Indigo Grand Petrol	2014	250000	100000	Petrol	Individual	Manual	First Owner	
14	Toyota Corolla Altis 1.8 VL CVT	2018	1650000	25000	Petrol	Dealer	Automatic	First Owner	
15	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner	
16	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner	
17	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner	
18	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner	
19	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner	
20	Maruti Alto LX BSIII	2007	140000	125000	Petrol	Individual	Manual	First Owner	
21	Hyundai Xcent 1.2 Kappa S	2016	550000	25000	Petrol	Individual	Manual	First Owner	
22	Tata Indigo Grand Petrol	2014	240000	60000	Petrol	Individual	Manual	Second Owner	
23	Hyundai Creta 1.6 VTVT S	2015	850000	25000	Petrol	Individual	Manual	First Owner	
24	Maruti Celerio Green VXI	2017	365000	78000	CNG	Individual	Manual	First Owner	
25	Chevrolet Sail 1.2 Base	2015	260000	35000	Petrol	Individual	Manual	First Owner	

Figure 2.2: Structured Data Visualisation

NOTES

- ❑ **Unstructured data:** This type of data lacks a predefined format like structured data, making it challenging to handle with traditional data management tools. Due to its unstructured nature, it is typically qualitative in nature, meaning it consists of descriptive and narrative information. Analysing unstructured data requires advanced analytics techniques such as Natural Language Processing (NLP) or sentiment analysis. Various sources of unstructured data for analysis are shown in Figure 2.3:

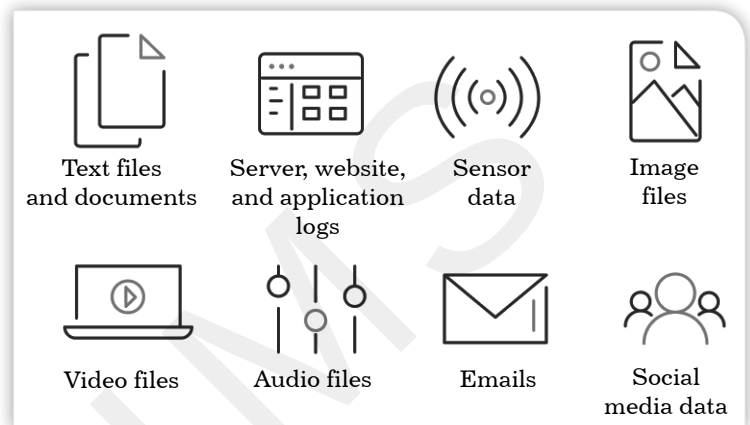


Figure 2.3: Sources of Unstructured Data

- ❑ **Semi-structured data:** Semi-structured data combines the characteristics of both structured and unstructured data. It includes information organised in a partial or flexible schema, making it not fully structured but still easier to analyse than completely unstructured data. Examples of semi-structured data include Extensible Markup Language (XML) and JavaScript Object Notation files (JSON), which are organised to some extent but also contain elements of unstructured data. Analysing semi-structured data typically requires a combination of traditional data management tools and advanced analytics techniques. Typical semi-structured data are shown in Figure 2.4:

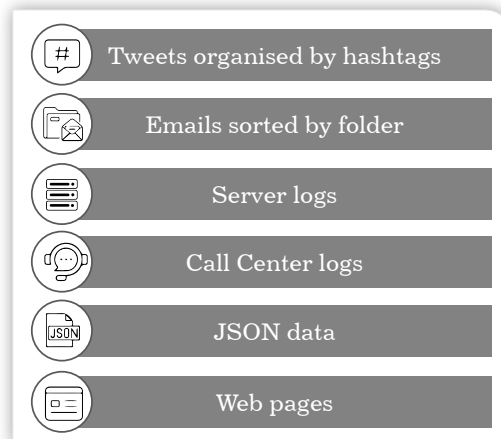


Figure 2.4: Semi-structured Data

- ❑ **Big Data:** Big Data refers to vast and complex data sets that cannot be handled with traditional data management tools. It includes a variety of data types, including structured, unstructured and semi-structured data. The primary challenge of analysing Big Data is its sheer volume, as the amount of data is too large to be analysed manually. Common sources of Big Data are shown in Figure 2.5:

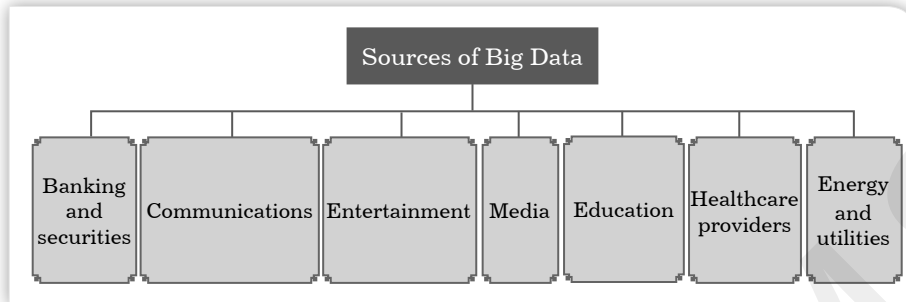


Figure 2.5: Sources of Big Data



NOTE

Data can be both quantitative (numerical) and qualitative (categorical), depending on the context and analysis objective. Recognising this duality allows for a richer and more comprehensive understanding of information.

The following formats can be used to utilise data in analytics:

- ❑ **Structured data:** Structured data, such as numbers and percentages, can be easily processed and analysed using traditional statistical techniques, such as regression or correlation analysis, to identify patterns, trends, or relationships in business performance, financial metrics, or customer data.
- ❑ **Unstructured data:** Unstructured data, such as text from emails, social media posts, or video content, lacks a predefined format, making it challenging to manage with conventional tools. Advanced analytics techniques, such as NLP or sentiment analysis, are used to extract insights from this data, helping businesses understand customer sentiment, brand perception or market trends.
- ❑ **Semi-structured data:** Semi-structured data, such as XML or JSON files, contains both structured and unstructured elements and is utilised in analytics by combining traditional data processing methods (for the structured portions) with advanced techniques (for the unstructured elements). This data type is commonly used in data integration tasks, web data extraction and log file analysis.

N O T E S

- ❑ **Big Data:** Big Data, which is a mix of structured, unstructured and semi-structured data, is used for large-scale analytics using advanced technologies such as Hadoop or Spark. Big Data analytics helps businesses to improve decision-making and forecast trends by analysing huge volumes of diverse data, such as customer behaviour, supply chain dynamics and market shifts.
- ❑ **Real-time data:** Unstructured and semi-structured data often come from sources such as social media feeds, Internet of Things (IoT) sensors, or email communications. This data is used in real-time analytics to monitor ongoing business activities, such as customer engagement, operational performance, or market developments, allowing businesses to respond quickly to changing conditions.
- ❑ **Integrated data:** In many cases, businesses combine structured, semi-structured, and unstructured data for a comprehensive view. For example, structured data from sales records can be combined with unstructured data from customer reviews to better understand product performance, leading to more informed marketing strategies or product development decisions.

It is very crucial to have clean data before analysis because of the following underlying reasons:

- ❑ **Accuracy of results:** Unclear data such as missing values, duplicate values, errors, or inconsistency can lead to inaccurate analysis and therefore uncertain insights and faulty conclusions that may influence decisions.
- ❑ **Efficiency:** Clean data reduces the necessity of manual correction of data during analysis. An error-free, well-organised data table streamlines analysis workflows and saves much time and effort.
- ❑ **Better decisions:** Clean, trusted data results in a more accurate analysis that creates a healthy base for good decision-making. This will lead to better business or research decisions that are based on real insights and not corrupted or biased information.
- ❑ **Simple data integration:** Clean data is much easier to merge and integrate from various sources. If the datasets have standardised formats and structures, fewer integration issues will be faced when they are combined for further analysis.
- ❑ **Compliance and security:** In some industries, business regulatory standards will mandate that data be kept clean and organised. Maintaining data clean has the benefits of reducing risks associated with the occurrence of data privacy violations, security breaches and compliance.

Understanding the various types of data is essential for effective analytics, but it is equally important to ensure that the data used for effective analytics is accurate and reliable. The next section focuses

on data cleansing which is a process for identifying and addressing errors or inconsistencies in datasets to enhance their quality and usability.



SELF ASSESSMENT QUESTIONS

4. Which type of data analysis involves numbers and percentages?
 - a. Structured data
 - b. Semi-structured data
 - c. Unstructured data
 - d. Big Data
5. Are files in the XML or JSON formats considered unstructured data? (True/False)
6. Semi-structured data is commonly used in data integration tasks, web data extraction and log file analysis. (True/False)



ACTIVITY

Examine various data types in business analytics. Utilising real-world examples, explore distinctions between the various types of data.

2.4 DATA CLEANSING: COMMON ISSUES WITH DATA

Data cleansing, also known as data cleaning or data scrubbing, involves identifying and correcting errors or inconsistencies within datasets to improve their accuracy. High-quality and error free data is critical for making informed decisions, conducting precise analysis, and maintaining the reliability of business operations. During the data cleansing process, several common challenges may arise, which require effective solutions to address them. The common issues concerning data are:

- ❑ **Missing values:** These occur when data entries are empty or null. It could be due to oversight, errors in data collection, or systems not capturing information. Strategies to handle missing values include deletion, imputation (filling in missing values based on statistical methods), or using algorithms that can handle missing data.
- ❑ **Duplicate entries:** These arise when the same data is recorded more than once within a dataset. They skew analysis by overrepresenting certain information. Identification, which

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involves comparing entries for similarity and then removing or consolidating duplicate records.

- ❑ **Inconsistent formatting:** It leads to issues when merging datasets or performing calculations. For instance, dates entered in various formats (e.g., MM/DD/YYYY, DD-MM-YYYY) or categorical data with different spellings (e.g., 'USA,' 'U.S.A,' 'United States'). Standardising formats ensures uniformity across the dataset.
- ❑ **Outliers:** These are the data points that significantly differ from other observations in a dataset. They can distort statistical analyses and model predictions. Identification involves statistical methods such as Z-score, Interquartile Range (IQR), or visualisation techniques like box plots. Deciding whether to remove, adjust, or keep outliers depends on the context of the analysis.
- ❑ **Inaccurate data:** Data entry errors, typos, or incorrect information can affect the integrity of a dataset. Verifying data against reliable sources or cross-referencing with other datasets helps identify and rectify inaccuracies.
- ❑ **Data imbalance:** The data might be imbalanced in classification problems, meaning one class significantly outnumbers the others. Balancing techniques such as oversampling, under sampling, or using specialised algorithms help mitigate this issue.
- ❑ **Data irrelevance:** It happens when some data do not make a major impact on the analysis. Identifying and removing irrelevant data can streamline the dataset and improve analysis accuracy and efficiency.

Having learned about the common issues of data during the cleansing process, we next focus on addressing missing data and the presence of outliers.

2.4.1 MISSING DATA

The term missing data in business analytics refers to the absence or incompleteness of data points in a dataset. Handling missing data is crucial, as it can significantly impact the accuracy and reliability of analytical models and interpretations. Understanding missing data in business analytics is essential as it significantly influences the accuracy of insights and decision-making.

The impact of missing data during analysis may result in:

- ❑ **Inaccurate analysis:** Missing data can lead to biased or incomplete analysis, which will affect the reliability of the outcome and conclusion drawn from the data.
- ❑ **Poor decision-making:** Decisions based on incomplete or inaccurate data may lead to a worse business strategy or missed opportunities.

- ❑ **Reduced model performance:** Predictive models and machine learning algorithms may underperform when trained on incomplete datasets, leading to less accurate predictions.
- ❑ **Loss of valuable insights:** Missing data can disguise important trends or patterns, preventing businesses from fully understanding customer behaviour, market dynamics or operational efficiency.
- ❑ **Decreased customer satisfaction:** Poor data quality may lead to customer dissatisfaction, especially in sectors like retail or service industries where customer preferences and behaviour are critical for personalised experiences.
- ❑ **Regulatory and compliance risks:** In industries with strict compliance requirements (e.g., finance, healthcare), missing or incomplete data can lead to legal risks, fines or regulatory penalties.

With an understanding of the impact of missing data, let us next explore the identification of outliers in datasets.

2.4.2 IDENTIFICATION OF OUTLIERS

Outliers are data points that significantly deviate from the majority of observations within a dataset. Detecting outliers is essential in business analytics, as they can distort statistical analyses and negatively impact model performance, potentially leading to incorrect conclusions. Ms Excel provides various methods for effectively identifying outliers.

Understanding outliers in business analytics is critical because these data points, which differ substantially from the norm, can skew analysis and decision-making. By identifying and managing outliers, businesses can gain more accurate insights.

There are two main types of outliers:

- ❑ **Univariate outliers:** These occur when a data point in a single variable stands out significantly from the others.
- ❑ **Multivariate outliers:** These exist when observations are extreme across multiple dimensions compared to other data points.

The impacts of outliers on data are as follows:

- ❑ **Distortion of statistical measures:** Outliers can twist averages (means) and variances, leading to inaccurate representations of data.
- ❑ **Misleading conclusions:** The presence of outliers can result in incorrect interpretations of trends and patterns in the data.
- ❑ **Reduced model effectiveness:** Outliers can negatively impact the performance of predictive models, making them less reliable for decision-making.

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- ❑ **Increased error rates:** Outliers may lead to higher error rates in statistical analyses, affecting overall data integrity.
- ❑ **Misleading visual representations:** Outliers can create misleading visual representations, such as charts and graphs, which may misinform stakeholders.

After identifying outliers, it is equally crucial to treat missing values effectively to ensure data integrity and analytical reliability which we will focus on next.

2.4.3 TREATING MISSING VALUES

Missing values in data are one of the common challenges encountered across various fields that rely on datasets. They can result from several factors, including human error during data collection and limitations in data gathering methods. Fortunately, there are strategies to manage missing data and mitigate its impact in an analysis. The two primary approaches are as follows:

- ❑ **Deletion:** This approach involves removing rows or columns that contain missing values. While this method is straightforward, it can create significant issues if a large portion of the data is missing. Excessive deletion may compromise the reliability of the conclusions.
- ❑ **Imputation:** This approach replaces missing values with estimates. There are several imputation techniques available, each with its advantages and disadvantages:
 - ◆ **Mean/median/mode imputation:** In this approach, the missing entries are replaced with the average (mean), middle value (median), or most frequently occurring value (mode) of the relevant column. Though quick and simple, this approach can introduce bias if the missing data is not randomly distributed.
 - ◆ **K-Nearest Neighbors (KNN) imputation:** This approach identifies the closest data points (neighbours) based on available features and uses their values to estimate the missing value. KNN is particularly effective when there is a large dataset where the missing values are dispersed.
 - ◆ **Model-based imputation:** This approach involves creating a statistical model to predict missing values based on other features in the dataset. While powerful, it requires a higher level of expertise and can be computationally intensive.

Perform the following steps to delete rows with missing data in Excel:

1. Open an Excel spreadsheet with the business data that contains missing values. Dataset can be downloaded from <https://www.kaggle.com/datasets/sticktogethertm/business-analysis-junior>.

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The data in Excel can be seen in Figure 2.6:

Order number	Client ID	Product code	Date of delivery	Delivery amount
97058	7121	494843	2019-01-24	9,565.00
2968	7167	111937	2019-01-29	18,907.00
2968	7167	218889	2019-01-29	54,132.00
45863	7136	495715	2019-02-07	28,023.00
45863	7136	495716	2019-02-07	59,120.00
45863	7136	495717	2019-02-07	45,829.00
45863	7136	495719	2019-02-07	42,980.00
45863	7136	495720	2019-02-07	4,028.00
45863	7136	495713	2019-02-07	38,903.00
45863	7136	495722	2019-02-07	33,661.00
45863	7136	495718	2019-02-07	1,937.00
45863	7136	495714	2019-02-07	41,604.00
45863	7136	495721	2019-02-07	30,900.00
40155	7131	497253	2019-02-18	65,313.00
40155	7131	497033	2019-02-18	30,649.00
40155	7131	497028	2019-02-18	57,336.00
40155	7131	497035	2019-02-18	20,574.00
40155	7131	497036	2019-02-18	12,677.00
40155	7131	497030	2019-02-18	24,098.00
40155	7131	497037	2019-02-18	5,790.00
40155	7131	497034	2019-02-18	35,737.00
40155	7131	497032	2019-02-18	43,414.00
40155	7131	497031	2019-02-18	11,964.00

Figure 2.6: Imported Data in Excel

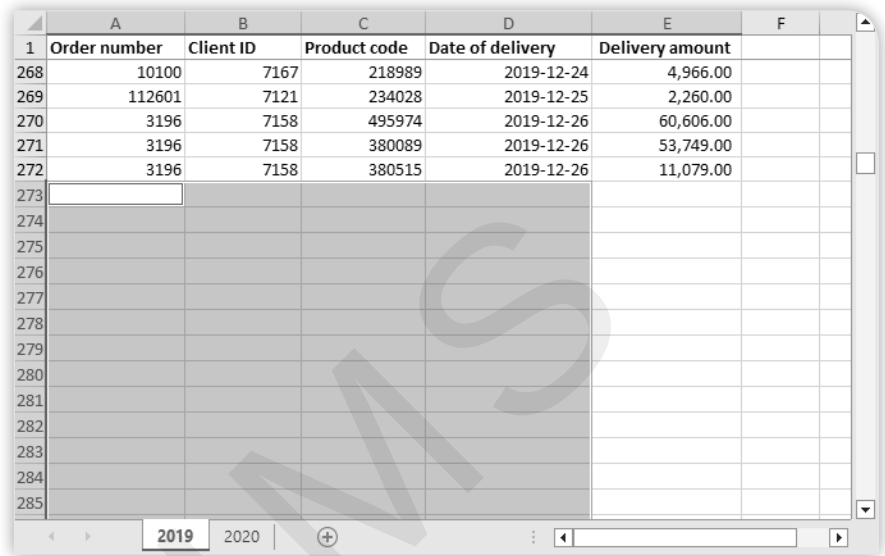
- Click and drag to highlight the entire data range or use Ctrl + A to select all.
- Press F5 or Ctrl + G, then click the Special button.
- Select Blanks and click OK button in the Go To Special dialog box. This will highlight all empty cells in the data, as shown in Figure 2.7:

Order number	Client ID	Product code	Date of delivery	Delivery amount
10100	7167	218989	2019-12-24	4,966.00
112601	7121	234028	2019-12-25	2,260.00
3196	7158	495974	2019-12-26	60,606.00
3196	7158	380089	2019-12-26	53,749.00
3196	7158	380515	2019-12-26	11,079.00
				46,417.00
				24,940.00
				4,512.00
				15,199.00
				5,868.00
				9,123.00
				16,642.00
				44,816.00
				37,776.00
				49,480.00
				35,916.00
				5,252.00
				13,391.00
				7,815.00

Figure 2.7: Highlighted Missing Data

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5. Right-click on any of the selected blank cells. From the context menu, select the Delete button. In the Delete dialog box, choose Entire Row and click the OK button. The selected row gets deleted, as shown in Figure 2.8:



	A	B	C	D	E	F
1	Order number	Client ID	Product code	Date of delivery	Delivery amount	
268	10100	7167	218989	2019-12-24	4,966.00	
269	112601	7121	234028	2019-12-25	2,260.00	
270	3196	7158	495974	2019-12-26	60,606.00	
271	3196	7158	380089	2019-12-26	53,749.00	
272	3196	7158	380515	2019-12-26	11,079.00	
273						
274						
275						
276						
277						
278						
279						
280						
281						
282						
283						
284						
285						

Figure 2.8: Treated Missing Data using Deletion Method

While data cleansing addresses critical issues like missing values, duplicates, and inconsistencies to enhance dataset quality, handling missing values through effective imputation is a pivotal step in ensuring data usability. The next section explores into commonly used imputation methods that enable analysts to fill gaps and maintain data integrity for more reliable analyses.



SELF ASSESSMENT QUESTIONS

7. Which of the following methods can be employed to handle missing data in a dataset using Excel?
 - a. Deletion method
 - b. Regression analysis
 - c. KNN
 - d. Model-based imputation
8. In data cleansing, which method helps in identifying and correcting outliers that significantly differ from other observations in a dataset?
 - a. Mean/Median/Mode imputation
 - b. Pairwise deletion
 - c. Z-score method
 - d. Regression imputation

9. Missing data does not lead to biased or incomplete analysis.
(True/False)



ACTIVITY

Learn to spot and solve common issues like duplicates and missing values using Excel. Practice cleaning data for accurate analysis.

2.5 IMPUTATION METHODS

Imputation methods in business analytics refer to techniques used to handle missing or incomplete data. In any dataset, missing values can occur due to various reasons, such as human error, data corruption, or simply because certain information was not collected. Imputation methods are crucial in maintaining data integrity and ensuring that analyses and models built on incomplete datasets are as accurate as possible.

The commonly used imputation methods are as follows:

- **Mean:** The arithmetic mean, also known as the average, is calculated by adding up the observations and dividing the total by the number of observations. The mean of a population is symbolised by the Greek letter μ , whereas the mean of a sample is denoted by $\bar{x}$.

If a population consists of N observations $x_1, x_2, x_3, \dots, x_N$, the population mean, μ , is computed as:

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

Whereas, if a sample consists of n observations $x_1, x_2, x_3, \dots, x_n$, denoted by $\bar{x}$, is computed as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

The arithmetic mean serves as an important measure of central tendency, offering a concise representation of the dataset's overall distribution for both populations and samples.

- **Median:** The median represents the central value in a dataset when arranged in ascending order. It divides the data into two equal halves, with half below and half above it. In cases with an odd number of observations, the median stands as the middle value. In scenarios with an even count of observations, the median

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becomes the average of the two middle values. This measure holds significance for ratio, interval and ordinal data. Unlike the mean, the median remains unaffected by outliers, offering a robust representation of central tendency.

- ❑ **Mode:** The mode represents the most commonly occurring observation. This measure becomes particularly valuable in datasets with a limited range of unique values. However, when datasets contain minimal repetition, the mode lacks significant practical utility. Identifying the mode involves recognising the value with the highest frequency in a frequency distribution or the tallest bar in a histogram. In the case of grouped data represented in frequency distributions or histograms, the mode corresponds to the group with the highest frequency.

Example: A retail company collects data on the monthly sales of a product over a year (12 months). The sales numbers (in units) for the 12 months are as follows:

Sales data: 150, 160, 155, 170, 165, 180, 190, 200, 180, 210, 220, 800

To calculate the mean sales, the company adds all the sales numbers together and divides by the number of months. That means:

$$\begin{aligned}\text{Mean} &= \frac{150 + 160 + 155 + 170 + 165 + 180 + 190 + 200 + 180 + 210 + 220 + 800}{12} \\ &= \frac{2780}{12} = 231.67\end{aligned}$$

The mean sales of 231.67 units are heavily influenced by the unusually high value of 800 in month 12, which skews the average upwards.

To calculate the median, the sales data must be sorted in ascending order, like:

150, 155, 160, 165, 170, 180, 180, 190, 200, 210, 220, 800.

With an even number of observations (12), the median is the average of the 6th and 7th values:

$$\text{Median} = \frac{180 + 180}{2} = 180$$

The mode is the most frequent value in the dataset. Here, the mode is 180, as it appears twice, more frequently than any other value in the data.

Now to impute data using mean, median and mode using Excel, perform the following steps:

1. Open the Excel dataset. Since there are no missing values in the dataset, create some missing values by deleting some random data from the Delivery amount column.

	A	B	C	D	E	F	G	H	I
1	Order number	Client ID	Product code	Date of delivery	Delivery amount				
2	97058	7121	494843	2019-01-24	9,565.00				
3	2968	7167	111937	2019-01-29	18,907.00				
4	2968	7167	218889	2019-01-29	54,132.00				
5	45863	7136	495715	2019-02-07					
6	45863	7136	495716	2019-02-07	59,120.00				
7	45863	7136	495717	2019-02-07	45,829.00				
8	45863	7136	495719	2019-02-07	42,980.00				
9	45863	7136	495720	2019-02-07					
10	45863	7136	495713	2019-02-07	38,903.00				
11	45863	7136	495722	2019-02-07	33,661.00				
12	45863	7136	495718	2019-02-07	1,937.00				
13	45863	7136	495714	2019-02-07					
14	45863	7136	495721	2019-02-07	30,900.00				
15	40155	7131	497253	2019-02-18	65,313.00				
16	40155	7131	497033	2019-02-18	30,649.00				

- Click the Data tab, and then click the Data Analysis tool. Select Descriptive Statistics from the Data Analysis dialog box, as shown in Figure 2.10:



	A	B	C	D	E	F	G	H	I
1	Order number	Client ID	Product code	Date of delivery	Delivery amount				
2	97058	7121	494843	2019-01-24	9,565.00				
3	2968	7167	111937	2019-01-29	18,907.00				
4	2968	7167	218889	2019-01-29	54,132.00				
5	45863	7136	495715	2019-02-07		Mean		35095.9777	
6	45863	7136	495716	2019-02-07	59,120.00	Standard Error		1160.634686	
7	45863	7136	495717	2019-02-07	45,829.00	Median		37466	
8	45863	7136	495719	2019-02-07	42,980.00	Mode		#N/A	
9	45863	7136	495720	2019-02-07	4,028.00	Standard Deviation		19035.8242	
10	45863	7136	495713	2019-02-07	38,903.00	Sample Variance		362362603	
11	45863	7136	495722	2019-02-07	33,661.00	Kurtosis		-1.239592043	
12	45863	7136	495718	2019-02-07	1,937.00	Skewness		-0.143214452	
13	45863	7136	495714	2019-02-07		Range		64316	
14	45863	7136	495721	2019-02-07	30,900.00	Minimum		1267	
15	40155	7131	497253	2019-02-18	65,313.00	Maximum		65583	
16	40155	7131	497033	2019-02-18	30,649.00	Sum		9440818	
17	40155	7131	497028	2019-02-18	57,336.00	Count		269	
18	40155	7131	497035	2019-02-18	20,574.00				1

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4. Copy and paste Delivery amount column in the columns of J, L and N. Rename the column labels as Delivery amount Mean, Delivery amount Median and Delivery amount Mode. As there is no mode value in the dataset, we will leave replacing the missing values Delivery amount Mode.
5. Select the Delivery amount Mean and press Ctrl + F in the keyboard. The Find and Replace dialog box opens. Select the Replace tab, and in place of Find What leave it blank, and in Replace with, enter the Mean value, as shown in Figure 2.12:

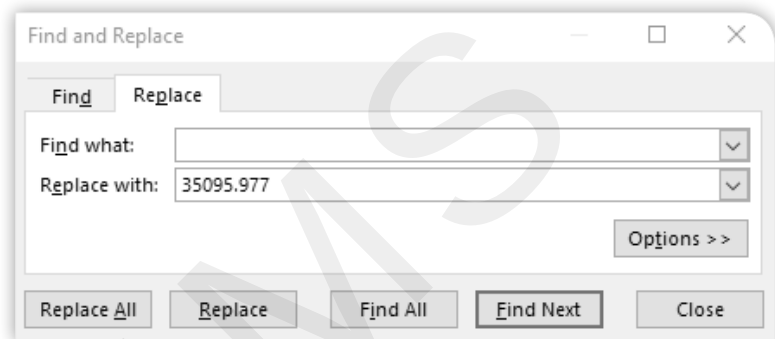


Figure 2.12: Replacing Mean Values in the Missing Data

6. Click on Replace All. You will find a prompt saying 'All Finished. We made _ replacement'. The value of replacement will vary with the number of missing values in the dataset. The mean replacements are shown in Figure 2.13:

F	G	H	I	J
				Delivery amount
				9,565.00
				18,907.00
				54,132.00
	Mean	35095.9777		35,095.98
	Standard Error	1160.634686		59,120.00
	Median	37466		45,829.00
	Mode	#N/A		42,980.00
				4,028.00
				38,903.00
				33,661.00
				1,937.00
				35,095.98
				30,900.00
				65,313.00
	Sum	9440818		30,649.00
	Count	269		57,336.00
				20,574.00
				12,677.00

Figure 2.13: Replaced Missing Values with Mean

7. Continue the above steps for Delivery amount Median. In place of the mean value in the Replace with box, enter the median value.

Having explored various imputation methods and their applications, it is crucial to understand how these methods can be further leveraged and communicated through effective data visualisation. While imputation techniques such as mean, median, and mode serve to address missing or incomplete data, visualisation provide a clear and impactful way to interpret and present the outcomes of these analyses. In the next section, we will explore different types of data visualisations, which allow us to communicate findings more effectively, facilitating better decision-making in business analytics.



SELF ASSESSMENT QUESTIONS

10. Which measure of central tendency is least affected by outliers in a dataset?
 - a. Mean
 - b. Median
 - c. Mode
 - d. All are equally affected
11. The mean is calculated differently for a population and a sample. (True/False)
12. The mode may not be particularly useful in datasets with minimal repetition of values. (True/False)



ACTIVITY

Practice to fill missing values using mean, median and mode by grasping how each method calculates central tendencies, accounting for outliers, and its impact on imputing absent data.

2.6 DATA VISUALISATION

The well-known saying ‘a picture is worth a thousand words’ is more relevant today than ever, especially in our information-driven world. Data visualisation involves presenting large amounts of data clearly to help extract insights that improve decision-making. In today’s competitive business environment, understanding large and complex data sets is not only beneficial but essential for survival. Furthermore, data visualisation plays a crucial role in building decision models and interpreting their results.

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2.6.1 COLUMN AND BAR CHARTS

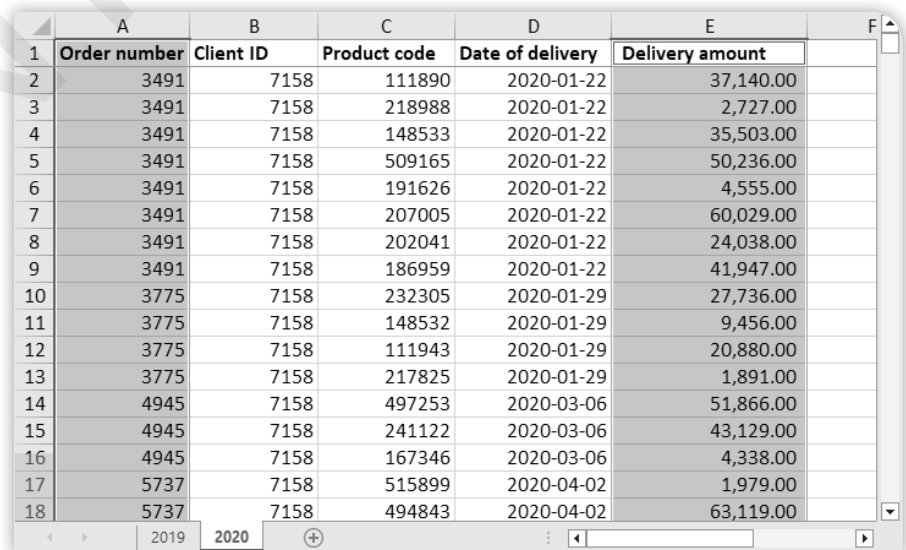
Column and bar charts are essential data visualisation tools that offer a simple yet powerful way to represent categorical data and facilitate comparisons between different categories. An overview of both types of charts is provided as follows:

- ❑ **Column charts:** These charts represent categorical data using vertical bars, with the height of each column corresponding to the value it represents. Column charts are particularly useful for comparing data points across different categories, such as sales figures or monthly revenue. They are widely used due to their ease of use and effectiveness in conveying information clearly.
- ❑ **Bar charts:** Similar to column charts but with horizontal bars, bar charts are equally effective for comparing categorical data. They are especially beneficial when dealing with long category labels or when space constraints are a factor. However, bar charts are not suitable for continuous data and can become cluttered if there are too many categories.

Both column and bar charts are commonly employed across business, finance, and marketing sectors to display categorical data clearly.

Perform the following steps to create column and bar charts in Excel:

1. Open a dataset using Excel. Sort the Order Number column by selecting Sort A to Z from the Sort & Filter tab from the Menu bar. Highlight the data you want to include in the chart (including headers if applicable), as shown in Figure 2.14:



	A	B	C	D	E	F
1	Order number	Client ID	Product code	Date of delivery	Delivery amount	
2	3491	7158	111890	2020-01-22	37,140.00	
3	3491	7158	218988	2020-01-22	2,727.00	
4	3491	7158	148533	2020-01-22	35,503.00	
5	3491	7158	509165	2020-01-22	50,236.00	
6	3491	7158	191626	2020-01-22	4,555.00	
7	3491	7158	207005	2020-01-22	60,029.00	
8	3491	7158	202041	2020-01-22	24,038.00	
9	3491	7158	186959	2020-01-22	41,947.00	
10	3775	7158	232305	2020-01-29	27,736.00	
11	3775	7158	148532	2020-01-29	9,456.00	
12	3775	7158	111943	2020-01-29	20,880.00	
13	3775	7158	217825	2020-01-29	1,891.00	
14	4945	7158	497253	2020-03-06	51,866.00	
15	4945	7158	241122	2020-03-06	43,129.00	
16	4945	7158	167346	2020-03-06	4,338.00	
17	5737	7158	515899	2020-04-02	1,979.00	
18	5737	7158	494843	2020-04-02	63,119.00	

Figure 2.14: Selecting Data for Visualisation

2. Select 2-D Bar from the Charts option in the Insert tab in the Menu bar, as shown in Figure 2.15:

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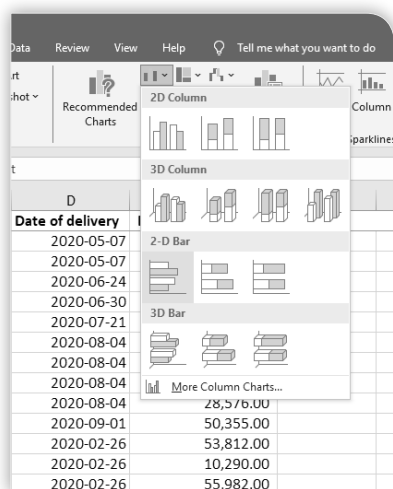


Figure 2.15: Selecting 2-D Bar Option

3. Use the Chart Tools on the ribbon to customise the design, layout and style once the chart is created, as shown in Figure 2.16:

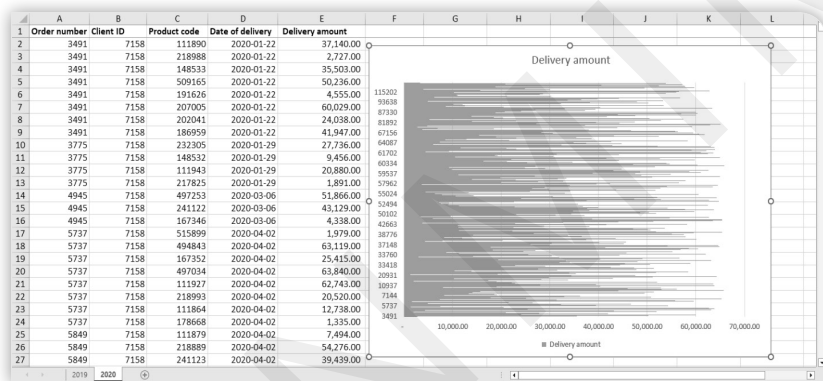


Figure 2.16: Customised 2-D Bar Chart

4. Continue to select 2-D Column from the Charts option in Insert tab in the Menu bar to create a column chart, as shown in Figure 2.17:

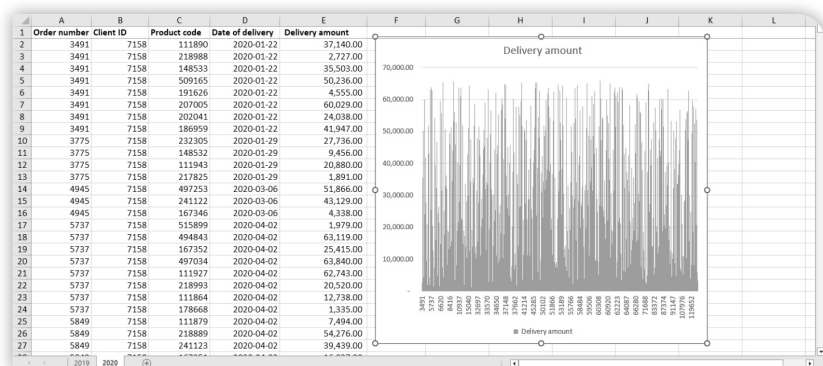


Figure 2.17: 2-D Column Chart in Excel

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2.6.2 LINE CHARTS

Line charts are an effective data visualisation tool used to display trends and changes over time by connecting data points with straight lines. Some of the key features and considerations for line charts are:

- ❑ A line chart is a type of data visualisation that displays data points connected by straight lines, forming a continuous line.
- ❑ Line charts use the horizontal axis to represent time or other continuous variables, and the vertical axis represents the corresponding data values.
- ❑ In line charts, the continuous line created by connecting data points helps to illustrate the overall trend and the relationship between data points.
- ❑ Line charts are effective for visualising data with many data points or data that changes smoothly over time.

Perform the following steps to create line charts using Excel:

1. Open the dataset using Excel. Sort the Order Number column by selecting Sort A to Z from the Sort & Filter tab from the Menu bar. Highlight the data you want to include in the chart (including headers if applicable). In this process, select the Date of delivery and Delivery amount.
2. Select 2-D Line from the Insert Line or Area Chart from the Chart option in the Insert tab in the Menu bar. The visualisation of the line chart as per the dataset is shown in 2.18:

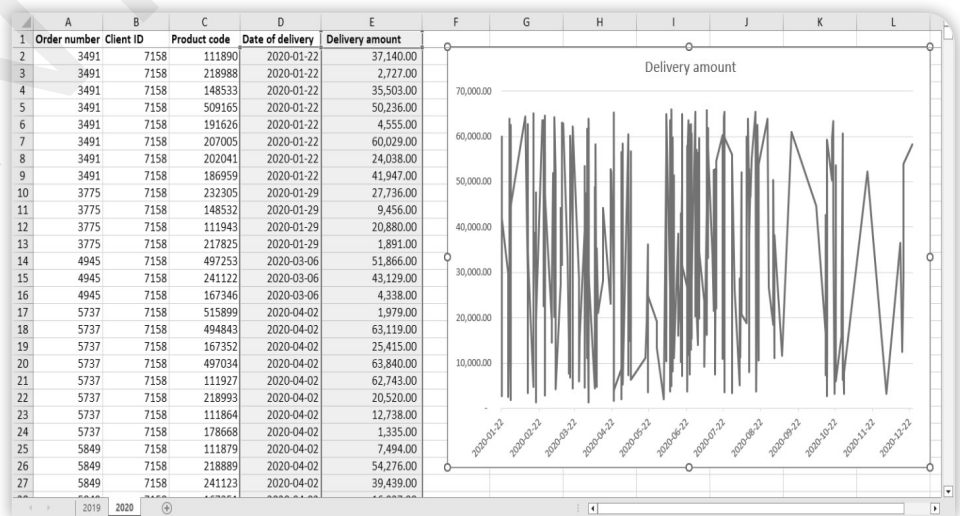


Figure 2.18: Line Chart in Excel

2.6.3 PIE CHARTS

Pie charts are the visual representation that demonstrates the relative proportions of different data categories in a single circular graph.

Some of the key features and considerations of pie charts are:

- ❑ Pie charts provide a simple and intuitive way to select the data source and the corresponding data fields to generate the pie chart automatically.
- ❑ Users can customise the appearance of pie charts by adjusting colours, labels and font styles.
- ❑ The Pie charts support interactivity, enabling users to drill down into specific slices for more detailed information or link to other reports and data sources.

Perform the following steps in Excel to create a pie chart:

1. Open the dataset using Excel. Sort the Order Number column by selecting Sort A to Z from the Sort & Filter tab from the Menu bar. Highlight the data you want to include in the chart (including headers if applicable). In this process, select the Date of delivery and Delivery amount.
2. Click the Subtotals option from Outline from the Data tab in the Menu bar. Check the options, as shown in Figure 2.19:

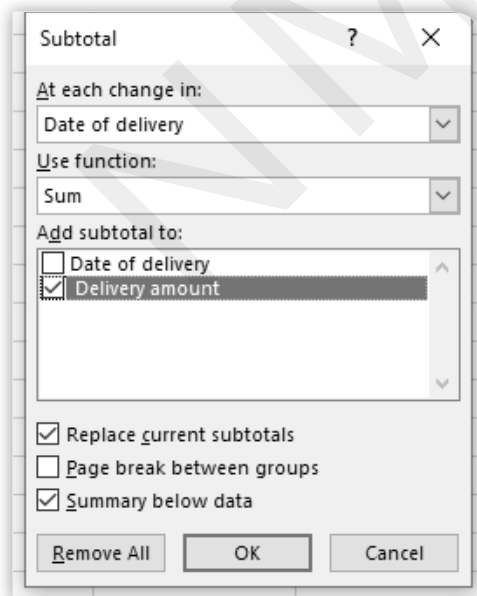


Figure 2.19: Creating Subtotal in Excel

3. Click the OK button. Excel would insert the subtotal automatically in the Date of delivery column.

NOTES

4. Select 3-D Pie from the Insert Pie or Doughnut Chart from the Chart option in the Insert tab in the Menu bar. In the Chart Filters, deselect the (Select All) option and only select the totals of each date, as shown in Figure 2.20:

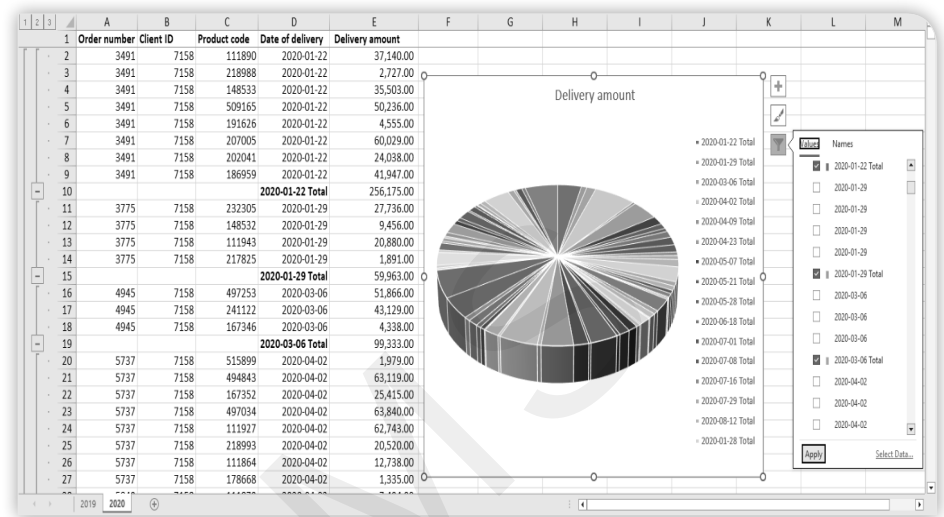


Figure 2.20: Creating Pie Chart in Excel

2.6.4 SCATTER CHARTS

Scatter charts are the graphical representations that are used to display the relationship between two or more numerical variables. Some of the key features and considerations of scatter charts are:

- ❑ A scatter chart helps identify patterns and trends in the data by clustering data points based on their positions on the graph.
- ❑ Scatter chart shows how some changes in one variable can affect another variable.
- ❑ Users can further enhance the scatter chart by introducing additional dimensions to the graph, such as colour-coded data points based on a categorical variable or sizing data points based on a third numerical variable.

Perform the following steps in Excel to create a scatter chart:

1. Open the dataset using Excel. Highlight the data you want to include in the chart (including headers if applicable). For this visualisation, select the Date of delivery and Delivery amount.
2. Select the Scatter chart from the Insert Scatter (X, Y) or Bubble Chart. The visualisation of Date of Delivery and Delivery amount is visualised as shown in Figure 2.21:

NOTES

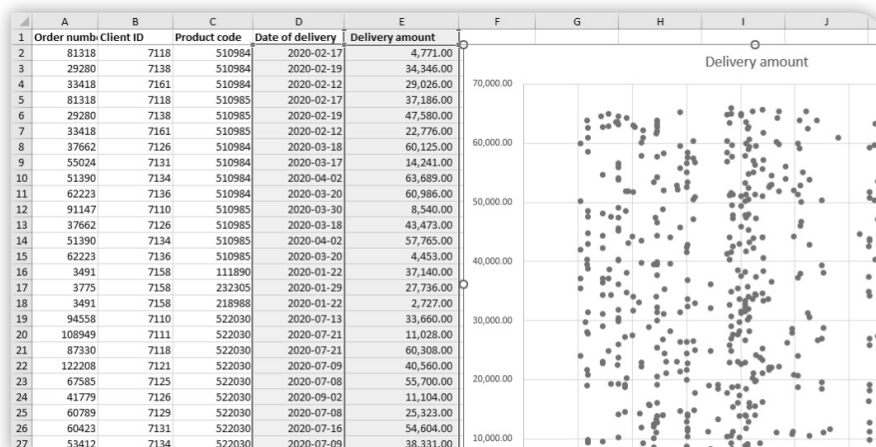


Figure 2.21: Visualising Scatter Chart in Excel



SELF ASSESSMENT QUESTIONS

13. Which of the following data visualisation types is best suited for representing discrete data and is widely used in various fields for its simplicity and efficiency?
 - a. Line chart
 - b. Pie chart
 - c. Bar chart
 - d. Scatter chart
14. Which data visualisation type is most effective for illustrating trends and changes over time and is ideal for time series analysis?
 - a. Bar chart
 - b. Scatter chart
 - c. Line chart
 - d. Pie chart
15. The scatter chart primarily displays the relationship between two numerical variables and does not support any form of interactivity. (True/False)



ACTIVITY

Create a data visualisation in Excel to illustrate the trend of the dataset by employing line chart and downloading the dataset from <https://www.kaggle.com/datasets/kandij/electric-production>.

2.7 SUMMARY

- ❑ Data guides decision-making and problem-solving by uncovering patterns and trends.
- ❑ The data collection process systematically gathers information from various sources, ensuring accuracy and reliability essential for research, analysis, and informed decision-making in diverse fields.

N O T E S

- ❑ Structured data is a type of data that is organised and easily managed using traditional data management tools such as spreadsheets, databases or tables.
- ❑ Unstructured data is the type of data that does not have a predefined format, making it difficult to manage using traditional data management tools.
- ❑ Examples of semi-structured data include XML and JSON files, which have some organisation but also contain elements of unstructured data.
- ❑ Big Data is a term used to describe large and complex data sets that cannot be processed using traditional data management tools.
- ❑ Data cleansing addresses issues such as missing values, outliers and inaccuracies, ensuring the reliability and accuracy of datasets for effective analysis.
- ❑ Imputation methods fill in missing data points with techniques like mean, median and mode.
- ❑ Data visualisation employs graphical representations to convey patterns, trends, and insights in a clear and understandable manner.



KEY WORDS

- ❑ **Analytics:** It involves the systematic analysis of data and statistics to derive meaningful insights.
- ❑ **Box plot:** A box plot is a graphical representation of data distribution, showing the median, quartiles, and potential outliers, helping to visualise data spread and skewness.
- ❑ **IQR:** The IQR is the range between the 25th and 75th percentiles, representing the middle 50% of a dataset and used to identify outliers.
- ❑ **Z-score:** A Z-score measures how many standard deviations a data point is from the mean, indicating its relative position within a dataset.

2.8 MULTIPLE CHOICE QUESTIONS

1. What is the primary goal of data preprocessing in business analytics?
 - a. To eliminate unnecessary data
 - b. To clean, transform and organise raw data for higher quality and consistency

- c. To generate insights directly from raw data
 - d. To automate business decisions
2. Which of the following is an example of primary data collection?
- a. Using information from online databases
 - b. Conducting surveys and questionnaires
 - c. Collecting data from published books
 - d. Reviewing data from past research studies
3. What is the primary characteristic of structured data?
- a. It is organised and can be easily managed using traditional tools.
 - b. It is complex and requires advanced tools to manage.
 - c. It has no predefined format.
 - d. It is partially organised but contains unstructured elements.
4. Unstructured data is typically analysed using which of the following methods?
- a. Statistical regression analysis
 - b. Advanced techniques like NLP
 - c. Simple aggregation methods
 - d. Traditional databases
5. Outliers in a dataset can affect the results in which of the following ways?
- a. Outliers can enhance the accuracy of statistical models.
 - b. Outliers have no impact on the analysis.
 - c. Outliers can distort statistical measures like the mean and variance.
 - d. Outliers can automatically be ignored during analysis.
6. Which of the following describes multivariate outliers?
- a. Outliers that occur when a data point stands out significantly in one variable.
 - b. Outliers that occur when data points deviate significantly across multiple variables.
 - c. Outliers that only exist in datasets with one variable.
 - d. Outliers that can be ignored as they do not impact data analysis.

N O T E S

7. In the context of data cleansing, how are duplicate entries typically handled?
 - a. By removing or consolidating them
 - b. By leaving them as they are
 - c. By modifying their values randomly
 - d. By replacing them with missing values
8. Which of the following charts is best suited for representing trends over time?
 - a. Pie chart
 - b. Column chart
 - c. Scatter chart
 - d. Line chart
9. When dealing with semi-structured data, which of the following is a common file format?
 - a. SQL
 - b. XML
 - c. CSV
 - d. PNG
10. What method can be used to handle missing values by replacing them with the most frequent occurrence in a column?
 - a. Mean imputation
 - b. Median imputation
 - c. Mode imputation
 - d. Deletion

2.9 DESCRIPTIVE QUESTIONS

1. How does data visualisation contribute to better decision-making in business analytics, and what advantages does it offer over raw data presentation?
2. Explain the significance of structured and unstructured data in business analytics.
3. What are the common issues associated with data that require cleansing, and how does data cleansing contribute to the reliability of analytics?
4. Explain Big Data in business analytics.
5. Explore the various imputation methods used in handling missing data.

2.10 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

- In a dataset analysing customer satisfaction across regions, you notice a considerable number of missing values in the 'FeedbackRating' column. Which imputation method would be most suitable to maintain the dataset's integrity without skewing the central tendency?
 - Mean imputation
 - Median imputation
 - Mode imputation
 - Multiple imputation
- While examining quarterly sales data for a retail chain, extreme values are observed in the 'Revenue' column, significantly deviating from the usual sales figures. What would be an appropriate visualisation method to identify and display these outliers?
 - Line chart
 - Scatter chart
 - Pie chart
 - Bar chart
- An analyst wants to represent the distribution of customer ages in a survey sample accurately. Which sampling method is a proportional representation of different age groups, reflecting the population's diversity?
 - Simple random sampling
 - Stratified sampling
 - Cluster sampling
 - Systematic sampling
- In a retail dataset, missing values are detected in the 'ProductReview Ratings.' Describe two potential consequences of imputing these missing values using the mean imputation method.
- A marketing team is comparing sales performance across three product categories using line charts. Explain how line charts could effectively highlight trends and differences among these categories.

2.11 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
Using Data in Analytics	1.	c. Ascertaining a business's contribution to shareholders
	2.	c. Books and academic journals

N O T E S

Topic	Q. No.	Answer
Types of Data	3.	False
	4.	a. Structured data
	5.	False
	6.	True
Data Cleansing: Common Issues with Data	7.	a. Deletion method
	8.	c. Z-Score method
	9.	False
Imputation Methods	10.	b. Median
	11.	True
	12.	True
Data Visualisation	13.	c. Bar chart
	14.	c. Line chart
	15.	False

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. To clean, transform and organise raw data for higher quality and consistency
2.	b. Conducting surveys and questionnaires
3.	a. It is organised and easily managed using traditional tools
4.	b. Advanced techniques like NLP
5.	c. Outliers can distort statistical measures like the mean and variance
6.	b. Outliers that occur when data points deviate significantly across multiple variables.
7.	a. By removing or consolidating them
8.	d. Line chart
9.	b. XML
10.	c. Mode imputation

HINTS FOR DESCRIPTIVE QUESTIONS

1. Data visualisation plays a pivotal role in business analytics by presenting complex data in a visually comprehensible manner. It enhances decision-making by empowering managers with improved analytical abilities, reducing dependence on IT professionals, and fostering better collaboration. Refer to Section 2.6

N O T E S

2. In business analytics, both structured and unstructured data play critical roles in providing insights and guiding decision-making. Each type of data has its own significance and requires different approaches for processing and analysis. Refer to Section 2.3
3. Data cleansing is essential for improving the quality of datasets in business analytics. Common issues include missing values, duplicate entries, inconsistent formatting, outliers, inaccurate data, imbalance and irrelevance. Refer to Section 2.4
4. Big Data refers to datasets that are extremely large, complex and varied, often encompassing structured, semi-structured and unstructured data types. Refer to Section 2.3
5. Imputation methods handle missing or incomplete data, ensuring accurate analyses. Mean imputation involves replacing missing values with the mean of observed values in that variable. Its strengths include simplicity and ease of implementation. Refer to Section 2.5

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	d. Multiple imputation
2.	b. Scatter chart
3.	b. Stratified sampling
4.	Mean imputation could distort the true variability in product ratings as it replaces missing values with a single value, potentially reducing the spread of ratings. The mean imputation might not account for underlying patterns or trends in ratings, leading to biased estimations and inaccurate analysis of customer sentiment.
5.	Line charts display trends over time or across categories. By plotting sales data for each product category over time, line charts can illustrate the comparative growth rates, seasonal trends or performance disparities among the categories visually. This aids in identifying patterns and making informed strategic decisions based on sales trends.

2.12 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

- ❑ Evans, J. R. (2016). Business Analytics (2nd ed.). Pearson.
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CASE STUDY

N O T E S

PREDICTING CUSTOMER CHURN WITH POPULATION AND SAMPLE DATA

Imagine you are a data analyst at a streaming service like Netflix. You are tasked with predicting customer churn (cancellations) to improve retention strategies. You have access to a massive dataset containing information about all Netflix subscribers (population data). This includes demographics, viewing habits, subscription plans and cancellation history.

CHALLENGE

Analysing the entire population data is computationally expensive and time-consuming. To gain insights quickly and efficiently, you need to leverage sample data. But how would you ensure your sample accurately represents the population and yields reliable predictions.

SOLUTION

❑ Stratified sampling visualisation

- ◆ Visualise the distribution of key characteristics such as age groups, location, or subscription types within the population dataset.
- ◆ Create visualisations that illustrate the proportional representation of each level in your sample selection process. This helps stakeholders understand how the sample mirrors the diversity of the overall customer base.

❑ Sample size calculation visualisation

- ◆ Use visual aids to demonstrate the impact of different sample sizes on the accuracy of predictions. Show graphs or charts illustrating how confidence levels and margins of error change with varying sample sizes.
- ◆ Develop interactive visuals that allow stakeholders to dynamically adjust sample sizes and observe the corresponding changes in prediction accuracy.

❑ Data cleaning and preprocessing visualisation

- ◆ Create before-and-after visualisations highlighting the effects of data cleaning on the dataset. Show how outliers or missing values are identified and treated.
- ◆ Use data quality dashboards or visuals to showcase the improved dataset's cleanliness and consistency, providing a clear understanding of data preparation steps.



Case Objective

The objective of this case study is to predict customer churn at Netflix by efficiently leveraging sample data, demonstrating benefits and emphasising key considerations.

CASE STUDY

☐ **Modelling and analysis visualisation**

- ◆ Visualise the performance metrics of churn prediction models (e.g., accuracy, precision, recall) through charts or graphs. Compare different models' performance visually to aid in selecting the most effective one.
- ◆ Develop interactive dashboards showing the factors influencing churn prediction, allowing users to explore and understand the model's insights.

☐ **Model validation and improvement visualisation**

- ◆ Create visuals that depict cross-validation techniques, illustrating how the model performs across different subsets of the population data.
- ◆ Use visualisations to track the models' performance over time as new data is incorporated, demonstrating how the model evolves and improves.

BENEFITS

- ☐ **Reduced cost and time:** Analysing a smaller sample saves time and computational resources compared to processing the entire population data.
- ☐ **Faster insights:** Sample-based analysis allows for quicker identification of churn patterns and trends.
- ☐ **Improved model accuracy:** Stratified sampling and proper sample size calculations ensure your sample accurately represents the population, leading to more reliable predictions.

REAL-WORLD EXAMPLE

Netflix has successfully implemented a churn prediction model based on sample data analysis. They identified factors such as inactive accounts, low engagement and high viewing costs as key churn drivers. This enabled them to implement targeted retention strategies, like personalised recommendations and special offers, to reduce customer cancellations.

CONCLUSION

- ☐ Choosing the right sampling technique and sample size is crucial for accurate predictions.
- ☐ Data quality and proper preprocessing are essential for reliable model building.
- ☐ Continuously monitor and improve your models to adapt to changing customer behaviour and market trends.

CASE STUDY

N O T E S

By effectively leveraging population and sample data, businesses can gain valuable insights into customer behaviour, optimise their operations, and achieve their goals.



QUESTIONS

1. How does stratified sampling improve sample representation?

(**Hint:** Stratified sampling ensures proportional representation of diverse population segments, reducing bias in the sample.)

2. Why is ongoing model monitoring crucial to churn prediction?

(**Hint:** Continuous model monitoring allows adaptation to evolving customer behaviour, enhancing the accuracy of churn predictions over time.)

3

C H A P T E R

INTRODUCTION TO PROBABILITY THEORY

CONTENTS

- 3.1 Introduction
- 3.2 Principles of Counting
 - 3.2.1 Permutations
 - 3.2.2 Combinations
 - Self Assessment Questions
 - Activity
- 3.3 Discrete Probability Distributions
 - 3.3.1 Probability Mass Function (PMF)
 - 3.3.2 Types of Discrete Probability Distributions
 - Self Assessment Questions
 - Activity
- 3.4 Continuous Probability Distributions
 - 3.4.1 Probability Density Function (PDF)
 - 3.4.2 Types of Continuous Probability Distributions
 - 3.4.3 Difference Between PDF and PMF
 - Self Assessment Questions
 - Activity
- 3.5 Summary
- 3.6 Multiple Choice Questions
- 3.7 Descriptive Questions
- 3.8 Higher Order Thinking Skills (HOTS) Questions
- 3.9 Answers and Hints
- 3.10 Suggested Readings & References

N O T E S

INTRODUCTORY CASELET

**Case Objective**

The objective of this caselet is to understand the distribution of defects and identify the key variables influencing the defect rates.

IMPROVING PRODUCT QUALITY THROUGH PROBABILITY DISTRIBUTION ANALYSIS

A manufacturing company was interested in quality improvement and reduction of defects in one of the critical components of the production process. In this company, they could not identify some specific causes of the defects and also needed improvement in certain areas. Probability distribution analysis was used here as it appeared to be a statistically sound tool for gaining insight into factors causing variation.

The manufacturing process was characterised by multiple variables that may make it challenging to separate some factors contributing to defects. Since the company had no significant historical data on defects, it could not easily establish a sound statistical model from which to analyse. The production environment dynamic changed either from variations of the machinery settings or from materials being used; thereby, requiring an adaptive strategy to deal with the differences.

A suitable amount of data detailing defects and process variables was collected. Preprocessing of data involved cleaning, followed by creating an organised dataset for further analysis. Distribution based probability models, such as a Poisson distribution, were applied to model the count of defects in order to understand patterns inherent in defect occurrences. The techniques in distribution fitting were used to determine which of the probability distributions represented best the data observed. The output from the probability distribution analysis enabled targeted improvements and implementation. Continuous monitoring and feedback loops ensured adaptation of the model to the changes in the production environment over time.

The company was able to drive down defects through targeted improvements derived from the probability distribution analysis. The predictive model for product quality helped identify the leading influences so that precautionary steps were taken to maintain very high standards. The adaptive approach meant the company was better placed to adapt to shifts in the production environment, which accelerated the culture of continuous improvement.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Describe the fundamental counting principle
- Understand discrete probability distribution
- Define continuous probability distribution

3.1 INTRODUCTION

In the previous chapter, we studied about the application of data and the process of cleaning data, such as missing value treatment. We discussed various imputation methods like mean, mode and median. We learnt different kinds of charts used for data visualisation, such as pie charts, scatter charts, bar charts, line charts and column charts.

Probability theory is a mathematical framework used to analyse random events and predict their likelihood. Some of the important aspects include counting principles, like addition and multiplication rules, permutations, and combinations, involved in determining the number of possible outcomes. There are two types of discrete probability distributions which are Binomial and Poisson distributions that describe both the probabilities of discrete variables per given value and coupled with continuous probability distributions- normal and exponential distributions for values that can take any number within a certain range.

In this chapter, you will be introduced to the principles of counting. Then you will learn about the concept of discrete probability distributions. Further, you will learn about the concept of continuous probability distributions.

3.2 PRINCIPLES OF COUNTING

The fundamental principle of counting is a technique employed to determine the overall number of possible outcomes for various events or choices. In simple language, it is like figuring out how many outcomes you can have when you make a series of choices. There are two main rules:

- ❑ **Multiplication rule:** If there are m ways something can happen and n ways another thing can happen, then there are $m \times n$ ways where both can happen together.

Example: If you have 3 choices for a shirt and 4 choices for pants, then you have $3 \times 4 = 12$ different outfits.

- ❑ **Addition rule:** If there are m ways something can happen and n ways another thing can happen independently of the first, then there are $m + n$ ways in which either one can happen.

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Example: If you can either take a bus ($m = 5$ ways) or a train ($n = 3$ ways) to work, then there are $5 + 3 = 8$ different ways to commute.

These rules help you count the total number of outcomes in different situations where there are multiple choices or events.

The fundamental counting principle is like a math shortcut that helps us figure out how many different outcomes we can get when making a series of independent choices. If there are n_1 ways to do one thing and n_2 ways to do another, then the total number of outcomes for both is $n_1 \times n_2$. It is a handy rule for counting all the possibilities when there are several steps or decisions involved.

The fundamental counting principle asserts that when one event or decision has 'a' possible outcomes or choices and another event has 'b' possible outcomes or choices, then the total number of distinct combinations of outcomes between the two is given by the product $a \times b$.

For instance, consider having 2 pairs of shoes and 3 pairs of socks. The number of ways you can wear them is determined by multiplying the choices for shoes (2) by the choices for socks (3). In this case, there are 2 pairs of shoes, each offering 3 choices of socks, resulting in a total of $2 \times 3 = 6$ possible combinations.

To illustrate, let A_1 and A_2 represent the 2 pairs of shoes, and B_1 , B_2 and B_3 represent the 3 pairs of socks, as shown in Figure 3.1:

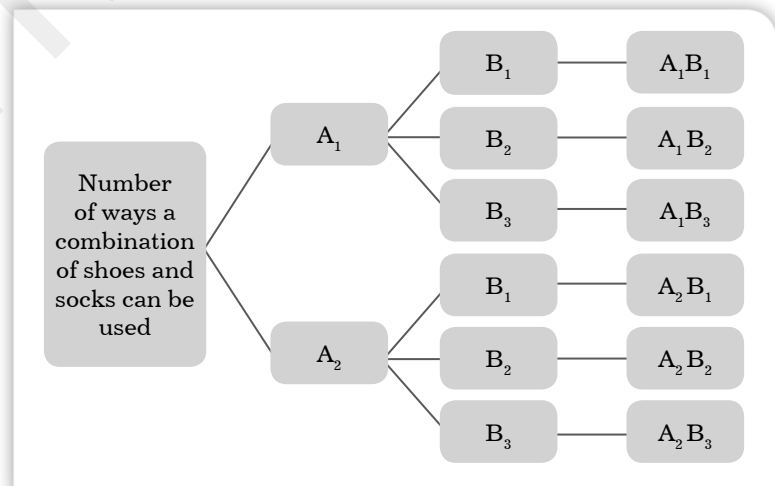


Figure 3.1: Six Ways of Wearing a Combination of Shoes and Socks

Counting rules such as permutations and combinations help build the foundation of probability, which makes possible outcomes measurable. Understanding these principles is essential to progress into complex probability distributions, where these counts play

a crucial role in defining possible scenarios. We will learn about permutations and combinations in the next sections in detail.

3.2.1 PERMUTATIONS

A permutation is an arrangement of objects in a specific order. The order in which the objects are arranged is crucial, meaning that different orders of the same objects are considered distinct permutations. The number of permutations of a set of objects is often denoted by $P(n, r)$ or ${}^n P_r$, where n is the total number of objects and r is the number of objects selected for the arrangement.

The formula for permutations is given by: $P(n, r) = \frac{n!}{(n-r)!}$, where

‘ $n!$ ’ (read as ‘ n factorial’) is the product of all positive integers up to n and r is the number of objects selected at a time.

Suppose you have 5 different coloured balls (red, blue, green, yellow and purple), and you want to arrange 3 of them in a specific order. The number of permutations (${}^5 P_3$) can be calculated using the formula:

$${}^5 P_3 = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = \frac{120}{2} = 60$$

So, there are 60 different ways you can arrange 3 balls out of 5 in a specific order.

There are many applications of permutation in business analytics such as:

- ❑ **Product placement:** When one wants to show company products in some sort of order in the advertisement or on a website, this would require permutations to determine how many possible combinations of that order are possible. The order in which the product appears could influence customer behaviour.
- ❑ **Advertisement scheduling:** In advertisement scheduling, marketing teams make use of permutations to determine when which advertisements should be displayed to consumers.
- ❑ **Sales sequencing:** The choice of which sequence of contact to have with potential customers or in which sequence to offer products for sales teams could often be associated with higher success rates. Permutations are used to examine all possible sequences and to determine the best course of action.
- ❑ **Revenue management:** Companies with limited inventory use permutations to analyse which combination of products and services can maximise revenue given different customer preferences.

N O T E S

- ❑ **Scenario planning:** Companies use permutations to build models of the different ways that events might play out over time. This is an important approach to forecasting in volatile markets because the sequence of events—the launching of new products, economic changes, and so on—can often create huge effects.

3.2.2 COMBINATIONS

A combination is such a selection of items from a set that has distinct members, from which order of selection does not matter. Suppose you have a collection, like fruits in a basket. If you want to make a combination of two fruits, you can pick any two fruits from the basket without caring to know the order in which you picked them. Thus, given you have apples and bananas, choosing apples first and then bananas is understood to be the same combination as choosing bananas first and then apples.

A combination is a selection of ‘r’ items from a set of ‘n’ items, where ‘n’ is the total number of items in the set. The selection is done without replacement and where order does not come into play or matter. The

formula to compute the number of combinations is: ${}^nC_r = \frac{n!}{r!(n-r)!}$

For instance, a committee of 5 members is to be formed from a group of 10 people. In how many ways can this be done?

Using the combination formula, we get:

$${}^{10}C_5 = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(5 \times 4 \times 3 \times 2 \times 1)(5 \times 4 \times 3 \times 2 \times 1)} = \frac{30240}{120} = 252$$

That means there are 252 ways to form a committee of 5 members from a group of 10 people.

The applications of combinations in business analytics are:

- ❑ **Product bundling and pricing strategies:** Many organisations are interested in knowing how many various product packages can be generated as bundles. For example, the number of three-pack bundles that can be formed by a retailer who would be offering 5 products can be determined using combinations.
- ❑ **Market basket analysis:** Combinations would help identify a set of products that are purchased together frequently; therefore, this supports decisions on appropriate cross-selling and upselling strategies. Transactional data analysis reveals frequent item sets that can be improved to manage inventory effectively and run targeted marketing campaigns.
- ❑ **Customer segmentation:** When conducting a multi-attribute segmentation of customers by criteria such as age, income and

preferences, combinations aid one in determining how many different customer groups may be created based on the selected attributes. Analysts use combinations to determine the number of ways customers may be categorised when a subset of attributes is chosen from the entire set.

- ❑ **Resource allocation:** Businesses always have to distribute resources, such as employees or a budget over various departments or projects. Combinations are used to determine the number of ways in which resources may be distributed without respect to their sequence. It is finding all possible combinations through a fixed set of resources among multiple projects or teams.
- ❑ **Project portfolio management:** Combinations are also used in attempts to determine which projects ought to be undertaken, given a set of constraints such as budget or resources, while trying to assess different portfolios of projects or initiatives. Combinations help in picking the best combination of projects that maximise value or returns in light of limited resources.

Counting principles, such as permutations and combinations, form the basis of discrete probability distributions by calculating possible outcomes. We will learn about discrete probability distributions in the next section.



SELF ASSESSMENT QUESTIONS

1. A lock has 4 digits (0-9). How many different passwords are possible if repetition of digits is allowed?
 - a. 10
 - b. 100
 - c. 1,000
 - d. 10,000
2. In a deck of playing cards, in how many ways can you choose two cards?
 - a. 52
 - b. 104
 - c. 1,326
 - d. 2,652
3. A chef wants to create a new dish using three different types of pasta, four kinds of vegetables and two sauces. If the chef can choose only one type of pasta, one vegetable and one sauce for the dish, how many different combinations of pasta, vegetables and sauces can be created?
 - a. 9
 - b. 12
 - c. 16
 - d. 24
4. The principle of counting can be applied to determine the number of ways to arrange a set of items when the order of arrangement is significant. (True/False)

N O T E S



ACTIVITY

You are tasked with creating a secure password for your online account. Your password must consist of 8 characters, and you decide to use a combination of uppercase letters, lowercase letters, numbers and special characters.

3.3 DISCRETE PROBABILITY DISTRIBUTIONS

A discrete probability distribution gives the probability of occurrence of each possible outcome for a set of distinct and countable events. In simple words, it assigns probabilities to individual outcomes of a random experiment, which may either be finite or infinite in possibilities.

Some components of the discrete probability distribution are as follows:

- ❑ **Sample space and events:** Every probability scenario has a set of possible outcomes, called the sample space. Groups within the sample space are called events, representing specific outcomes or combinations of outcomes.
- ❑ **Probability mass function:** The probability mass function is the possibility of assuming some value that a discrete random variable may assume. It maps to each value the chance of it happening such that probabilities add up to 1 across all values.
- ❑ **Random variables:** A random variable is a number that corresponds to each outcome in the sample space. It links an outcome to a numerical value, so we can represent the uncertainty of which outcome will happen with a variable that has a range of values.
- ❑ **Cumulative distribution function:** This function gives the probability that a random variable is less than or equal to a certain value. It shows how the probabilities of the random variable add up for each value.
- ❑ **Expected value and variance:** The expected value of a discrete random variable is its mean and is calculated as the average of its possible values weighted by the probabilities given by the PMF. Variance is how spread out the values in the distribution are.
- ❑ **Distribution parameters:** Most discrete probability distributions have parameters that affect their shape and characteristics. Some examples are the Binomial, Poisson and Hypergeometric distributions.

Discrete probability distributions are defined by the PMF, with various types such as Binomial and Poisson distributions which are discussed in the next sections.

3.3.1 PROBABILITY MASS FUNCTION (PMF)

The PMF of a discrete random variable is X a function that takes each possible value x of X and returns the probability $p(X = x)$. For any value of the random variable, the probability of occurrence is non-negative and is at most one. While the PMF will be zero for some values of x , it will be non-negative for all others. However, the sum of the probabilities over all possible values of X must equal 1, that is,

$$\sum P(X = x) = 1.$$

It is typically used to specify a discrete probability distribution; in other words, the PMF is a function that corresponds discrete events to the probabilities assigned to those events happening. The term ‘mass’ shows that the probabilities are concentrated on discrete events.

Suppose a fair coin is tossed two times and the sample space $S = [HH, HT, TH, TT]$. Now how many heads are to be counted. Let X be the random variable that indicates how many heads are obtained. The possible values of X are 0, 1, 2. The probability that X equals 1 is 0.5. Thus, it can be concluded that the PMF of X at 1 will be 0.5.

The properties of PMF are:

- ❑ **Non-negativity:** The probability for any specific value is always non-negative, that means, $P(X = x) \geq 0$
- ❑ **Sum of probabilities:** The sum of probabilities for all possible values of X is equal to 1, that means, $\sum P(X = x) = 1$

3.3.2 TYPES OF DISCRETE PROBABILITY DISTRIBUTIONS

Probability distribution refers to the mathematical function that depicts the probability of different possible outcomes resulting from a random experiment. Understanding these distributions is important in statistics, finance and science as they give mathematical reasoning for predicting and analysing random events. The selection of a distribution is entirely dependent upon the characteristics of the problem under study and the nature of the random variable under investigation.

Some common types of discrete probability distributions are as follows:

- ❑ **Uniform distribution:** A uniform discrete distribution is one in which all possible results have equal probabilities. All the possible results are, therefore, equally likely to occur. The distribution is ‘discrete’ because it really consists of distinct, separate outcomes. For a uniform discrete distribution with n possible outcomes, the mass function assigns the same probability to each outcome.

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The PMF is defined as: $P(X = x) = \frac{1}{n}$

Where:

$P(X = x)$ is the probability of the random variable X taking on the value

n is the total number of possible outcomes

- **Bernoulli distribution:** This is a discrete probability distribution in which every trial performed asks a question that can be answered only in the yes or no. That is, the random variable can either be 1 with probability p or it can be 0 with probability $(1-p)$. Such an experiment is termed as Bernoulli trial. A pass or fail exam can be modelled by a Bernoulli distribution.

If we have a Binomial distribution with $n=1$, then that is known as a Bernoulli distribution. Because this distribution is very intuitive, it can actually be treated as the roots for the construction of more complicated distributions. We can use the Bernoulli distribution to calculate the records of events that could only have two possible results, like success or failure.

The PMF of Bernoulli distribution can be expressed as:

$$f(x, p) = p^x (1-p)^{(1-x)}, x \in \{0, 1\}$$

Where:

$x = 0, 1, 2, 3, 4, \dots$

p = probability of success in a single experiment

- **Binomial distribution:** The Binomial distribution is a discrete probability distribution that explains the number of successes in a fixed number of independent and identically distributed Bernoulli trials. In each trial, either success occurs with probability p or failure with the probability q such that $q=1-p$. A series of such results is a Bernoulli process and is sometimes referred to as a Bernoulli trial or Bernoulli experiment if one single test for success or failure is at issue. The Binomial distribution is effectively a Bernoulli distribution if the number of trials is specified to $n=1$, or one trial. The underlying Binomial distribution is the basis for the very common binomial test employed in the search for statistical significance.

A Binomial distribution of any random variable X can be stated as: $P(x:n, p) = {}^n C_x p^x (1-p)^{n-x}$

Where:

n is the number of experiments

$x = 0, 1, 2, 3, 4, \dots$

p = probability of success in a single experiment

Table 3.1 shows the difference between Bernoulli and Binomial distributions:

TABLE 3.1: DIFFERENCE BETWEEN BERNOULLI DISTRIBUTION AND BINOMIAL DISTRIBUTION

Bernoulli Distribution	Binomial Distribution
The Bernoulli distribution is applied when we aim to represent the result of a singular trial in an event.	When the goal is to characterise the results of numerous trials of an event, the Binomial distribution is employed.
It is denoted as $X \sim \text{Bernoulli}(p)$, where p signifies the probability of success.	It is expressed as $X \sim \text{Binomial}(n, p)$, with n representing the number of trials.
The mean, denoted as $E[X]$, is equal to p .	The mean, represented as $E[X]$, is given by the product of n and p .
The variance, denoted as $\text{Var}[X]$, is equal to p times $(1 - p)$.	The variance of a random variable X , denoted as $\text{Var}[X]$, is equal to the product of the probability of success (p) and the probability of failure $(1 - p)$, multiplied by the number of trials (n).
Example: If there is an 80% probability of passing and a 20% probability of failing an exam, the Bernoulli distribution is suitable for modelling the outcomes of success (passing) or failure in this examination.	Example: Given an 80% probability of passing and a 20% probability of failing an exam, when determining the likelihood of a student passing exactly 4 out of 5 exams, the Binomial distribution is employed.

- **Poisson distribution:** A Poisson distribution is a discrete probability distribution that gives the number of times an event happens (k) within a given interval of time or space. The Poisson distribution has only one parameter λ (lambda) which is the mean number of events. 'Events' could be anything from disease cases to customer purchases to meteor strikes. The time or space interval can be described as some time or space period—for example, 10 days or 5 square inches. A Poisson distribution is applied when:

- ◆ The average frequency in a given space or time interval is known, and its value is denoted by λ (lambda).
- ◆ Events are independent and random; this tends to mean that the probability of one event does not depend on the other event.

The Poisson distribution is a common application in business analytics to model the number of occurrences of an event within a set time or space frame where those events occur at some known constant rate and are mutually independent.

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When applying the different types of discrete probability distributions in real-world scenarios, their constraints and assumptions need to be understood as each model works under specific conditions, which, if not met, may affect the accuracy of predictions. The constraints and assumptions of the different types of discrete probability distributions are as follows:

- ❑ **Uniform distribution:** There are two main assumptions in uniform distribution. They are as follows:
 - ◆ **Equal likelihood:** The uniform distribution assumes that all of the possible outcomes are equally probable. Its applicability is confined to circumstances in which each of the outcomes really does have an equal chance of occurring, such as rolling a fair die.
 - ◆ **Discrete outcomes:** The uniform distribution here is discrete, which implies that the outcomes have to be distinct and nonoverlapping. If the outcomes vary in their probability, for example, in a biased coin flip, the uniform model is not applicable.
- ❑ **Binomial distribution:** There are two main assumptions in Binomial distribution. They are as follows:
 - ◆ **Fixed number of trials:** The Binomial distribution requires a fixed number of independent trials, each with the same probability of success. This constraint makes it less suitable for variable or open-ended trial counts, such as monitoring defect rates over an indefinite period.
 - ◆ **Binary outcomes:** Each trial must result in one of two possible outcomes (success or failure). This model is inapplicable in cases with more than two outcomes unless they can be consolidated into binary categories.
 - ◆ **Independence of trials:** The Binomial distribution assumes that each trial is independent of others. In real-world cases where outcomes may influence one another (e.g., network effects in marketing), this assumption may not hold.
- ❑ **Bernoulli distribution:** There are two main assumptions in Bernoulli distribution. They are as follows:
 - ◆ **Single trial:** The Bernoulli distribution is limited to a single trial scenario. For multiple trials, the Binomial distribution is more appropriate.
 - ◆ **Binary nature:** Like the Binomial distribution, it requires only two possible outcomes for the event.

- ◆ **Constant probability:** The probability of success must remain constant. If the probability of success changes across trials (for example, due to learning effects), the Bernoulli distribution's applicability is restricted.
- **Poisson distribution:** There are two main assumptions in Poisson distribution. They are as follows:
 - ◆ **Independence of events:** The Poisson distribution assumes that events occur independently over a specified time or space interval. If events influence one another (e.g., customer visits influenced by sales events), the model's predictions could be inaccurate.
 - ◆ **Constant average rate (λ):** This distribution relies on a known, constant average rate of occurrence (λ) within each interval. Situations, where this rate fluctuates (like during seasonal sales), may not fit the Poisson model well.
 - ◆ **Rare events:** The Poisson distribution is best suited for modelling rare events within a given interval. In cases with a high frequency of occurrences or where the events are not discrete, alternative models may offer better accuracy.

Building on discrete probability distributions, continuous probability distributions further expand the concept, using PDF to model outcomes for continuous random variables, highlighting key differences with PMFs provided in the next section.



SELF ASSESSMENT QUESTIONS

5. You are counting the number of emails received per hour. Which probability distribution is appropriate for modelling the number of rare events occurring in a fixed interval of time?
 - a. Negative Binomial distribution
 - b. Poisson distribution
 - c. Uniform distribution
 - d. Binomial distribution
6. In a Bernoulli distribution, what is the mean (μ) of the distribution if the probability of success is:
 - a. 0.4
 - b. 0.6
 - c. 0.16
 - d. 0.24

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7. You are rolling a fair six-sided die. Let X be the random variable representing the outcome. What type of probability distribution does this scenario represent?
 - a. Uniform distribution
 - b. Binomial distribution
 - c. Poisson distribution
 - d. Negative binomial distribution
8. The multinomial distribution is appropriate for scenarios with a fixed number of trials and two possible outcomes. (True/ False)



ACTIVITY

You have a standard deck of 52 playing cards, including four suits (hearts, diamonds, clubs, spades) with cards numbered from 2 to 10 and face cards (jack, queen, king) along with an ace. Define a discrete random variable X representing the number on a randomly drawn card. Determine the probability distribution for X .

3.4 CONTINUOUS PROBABILITY DISTRIBUTIONS

A continuous probability distribution is used to model the probability of different outcomes in scenarios where the random variable can take on an infinite number of possible values within a given range, such as any real number between two points. In contrast to discrete probability distributions, where the variable can only assume distinct, or separate values; continuous distributions involve variables that can occupy any point in a continuous interval.

Continuous probability distributions are fundamental in statistics and probability theory, playing a crucial role in various scientific, engineering and financial applications. Understanding and working with continuous distributions allow researchers and analysts to model and analyse real-world phenomena that involve continuous measurements and uncertainties.

Some key concepts of continuous probability distribution are as follows:

- ❑ **Continuous random variables:** In continuous probability, we encounter random variables that can assume any value within a given range. These variables are often associated with measurements, such as time, distance, or weight.

- ❑ **Probability density function:** Unlike the PMF in discrete distributions, the PDF does not provide the probability of a specific point but rather the probability density over a range of values. Integrating the PDF over a range yields the probability of the variable falling within that range.
- ❑ **Cumulative distribution function:** The cumulative distribution function for continuous distributions is an integral part of understanding the cumulative probability up to a certain point. It gives us insights into the likelihood of a random variable taking on values less than or equal to a specific point.
- ❑ **Expected value and variance:** Just as in discrete probability, continuous distributions have measures that describe their central tendency and spread. The expected value (mean) and variance provide a summary of the location and variability of the distribution.

3.4.1 PROBABILITY DENSITY FUNCTION (PDF)

The Probability Density Function (PDF) is a fundamental concept in statistics as it shows the probability of a particular value being taken on by a random variable. It is mostly used in continuous probability distributions, where the variable can take any value in a certain range instead of discrete values. For a continuous random variable X , the PDF(x) has the following properties:

- ❑ **Non-negative:** The PDF is always non-negative for all x :

$$f(x) \geq 0 \text{ for all } x$$

- ❑ **Total probability is 1:** The total area under the PDF curve, representing the total probability of all possible outcomes, must equal 1:

$$\int_{-\infty}^{\infty} f(x) \, dx = 1$$

This property ensures that the random variable with certainty assumes some value from the entire range.

- ❑ **Probability over an interval:** The probability that the random variable X lies between two values a and b is given by the integral of the PDF over that interval:

$$P(a \leq X \leq b) = \int_a^b f(x) \, dx$$

The value of the PDF at a particular point cannot be identified with a probability (because the probability would be extremely small), while the integral over an interval provides the probability that X will fall into that interval.

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3.4.2 TYPES OF CONTINUOUS PROBABILITY DISTRIBUTIONS

There are several types of continuous probability distributions, each with its own characteristics and applications. Refer to Figure 3.2 to get a brief idea about the visualisation of the continuous probability distributions graphically:

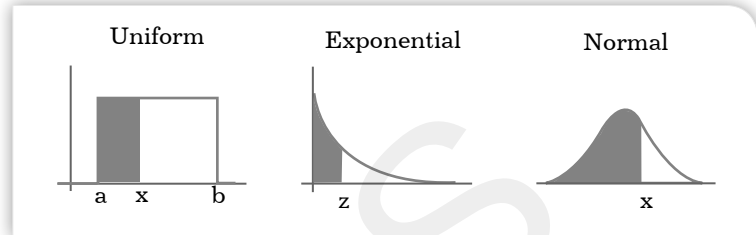


Figure 3.2: Types of Continuous Probability Distributions

The common types of continuous probability distributions as shown in Figure 3.2 are:

- ❑ **Uniform distribution:** Uniform distribution is that probability distribution in which all the outcomes lying within a given range have an equal chance of occurrence. This distribution is termed as 'uniform' because its PDF is constant over the range, resulting in the form of a flat curve or uniform shape. It is a continuous distribution for real numbers and a discrete distribution for integers. The PDF of the continuous uniform distribution and the PMF of the discrete version also reflect equal likelihood of all values within the defined range. The graphical representation of uniform distribution is shown in Figure 3.3:

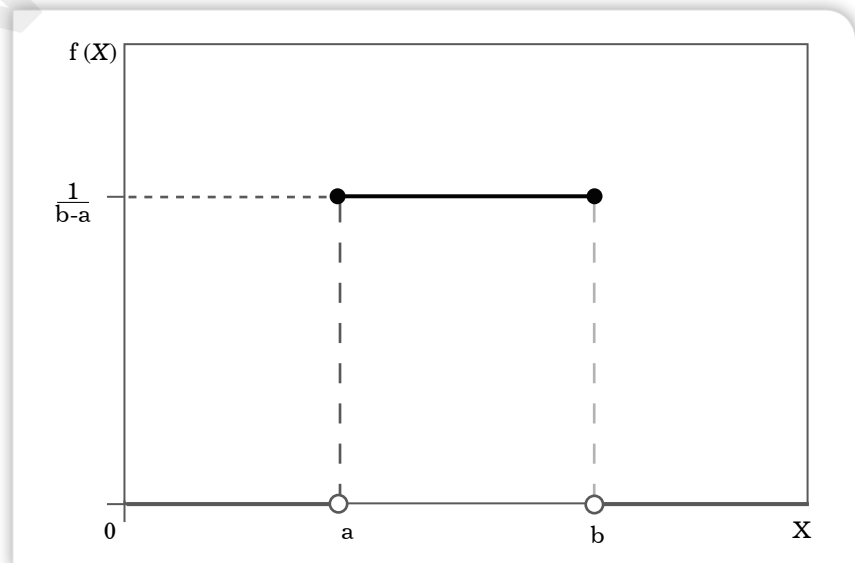


Figure 3.3: Uniform Distribution

In a uniform distribution, the chance of each value occurring is equal. When graphed, the distribution looks like the same height for every possible outcome and is rectangular; therefore, it can sometimes be referred to as a 'rectangular distribution.' With the drawing of a card from a deck of cards, the chances of drawing any suit—for example, heart, spade, diamond, or club are equal, with a $1/4$ (or 25%) chance of each of these possibilities.

An example of a continuous uniform distribution is selecting a random value between two points, such as any decimal value between 0 and 1. In this case, each value within the range 0 to 1 has an equal chance of being chosen, and the PDF is constant across the interval, giving a rectangular shape when graphed. On a graph, such a distribution looks like a horizontal line where each outcome sits on the x-axis and the value is on the y-axis.

Some of the common applications of uniform distribution in business analytics are as follows:

- ◆ **Random sampling:** A uniform distribution ensures that all members of a population have an equal probability of being selected when sampling. As a result, it makes it possible for customer perceptions or behaviour to be objectively depicted in surveys and market research.
- ◆ **Quality control:** Uniform distribution can be used for manufacturing to replicate the range of product dimensions or attributes with predetermined bounds. This helps to define allowable limits for product attributes, leading to consistency in production quality.
- ◆ **Marketing strategies:** In case all the channels have the same possibility of response, then uniform distribution is permitted in the promotional strategies. Equally on the basis of the probability of the response, a firm can measure which marketing channel is performing better and which channel it will need to change in its marketing mix.
- ◆ **Game theory and decision-making:** The uniform distribution can be applied to the modelling of uncertainty in decision-making situations whereby each outcome is given an equal probability of occurrence. This would be an example for the business to analyse different strategies and understand the possible risks in a competitive environment.
- ◆ **Pricing strategies:** In new product testing, uniform distribution models the response from customers over a range of price points where all prices are equally probable. This aids in the establishment of the optimum revenue maximising pricing strategy under equal-probability assumptions.
- **Exponential distribution:** This is a continuous probability distribution that describes the time elapsed between events in a Poisson process, where events occur continuously and

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independently at an average constant rate. It may be used to model the time until an event occurs, that is, the time elapsed between one event and the next one, e.g., the time when a radioactive atom decays, the time when a customer comes to a service point, or the time until a piece of equipment fails. The PDF of the exponential distribution, applicable for $x \geq 0$, is given as: $f(x|\lambda) = \lambda e^{-\lambda x}$

Where:

x is the random variable that describes the time until an event takes place

λ (lambda) is the rate parameter, the average number of events per unit time

A rough plot of the exponential distribution is given in Figure 3.4:

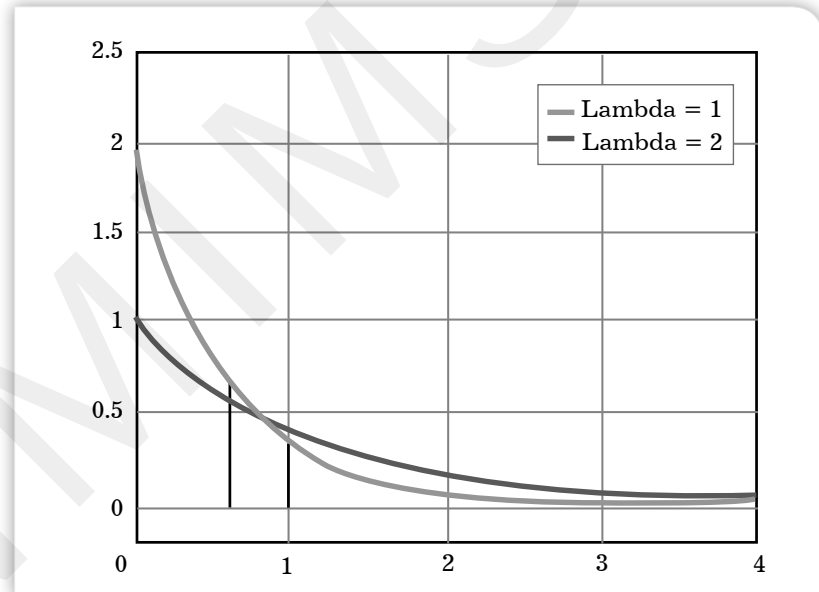


Figure 3.4: Exponential Distribution Plot

Some of the common applications of exponential distribution are as follows:

- ◆ **Queuing theory:** It is used to model the time between arrivals of customers in a queue or the time between service completions.
- ◆ **Reliability engineering:** The exponential distribution is often used to model the time until failure of a component or system with a constant failure rate.
- ◆ **Telecommunications:** In the study of telecommunication systems, the exponential distribution is used to model the time between arrivals of requests for service.
- ◆ **Physics and radioactive decay:** The exponential distribution is used to model the time until a radioactive atom undergoes decay.

◆ **Economics:** In economic models, the exponential distribution can be applied to model the time until an economic event occurs.

□ **Normal distribution (Gaussian distribution):** A normal distribution, also known as a Gaussian distribution, is a continuous probability distribution that is symmetric around its mean. The distribution is bell-shaped as shown in Figure 3.2, where most data points cluster around the mean and tails go off to infinity in both directions. Normal distributions apply very frequently in statistics and other fields because of mathematical properties and because many phenomena within the natural world exhibit a form of distribution that closely approximates normality. These distributions are found in many applications of hypothesis testing and for other estimations such as confidence intervals, among a wide range of analyses that involve statistics.

The Z-score quantifies the number of standard deviations that a specific data point deviates from the mean. It is computed as follows: $Z = \frac{x - \mu}{\sigma}$,

Where:

x is the data point,

μ is the mean of distribution, and

σ is the standard deviation.

The normal distribution has several common applications in business analytics that include:

- ◆ **Sales forecasting:** The use of historical sales data for companies is one method of forecasting future sales performance using normal distribution. Assuming that most cases are normally distributed, it becomes easier to be able to predict the demand, plan the inventory and allocate the resources more effectively. Assuming normal distribution, businesses are able to determine confidence intervals for expected sales and detect deviations such as anomalously high or low demand.
- ◆ **Risk management:** Banks and businesses estimate risks through return-loss modelling based on the assumption, most of the time, of normal distribution for returns over some period of time. Businesses make use of the normal distribution to model and measure their risk and therefore calculate Value at Risk (VaR) and decisions regarding investments or strategic planning based on uncertainty.
- ◆ **Market research and customer analytics:** Normal distribution describes the scores of customer satisfaction, any other results of the survey, or even the response times in a market research exercise. Businesses rely on normal distribution to analyse

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the behaviour, preferences, and feedback of customers to come up with retention strategies and strategies meant to drive improved customer services.

- ♦ **Pricing strategies:** Businesses use normal distribution in the analysis of pricing data for determining elasticity of demand. All the given settings optimise the pricing strategy by understanding the degree to which price changes impact the level of demand, assuming a normally distributed behaviour.

3.4.3 DIFFERENCE BETWEEN PDF AND PMF

The PMF and PDF are two distinct types of functions used in probability and statistics to describe distributions of random variables. Table 3.2 shows the difference between PDF and PMF:

TABLE 3.2: DIFFERENCE BETWEEN PDF AND PMF

	Probability Density Function (PDF)	Probability Mass Function (PMF)
Type of random variable	The PDF is used for continuous random variables, which can take any value within a range.	The PMF is used for discrete random variables, which can only take distinct, separate values.
Denotation symbol	It is often denoted as $f(x)$, where x represents a specific value of the continuous random variable.	It is often denoted as $P(X=x)$, where X represents the discrete random variable and x is a specific value it can take.
Probability at specific points	The probability of a continuous random variable taking on a specific point value is essentially zero. This is because the area under the curve over a single point is zero.	The PMF gives the probability that the discrete random variable takes on a specific value. The probability is non-zero for each distinct value the variable can assume.
Interpretation	The area under the PDF curve represents the probability that the continuous random variable falls within a specific interval. The probability is calculated as the integral of the PDF over that interval.	For a PMF, the probability of a discrete random variable falling within a specific interval (between two points) is the sum of the probabilities at those individual points. In contrast, for a PDF, the probability of a continuous random variable falling within an interval is represented by the area under the PDF curve over that interval.

**SELF ASSESSMENT QUESTIONS**

9. What is a characteristic feature of the normal distribution?
 - a. Heavy tails
 - b. Undefined mean
 - c. Symmetrical shape
 - d. Defined variance
10. Which distribution is used to model the time until a specified number of events occur in a Poisson process?
 - a. Normal distribution
 - b. Exponential distribution
 - c. Beta distribution
 - d. Uniform distribution
11. The area under the PDF curve for a continuous probability distribution represents the probability of a random variable falling within a certain range. (True/False)
12. The sum of the probabilities in the PMF for a discrete distribution is always equal to 1. (True/False)

**ACTIVITY**

You are tasked with measuring the heights of students in your class. Heights are recorded in inches, and you want to analyse the continuous probability distribution of the height variable. Define a continuous random variable X representing the height of a randomly selected student. Define the PDF for X , representing the height of a randomly selected student.

3.5 SUMMARY

- ❑ The fundamental principle of counting is a technique employed to determine the overall number of possible outcomes for various events or choices. A permutation is an arrangement of objects in a specific order.
- ❑ A combination is a way of choosing or arranging things without considering the order.
- ❑ A discrete probability distribution describes the likelihood of occurrence of each possible outcome in a set of distinct and countable events.
- ❑ A random variable is a numerical quantity associated with each outcome in the sample space.

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- ❑ The PMF of a discrete random variable X is a function that describes the probability of each possible outcome x in the sample space.
- ❑ Some common types of discrete probability distributions are uniform distribution, Binomial distribution, Poisson distribution, etc.
- ❑ A continuous probability distribution is a statistical concept used to model the probability of different outcomes in a scenario where the random variable can take on an infinite number of possible values within a given range.
- ❑ The PDF, denoted as $f(x)$, describes the probability distribution of a continuous random variable.
- ❑ The PDF is for continuous random variables, and PMF is for discrete random variables.
- ❑ The PDF deals with intervals and has zero probabilities at specific points, while PMF deals with specific points and the probabilities are non-zero at those points.



KEY WORDS

- ❑ **Combination:** It is a selection of objects without considering the order. The number of combinations of n elements is denoted as $C(n,r)$.
- ❑ **Discrete random variable:** It is a variable that can take on a countable number of distinct values.
- ❑ **Expected value (Mean):** It is the average value of a random variable, calculated by summing each possible value multiplied by its probability.
- ❑ **Factorial:** It is the product of all positive integers up to a given number denoted by $n!$, where n is a positive integer.
- ❑ **Fundamental counting principle:** It is the principle that states that if there are n_1 ways to do one thing and n_2 ways to do another, then there are $n_1 \times n_2$ ways to do both.
- ❑ **Mode:** It is the value(s) of a random variable with the highest probability in a probability distribution.
- ❑ **Permutation:** It is an arrangement of objects in a specific order. The number of permutations of r elements taken from a set of n elements is denoted as $P(n,r)$.
- ❑ **Probability distribution:** It is a function that describes the likelihood of obtaining the possible values that a random variable can take.
- ❑ **PMF:** It is a function that gives the probability of a discrete random variable taking on a specific value.

- ❑ **Skewness:** It is a measure of the asymmetry of a probability distribution.
- ❑ **Standard deviation:** It is the square root of the variance, providing a measure of the average distance between each value and the mean.
- ❑ **Variance:** It is a measure of the spread or dispersion of a probability distribution.

3.6 MULTIPLE CHOICE QUESTIONS

1. The fundamental principle of counting states that if there are 'm' ways to do one thing and 'n' ways to do another thing, then the total number of ways to do both is:
 - a. $m + n$
 - b. $m \times n$
 - c. m / n
 - d. $m - n$
2. In which of the following situations is a combination used instead of a permutation?
 - a. Assigning specific seats to students in a classroom
 - b. Arranging books on a shelf in a particular order
 - c. Selecting a team of 5 members from a group of 10 people
 - d. Ordering items in an online shopping cart
3. What does the PMF describe?
 - a. The probability of a continuous random variable.
 - b. The cumulative probability up to a certain value.
 - c. The probability associated with each possible value of a discrete random variable.
 - d. The total number of possible outcomes.
4. The Bernoulli distribution is a special case of the Binomial distribution where:
 - a. $n = 2$
 - b. $n = 1$
 - c. $p = 0.5$
 - d. $p = 1$
5. Which of the following is not a key characteristic of a discrete probability distribution?
 - a. PMF
 - b. Sample space and events
 - c. Cumulative Distribution Function (CDF)
 - d. Normal distribution

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6. A Poisson distribution is most appropriate for modelling which type of situation?
 - a. The number of defective items in a batch of products.
 - b. The probability of heads or tails in a coin flip.
 - c. The number of website visitors in a given hour.
 - d. The number of students in a class.
7. Which of the following is true for the PDF of a continuous random variable?
 - a. The PDF provides the exact probability of the random variable assuming a specific value.
 - b. The PDF is always non-negative and integrates to 1 over its entire range.
 - c. The PDF gives the cumulative probability up to a specific point.
 - d. The PDF is only applicable to discrete random variables.
8. In which type of probability distribution do all outcomes within a given range have equal probability of occurrence?
 - a. Exponential distribution b. Normal distribution
 - c. Uniform distribution d. Poisson distribution
9. The time between events in a Poisson process is best modelled using which probability distribution?
 - a. Uniform distribution b. Exponential distribution
 - c. Normal distribution d. Binomial distribution
10. What is the shape of the probability distribution curve for a normally distributed variable?
 - a. Rectangular b. Exponential
 - c. Bell-shaped d. Triangular

3.7 DESCRIPTIVE QUESTIONS

1. Differentiate between PMF and PDF with the help of an example.
2. What are the applications of permutations and combinations?
3. What are the differences between Binomial and Bernoulli distribution?
4. List two properties of a valid PDF.
5. How is permutation different from the fundamental principle of counting? Explain with the help of an example.

3.8

HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. You have a collection of six different books, each from a unique genre. You want to arrange these books on a shelf. Considering the rule that Books A and B must always be next to each other, determine the total number of arrangements.
2. You have a modified deck of cards where certain cards are marked with specific symbols. The deck consists of 36 cards with four suits (hearts, diamonds, clubs, spades) and numbered cards 2 through 9, each with a unique symbol. The marked cards include one heart, one diamond, one club and one spade, each with the symbol 'M.' Define a discrete random variable X representing the number on a randomly drawn card. Determine the probability distribution for X considering both numbered cards and marked cards.
3. You are tasked with modelling the arrival times of an elevator in a building. The continuous random variable X represents the time (in minutes) a person has to wait for the elevator. Define a continuous PDF for the random variable X representing the arrival times of the elevator. Choose a suitable distribution for this scenario.
4. Consider a random variable X representing the number of heads obtained when tossing a fair coin three times. Which of the following represents the PMF for X ?
 - a. $P(X=0) = \frac{1}{8}, P(X=1) = \frac{3}{8}, P(X=2) = \frac{3}{8}, P(X=3) = \frac{1}{8}$
 - b. $f(x) = \frac{1}{4}$ for $x = 0, 1, 2, 3$
 - c. $P(X=0) = \frac{1}{2}, P(X=1) = \frac{1}{4}, P(X=2) = \frac{1}{8}, P(X=3) = \frac{1}{8}$
 - d. $f(x) = \frac{1}{2}$ for $x = 0, 1; f(x) = \frac{1}{4}$ for $x = 2; f(x) = \frac{1}{8}$ for $x = 3$
5. Consider a scenario where you are conducting a series of independent experiments, each with a probability of success p . Let X be the random variable representing the number of successes in n trials, following a Binomial distribution: $P(X=k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$. Which of the following represents the combination notation $\binom{n}{k}$?
 - a. $\frac{n!}{k!(n-k)!}$

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$$\text{b. } \frac{k!}{n!(n-k)!}$$

$$\text{c. } \frac{n!}{k!}$$

$$\text{d. } \frac{n!}{(n-k)!}$$

3.9 ANSWERS AND HINTS**ANSWERS FOR SELF ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Principles of Counting	1.	d. 10,000
	2.	c. 1326
	3.	d. 24
	4.	False
Discrete Probability Distributions	5.	b. Poisson distribution
	6.	a. 0.4
	7.	a. Uniform distribution
	8.	False
Continuous Probability Distributions	9.	c. Symmetric shape
	10.	b. Exponential distribution
	11.	True
	12.	True

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. $m \times n$
2.	c. Selecting a team of 5 members from a group of 10 people
3.	c. The probability associated with each possible value of a discrete random variable.
4.	b. $n = 1$
5.	d. Normal distribution
6.	c. The number of website visitors in a given hour.
7.	b. The PDF is always non-negative and integrates to 1 over its entire range.

Q. No.	Answer
8.	c. Uniform distribution
9.	b. Exponential distribution
10.	c. Bell-shaped

HINTS FOR DESCRIPTIVE QUESTIONS

1. The PDF is employed for continuous random variables, which can assume any value within a given range. On the other hand, the PMF is utilised for discrete random variables, which can only take distinct and separate values. It is important to note that the probability of a continuous random variable attaining a specific point value is essentially zero. In contrast, the PMF provides the probability associated with the discrete random variable taking on a particular value. Refer to Section 3.4
2. Permutations find practical applications in diverse fields, including but not limited to product placement, advertisement scheduling, sales sequencing, revenue management and scenario planning. On the other hand, combinations are utilised in various practical scenarios such as product bundling and pricing strategies, market basket analysis, customer segmentation, resource allocation and project portfolio management. Refer to Section 3.2
3. A Bernoulli distribution is characterised by a discrete probability distribution where the random variable has only two possible outcomes. In a Bernoulli trial, a random variable assuming the value of 1 signifies a success. The Binomial distribution, another discrete probability distribution, explains the count of successes in a predetermined number of independent and identically distributed Bernoulli trials. Refer to Section 3.3
4. A valid PDF for a continuous random variable must satisfy the following two properties: The PDF must be non-negative for all possible values of the random variable. Mathematically, for any real number x , the PDF $f(x)$ should be greater than or equal to zero: $f(x) \geq 0$. The total area under the probability density curve over the entire range of possible values must be equal to 1. Refer to Section 3.4
5. The fundamental principle of counting, also known as the counting rule or the multiplication rule, is a basic combinatorial principle used to determine the total number of outcomes in a sequence of events. A permutation is a specific arrangement or ordering of a set of elements. Refer to Section 3.2

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ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.

Answer

1. Arrangements with Books A and B together:
Treat Books A and B as a single entity (AB) when arranging.
Now, you have 5 entities to arrange: A, B, C, D, E, F
Total arrangements with AB together = $5!$, that means
 $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

2. Identify the probability of each outcome $X = x$, considering the modified deck:

X (Number on card)	2	3	4	5	6	7	8	9	M
P(X=x)	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{4}{36}$

3. Choose a suitable continuous distribution for elevator arrival times, such as the exponential distribution. The PDF for the exponential distribution is given by: $f(x, \lambda) = \lambda e^{-\lambda x}$.

4. a. $P(X=0) = \frac{1}{8}, P(X=1) = \frac{3}{8}, P(X=2) = \frac{3}{8}, P(X=3) = \frac{1}{8}$

5. a. $\frac{n!}{k!(n-k)!}$

3.10 SUGGESTED READINGS & REFERENCES**SUGGESTED READINGS**

- ❑ *Business analytics* (2023). S.l.: CENGAGE LEARNING.
- ❑ *Camm, J.D. et al. (2024) Business analytics: Descriptive, predictive, prescriptive*. Boston, MA: Cengage Learning.
- ❑ *Business analytics* (2023a). S.l.: MCGRAW-HILL EDUCATION.

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- ❑ Britannica, T. Editors of. (n.d.). *Permutation*. Encyclopedia Britannica. <https://www.britannica.com/science/permutation>
- ❑ *Probability Mass Function | PMF* (n.d.). Retrieved from https://www.probabilitycourse.com/chapter3/3_1_3_pmf.php
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CASE STUDY

N O T E S

PREDICTING FLIGHT DELAYS USING PROBABILITY THEORY

SCENARIO

You are an airline data analyst tasked with developing a model to predict flight delays. Knowing which flights are likely to be delayed can help airlines improve operational efficiency, minimise passenger inconvenience and optimise resource allocation.

DATA AND TOOLS

- ❑ Historical flight data, including departure/arrival times, origin/destination airports, weather conditions, aircraft type, maintenance records, etc.
- ❑ Statistical and machine learning tools like Python libraries (e.g., pandas, scikit-learn)

CHALLENGES

- ❑ Data quality and completeness are crucial for model accuracy.
- ❑ Real-time data integration and model updates are necessary.
- ❑ Accounting for external factors like unexpected events or air traffic control decisions.

APPROACH

1. **Data cleaning and preprocessing:** Organise and clean the historical data, handle missing values and encode categorical variables.
2. **Feature engineering:** Create new features based on existing data that might be relevant to predicting delays (e.g., flight duration, distance between airports, average delay at origin/destination airports).
3. **Exploratory Data Analysis (EDA):** Analyse the data to understand patterns and relationships between features and delays. Visualise delay distribution per day, month, year, airlines, weather conditions, etc. Identify potential factors contributing to delays.
4. **Model building and training:** Choose appropriate machine learning models for predicting delays (e.g., Logistic Regression, Random Forest, Gradient Boosting). Train the models on the prepared data, evaluating their performance using metrics such as accuracy, precision, recall and AUC-ROC.



Case Objective

The objective of this case study is to develop a probability theory-based model for predicting flight delays. This initiative aims to enhance airline operational efficiency, minimise passenger inconvenience and optimise resource allocation for improved overall performance.

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CASE STUDY

5. **Model evaluation and improvement:** Fine-tune the models and select the best performing one based on validation and testing datasets. Analyse model errors and identify areas for improvement.

OUTCOMES

- ❑ The model might predict that flights with a specific aircraft type or departing during bad weather have a higher probability of delay.
- ❑ Airlines can prioritise de-icing or equipment maintenance for flights identified as high-risk.
- ❑ Passengers can be informed about potential delays and adjust travel plans accordingly.

BENEFITS

- ❑ Reduced flight delays and improved on-time performance
- ❑ Enhanced passenger satisfaction and brand reputation
- ❑ Optimised resource allocation for ground crew and maintenance personnel
- ❑ Data-driven decision-making for airlines

**QUESTIONS**

1. What role does feature engineering play in improving flight delay predictions?

(**Hint:** Feature engineering allows for the creation of new indicators from existing data, potentially revealing relationships that directly impact flight delays.)

2. How can airlines leverage the model's predictions to enhance operational efficiency?

(**Hint:** By identifying high-risk flights, airlines can strategically allocate resources, prioritise maintenance, and inform passengers, minimising delays.)

4

C H A P T E R

STATISTICAL INFERENCE

CONTENTS

- 4.1 Introduction
- 4.2 Hypothesis Testing
 - 4.2.1 p-values and High p-values
 - 4.2.2 One-Sample Hypothesis Tests
 - 4.2.3 Two-Sample Hypothesis Tests
 - 4.2.4 Types of Errors
 - Self Assessment Questions
 - Activity
- 4.3 Chi-square Test
 - 4.3.1 Chi-square Test for Independence
 - 4.3.2 Chi-square Goodness of Fit Test
 - Self Assessment Questions
 - Activity
- 4.4 Summary
- 4.5 Multiple Choice Questions
- 4.6 Descriptive Questions
- 4.7 Higher Order Thinking Skills (HOTS) Questions
- 4.8 Answers and Hints
- 4.9 Suggested Readings & References

N O T E S

**Case Objective**

The objective of this caselet is to apply statistical inference to identify factors causing customer dissatisfaction and allow strategic interventions for improved satisfaction and customer loyalty.

INTRODUCTORY CASELET**ENHANCING CUSTOMER SATISFACTION THROUGH STATISTICAL INFERENCE**

Mehta E-commerce is an online retailing major that is dealing with complaints of customer dissatisfaction, noticeable from the decrease in ratings provided by the customers and increasing negative feedback. Managers assume it will have a significant impact on customer retention as well as overall business performance. Therefore, the company applied statistical inference techniques to understand factors affecting customer satisfaction and effective solutions for them.

The managers, at first, were not able to determine what factors, such as the quality of the products, the speed of delivery times and the support provided to customers, play the most important roles in the satisfaction of a customer. The company hoped to find a balance between gathering adequate information to create meaningful results without becoming too taxing on the customer with frequent surveys. Further, statistical analysis had to be performed to make the right conclusions that can be obtained from the data gathered. The company further analysed why some customers are dissatisfied. This included identification of the key areas that were being given major causes of the poor customer experience at the company's operations. It also established a system of continuous monitoring of customer satisfaction metrics and applied statistical process control techniques to detect shifts or trends in the trend of satisfaction levels so that interventions may be sought in a timely manner.

The use of causal analysis allowed the company to closely understand the underlying reasons for the dissatisfaction of customers. Strategies that were implemented culminated into highly visible improvements in customer satisfaction ratings that led to a highly improved level of customer loyalty and word-of-mouth publicity. Understanding the statistical significance of various factors helped Mehta E-commerce to effectively allocate its resources to critical areas affecting customer satisfaction.



LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Understand the fundamentals and processes of testing hypotheses
- Apply the Chi-square test using datasets in Excel

4.1 INTRODUCTION

In the previous chapter, we discussed the fundamental counting principle, along with discrete and continuous probability distributions and their types.

Inferential statistics is a statistical branch dedicated to concluding entire populations based on representative data samples. It enables predictions and inferences about larger groups from analysed subsets to generalise findings from samples into whole populations. Inferential statistics use statistical methods to conclude, predict or deduce conclusions about a larger population, relying on appropriate sample selection. This includes methods like the Chi-square test for independence and the Chi-square goodness of fit test, which are essential tools in determining relationships between variables and assessing how well the data fits a given distribution.

Inferential statistics include methods like hypothesis testing, confidence intervals and regression analysis. Hypothesis testing tests the evidence against a known null hypothesis, while confidence intervals give an idea of the likely range for a population parameter. On the other hand, regression analysis examines the relationship between variables towards predictions or estimations.

Most often a student of statistics in an MBA programme would wonder, why should one study the subject of statistics, especially in the age of software tools? A simple answer would be that statistics is the base for business decisions and is subconsciously applied in daily routine. Consider a new grocery retail outlet has opened and the owner claims that the grains he stores are the best. How do you either reject or not reject his claim? A school going student says that the burden of education has increased. Do we really agree with this statement or do we disagree with this statement? A luxury bus manufacturer claims that the new line of luxury buses is better than the earlier line of the same product. Do we take the claim at face-value or we would like to take a ride first before we draw any conclusions? In most of these cases, relying on the gut feel is not the right process. We need proper irrefutable scientific evidence which would address all possibilities. If we move from the daily life examples and consider business examples like rejecting a lot or not rejecting a lot. The supplier would always feel that he has been wronged if his products get rejected. In worst cases these kinds of disputes might land up

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in arbitration courts, where the need for scientific evidence is very important. Testing of hypothesis is one of the important inferential statistics applications. The process of hypothesis testing commences with stating the null hypothesis and the alternate hypothesis. Null hypothesis is a statement of a known fact or something that does not need proof or something that is “as is”. The alternate hypothesis on the contrary is a bold statement, that requires justification and proof. “Sachin Tendulkar is a great cricketer” is a statement that would not require any proof. However, a statement “Sachin Tendulkar is not a great cricketer” would require substantial proof.

In this unit, we will first discuss the hypothesis testing process, followed by a discussion on hypothesis testing with the p-value and subsequently solving multiple questions. The chapter also discusses the Chi-square test.

4.2 HYPOTHESIS TESTING

Hypothesis testing is a technique of creating inferences from sample data considering the hypothesised values for population parameters. This type of process includes the designing of hypotheses, collecting and analysing data and then outlining conclusions according to the population based on the data provided by the sample. Hypothesis testing is used to conclude characteristics of populations from sample data across various domains of application, including psychology, biology, economics and so on.

Hypothesis testing is a statistical method used to make inferences about a population based on sample data. Although the process is focused on the population, it is often impractical to study the entire population, so sample data are used to assess hypotheses about population parameters. The goal of hypothesis testing is to determine whether there is enough evidence in the sample to support a particular claim about the population. The null hypothesis (H_0) and the alternative hypothesis (H_1) represent statements about the population. The null hypothesis often represents a position of “no effect” or “no difference,” while the alternative hypothesis suggests the presence of an effect or difference.

In hypothesis testing, Greek symbols are typically used to represent population parameters, such as μ for the population mean, σ for the population standard deviation, and p for the population proportion. Roman symbols are used to represent sample statistics, such as $\bar{x}$ for the sample mean, s for the sample standard deviation, and $\hat{p}$ for the sample proportion. The burden of proof in hypothesis testing lies with the alternative hypothesis (H_1). We assume the null hypothesis (H_0) is true by default and look for evidence to reject it in favor of the alternative hypothesis. The null hypothesis is generally considered a claim that requires no justification, as it often represents a status quo or a hypothesis of no effect. If the sample data provide sufficient evidence against the null hypothesis, we reject it in favor of the

alternative hypothesis. If the sample evidence is insufficient, we fail to reject the null hypothesis, meaning we do not have enough evidence to support the claim in the alternative hypothesis.

There are essentially five steps in the entire process of hypothesis testing and it is necessary to understand the significance of each of these steps. The steps are discussed below.

Step 1: This step involves formulating the null hypothesis, H_0 , and the alternate hypothesis, H_1 . Some authors refer to the alternate hypothesis as H_a ; however, it is preferable to use H_1 since the null hypothesis includes “0”.

An essential aspect of the null hypothesis is the use of the “=” sign, which signifies an “as-is, where-is” condition. While some authors utilise notations like “ $\geq$ ” or “ $\leq$,” these are not incorrect but may complicate understanding. For clarity and simplicity, the use of “=” is sufficient.

For the alternate hypothesis, the symbols “ $\neq$ ” or similar ones are employed as necessary. The distinction between the null hypothesis and the alternate hypothesis lies primarily in the equality sign, with other aspects remaining unchanged. Hypothesis testing always centres on the null hypothesis, and the primary objective is to either “reject” the null hypothesis or “not reject” it. The alternate hypothesis is “accepted” only if the null hypothesis is rejected.

In simpler terms, it is assumed that any difference between the null hypothesis value and the observed sample value is due to random or chance events. This does not imply an error. However, if the test reveals that the difference is significant, it is concluded that the difference is unlikely to have occurred by chance. In such cases, the null hypothesis is rejected.

Step 2: At this stage, we select the level of significance, α , (in simple words the chance of making an error) with the default α being 0.05. Note that $\alpha = 1 - \text{confidence interval}$ and when we say α is 5% or 0.05, we are saying 95% confidence interval.

Step 3: This step involves identifying the test statistic and setting the decision rule. Remember any test derives meaning only when you have set the pass/fail threshold. At this stage, we will select the appropriate test statistics (z , t , χ^2 , F) and decide on the “reject”/ “do not reject” threshold. The test can be a two-tailed test (if H_1 has “ $\neq$ ” relation) or a one-tailed test (left-tail if H_1 has “ $<$ ” relation, or right-tail if H_1 has “ $>$ ” relation). Identifying the test statistic and setting the decision rule is crucial. Any statistical test has meaning only when a pass/fail threshold (decision criterion) is established. At this stage, the appropriate test statistic (e.g., z , t , χ^2 , F) is selected and the “reject” or “do not reject” threshold is determined. The type of test such as two-tailed, left-tailed or right-tailed depends on the alternate hypothesis (H_1) as shown below:

□ A two-tailed test is used if H_1 includes a “ $\neq$ ” relationship.

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- ❑ A left-tailed test is used if H_1 includes a “<” relationship.
- ❑ A right-tailed test is used if H_1 includes a “>” relationship.

Additionally, care must be taken to differentiate between actual outcomes and conclusions drawn from the test. For example:

- ❑ **Actual hypothesis:** False
- ❑ **Conclusion based on test:** True

Figure 4.1 illustrates the “Reject” and “Do Not Reject” areas for a two-tailed test when H_1 has a “ $\neq$ ” relationship.

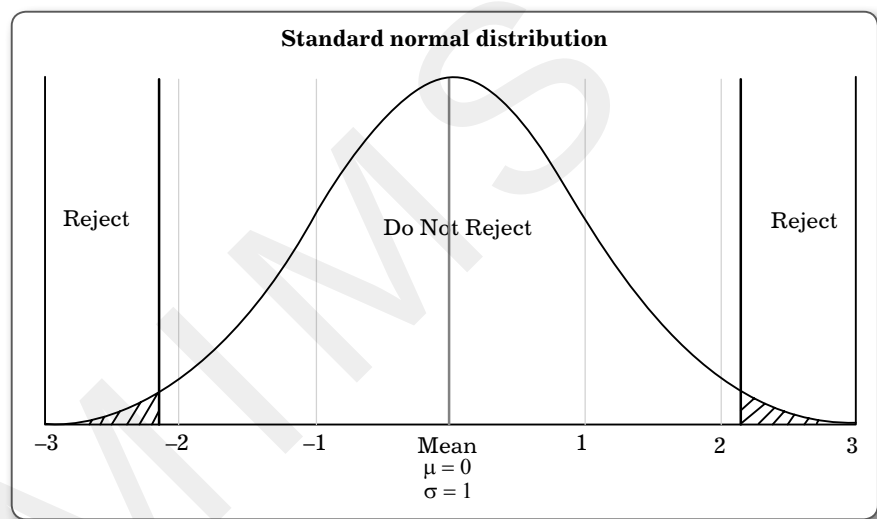


Figure 4.1: “Reject” and “Do Not Reject” Areas When H_1 has “ $\neq$ ” Relationship

Figure 4.2 illustrates the “Reject” and “Do Not Reject” areas for a two-tailed test when H_1 has a “<” relationship.

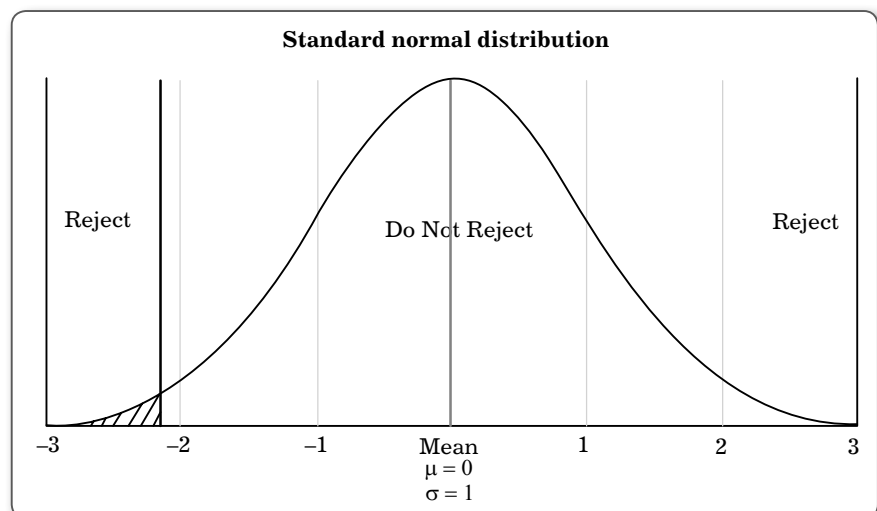


Figure 4.2: “Reject” and “Do Not Reject” Areas When H_1 has “<” Relationship

Once the reject area is correctly identified, the table value of the statistic becomes the critical value, signifying “reject” and “do not reject” statistic values. We can obtain the z_{critical} values or the t_{critical} values or the χ^2_{critical} values or the F_{critical} values from the respective tables. Alternately, all these values can be obtained by using the appropriate excel function.

Step 4: The next logical step in the process is to perform the test. With this we mean that we find out the test statistic value using the sample observations and the expected population parameter, as specified by the null hypothesis. This is then checked vis-à-vis the critical values, identified in step 3 earlier. If the calculated value of the statistic falls in the reject area, we reject the null hypothesis. We do not reject the null hypothesis when the calculated value of the statistic is in the do not reject area. These conclusions based on scientific evidence and proven statistical techniques cannot be challenged.

Step 5: Based on whether the null hypothesis is rejected or not, further subject related conclusions are drawn. Technically, step 5 is very important because the process in steps 1 to 4 are mechanical while step 5 requires application of mind. Further, multiple excel function can perform steps 1 to 4, but step 5 requires managerial intellect and intervention.

Testing of Hypothesis – p-Value Approach

At times, just knowing whether a null hypothesis is rejected or not is insufficient. If there is a question, whether you have sufficient confidence to reject a null hypothesis, it would be difficult to answer without more information. Consider two cases of students appearing in an exam where the minimum passing marks is 50. Student A get 51 marks and is declared “pass” while student B gets 48 marks and is declared “fail”. When would there be more concern? When a student just clears the passing hurdle or when the student falls just short of the passing hurdle? Now, consider a third case of a student C who has received 20 marks and is declared “failed”. Will there be any doubt in our minds to fail the student C? you will agree that out of the three students the results of student B would be putting a question mark in our minds, like whether the total of marks was correct, whether the assessments were fair or whether the examiner was correct in failing the student by 2 marks. Likewise, when we reject a null hypothesis, there might be questions on authenticity of data or the correctness of the sampling process because to “reject” is a major decision or a bold decision, whereas to “not reject” the null hypothesis is no big deal after all. The p-value approach firstly tells us whether to reject or not reject the null hypothesis. The p-value method for hypothesis testing involves calculating this probability to assess whether there is sufficient evidence to reject the null hypothesis. A p-value represents the probability of observing the data, or something more extreme, assuming the null hypothesis is true. It serves as an alternative to

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using rejection criteria and indicates the smallest level of significance at which the null hypothesis would be rejected; a lower p-value suggests stronger evidence against the null hypothesis in favour of the alternative hypothesis.

Further, it gives us the confidence to reject or tells us that perhaps there is a requirement to redo the process again.

Reject H_0 : If p-value is $\leq \alpha$

Do not reject H_0 : If p-value is $> \alpha$

A third advantage is that most software will automatically give you the p-value and all that you need to do is to compare this p-value with level of significance, α , to draw your conclusions. Figure 4.3 shows the statistical significance of p-values:

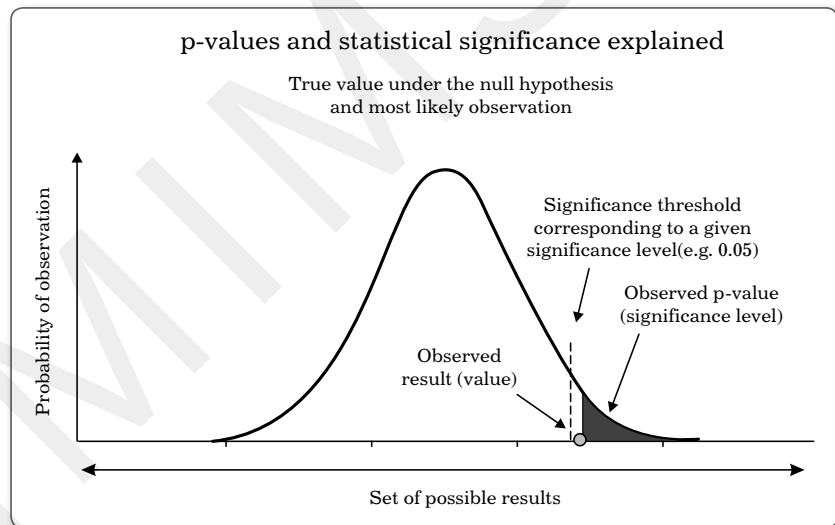


Figure 4.3: p-values and Their Statistical Significance

There are essentially 5 steps involved in testing of hypothesis using the p-value approach, as follows:

- ❑ **Step 1:** Define the null (H_0) and the alternate (H_1) hypothesis.
- ❑ **Step 2:** Define the level of significance, α .
- ❑ **Step 3:** Decide the test and identify the critical value for the test.
- ❑ **Step 4:** Calculate the test statistic and compare it with the critical value.
- ❑ **Step 5:** Draw inferences based on whether the null hypothesis was rejected or not rejected

Some of the most used hypothesis tests are as follows:

- ❑ **z-test:** A z-test is implemented in a hypothesis test to confirm whether a specific outcome or an association is statistically significant. A z-test simply checks whether the two means are

equal, assuming equality in the null hypothesis. A z-test is applied in a case when the population standard deviation is known, and its sample size is > 30 .

Example: A company seeks to reorganise work hours and introduce a new schedule to enhance productivity among its employees. The company will want to know whether the mean productivity before and after the schedule change is significantly different. By carrying out a z-test, the company can compare the mean productivity levels before and after adopting the new schedule. The z-test can determine whether the change in productivity is statistically significant, provided that their sample size is large enough, and they are aware of their population standard deviation, thus confirming the validity of their schedule change.

- **t-test:** The method of comparing the means between two groups is called the t-test. It is a very common procedure which is carried out in hypothesis testing towards identifying differences or the effects of a treatment on a population. This is the preferred procedure if one needs to compare one population means to a standard or means between two populations and especially when the standard deviation is unknown and the sample size is small ($n \leq 30$). It is especially useful in the analysis of differences between means of independent samples. The t-test is a t-distribution, which is dependent on degrees of freedom representing the number of independent observations in a set.

The three different types of t-tests that are used in hypothesis testing are:

- ◆ **One-sample t-test:** This t-test is simply used to determine if the sample mean is different from the population mean.
- ◆ **Independent two-sample t-test:** This is a type of t-test for testing the difference in means of two independent events.
- ◆ **Paired sample t-test:** This type of t-test compares the difference of means of the same group at two different times, often taken before and after treatment or intervention.

Example: A university wants to know if a new instructional methodology really increases test scores. To check this, they select a random sample of 25 students and record their test score before and after starting the new teaching method. A paired sample t-test is used to compare the mean test scores before and after applying the new teaching method since the sample size is small ($n < 30$) and the population standard deviation is unknown. By analysing the differences in these means, the university evaluates whether the improvement in the test scores is statistically significant.

- **f-test:** f-test is related to testing of hypothesis for two samples drawn from two different populations and their equality.

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The critical values of the f-test are obtained from the F table and F distribution.

Example: A manufacturing company uses two different machines to make a product. To check whether there is a significant difference in the product quality variance between the two machines, the company decided to employ the f-test in the hypothesis testing time. The f-test will help the company to compare the sample variances of product quality produced by each machine. The null hypothesis in this case would be that the variances should be equal. The f-test will then allow the company to determine whether there is a statistically significant difference in variance, which will assist in decisions about quality control and possibly in identifying the superior machine in terms of consistency of the product.

Some of the most common applications of hypothesis testing in business analytics are as follows:

- ❑ **Market research:** Businesses utilise hypothesis testing to determine the efficacy of marketing strategies such as advertising campaigns or pricing changes. Companies test certain hypotheses involving the consumer behaviours and sales performances and then can take informed decisions regarding the best utilisation of resources and strategy adjustments for maximum impact.
- ❑ **A/B testing:** This is an application where two versions of the same product or service are compared to find out which version is better, according to the metrics, for instance, conversion rates or customers' engagement. A/B testing helps businesses optimise web designs, email campaigns and features of a product using statistical evidence to determine which feature resonates best with the audience.
- ❑ **Quality control in manufacture:** Hypothesis testing is used to decide whether the quality requirements have been met or not through a comparison of the means for batches of different production processes if there are any variations which might reduce the quality of products manufactured.
- ❑ **Employee performance analysis:** Organisations apply the hypothesis testing technique by examining whether the effectiveness of the training programme can be measured by analysing the employee performance metrics both before and after the employee training. This enables an organisation to determine whether the type of training provided in an organisation results in further skilled and productive work; thereby, validating the expenditure on employee development.
- ❑ **Customer satisfaction:** Using survey data and hypothesis testing allows firms to decide whether the differences are statistically significant in the areas of customer satisfaction for specific demographics or product lines. This may help them identify

specific areas to focus their efforts on improvement and increasing retention among customers that will respond better to targeted changes.

4.2.1 p-VALUES AND HIGH p-VALUES

A p-value represents the probability of observing the data, or something more extreme, assuming the null hypothesis is true. It serves as an alternative to using rejection criteria and indicates the smallest level of significance at which the null hypothesis would be rejected; a lower p-value suggests stronger evidence against the null hypothesis in favour of the alternative hypothesis.

The p-value method for hypothesis testing involves calculating this probability to assess whether there is sufficient evidence to reject the null hypothesis. The result depends largely on the test statistic, which summarises the sample data in relation to the hypothesis. The null hypothesis is the initial assumption about a population or data-generating process, while the alternative hypothesis suggests that the population parameter differs from the value proposed in the null hypothesis.

Table 4.1 shows the relation between the p-value and the inference based on the p-value:

TABLE 4.1: INFERENCE BASED ON THE p-VALUE

p-value	Inference
$0.01 \leq p\text{-value} \leq \alpha$	Somewhat confident to reject H_0 .
$0.001 \leq p\text{-value} < 0.01$	Very confident to reject H_0 .
$p\text{-value} < 0.001$	Extremely confident to reject H_0 .

A high p-value, between 0.5 and 1.0, means that it is more likely that the results occurred by random chance or that the difference is not statistically significant in the case of a hypothesis test.

Example of a low p-value: You are conducting a study to test whether a new drug is effective in reducing blood pressure. The null hypothesis (H_0) might be that the drug has no effect and (H_1) is that the drug does have an effect.

If, after conducting the study, a p-value of 0.02 is obtained, this low p-value indicates that there is strong evidence against the null hypothesis. In this case, the null hypothesis can be rejected and it can be concluded that the drug has a significant effect on reducing blood pressure.

Example of a high p-value: Continuing with the same example, suppose you obtain a p-value of 0.35. This high p-value suggests that the observed data is not inconsistent with the null hypothesis. You might fail to reject the null hypothesis and conclude that there is not

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enough evidence to support the idea that the drug has a significant effect on reducing blood pressure. It is important to note that the interpretation of “high” or “low” p-values depends on the chosen significance level (often denoted as alpha, commonly set at 0.05).

If the obtained p-value is greater than or equal to the chosen alpha level, you typically fail to reject the null hypothesis. Researchers often use a significance level to make decisions about whether to reject or not reject the null hypothesis based on the p-value.

For example, if $\alpha = 0.05$, we are willing to accept a 5% chance of rejecting a true null hypothesis, i.e., there's a 5% risk of falsely concluding an effect or difference exists when it does not. Decreasing the α level, say to 0.01, reduces the risk of Type I error but also renders the test more stringent and requires stronger evidence to reject H_0 .

The power of a hypothesis test, often denoted as $1 - \beta$, is the probability that the test will correctly reject a false null hypothesis. In other words, it is the likelihood of avoiding a Type II error (failing to reject H_0 when it is actually false). A higher power increases confidence in detecting a true effect when one exists, which is particularly important in studies with serious implications, such as clinical trials. The power is influenced by several factors, such as:

- ❑ **Significance level (α):** A higher α increases power but also raises the chance of a Type I error.
- ❑ **Sample size:** Larger samples provide more information, thus increasing power.
- ❑ **Effect size:** Tests are more powerful when the actual difference from the hypothesised value is large.
- ❑ **Population variability:** Lower variability in the population leads to increased power, as the effect becomes easier to detect.

Balancing Type I and Type II errors requires careful selection of α and consideration of the study's sample size. Reducing α lowers the risk of a Type I error but may also reduce power, increasing the risk of a Type II error. Researchers aim to minimise both errors by choosing parameters that reflect the study's goals and the consequences of making either type of error. The significance level helps manage the risk of Type I errors, while power enhances the test's sensitivity to detect real effects, thereby providing a comprehensive framework for reliable decision-making in hypothesis testing.

4.2.2 ONE-SAMPLE HYPOTHESIS TESTS

One-sample hypothesis tests assess whether a population parameter (such as the mean, proportion or variance) for a single sample significantly differs from a known value or population parameter. For example, we can use a t-test or z-test to determine if the sample mean differs from a known population mean. A t-test is typically used when

the sample sizes are small and the population standard deviations are unknown, while a z-test is used when the sample size is large and the population standard deviations are known. Similarly, for one-sample hypothesis test of variance, we can use a Chi-square test to compare the variance of a single sample to a known population variance. For a two-sample hypothesis test of variance, we can use an f-test to compare the variances of two independent samples.

The most common one-sample hypothesis tests are as follows:

- **One-sample t-test:** These types of one-sample hypothesis tests are used when testing the mean of a continuous variable for a normally distributed population, especially when the population standard deviation is unknown or the sample size is small. The

formula of calculating the t-statistic in a t-test is: $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

Where:

t = t-statistic in the t-test

$\bar{x}$ = sample mean

μ = hypothesised mean

s = sample standard deviation

n = sample count

- **One-sample z-test:** It is almost identical to the former except for the difference in its applicability to populations with known population standard deviation and accompanying large samples. The formula for calculating the z-statistic in a z-test is:

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ (when population standard deviation is known)}$$

$$z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \text{ (when population standard deviation is not given and } n > 30 \text{)}$$

Where:

$\bar{X}$ = sample mean

μ = population mean (hypothesised)

σ = population standard deviation

n = sample size

s = sample standard deviation

- **One-sample proportion test:** These types of tests are applicable when testing proportions, for example, the proportion of a population bearing a certain characteristic. The formula of

calculating the proportion test is: $z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

NOTES

Where:

$\hat{p}$ = sample proportion (observed proportion of successes)

p_0 = hypothesised population proportion

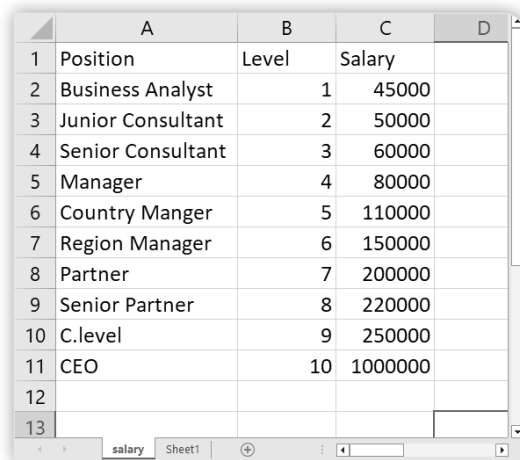
n = sample size

The assumptions for the one-sample hypothesis test include the following:

- ❑ **Continuous data:** It should be measured in terms of continuous scale; the variable may lie on any value of scale. A discrete or a categorical variable, for instance, counts and yes/no type responses cannot be measured.
- ❑ **Random sampling:** The sample should be bias-free by randomly selected from the population. This means that every member of a population has an equal chance of being included in the sample.
- ❑ **Independence of observations:** The values need to be independent of one another in the sample. This means that one observation should not affect or influence another.
- ❑ **Normality:** The population should be approximately normally distributed for the variable. This is because the foundation of the t-distribution in the t-test is the normal distribution.
- ❑ **No outliers:** The data are not extreme outliers because they exert an undue influence on the outcome of the t-test.

Perform the following steps to calculate the one-sample hypothesis test in Excel:

1. Open Excel spreadsheet with the business data that contains missing values. The dataset can be downloaded from the following website: <https://www.kaggle.com/datasets/taseermehboob9/salary-dataset-of-business-levels?select=salary.csv>. The data in Excel can be seen in Figure 4.4:



	A	B	C	D
1	Position	Level	Salary	
2	Business Analyst	1	45000	
3	Junior Consultant	2	50000	
4	Senior Consultant	3	60000	
5	Manager	4	80000	
6	Country Manger	5	110000	
7	Region Manager	6	150000	
8	Partner	7	200000	
9	Senior Partner	8	220000	
10	C.level	9	250000	
11	CEO	10	1000000	
12				
13				

Figure 4.4: Salary Data in Excel

NOTES

2. Enter parameters such as Mean ($\bar{X}$), Standard Deviation (s), Count (n), Standard Error of Mean (SEM), df, Hypothesised Mean (μ), t-statistic and p-value as shown in Figure 4.5:

	A	B	C	D	E
1	Position	Level	Salary		
2	Business Analyst	1	45000		Mean ($\bar{X}$)
3	Junior Consultant	2	50000		Standard Deviation (s)
4	Senior Consultant	3	60000		Count (n)
5	Manager	4	80000		
6	Country Manger	5	110000		Standard Error of Mean (SEM)
7	Region Manager	6	150000		Degress of Freedom (df)
8	Partner	7	200000		Hypothesised Mean (μ)
9	Senior Partner	8	220000		
10	C.level	9	250000		
11	CEO	10	1000000		t-statistic
12					p-value
13					
14					
15					
16					
17					

Figure 4.5: Entering Parameters for t-test in Excel

3. Write the formula =AVERAGE(C2:C11) to calculate the mean of the salaries in the dataset as shown in Figure 4.6:

	A	B	C	D	E	F	G
1	Position	Level	Salary				
2	Business Analyst	1	45000		Mean ($\bar{X}$)	216500	
3	Junior Consultant	2	50000		Standard Deviation (s)		
4	Senior Consultant	3	60000		Count (n)		
5	Manager	4	80000				
6	Country Manger	5	110000		Standard Error of Mean (SEM)		
7	Region Manager	6	150000		Degress of Freedom (df)		

Figure 4.6: Calculating the Mean

4. Calculate the rest of the parameters in the Excel sheet by using the following formulas:
- For Standard Deviation (s):** =STDEV.S(C2:C11)
 - For Count (n):** =COUNT(C2:C11)
 - For Standard Error of Mean (SEM):** =F3/SQRT(F4)
 - For df:** =F4-1
 - For Hypothesised Mean (μ):** Let the hypothesised mean be 80,000, which is the average salary in the company.

NOTES

The output of the parameters is shown in Figure 4.7:

	A	B	C	D	E	F	G	H	I
1	Position	Level	Salary						
2	Business Analyst	1	45000		Mean ($\bar{X}$)	216500			
3	Junior Consultant	2	50000		Standard Deviation (s)	285054			
4	Senior Consultant	3	60000		Count (n)	10			
5	Manager	4	80000						
6	Country Manger	5	110000		Standard Error of Mean (SEM)	90142			
7	Region Manager	6	150000		Degress of Freedom (df)	9			
8	Partner	7	200000		Hypothesised Mean (μ)	80000			
9	Senior Partner	8	220000						
10	C.level	9	250000						
11	CEO	10	1000000		t-statistic				
12					p-value				
13									
14									
15									
16									
17									

Figure 4.7: Output of the Parameters

- Enter the formula $= (F2 - F8) / F6$ in cell F11 to calculate the t-statistic from the values in the dataset. The p-value is calculated using the formula $= TDIST(F11, F7, 2)$. Since the data is not hypothesised as greater or lower than the company average and the requirement is to only see if there is any significant difference between the hypothesised mean and the data, the two-tailed distribution is selected in place of one-tailed distribution. The output of the t-statistic and p-value is shown in Figure 4.8:

	A	B	C	D	E	F	G	H	I
1	Position	Level	Salary						
2	Business Analyst	1	45000		Mean ($\bar{X}$)	216500			
3	Junior Consultant	2	50000		Standard Deviation (s)	285054			
4	Senior Consultant	3	60000		Count (n)	10			
5	Manager	4	80000						
6	Country Manger	5	110000		Standard Error of Mean (SEM)	90142			
7	Region Manager	6	150000		Degress of Freedom (df)	9			
8	Partner	7	200000		Hypothesised Mean (μ)	80000			
9	Senior Partner	8	220000						
10	C.level	9	250000						
11	CEO	10	1000000		t-statistic	1.5143			
12					p-value	0.16425			
13									

Figure 4.8: t-statistic and p-value Output

If we assume that the significance level (α) is 0.05 and the calculated p-value from the dataset is more than α , the hypothesis is not rejected. This means that there is insufficient proof to conclude that the average salary in the dataset significantly differs from the hypothesised mean of 80,000.

4.2.3 TWO-SAMPLE HYPOTHESIS TESTS

Two-sample hypothesis tests are statistical tests in which two independent groups' means are compared to know whether there is a statistically significant difference between them or not. These tests, therefore, are widely utilised in research and experimental studies to know whether an observed difference between two groups is because of a real effect or only due to random chance.

There are different kinds of two-sample hypothesis tests which depend on the nature of the data and the research question. The most common kinds of two-sample hypothesis tests are:

- ❑ **Independent samples t-test:** This is a test that compares the means of two different groups in order to determine whether or not there is a statistically significant difference between them.
- ❑ **Two-sample z-test:** This is identical to the t-test but where the sample size is large and population standard deviations are known.
- ❑ **Paired samples t-test:** It is a test used to compare two related groups which usually are before and after treatment or matched pairs.
- ❑ **Mann-Whitney U test (Wilcoxon Rank-Sum test):** This non-parametric test compares the distributions of two independent groups when the assumption of normality is not satisfied or when the data are ordinal or not normally distributed. The test is used to compare the distributions of two independent groups when the assumption of normality is not satisfied or when the data are ordinal or not normally distributed.
- ❑ **f-test:** This test compares the variances of two independent samples to determine if they are significantly different.
- ❑ **Chi-square test:** This test is used to determine whether the variance of a single sample is significantly different from a known population variance.

The assumptions of two-sample hypothesis tests include the following:

- ❑ **Independence:** Observations within each group must not depend on each other. One member's score must not influence another's score. Similarly, observations from groups are independent. That is, scores in one group should not influence that of the other.
- ❑ **Random sampling:** A sample must be randomly taken from its population. A sample is therefore representative in its own group.
- ❑ **Normality:** The data in each group should be normally distributed. This permits the use of the t-distribution for calculations. However, the t-test is fairly robust to moderate deviations from normality, especially with larger sample sizes.

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- ❑ **Continuity:** Continuous data like height or weight are ideal. Some tests tolerate ordinal or Likert-scale data with certain conditions. It is advisable to avoid using nominal (categorical) data with two-sample t-tests.
- ❑ **Homogeneity of variance:** This is that the groups have roughly equal variances (spreads). Ideally, both groups should have similar standard deviations. However, the t-test is also reasonably robust against modest violations of this assumption, especially with larger samples.

4.2.4 TYPES OF ERRORS

When performing a hypothesis test, there are some chances of two possible decisions, which are reject the null hypothesis or fail to reject the null hypothesis. Since hypothesis testing utilises data from a sample to make an inference about a population, the population constraints are not known. Therefore, in most cases, it cannot be known whether the inference is correct or incorrect.

There can be four possible outcomes because of statistical tests. The first outcome is that the hypothesis is true, but due to the statistical process we are labelling it as false. It is an error termed as “Type – I” error. The second outcome is that the hypothesis is true, and after the statistical process we are labelling it as true. It is a correct combination. The third outcome is that the hypothesis is false, and our conclusion after the statistical process is that it is false. It is a correct combination. The last outcome is that the hypothesis is false, but after the statistical process we are labelling it as true. It is an error, termed as Type II error or the β error. All these combinations are pictorially shown in Figure 4.9:

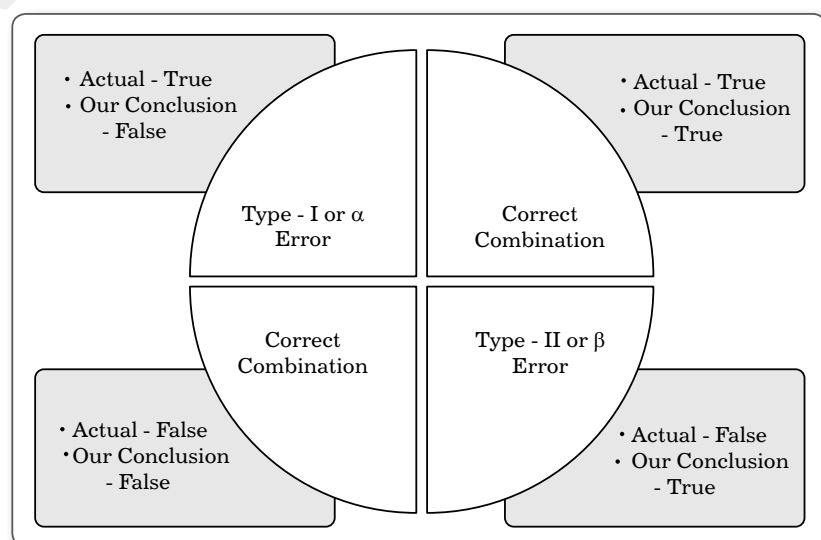


Figure 4.9: Combination of Type I and Type II Errors

In simple words,

- ❑ **Type I error:** Rejecting H_0 when H_0 is true, denoted by α (“alpha”) and commonly set at 0.05.
- ❑ **Type II error:** Failing to reject H_0 when H_0 is false, denoted by β (“beta”).

For example, a group of culinary arts students started comparing two methods for cooking asparagus: traditional steaming and a new frying method. They want to know if clients of their school restaurant prefer their new frying method over the traditional steaming method. A sample of clients was given asparagus prepared using each method and was asked to select their preference. A statistical analysis is implemented to determine if more than 50 percent of participant clients prefer the new frying method over the traditional steaming method.

Type I error occurs if they reject the null hypothesis and conclude that their new frying method is preferred when in reality it is not. This may occur if, by random sampling error, they happen to get a sample that prefers the new frying method more than the overall population does. If this does occur, the consequence is that the students will have an incorrect belief that their new method of frying asparagus is superior to the traditional method of steaming.

Type II error occurs if they fail to reject the null hypothesis and conclude that their new method is not superior when in reality it is. If this does occur, the consequence is that the students will have an incorrect belief that their new method is not superior to the traditional method when in reality it is.

The balance between Type I and Type II errors is often controlled by adjusting the significance level (alpha) and the power of the test. Researchers aim to find a balance that minimises both types of errors based on the specific context and consequences of each type of error.

After learning about the hypothesis testing, we will learn about Chi-square tests in the next section.



SELF ASSESSMENT QUESTIONS

1. Which statement best describes the null hypothesis?
 - a. It represents the researcher’s expectation.
 - b. It is a statement of a significant difference or effect.
 - c. It is a default assumption of no difference or no effect.
 - d. It is denoted as H_1 .

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2. When is the two-sample z-test preferred over the two-sample t-test?
 - a. When the sample size is large and population standard deviations are known.
 - b. When the sample sizes are small.
 - c. When the variances of the groups are equal.
 - d. When the data is not normally distributed.
3. In a legal trial, if the court incorrectly convicts an innocent person, what type of error is this analogous to in hypothesis testing?
 - a. Type I error
 - b. Type II error
 - c. False positive
 - d. True positive
4. A high p-value indicates strong evidence against the null hypothesis. (True/False)



ACTIVITY

Imagine you are a quality control manager in a manufacturing plant producing light bulbs. The production process is designed to produce bulbs with an average lifespan of 1000 hours. However, you suspect that a recent change in the manufacturing process might have affected the lifespan of the bulbs. Use the collected data to conduct a hypothesis test. Calculate the test statistic and determine whether to reject or fail to reject the null hypothesis.

4.3 CHI-SQUARE TEST

The Chi-square test (χ^2 test) is highly versatile, used for both categorical and numerical data. It includes the test of independence (examining relationships between variables), goodness of fit (comparing observed and expected distributions) and test of homogeneity (analysing distributions across groups). Additionally, it can test population variance in a single sample, making it valuable across diverse applications.

One common χ^2 test is for analysing the contingency tables (also known as crosstabulation or crosstab), especially when the sample sizes are large. In this test, the observations are categorised into equally exclusive groups. Under the null hypothesis, which states that there are no differences between the classes in the population, the test statistic computed from the observations follows a χ^2 frequency distribution. The test aims to assess how likely the observed frequencies are if the null hypothesis is true. This distribution of

test statistics is normal when the observations are independent. Chi-square tests can also assess the independence of paired variables by analysing observations of the pairs. There are two main types of Chi-square tests which we will discuss further in the next sections. They are as follows:

- ❑ Chi-square test for independence
- ❑ Chi-square goodness of fit test

4.3.1 CHI-SQUARE TEST FOR INDEPENDENCE

The Chi-square test for independence is used to determine if there is a significant association between two categorical variables. This test helps evaluate whether the categories of one variable are associated with the categories of the other. The Chi-square test of independence compares the observed data to what would be expected if there were no association between the variables. When two variables are correlated, the likelihood of a particular value for one variable depends on the value of the other variable.

Let us apply the Chi-square test for independence in a real-life scenario, where a retail company wants to understand if there is a relationship between customers' age groups and the types of products they purchase. The company collects data on customer purchases over a month and categorises customers into three age groups:

- ❑ Young (18-30 years)
- ❑ Middle-aged (31-50 years)
- ❑ Senior (51+ years)

The products are grouped into three main categories:

- ❑ Electronics
- ❑ Clothing
- ❑ Home Goods

The data is then summarised in a contingency table, as shown in Table 4.2:

TABLE 4.2: CONTINGENCY TABLE (O)				
	Electronics	Clothing	Home Goods	Total
Young	35	20	15	70
Middle-aged	50	25	25	100
Senior	20	35	25	80
Total	105	80	65	250

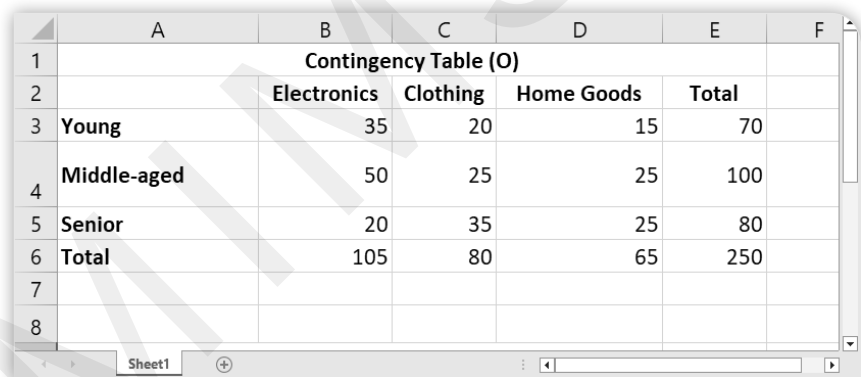
NOTES

Set the α level at 0.05. The null and alternative hypotheses are defined as:

- ❑ **Null hypothesis (H_0):** There is no association between customers' age groups and the types of products they purchase (age and product type are independent).
- ❑ **Alternative hypothesis (H_1):** There is an association between customers' age groups and the types of products they purchase (age and product type are not independent).

Perform the following steps to find the p-value for the Chi-square test for independence in Excel:

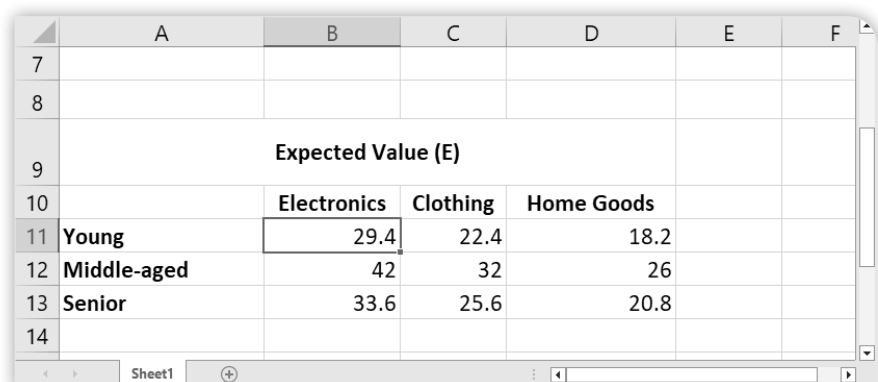
1. Enter the observed values into the cells in Excel for each combination of age group and product category, as shown in Figure 4.10:



	A	B	C	D	E	F
1	Contingency Table (O)					
2		Electronics	Clothing	Home Goods	Total	
3	Young	35	20	15	70	
4	Middle-aged	50	25	25	100	
5	Senior	20	35	25	80	
6	Total	105	80	65	250	
7						
8						

Figure 4.10: Contingency Table in Excel

2. Create another table named Expected Value (E) below the Contingency Table (O). Enter the product category and the age group similar to the contingency table.
3. Enter the formula $= (E3*B6)/E6$ in cell B11 to calculate the expected value of the Young age group with Electronics category. Fill the other cells using corresponding cell numbers. The filled Expected Value (E) table is shown in Figure 4.11:

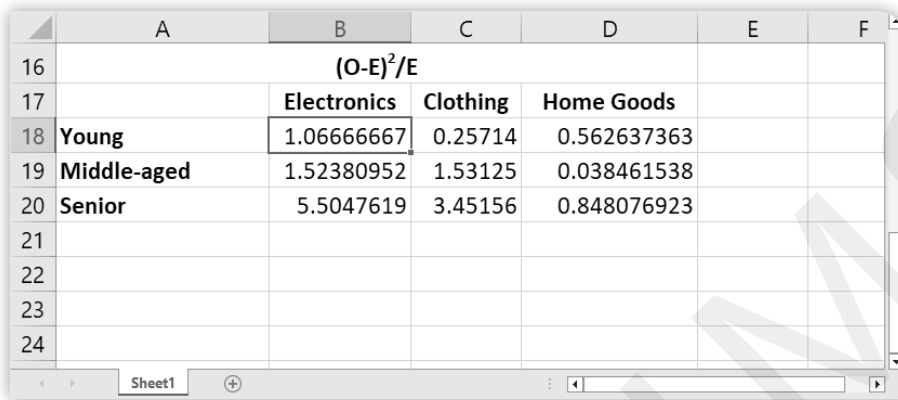


	A	B	C	D	E	F
7						
8						
9	Expected Value (E)					
10		Electronics	Clothing	Home Goods		
11	Young	29.4	22.4	18.2		
12	Middle-aged	42	32	26		
13	Senior	33.6	25.6	20.8		
14						

Figure 4.11: Expected Value Table

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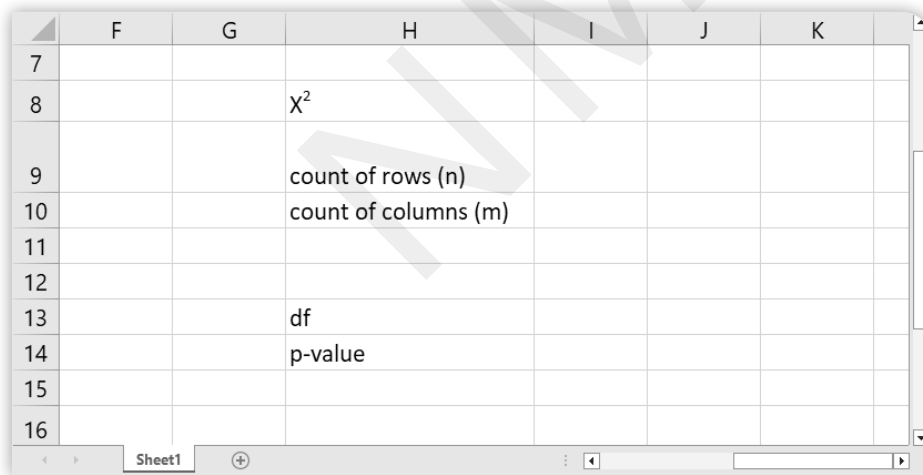
4. Create another table named $(O-E)^2/E$ below the Expected Value (E). Enter the product category and the age group similar to the contingency table.
5. Enter the formula $= (B3-B11)^2/B11$ in cell B18 to calculate the relation between the observed values and the expected values of the Young age group with Electronics category. Fill the other cells using corresponding cell numbers. The filled $(O-E)^2/E$ table is shown in Figure 4.12:



	$(O-E)^2/E$		
	Electronics	Clothing	Home Goods
Young	1.06666667	0.25714	0.562637363
Middle-aged	1.52380952	1.53125	0.038461538
Senior	5.5047619	3.45156	0.848076923

Figure 4.12: Calculating $(O-E)^2/E$ in Excel

6. Enter the Chi-square, count of rows (n), count of columns (m), df and p-values, as shown in Figure 4.13:



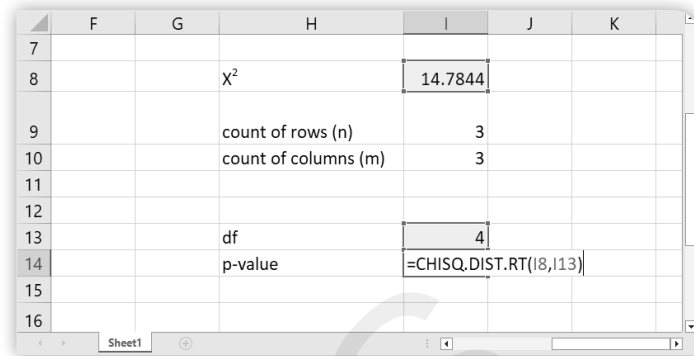
	F	G	H	I	J	K
7						
8			χ^2			
9			count of rows (n)			
10			count of columns (m)			
11						
12						
13			df			
14			p-value			
15						
16						

Figure 4.13: Creating Parameter Column in Excel

7. Enter the formula $= \text{SUM}(B18:D20)$ in the I8 cell to calculate the Chi-squared values. Enter the values of the count of rows (n) and the count of columns (m) to quantify the number of rows and columns in cells I9 and I10, respectively.
8. Enter the formula $= (I9-1)*(I10-1)$ to calculate the df in cell I13. Enter the formula $= \text{CHISQ.DIST.RT}(I8, I13)$ to calculate the p-value.

NOTES

The calculation of p-value is shown in Figure 4.14, which provides us with an answer that p-value = 0.005169994.



	F	G	H	I	J	K
7						
8			χ^2	14.7844		
9			count of rows (n)	3		
10			count of columns (m)	3		
11						
12			df	4		
13			p-value	=CHISQ.DIST.RT(I8,I13)		
14						
15						
16						

Figure 4.14: Calculating the df and p-value in Excel

Since the p-value is 0.005169994, which is less than α , we reject the null hypothesis (H_0). This shows that there is an association between customers' age groups and the types of products they purchase.

4.3.2 CHI-SQUARE GOODNESS OF FIT TEST

The Chi-square goodness of fit test is a statistical test used to determine if there is a significant difference between the expected and observed frequency distributions in a categorical data set. It is commonly applied when you want to assess whether the observed categorical data fits a theoretical distribution or expected values.

The Chi-square goodness of fit test is commonly used in areas such as quality control, market research and genetics to assess whether observed frequencies match expected frequencies. Figure 4.15 shows a Chi-square distribution with five degrees of freedom:

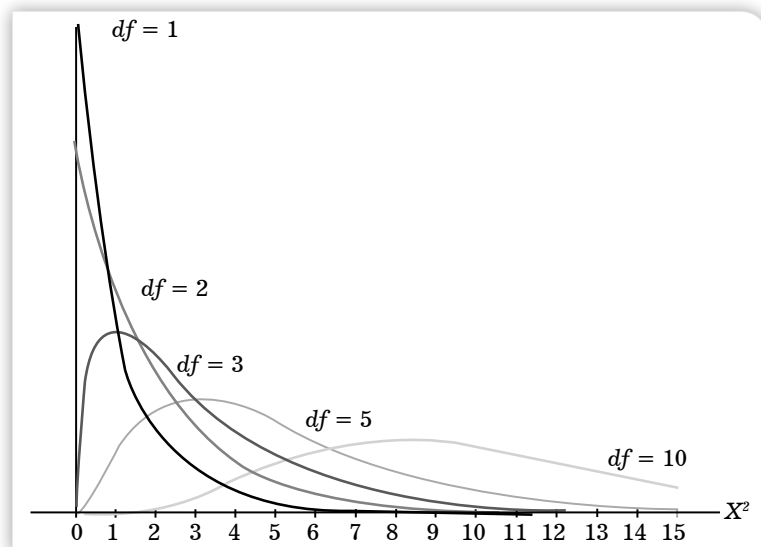


Figure 4.15: Chi-square Distribution with Five Degrees of Freedom

Example: Suppose you are conducting a study on the colour preferences of a population and you have a hypothesis about the expected distribution of colour preferences. The observed and expected frequencies for four colour categories: Red, Blue, Green and Yellow are shown in Table 4.3:

TABLE 4.3: OBSERVED AND EXPECTED FREQUENCIES OF SOME COLOURS

Colour	Observed Frequencies (O_i)	Expected Frequencies (E_i)
Red	25	30
Blue	30	25
Green	20	25
Yellow	25	20

The steps to perform the Chi-square goodness of fit test are:

1. **Formulate hypotheses:** The first step is to formulate a hypothesis based on the frequencies. The hypotheses can be written as:
 - ♦ **Null hypothesis (H_0):** The observed colour preferences are consistent with the expected colour preferences.
 - ♦ **Alternative hypothesis (H_a):** The observed colour preferences are not consistent with the expected colour preferences.
2. **Choose significance level:** Let us use a significance level (α) of 0.05.
3. **Determine expected frequencies:** We already have the expected frequencies based on the hypothesis.
4. **Calculate the test statistic:**

$$\chi^2_{\text{Red}} = \frac{(25-30)^2}{30} = 0.8333$$

$$\chi^2_{\text{Blue}} = \frac{(30-25)^2}{25} = 1$$

$$\chi^2_{\text{Green}} = \frac{(20-25)^2}{25} = 1$$

$$\chi^2_{\text{Yellow}} = \frac{(25-20)^2}{20} = 1.25$$

Therefore, $\chi^2 = 4.0833$

5. **Degrees of freedom (df):** To calculate the df , we can use the following formula: number of categories $- 1 = 4 - 1 = 3$
6. **Interpretation:** Now, if the significance level is set at, say, $\alpha = 0.05$ (commonly used), we would compare our calculated χ^2 value to the critical value from the Chi-square distribution table

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with $df = 3$ at $\alpha = 0.05$ to determine whether to reject the null hypothesis.

As the computed χ^2 value surpasses the critical value, we reject the null hypothesis, indicating a significant difference between the observed and expected frequencies.



SELF ASSESSMENT QUESTIONS

5. The Chi-square goodness of fit test is used for:
 - a. Comparing two categorical variables
 - b. Assessing the independence of variables
 - c. Testing the fit of observed data to expected frequencies
 - d. Comparing means of different groups
6. How are expected frequencies calculated in the Chi-square test for independence?
 - a. Based on the assumption of dependence between variables
 - b. Assuming a normal distribution of data
 - c. Calculated randomly
 - d. Assuming the variables are independent
7. What type of statistical test is most appropriate for analysing the relationship between gender and car colour preference in this scenario?
 - a. t-test
 - b. Chi-square test
 - c. ANOVA
 - d. Regression analysis
8. The Chi-square test assumes that the data in the contingency table is independent and identically distributed. (True/False)



ACTIVITY

Collect data from a sample of customers, including their subscription status (Subscribed/Not Subscribed) and their preferences for product categories (e.g., Electronics, Clothing, Home Goods). Organise the data into a contingency table with rows representing subscription status and columns representing product categories. Calculate the expected frequencies and perform the Chi-square test.

4.4 SUMMARY

- ❑ Inferential statistics is a branch of statistics that involves using sample data to generalise, make predictions or make decisions about a larger population.
- ❑ Unlike descriptive statistics, which focuses on summarising data, inferential statistics is concerned with drawing conclusions about the underlying population and quantifying the uncertainty in these conclusions.
- ❑ Statistical hypothesis involves statements about population parameters and are formulated to be testable using statistical methods.
- ❑ In hypothesis testing, the null hypothesis serves as the basis for statistical analysis, representing the status quo or default assumption, while the alternate hypothesis contrasts with it, proposing that there is a significant difference, effect or relationship in the population.
- ❑ One-sample hypothesis tests are used to make inferences about a population parameter, such as the mean or variance, based on data from a single sample.
- ❑ The two-sample hypothesis tests compare two independent groups to determine if there is a significant difference in their population parameters, such as means or variances.
- ❑ Both one-sample and two-sample hypothesis tests are crucial for analysing and interpreting statistical differences.
- ❑ The types of errors in hypothesis testing are to evaluate which errors are being committed in a statistical test.
- ❑ A Type I Error (α) occurs when the null hypothesis is incorrectly rejected despite being true, leading to a false positive conclusion.
- ❑ A Type II Error (β) occurs when the null hypothesis is not rejected even though it is false, resulting in a false negative conclusion.
- ❑ Both Type I and Type II errors highlight the trade-off between sensitivity and specificity in hypothesis testing.
- ❑ The p-value is a crucial concept in statistics, particularly in hypothesis testing. Researchers start by formulating a null hypothesis, assuming no significant effect or difference in the population.
- ❑ The p-value represents the probability of obtaining results as extreme as those observed, assuming the null hypothesis is true.
- ❑ A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, suggesting a significant effect or difference, while a larger p-value suggests there is not enough evidence to reject the null hypothesis.

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- ❑ The Chi-square test for independence serves the purpose of establishing whether there exists a notable association between two categorical variables.



KEY WORDS

- ❑ **A/B testing:** A/B testing, also known as split testing, is a randomised technique in which two or more variations of a variable are simultaneously introduced to different segments of users to determine which version performs better.
- ❑ **Contingency table:** A contingency table is a relationship between two or more categorical variables, which are shown in a tabular format.
- ❑ **Degrees of freedom:** It is a theory that is used in different statistical tests to symbolise the amount of freedom available for variability within a set of values.
- ❑ **Hypothesis:** It is a statement or a proposition that is put forward to explain a phenomenon or to provide a basis for further investigation.
- ❑ **Population:** It refers to the entire group of individuals or instances about whom the researcher is interested in making inferences.
- ❑ **p-value:** The p-value, or probability value, is a statistical measure that helps determine the strength of evidence against the null hypothesis in a hypothesis test. It indicates the probability of observing the test results, or something more extreme, assuming the null hypothesis is true.
- ❑ **Sample:** It is a subset of the population selected for analysis.
- ❑ **Statistics:** Statistics is a branch of mathematics that involves collecting, analysing, interpreting, presenting and organising data.
- ❑ **t-statistic:** It is a value calculated in a t-test to define whether there is a major difference between the sample mean and the population mean in a one-sample t-test, between two sample means in a two-sample t-test.

4.5 MULTIPLE CHOICE QUESTIONS

1. What is the purpose of hypothesis testing?
 - a. To estimate population parameters
 - b. To make inferences about population parameters based on sample data

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- c. To find descriptive statistics
 - d. To visualise data distributions
2. In hypothesis testing, what does the null hypothesis (H_0) represent?
- a. There is a significant effect or difference
 - b. The hypothesis researchers aim to support
 - c. No effect or no difference in the population
 - d. The samples mean is equal to the population mean
3. Which of the following tests is used while comparing two independent sample means, assuming equal variances?
- a. One-sample t-test
 - b. Independent samples t-test
 - c. Paired samples t-test
 - d. Chi-square test
4. What kind of error is made if we reject the null hypothesis when it is actually true?
- a. Type I error
 - b. Type II error
 - c. Power error
 - d. Sampling error
5. Which of the following statements is true about the p-value in hypothesis testing?
- a. It represents the probability that the null hypothesis is true.
 - b. A low p-value suggests stronger evidence against the null hypothesis.
 - c. A high p-value indicates a significant result.
 - d. A p-value above 0.05 always means the null hypothesis is rejected.
6. When would a researcher typically use a Chi-square test for independence?
- a. To compare two means
 - b. To examine the relationship between two categorical variables
 - c. To measure variance between groups
 - d. To calculate the t-statistic for small samples

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7. A researcher sets a significance level (α) of 0.01. What does this value represent, and how does it influence the risk of Type I errors in hypothesis testing?
 - a. Probability of Type I error
 - b. Probability of Type II error
 - c. Acceptable risk of Type I error
 - d. Acceptable risk of Type II error
8. In which situation is a paired samples t-test most appropriate?
 - a. Comparing the mean of a sample with a known population mean
 - b. Comparing two independent groups with different treatments
 - c. Comparing the same group's scores before and after a treatment
 - d. Examining relationships between two categorical variables
9. Which assumption must be met for a one-sample t-test to be valid?
 - a. Sample data must be categorical
 - b. Sample size must be greater than 100
 - c. Population should be approximately normally distributed
 - d. Two samples must be independent
10. What is indicated by a p-value of 0.001 in a hypothesis test?
 - a. There is no evidence against the null hypothesis
 - b. Weak evidence against the null hypothesis
 - c. Strong evidence against the null hypothesis
 - d. Very strong evidence against the null hypothesis

4.6 DESCRIPTIVE QUESTIONS

1. Define the terms p-values and high p-values.
2. What is the null hypothesis in a Chi-square goodness of fit test?
3. Describe the contingency table with the help of an example.
4. Discuss the different types of hypothesis testing.
5. Explain Type I and Type II errors in the context of hypothesis testing.

4.7 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. A researcher sets a significance level (α) of 0.01. What does this value represent, and how does it influence the risk of Type I errors in hypothesis testing?
 - a. Probability of Type I error
 - b. Probability of Type II error
 - c. Acceptable risk of Type I error
 - d. Acceptable risk of Type II error
2. When conducting a one-tailed hypothesis test, what factors should a researcher consider while deciding on the direction of the test?
 - a. Prior knowledge and theoretical considerations
 - b. Sample size and effect size
 - c. Significance level and power
 - d. Type I and Type II errors
3. In a Chi-square goodness of fit test, if the observed and expected frequencies significantly differ, what conclusion can be drawn about the fit of the observed data to the expected distribution?
 - a. Strong evidence of a good fit
 - b. Strong evidence of a poor fit
 - c. Weak evidence of a poor fit
 - d. No evidence of a good fit
4. Imagine you are conducting a hypothesis test to compare the mean scores of two different groups. How would the choice of significance level (α) impact your decision-making process, and what factors should you consider when selecting an appropriate α level?
5. Consider a situation where a hypothesis test fails to reject the null hypothesis. What are some potential reasons for this outcome?

4.8 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
Hypothesis Testing	1. c.	It is a default assumption of no difference or no effect.

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Topic	Q. No.	Answer
	2.	a. When the sample size is large and population standard deviations are known.
	3.	b. Type II error
	4.	False
Chi-square Test	5.	c. Testing the fit of observed data to expected frequencies
	6.	d. Assuming the variables are independent
	7.	b. Chi-square test
	8.	True

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. To make inferences about population parameters based on sample data
2.	c. No effect or no difference in the population
3.	b. Independent samples t-test
4.	a. Type I error
5.	b. A low p-value suggests stronger evidence against the null hypothesis.
6.	b. To examine the relationship between two categorical variables
7.	c. Acceptable risk of Type I error
8.	c. Comparing the same group's scores before and after a treatment
9.	c. Population should be approximately normally distributed
10.	d. Very strong evidence against the null hypothesis

HINTS FOR DESCRIPTIVE QUESTIONS

1. The p-value is a crucial concept in statistics, particularly in the context of hypothesis testing. A high p-value suggests that the observed results are relatively likely to occur by random chance and there is not sufficient evidence to reject the null hypothesis. Refer to Section 4.2
2. In a Chi-square goodness of fit test, the null hypothesis (H_0) asserts that there is no significant difference between the observed frequencies in each category and the expected frequencies that would be expected under a specific distribution or hypothesis. Refer to Section 4.3

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3. A contingency table, also known as a crosstabulation or crosstab, is a tabular interpretation of the relationship between two or more categorical variables. Refer to Section 4.3
4. t-test, z-test and f-test are statistical tests used for hypothesis testing in different scenarios. Each test serves a specific purpose and is applicable under certain conditions. t-tests are used to evaluate the means of two groups. To check whether a discovery or relationship is statistically significant, hypothesis testing uses a z-test. An f-test is used to compare two populations' variances. Refer to Section 4.2
5. Type I error, also known to as false positive, happens when the null hypothesis is incorrectly rejected despite being true. Type II error, known as a false negative, only happens when the null hypothesis is not rejected even though it is false. These errors, also termed the second kind of error or false negative, are represented by β (beta). Refer to Section 4.2

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	c. Acceptable risk of Type I error
2.	a. Prior knowledge and theoretical considerations
3.	b. Strong evidence of a poor fit
4.	When a hypothesis test fails to reject the null hypothesis, it suggests that there is insufficient evidence in the sample data to support the alternative hypothesis. Several potential reasons can contribute to this outcome. Firstly, the effect size may be smaller than anticipated, making it challenging to detect a significant difference between groups or conditions. Another factor is the choice of significance level (α). If a stringent significance level, such as 0.01, is chosen, the test becomes less sensitive, making it harder to reject the null hypothesis.
5.	The choice of significance level (α) plays a critical role in the decision-making process during hypothesis testing when comparing the mean scores of two different groups. The significance level represents the threshold for determining whether the observed results are statistically significant or occurred by chance. A common value for (α) is 0.05, indicating a 5 per cent chance of making a Type I error (rejecting a true null hypothesis).

N O T E S

4.9 SUGGESTED READINGS & REFERENCES**SUGGESTED READINGS**

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CASE STUDY

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ENHANCING HELP DESK SERVICE EFFICIENCY THROUGH HYPOTHESIS TESTING AND BUSINESS ANALYTICS

SCENARIO

In the competitive landscape of today's business environment, efficient and responsive customer service is a key factor in retaining customers and maintaining a positive brand image. The Help Desk service, responsible for handling customer queries and technical support, identified a need to streamline its operations. The management suspected that implementing changes in certain aspects of the service could lead to improvements in response time and customer satisfaction.

PROCESS

The following steps were performed by using hypothesis testing to identify areas of improvement and implement changes:

1. **Formulating hypotheses:** The first step is to formulate a hypothesis based on the frequencies. The hypothesis can be written as:
 - a. **Null hypothesis (H_0):** There is no significant difference in response time or customer satisfaction before and after implementing changes.
 - b. **Alternative hypothesis (H_1):** There is a significant improvement in response time and customer satisfaction after implementing changes.
2. **Data collection:** The next step is to collect the corresponding data from various sources. Data collection involves:
 - ◆ Collecting historical data on response times, customer feedback and resolution rates.
 - ◆ Obtaining customer satisfaction survey results before and after the proposed changes.
3. **Statistical analysis:** The next step is to perform statistical analysis based on the data. This includes:
 - ◆ Conducting hypothesis tests on response time data using a suitable statistical method (e.g., paired t-test for comparing means).
 - ◆ Analysing customer feedback data to identify patterns or trends.
4. **Business analytics:** The next step is to employ business analytics tools to further analyse data and extract insights. This involves:
 - ◆ Employing business analytics tools to visualise data and identify correlations.



Case Objective

The objective of this case study is to use hypothesis testing and business analytics to identify areas of improvement and implement changes to enhance overall Help Desk service efficiency.

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CASE STUDY

- ◆ Tracking Key Performance Indicators (KPIs) such as ticket resolution time, customer satisfaction scores and ticket closure rates.
- 5. **Implementation of changes:** Based on the results of the hypothesis tests and analytics, the next step is to implement targeted changes. The changes could include process optimisations, additional training for Help Desk agents or the introduction of new technology.
- 6. **Monitoring and evaluation:** The final step is to continuously monitor Help Desk performance and assess the impact of implemented changes. This involves:
 - ◆ Monitoring Help Desk performance after implementing changes.
 - ◆ Conducting follow-up surveys and collecting feedback to assess customer satisfaction.

RESULTS

- ❑ Hypothesis testing indicated a statistically significant improvement in response times after implementing changes.
- ❑ Business analytics revealed a positive correlation between agent training and customer satisfaction.
- ❑ Customer satisfaction scores increased by a noticeable margin.

CONCLUSION

The combination of hypothesis testing and business analytics allowed the Help Desk service to make data-driven decisions, leading to concrete improvements in response times and customer satisfaction. The iterative process of analysis, implementation and evaluation fostered a culture of continuous improvement within the Help Desk, ensuring sustained enhancements in service efficiency.

**QUESTIONS**

1. What was the null hypothesis in the Help Desk case study?
(**Hint:** The null hypothesis assumes no change in response time or customer satisfaction before and after implementing changes.)
2. Which statistical test could be used to compare response times before and after implementing changes?
(**Hint:** A paired t-test is used to compare response times before and after implementing changes.)

5

C H A P T E R

INTRODUCTION TO DATA MINING

CONTENTS

- 5.1 Introduction
- 5.2 Data Mining
 - 5.2.1 Data Mining Architecture
 - 5.2.2 Scope of Data Mining
 - 5.2.3 Ethical Considerations and Challenges in Data Mining
 - Self Assessment Questions
 - Activity
- 5.3 Data Mining Techniques
 - 5.3.1 Supervised Learning
 - 5.3.2 Unsupervised Learning
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 - 5.4.1 Association Rule Mining
 - 5.4.2 Clustering
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- 5.5 Summary
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- 5.7 Descriptive Questions
- 5.8 Higher Order Thinking Skills (HOTS) Questions
- 5.9 Answers and Hints
- 5.10 Suggested Readings & References

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**Case Objective**

The objective of this caselet is to find the perfect location for setting up stores through the use of data mining.

INTRODUCTORY CASELET**DETERMINING THE PERFECT LOCATION FOR SETTING UP STORES USING DATA MINING**

Starbucks is one of the leading coffee shops with multiple branches around the world. Starbucks employs in data mining to determine the perfect locations for setting up its stores. Tactics of data mining and modelling assist the numerous Starbucks locations within proximity.

They try to analyse the data based on the locations, the population composition of the area, and traffic in the streets around to predict whether setting up a store there will be successful or not. Starbucks seeks assistance from a data platform named ArcGIS, developed by Esri. They help them gather all the necessary information about the concerned location, demographic structure, the presence of customer homes around, work and other outings. All these data supplement the monitoring and boosting of their sales.

This particular company Esri gathers a lot of data from Starbucks and, after ingrained analysis, positions them on platforms that are easily understandable for the employees there.

(Source: <https://www.bizprospex.com/understanding-data-mining-with-the-help-of-case-studies-on-data-mining-in-market-analysis/>)



LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Define the concept of data mining
- Explain the different types of data mining techniques
- Describe the different types of unsupervised learning techniques

5.1 INTRODUCTION

In the previous chapter, you studied the concept of statistical inference, including hypothesis testing.

Data mining uses statistical analysis, machine learning, and database systems to highlight trends and patterns within the data using techniques like supervised and unsupervised learning. Then this data can be used to make fact-based decisions and find practical solutions to business problems. Data mining targets applications that could have a lot of business value, for example, customer profiling, segmentation marketing basket analysis, anomaly detection and predictive modeling. There are many data mining tools and technologies out there that enable you to collect large amounts of information from a variety of industries, such as finance, healthcare, retail, or telecommunications.

This chapter starts by discussing data mining, its process and scopes. After this, the chapter explains data mining techniques. At the end of the chapter, it focuses on unsupervised learning techniques.

5.2 DATA MINING

Data mining is the process of discovering patterns, trends, and useful information from large datasets using statistical, mathematical and computational techniques. It involves extracting meaningful insights from raw data, often stored in databases, to support decision-making and predictive analysis.

Some of the advantages of data mining are as follows:

- ❑ **Decision-making:** Data mining offers detection and understanding of patterns that contribute to data-driven decisions in an organisation.
- ❑ **Predictive analysis:** It assists in projecting trends and behavioural changes; therefore, organisations do not wait for an occurrence to act.
- ❑ **Customer insights:** The business's analytic results provide information that cuts across customers, leading to better targeting of customer preferences and services.

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- ❑ **Operational efficiency:** It enables the detection of the cause of waste in processes, enhancing efficiency and lowering operational costs.
- ❑ **Fraud detection:** This has the ability to reveal trends that are potentially fraudulent in nature and therefore limit the loss in finances.

Some of the disadvantages of data mining are as follows:

- ❑ **Privacy concerns:** The collection of personal information is a key element in most data mining operations, which raises some moral and legal questions concerning privacy.
- ❑ **Data quality issues:** The veracity of the findings from data mining processes is directly proportional to the fidelity of the input data; inaccurate data may result in incorrect conclusions.
- ❑ **Complexity:** The methodological approaches and algorithms that are applied in data mining tend to be complicated and technical expertise is required.
- ❑ **Overfitting:** The models tend to fit too well to certain specific datasets and thus fail on new data.
- ❑ **High costs:** The acquisition and setting up of data mining equipment and systems is expensive, especially for small and medium-scale enterprises.

After discussing the concept of data mining along with its advantages and disadvantages, we will focus on the data mining architecture and the ethical considerations while performing the data mining process in the following sections.

5.2.1 DATA MINING ARCHITECTURE

Data mining architecture means the general design and structure of a data mining system. A data mining architecture usually includes several key components, which work together to perform data mining tasks and extract useful insights and information from data. Some of the key components of a typical data mining architecture include:

- ❑ **Data sources:** Data sources are the origins of data that are used in data mining. These may include structured and unstructured data from databases, files, IoT devices and other sources. Data sources supply the raw data that are used in data mining and can be processed, cleaned, and transformed to design a usable dataset for analysis.
- ❑ **Data preprocessing:** Data preprocessing is the process of preparing data for analysis. This usually involves cleaning and transforming the data to eliminate errors, inconsistencies, and other irrelevant information and to make it fitting for analysis. Data preprocessing ensures that the data is of high quality and is ready for analysis.

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- ❑ **Data mining algorithms:** Data mining algorithms are the algorithms and models that are used to perform data mining. These algorithms usually include supervised and unsupervised learning algorithms, such as regression, classification and clustering, as well as more specialised algorithms for specific tasks, such as association rule mining and anomaly detection. Data mining algorithms are applied to the data to extract useful insights and information from it.
- ❑ **Data visualisation:** Data visualisation is the process of portraying data and insights clearly and effectively, typically using charts, graphs, or other visualisations. Data visualisation allows data miners to communicate their findings and insights to others in a way that is easy to understand and interpret.

In recent years, the availability of large datasets and the growth of cloud computing and Big Data technologies have made data mining even more powerful and widely used. Today, data mining is a crucial tool for many organisations and industries and is used to extract valuable insights and information from datasets in a wide range of domains. The data mining process usually follows the following steps:

1. **Problem definition:** This is the stage when the objective or goal of a data mining project is clearly stated. This also entails looking at the business problem or opportunity that the project is about and the level of analysis that will be carried out.
2. **Data collection:** This entails identifying and gathering relevant material about the study from other sources, such as databases, books in soft copy and hard copy, and global networks among others. This could involve Factors for Future Evaluation (FFE) of data from within the place systems or even capturing fresh data.
3. **Data preprocessing:** At this stage, any raw data that has been collected goes through a transformation ready for it to undergo analysis using various methods of data cleaning (deleting duplicates, correcting errors), data transformation (scaling, standardising) and data fusion (combine different datasets) among other techniques from different sources. This stage is crucial to ensure quality control of the data.
4. **Data exploration:** Involving the data and its features in some Exploratory Data Analysis (EDA) activity. These are things like drawing some graphs for the data, computing some summary statistics, or looking for trends or seasonality, if it exists.
5. **Modelling:** Implementing appropriate approaches in extracting data, such as classification, clustering, regression, or any including constructing the relationship models as per the dependent variable(s) and independent variables.

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- 6. Deployment:** Enact the data mining models within the business model practices or systems. This could require placing them into already standing architectures to reinforce their ability to offer intelligence solutions from the given data or even constructing new systems.

Perform the following steps for data mining in Excel:

1. Open the Excel spreadsheet with the business data that contains a Dataset of Products as shown in Figure 5.1:

	Product	Region	Sales Amount	Month
1	Laptop	North	15000	Jan
2	Phone	South	12000	Feb
3	Laptop	West	18000	Mar
4	Tablet	North	14000	Jan
5	Phone	West	10000	Feb

Figure 5.1: Dataset of Products

2. Select the dataset. Click the Remove Duplicates option from the Data in Menu bar. Check the Product option from the Columns and then click the OK button as shown in Figure 5.2:

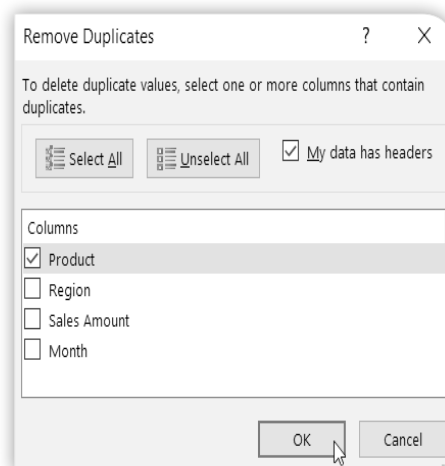


Figure 5.2: Remove Duplicates Dialog Box

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The output after removing the duplicate values is shown in Figure 5.3:

	Product	Region	Sales Amount	Month
1	Laptop	North	15000	Jan
2	Phone	South	12000	Feb
3	Tablet	North	14000	Jan

Figure 5.3: Dataset after Removing Duplicate Values

3. Select the column of Region and Sales Amount. Click the Insert button and select the Pie chart to explore the total sales by product and region. The output of this step is shown in Figure 5.4:

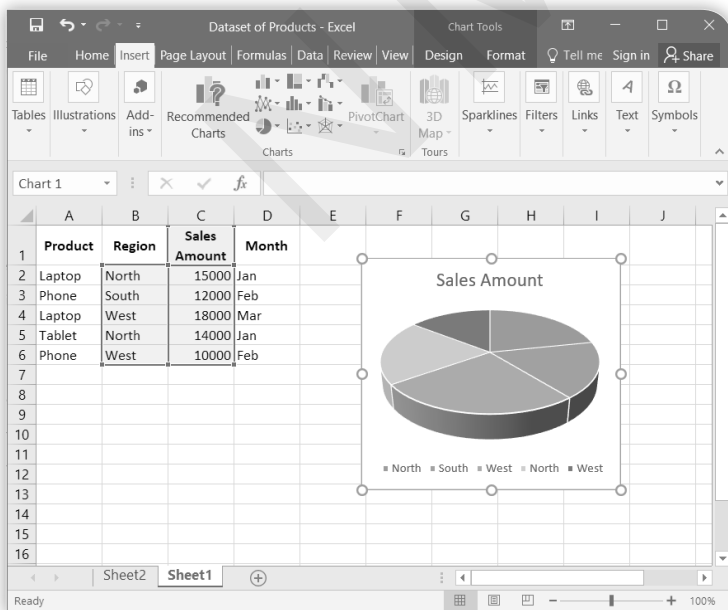
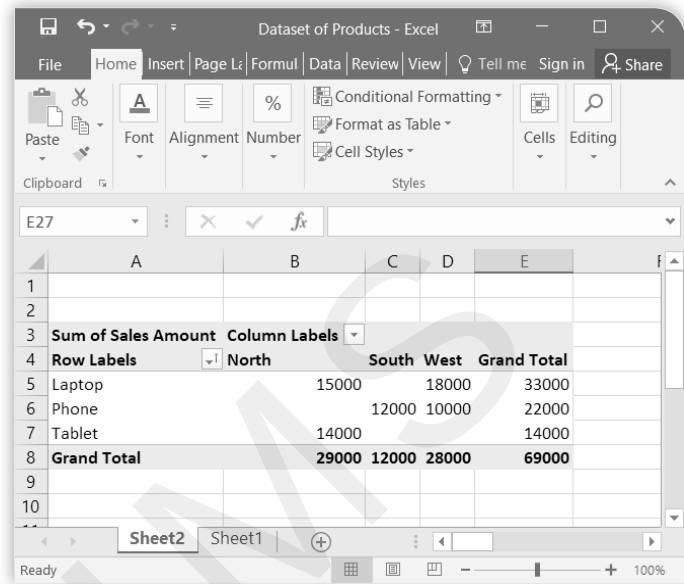


Figure 5.4: Total Sales by Product and Region

4. Select the dataset or use Ctrl + A to select all. Insert a Pivot Table for product-wise sales comparison and drag Product to

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Rows and Sales Amount to values in a pivot table to evaluate which product has the highest/lowest sales as shown in Figure 5.5:



Row Labels	South	West	Grand Total
Laptop	15000	18000	33000
Phone	12000	10000	22000
Tablet	14000		14000
Grand Total	29000	12000	28000

Figure 5.5: Pivot Table to Evaluate the Highest/Lowest Sales

5. Select the Pivot Table and go to Insert → Slicer. An Insert Slicer dialog box appears. Select the Sales Amount for dynamic filtering as shown in Figure 5.6:

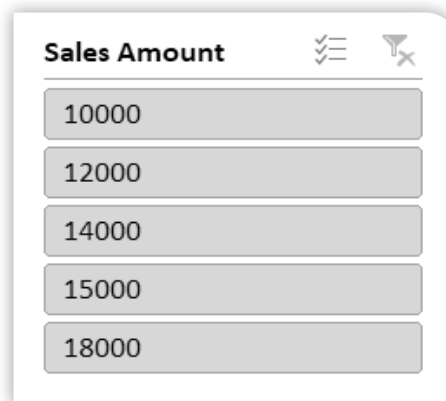


Figure 5.6: Sales Amount Slicer

5.2.2 SCOPE OF DATA MINING

The scope of data mining is defined as a model-based approach for sifting through voluminous amounts of data to uncover previously unseen structures aimed at enhancing the course of decision-making. Business analytics has a wider coverage and thus enables organisations to get meaningful information even from large volumes of both structured and unstructured data.

The following are some of the specific areas that have data mining applications:

- ❑ **Customer Relationship Management (CRM):** Market segmentations can be analysed and empirical data changes of prospective customers, demand trends, or others can be studied with CRM solutions that have, of course, a data analytics component based on models and analysis of the behaviour. Analytics of clients offers companies the opportunity to perform customer segmentation. In this way, an organisation can come up with suitable marketing strategies, and improve customer loyalty and satisfaction by providing tailored services, etc.
- ❑ **Market basket analysis:** Market basket analysis explains which products tend to be bought with each other. This supports businesses in identifying cross-sell opportunities, product packaging activity and stocking levels. This application assists retailers in increasing sales revenue by advertising the appropriate merchandise depending on the customers' purchasing habits.
- ❑ **Fraud detection:** Data mining can be used to detect fraud by identifying unusual or uncommon transactions, particularly in banking, insurance and e-commerce. With the help of progressive algorithms, a company is able to mitigate losses and secure its clients' information by undertaking fraud detection in real-time.
- ❑ **Risk management:** In accordance with the historical data, risk is assessed with the help of data mining. In finance and insurance, it is deployed to decide whether an individual or a business is a good candidate for credit, how likely they are to default on that credit, and what risks they present regarding the investments. Also, it allows enterprises to forecast potential threats and take steps against them.
- ❑ **Predictive analytics:** During the process of data mining, forecasting trends and attributes of objects and events in the future is less complicated thanks to the use of predictive models. This makes it possible to undertake one of the most important applications — anticipating customer demands and evaluating the orders and dynamics of particular markets enabling a firm to take timely actions and avert losses due to unforeseen circumstances.
- ❑ **Supply chain optimisation:** Data mining is a technique, which can improve the supply chain by managing the associated operations, such as procurement, logistics and reduction of operational costs, especially extended supply chains. It involves forecasting the depletion of stocks, delays in shipments and even managing the stocks in the company range, improving their operational efficiency and minimises their operational expenditure.
- ❑ **Healthcare analytics:** Predicting ill health among patients and better treatment and planning of health services in the country are some of the aspects of data mining in health. This facilitates

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early detection of cases, individualised management of patients, and efficient organisation of health services by finding insightful correlations in the health databases.

- ❑ **Social media analytics:** Organisations can analyse data from social networks to assess attitudes, acceptance of brands and their dynamics. Monitoring such posts, comments, interactions, and also the quantity of these, allows for adjusting each company's marketing policies and strategies and enhancing its branding.
- ❑ **Product development:** Data mining provides a means for organisations to uncover unsatisfied consumer needs within a certain market and fill that need with a new or modified product. Enterprises learn from consumer evaluations, feedback, and preferences in a bid to come up with their products, an approach that enhances market creativity and competitiveness.
- ❑ **Human resource analytics:** In business and management, particularly in human resources, data mining can be employed to evaluate an individual's performance, forecast the rate of employee turnover and enhance the recruitment process. If employers can identify the features and attributes of the top earners in the company, they can even improve the processes of hiring, training and managing human resources.

5.2.3 ETHICAL CONSIDERATIONS AND CHALLENGES IN DATA MINING

Ethical considerations in data mining revolve around privacy, consent, and data security. As organisations extract valuable insights from large datasets, there is a growing concern about the misuse of personal information. Privacy breaches, like the Aadhar data breach in the year 2018 or the Dominos India data breach in 2021, highlight the risks of unauthorised data access. Another challenge is obtaining informed consent from individuals whose data is being collected, especially when they may not fully understand how their information will be used. Additionally, data mining can lead to biased or unfair outcomes if the algorithms used are not carefully designed. To address these challenges, organisations must ensure transparency, minimise data collection, anonymise sensitive information, and adhere to strict ethical guidelines to protect users' rights.

There are some barriers to ethical data mining and some tactics to deal with such barriers, among which there are:

- ❑ **Convenience vs. privacy:** Effective analytic strategies that reduce data mining run the risk of having to deal with the boundaries of privacy. All AI and related technologies require increasingly bigger datasets to operate effectively, which is the same case that undermines the confidentiality of the data. Organisations have to either sacrifice service delivery convenience or deal with patrons' privacy.

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- ❑ **Data privacy and confidentiality:** The activity of data mining typically is associated with pattern recognition within huge amounts of information, some of which are considered private. Therefore, data protection poses a serious ethical challenge. Consumers' data must be protected from unauthorised access or use as provided by various laws, regulations and guidelines, including but not limited to the General Data Protection Regulation (GDPR). In most western societies that have witnessed recent developments in social infrastructure that favour massive surveillance of the citizens, let alone any sensitive personal data, this multipronged development will necessarily curtail the individual's right to privacy which will in turn lead to breaches of such sensitive data.
- ❑ **Informed consent:** Informed consent is the process of getting permission from a person or organisation before gathering or using their data. This idea addresses the concerns of the data subjects as to what and how their data would be used, stored and protected. Concerning business analytics, informed consent is vital in preserving transparency and privacy, as well as user confidence.
- ❑ **Data ownership and control:** Ethical issues in data mining include questions of data ownership and control. This means that an organisation should acknowledge the data ownership of any data providers, including individual users, businesses or government. The organisation retains responsibility for protecting against unauthorised third parties' use or sale of data without data owners' consent.
- ❑ **Bias and discrimination:** Data mining processes frequently involve algorithms that are prone to bias when trained on data which is skewed or represents only a narrow segment of the user group. Such biases can facilitate discrimination in areas such as employment, credit assessment or even marketing practices. In particular, problematic models increase negative expectations against particular groups or disallow their usability entirely.
- ❑ **Data quality and accuracy:** When dealing with ethical data mining, there is a requirement for data to be accurate and of high quality. This is because low-quality data is prone to wrong conclusions and cross-sections which can be harmful to people and firms. One of the great difficulties encountered during the process of data mining, more so during data collection is ensuring that the information collected is as accurate, relevant and free from any errors as possible.
- ❑ **Misuse of data insights:** Data mining can produce results that are taken to the extreme to target potential consumers of a business or to unduly discriminate against unsuspecting audiences through tactics such as price optimisation or extremely targeted

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advertisements. Companies should evaluate the consequences of acting upon such relevant information in an informed society.

After learning about data mining, we will study the various techniques used in mining in the next section.



SELF ASSESSMENT QUESTIONS

1. Which of the following perspectives constitutes a primary ethical concern in data mining?
 - a. Data quality and accuracy
 - b. Increased revenue generation
 - c. Market expansion
 - d. Customer retention strategies
2. Which of the following applications assists businesses in determining the opportunities for cross-selling by examining the purchase behaviour of a customer?
 - a. Customer relationship management
 - b. Market basket analysis
 - c. Fraud detection
 - d. Predictive analytics
3. Data mining is solely focused on extracting patterns from structured datasets and does not involve text, web, audio or video data. (True/False)



ACTIVITY

Analyse a dataset from a retail company to identify customer purchasing patterns. Use data mining techniques to segment customers based on their buying behaviour, exploring associations between products and identifying trends.

5.3 DATA MINING TECHNIQUES

In business analytics, organisations use data mining techniques for insight generation. In this regard, data mining techniques include classification methods, clustering techniques, regression methods and association rules learning and each has its function for data patterns and the different questions being asked. Classification methods help to assign items of interest to the specified categories, whereas clustering methods do not use any prior labelling and aim to find motivated groups in the data. On the other hand, regression techniques allow working with incompressible outcomes, where a

forecast is being made based on the analysis of previous trends. Then, association rule mining is concerned with the possible correlation of variables which could help to determine how best to sell products or manage stocks. When these data mining techniques come into play, organisations can better their analysing processes, improve their activities and support their strategic efforts that, in the end, result in better performance and a sustainable edge over competitors in the marketplace.

5.3.1 SUPERVISED LEARNING

Supervised learning is a category of machine learning in which an algorithm is trained with the assistance of a labelled dataset. In this method, every input is associated with the output in the form of a label. The objective of the algorithm is to understand how to determine the appropriate output for new inputs that have not been encountered earlier. The reason this approach is termed “supervised learning” is that training this algorithm requires the presence of an expert, just like a teacher who provides the right answers within the context.

Supervised learning mostly aims at bridging the gap between the predictions made by the algorithm and the labels provided to the training data. The training lowers such discrepancies between the predicted labels and the true labels using a loss function. The algorithm aids in the reduction of this loss through modification of its parameters. This process is called the optimisation of the algorithm.

Some examples of algorithms performed under supervised learning are predicting non-categorical values using linear regression or predicting categorical values using classification techniques such as SVMs, decision trees or neural networks. Different algorithms are implemented for different kinds of data and problems. A supervised machine learning pipeline including raw data input, features, outputs, the ML model and model parameters, and prediction outputs to classify whether a customer will remain or leave is shown in Figure 5.7:

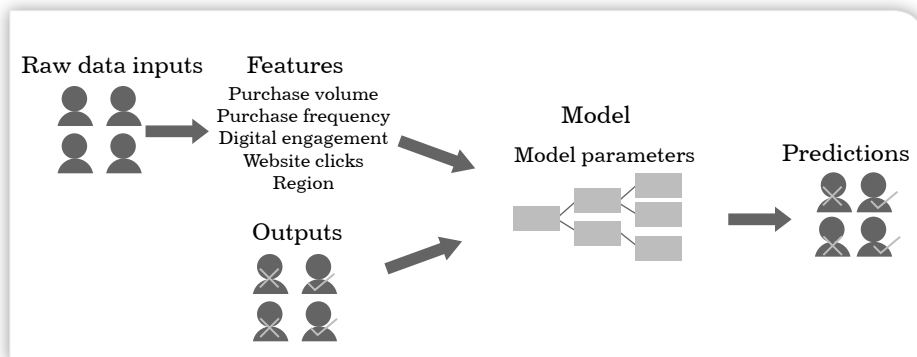


Figure 5.7: Working of Supervised Learning

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There are numerous implementations of supervised learning, as follows:

- ❑ **Image and object recognition:** In image recognition, a model teaches itself to classify objects or scenes using a dataset of labelled images. It can be used for facial recognition, autonomous cars recognising road signs, or in manufacturing, quality control inspection.
- ❑ **Speech recognition:** Supervised learning is employed to develop models that can detect segments of speech, in the real-time, which makes possible intelligent systems that incorporate turned speech, such as Siri, Alexa and others, transcription or video voice-over services and command by voice operations.
- ❑ **Medical diagnosis:** Supervised learning is employed in the health sector to forecast sickness using healthcare records, MRI scans or DNA analysis. For instance, predicting the chances of the occurrence of diabetes or cancer using patient data.
- ❑ **Credit scoring:** Financial entities also used supervised learning by preparing raw historical data inclusive of credit score, income and other relevant characteristics.
- ❑ **Financial fraud detection:** Supervised learning also prevents the occurrence of frauds and seeks to understand certain behavioural patterns based on a classification of the user into either normal or fraudulent transactions thanks to a database of non-fraudulent transactions.
- ❑ **Recommendation systems:** Supervising learning in recommendation systems is also done to estimate the possible choice of the user towards certain items using the extended history of the user's actions. This is why these systems are well-known for use on such websites as Netflix and Amazon, whereby users can easily access and enjoy content or products that are customised for them.

The following constraints are present in the supervised machine learning paradigm:

- ❑ **Labelling data:** One major challenge in supervised machine learning is the necessity to have annotated examples. This entails creating and/or obtaining annotated resources, which can prove to be difficult and costly, especially in some domains such as medicine which depend on specialists' evaluations.
- ❑ **Overfitting and underfitting:** The scale of overfitting and underfitting should be moderated. Overfitting is when a model does well on the training data because it competently learned the 'noise' instead of the useful signal, thus it performs poorly on new unseen data. Underfitting is when a model is too complex as

it proposes a solution that even goes above the highest pattern present.

- ❑ **Bias and fairness:** Supervised models may learn from the data and even become biased from it. Therefore, when applying these models, the outcomes can be biased and unfair. There is always fairness and biases overcoming, which is even worse for applications that have an ethical aspect.

Supervised learning is a machine learning approach where the model is trained on labelled data, meaning the input data is paired with the correct output. There are two main types of supervised learning which are classification and regression as shown in Figure 5.8:

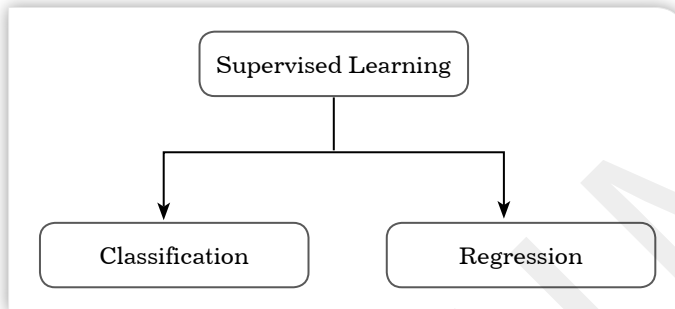


Figure 5.8: Types of Supervised Learning

The types of supervised learning as shown in Figure 5.8 are discussed below:

- ❑ **Classification:** In classification, the task is to predict discrete labels or categories from the input data. For example, predicting whether an email is spam or not based on its content is a classification problem. Classification models can have binary or multiple classes, and common algorithms include decision trees, Support Vector Machines (SVM) and neural networks.
- ❑ **Regression:** In regression, the goal is to predict a continuous value based on input data. For instance, predicting house prices based on features like square footage, number of rooms, and location is a regression problem. Popular regression algorithms include linear regression, polynomial regression and decision trees.

The following are some of the common supervised learning algorithms:

- ❑ **Linear regression:** Linear regression is one of the core algorithms applied to evaluate and predict the values for the continuous output based on the linear connection between the dependent and the independent variables.
- ❑ **Logistic regression:** This is another popular algorithm used for the analysis of binary classifications. It assumes that the

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probability of an object belonging to a certain class can be determined which is very useful in situations like determining if a certain transaction is fraudulent or engaged in spam activities. The calculated probability at this stage is transformed using a logistic function, which confines the values to the interval zero and one.

- ❑ **Decision trees:** Any decision process and its outcomes can be represented with decision trees that are n-dimensional models drawn in a tree-like manner. Classification and regression tasks can use these structures as they answer several questions outlining the process of a decision and where the progression of thought can be clearly illustrated.
- ❑ **Random forest:** Random forests are developed as a solution to the limitations of a single model and are ensembles of decision trees that work by mitigating the risk of overfitting while improving the accuracy of prediction. Each tree is fit to a different portion of the features and data, and the final answer is then the summation of all the tree forecasts in the forest.
- ❑ **Support Vector Machine (SVM):** SVM is a unique algorithm, which builds a hyperplane to facilitate classification and regression in an n-dimensional space. They find support vectors, which are the data points that have a serious effect on the placement of the hyperplane and help to separate the different classes to the fullest potential.
- ❑ **K-Nearest Neighbors (KNN):** KNN determines is also known as the classification d at a point P based on the neighbourhood of k points within some vicinity of P . It is a very simple method that is applicable in regression as well as classification problems, even though its performance is again sensitive to the presence of noisy data, and the choice of ' k ' is crucial.

5.3.2 UNSUPERVISED LEARNING

Unsupervised learning is the process of training an algorithm on unlabelled data to identify and learn patterns, relationships, or structures within the data. There are no output labels within the training data that can be referred to as 'correct,' unlike in supervised learning.

In unsupervised learning, the algorithm analyses the data and makes sense of its organisation on its own without assistance. Some common methods are clustering (putting similar data points together) and dimensionality reduction (keeping the important aspects of the data while making it less complex).

The working of unsupervised learning is shown in Figure 5.9:

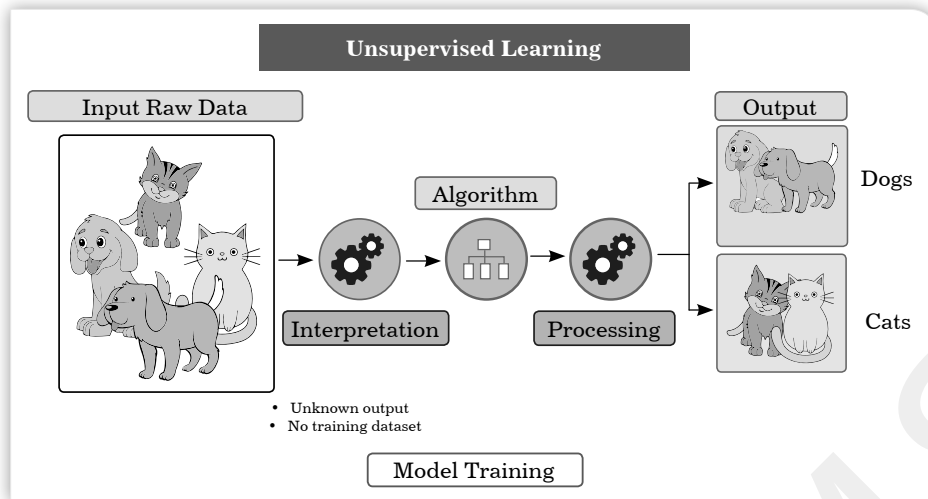


Figure 5.9: Working of Unsupervised Learning

Some of the applications of unsupervised learning are as follows:

- ❑ **Customer segmentation:** One of the most beneficial applications of unsupervised learning within business analytics is customer segmentation. Organisations try to group customers according to their buying behaviours, demography, and preferences so that the marketing plans, customer service, and product recommendations can be improved. This kind of segmentation makes it possible for businesses to run campaigns that focus on psychographic and demographic groups of customers, resulting in more customers returning and contributing more revenue.
- ❑ **Market basket analysis:** Unsupervised learning techniques may also be applied to conduct market basket analysis which is the study of purchasing behaviour to find the relationship between products. Using transaction data, businesses can find out which of their products are often bought together, and hence, assist them in product positioning, cross-selling and marketing.
- ❑ **Fraud detection:** Effective learning strategies are very essential, particularly in the financial and insurance industries, to implement fraud prevention measures. By monitoring changes in trends of historical transaction data, organisations can place suspicious points of transactions and make them real-time. Such forward action reduces any possible losses and, more importantly, helps maintain consumer confidence.
- ❑ **Recommendation systems:** Unsupervised learning techniques are also useful in creating recommendation systems that make suggestions based on the study of an individual's activities. This may be achieved by segmenting the users and looking for the patterns in their clustering based on user preference. Thus, such

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techniques aim at the overall performance of the system and maximise the focus on appropriate users.

- ❑ **Employee performance and retention analysis:** Human resources within a company can utilise unsupervised learning in sifting through the data on the performance of the employees, trends on attrition clumps, and improving the strategising of human resources. The employees are determined by making cluster diplomacy based on their performance scores.

After learning about supervised learning in this section, we will learn about unsupervised learning in the next one.



SELF ASSESSMENT QUESTIONS

4. Which technique is commonly used in unsupervised learning for grouping similar data points?
 - a. Linear regression
 - b. Classification
 - c. Clustering
 - d. Decision trees
5. What type of problem is addressed by regression in supervised learning?
 - a. Predicting discrete labels
 - b. Clustering data points
 - c. Predicting continuous values
 - d. Identifying outliers
6. Overfitting occurs when a model performs poorly on training data because it has learned the 'noise' instead of the useful signal. (True/False)



ACTIVITY

Examine different data mining techniques in business analytics. Analyse real-world case studies to compare and contrast techniques like classification, regression, clustering and association rule mining. Discuss how each method is applied to solve specific business problems, emphasising their advantages and limitations in deriving actionable insights from data.

5.4 UNSUPERVISED LEARNING TECHNIQUES

Unsupervised learning techniques are a crucial component of data analysis in business analytics, enabling organisations to extract insights from unlabelled data without predefined categories or

outcomes. Unlike supervised learning, which relies on labelled datasets to train models, unsupervised learning identifies hidden patterns and structures within data, making it particularly useful for exploratory analysis. Techniques such as clustering, dimensionality reduction, and anomaly detection help businesses uncover customer segments, reduce the complexity of datasets and identify unusual behaviours or trends. By leveraging unsupervised learning, organisations can make data-driven decisions, enhance customer experiences, and optimise operations, paving the way for more strategies that are informed and innovative in a competitive market. The two main techniques of unsupervised learning which are clustering and association rule mining are shown in Figure 5.10:

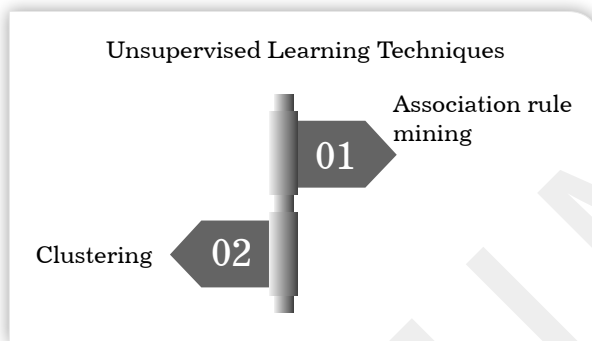


Figure 5.10: Unsupervised Learning Techniques

In the next sections, we will learn about the techniques of unsupervised learning in details.

5.4.1 ASSOCIATION RULE MINING

Association rule mining, as a data mining technique, helps in discovering interesting relations and patterns in large amounts of data. Mostly used in market basket analysis, association rule mining helps organisations comprehend their customers better by showing how items or products are likely to be associated. This can then make way for better marketing campaigns, optimal stocking of goods, and greater enhancement of customer service systems.

The key metrics of association rule mining are as follows:

- ❑ **Support:** The support of an item set is defined as the number of transactions present in the database, including that item as a percentage of the total transaction number in the database. It shows the rate of occurrence. When the support value is high, it can be interpreted that the groups of items appear in transactions more often.

$$\text{Support}(A) = \frac{\text{Total number of transactions}}{\text{Number of transactions containing } A}$$

NOTES

- ❑ **Confidence:** Confidence is the likelihood of finding an item B , in cases where there is an item. It shows the degree of strength of the association rule.

$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A)}{\text{Support}(A \cup B)}$$

- ❑ **Lift:** The measure explains how many times more probable the purchase of an item B is when an item A is purchased rather than the baseline probability of purchasing item B . A lift index that is more than one provides evidence of a direct relationship existing between A and B .

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Support}(A) \times \text{Support}(B)}{\text{Support}(A \cup B)}$$

Some of the few techniques of association rule mining are as follows:

- ❑ **Apriori algorithm:** A well-known algorithm that finds frequent item sets and produces association rules based on the support and confidence levels.
- ❑ **FP-Growth (Frequent Pattern Growth):** A more advanced and faster version of the Apriori algorithm where no candidate item sets are ever explicitly generated as a tree is used to discover all the frequent patterns.

5.4.2 CLUSTERING

Clustering is the process of classifying data in groups (clusters) in such a way that objects in a single group are more similar to each other than to objects from other groups. This similarity may be expressed in various forms, including distance measures (Euclidean, Manhattan, etc.) or statistical characteristics.

Clustering is one of the prime techniques in unsupervised learning, where the goal is to separate a set of objects into clusters where members of each cluster are similar based on the attributes of the objects. This strong and efficient analytical technique has found a great application in business analytics for analysing huge datasets to obtain useful information. With an understanding of the grouping present in the data, businesses can strategise effectively, improve customer service and optimise processes.

CLASSIFICATION OF CLUSTERING

A clustering algorithm's general approach can be put into the following subcategories:

- ❑ **Partitioning methods:** These algorithms split a dataset into distinct, non-overlapping subgroups. The best-known of these

algorithms is K-means, which tries to minimise the variance in clusters of K numbers.

- ❑ **Hierarchical methods:** In hierarchical clustering, a tree structure of the clusters is created, either by merging downwards (agglomerative) or splitting upwards (divisive). This helps to reveal more about the organisation of the data.
- ❑ **Model-based methods:** In model-based methods, it is assumed that the data we observe are a mixture of a fixed number of unknown mechanisms of probability distribution.

Some of the well-known clustering techniques are as follows:

- ❑ **K-means clustering:** Groups the data into a K feature closely associated set. This method is very often used for dividing clients in marketing and for audience analysis.
- ❑ **Hierarchical clustering:** This method creates a tree-like structure of clusters (agglomerative or divisive), enabling the user to analyse multiple layers of data. This is beneficial for grasping the different relationships of data.
- ❑ **Density-Based Spatial Clustering of Applications with Noise (DBSCAN):** This approach categorises the data based on distance and hence helps to discover clusters of different shapes, especially in the case of geospatial data.



SELF ASSESSMENT QUESTIONS

7. Which algorithm is known for finding frequent item sets in association rule mining?
 - a. K-means
 - b. FP-Growth
 - c. Naive Bayes
 - d. Decision tree
8. Which clustering method is known for using point density to form clusters?
 - a. K-means
 - b. Hierarchical clustering
 - c. DBSCAN
 - d. Apriori algorithm



ACTIVITY

Analyse customer purchasing behaviour using clustering techniques in business analytics. Utilise a dataset of customer transactions to segment customers into distinct groups based on their purchasing patterns. Discuss how these clusters can inform targeted marketing strategies and improve customer retention by tailoring offers to specific segments.

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5.5 SUMMARY

- ❑ Data mining is the process of extracting information and patterns from massive collections of data.
- ❑ The process of data mining is defined as a model-based approach for sifting through voluminous amounts of data to uncover previously unseen structures aimed at enhancing the course of decision-making.
- ❑ In business analytics, organisations use data mining techniques for purposes of insight generation.
- ❑ Supervised learning is a machine learning approach where the model is trained on labelled data, meaning the input data is paired with the correct output.
- ❑ Linear regression is one of the core algorithms applied to evaluate and predict the values for the continuous output based on the linear connection between the dependent and the independent variables.
- ❑ SVM is a unique algorithm, which builds a hyperplane to facilitate classification and regression in an n-dimensional space.
- ❑ Unsupervised learning is the process of training an algorithm on unlabelled data to identify and learn patterns, relationships, or structures within the data.
- ❑ Association rule mining as a data mining technique helps in discovering interesting relations and patterns in large amounts of data.
- ❑ Clustering is the process of classifying data in groups (clusters) in such a way that objects in a single group are more similar to each other than to objects from other groups.



KEY WORDS

- ❑ **Business analytics:** Business analytics is the application of advanced analytical techniques using data, statistical models, and machine learning technology to extract insight, make predictions, and support business decision-making through the use of data.
- ❑ **Data warehouses:** Data warehouses work as repositories to hold large amounts of data from multiple sources that are often useful in running the business. In other words, data is formatted and made readily available and easy to use for data analysis, reporting, as well as querying.
- ❑ **EDA:** The term EDA encompasses the techniques employed to analyse any dataset in order to describe its basic features, usually with the help of visualisations like histograms, scatter plots, or box plots.

- ❑ **GDPR:** The GDPR is a regulation that sheds light on data protection and privacy in the European Union. It provides strict rules regarding how organisations process, regulate, or handle personal information.

5.6 MULTIPLE CHOICE QUESTIONS

1. Which technique is commonly used for improving supply chain operations through data mining?
 - a. Risk management
 - b. CRM
 - c. Predictive analytics
 - d. Fraud detection
2. Which of the following describes the process of transforming raw data into a usable format?
 - a. Data exploration
 - b. Data collection
 - c. Modelling
 - d. Data preprocessing
3. In data preprocessing, which of the following techniques is not commonly used?
 - a. Data cleaning
 - b. Data transformation
 - c. Data visualisation
 - d. Data fusion
4. What ethical issue is commonly associated with data mining?
 - a. Data storage
 - b. Informed consent
 - c. Data compression
 - d. Data visualisation
5. Which of the following is a classification method in data mining?
 - a. Decision trees
 - b. Regression
 - c. Clustering
 - d. Association rules

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6. Which of the following algorithms is not commonly used in supervised learning?
 - a. Linear regression
 - b. K-nearest neighbours
 - c. Support vector machine
 - d. K-means clustering
7. Which algorithm is specifically used for binary classification problems?
 - a. K-nearest neighbours
 - b. Linear regression
 - c. Logistic regression
 - d. Decision trees
8. What is the main purpose of data mining techniques in business analytics?
 - a. To create new data
 - b. To generate insights from data
 - c. To clean data
 - d. To store data
9. Which algorithm is known for finding frequent item sets and producing association rules?
 - a. DBSCAN
 - b. K-means
 - c. Apriori
 - d. FP-Growth
10. Which method of clustering involves creating a tree structure of clusters?
 - a. Partitioning methods
 - b. Hierarchical methods
 - c. Density-based methods
 - d. Model-based methods

5.7 DESCRIPTIVE QUESTIONS

1. Describe the fundamental concepts of data mining and its applications in discovering patterns and insights from large datasets.
2. What are the key steps in the data mining process, and how do they contribute to effective data analysis?

3. What is the scope of data mining in various industries, and how does it enhance decision-making?
4. What is unsupervised learning, and in what scenarios is it particularly useful?
5. What role does clustering play in data mining, and how can it be utilised to enhance business intelligence?

5.8 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. Globus Bank is exploring the use of data mining to improve customer experience and operational efficiency. They want to focus on fraud detection, customer segmentation and credit scoring. How does the scope of data mining align with the bank's goals, and what additional areas might it explore for business optimisation?
2. At Stellar Inc., a retail chain, the data analytics team is working on analysing customer transactions to predict which products customers are likely to buy together. However, the initial results seem too broad to provide actionable insights. The team needs to refine the process. Which data mining techniques can the team use to effectively identify product combinations, and how can the data mining process help them refine their results?
3. A ride-hailing company wants to determine peak areas where demand surges during specific times. How can clustering be applied here?
4. A supermarket wants to analyse product sales to determine frequently co-purchased items and create effective cross-selling strategies. What metric is most important when evaluating association rules?
 - a. Recall
 - b. Support
 - c. Accuracy
 - d. Mean squared error
5. A retail company wants to increase sales by targeting specific customer segments. They have customer purchase history and demographic data. What data mining technique should they apply to identify these segments?
 - a. Clustering
 - b. Supervised learning
 - c. Association
 - d. Regression

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5.9 ANSWERS AND HINTS**ANSWERS FOR SELF ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Data Mining	1.	a. Data quality and accuracy
	2.	b. Market basket analysis
	3.	False
Data Mining Techniques	4.	c. Clustering
	5.	c. Predicting continuous values
	6.	False
Unsupervised Learning Techniques	7.	b. FP-Growth
	8.	c. DBSCAN

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	c. Predictive analytics
2.	d. Data preprocessing
3.	c. Data visualisation
4.	b. Informed consent
5.	a. Decision trees
6.	d. K-means clustering
7.	c. Logistic regression
8.	b. To generate insights from data
9.	c. Apriori
10.	b. Hierarchical methods

HINTS FOR DESCRIPTIVE QUESTIONS

1. The term data mining refers to a method of obtaining relevant and beneficial information from datasets. In this case, it means visualising or interpreting the data with the help of predictive analysis to derive its trends and patterns. This technique is used for a range of applications such as target marketing, crime detection, sales forecasting and systems advice. Refer to Section 5.2
2. The overall structure of the data mining involves five distinct activities which are: problem statement, data collection and preparation, data analysis, pattern evaluation and interpretation

of the results. Each stage is in line with the objectives of the organisation enabling productive information to be obtained without unnecessary disturbances. Refer to Section 5.3

3. Data mining is one of the important aspects of the decision-making process in the public health, financial, retail, or industrial sectors. Some examples include health predictions, financial risk management, marketing analysis and industry optimisation. Refer to Section 5.2
4. Unsupervised learning creates models in the absence of labelled data, resulting in pattern recognition and structure detection in data, which are well-suited for exploratory processes such as customer segmentation and fraud detection. Refer to Section 5.4
5. Clusters consist of objects that are similar to each other and help businesses drive trends, improve focused marketing and manage stock levels more effectively. It also aids in new product development, risk management and improving operational effectiveness. Refer to Section 5.4

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	The scope aligns with fraud detection using classification techniques, customer segmentation via clustering, and credit scoring through supervised learning models. Additionally, the bank could explore churn prediction, loan default analysis and personalised marketing to enhance operations further.
2.	The team can use the association rule mining technique to identify product combinations (e.g., market basket analysis). Refining the process involves data cleaning, transformation and selecting relevant features. Steps like data pre-processing, pattern evaluation, and iterative improvement are essential to filter meaningful associations.
3.	Using clustering algorithms like K-means, the company can group geographical locations based on demand patterns. This helps in optimising driver allocation and dynamic pricing strategies to meet demand efficiently during peak hours.
4.	b. Support
5.	a. Clustering

N O T E S

5.10 SUGGESTED READINGS & REFERENCES**SUGGESTED READINGS**

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CASE STUDY

N O T E S

PREDICTING HOSPITAL READMISSIONS WITH DATA MINING

SCENARIO

A significant healthcare organisation seeks to improve patient care by minimising the readmission rate of patients within 30 days post-discharge. They believe that data mining techniques can be used to identify patients at high risk of readmission, so they can target them with preventive interventions.

DATA

There are extensive patient records available at the hospital. Such records include:

- ❑ Although the patients' demographic can be contained, specific details such as age and gender are given.
- ❑ Also includes evidence of medical and surgical diagnoses as well as details of drugs for current or previous medications and procedures done.
- ❑ These include household income, level of education, and even housing characteristics.
- ❑ Hospital discharge (e.g., length of stay, admission diagnosis, discharge medications)

DATA MINING TECHNIQUES

The following are the techniques used by the hospital:

- ❑ **Clustering:** Clustering algorithms allow the hospital to group patients who have similar characteristics. This helped them to determine which segments of patients were likely to be readmitted for various reasons. For example, one cluster contained older patients suffering from multiple chronic illnesses, while another experienced younger patients with a history of drug abuse.
- ❑ **Classification:** Classification algorithms were also used in the hospital to create a prediction model indicating the likelihood of a patient being readmitted within thirty days after discharge. This model can then help in identifying patients who are at high risk and availing preventive measures to them.
- ❑ **Association rule mining:** Association rule mining is developed by the hospital to examine aspects of their data that provide a greater understanding of why certain groups of patients were readmitted more frequently than others. For example, they noted that a patient who is discharged on a Friday is more likely to be readmitted than one who is discharged on any other day of the week.



Case Objective

The aim of this case study is to apply the data mining methodology in order to find patients that have a high risk of 30-day readmission, thus enabling interventions and improvement of health care outcomes while dealing with data quality and ethical issues.

CASE STUDY

CHALLENGE

During the process, the following challenges were experienced:

- ❑ **Data quality:** Ensuring that proper quality data is utilised for data mining was a major concern. It also focused on the problem of inconsistencies or missing data to avoid presenting findings that cannot be corroborated by the analysis done on the other variables.
- ❑ **Model interpretability:** Emphasising the importance of understanding how the data mining model operates ensures clinicians have confidence in it, making it valuable for informed clinical management decisions.
- ❑ **Ethical issues:** While employing data mining in medicine, several ethical issues were considered. The use of the data was ensured to be within the limits that do not lead to any form of bias against a particular race or ethnicity in the patient dataset.

CONCLUSION

Data mining technologies can be of great assistance in enhancing the quality and effectiveness of healthcare services. In such a way that, through the help of data mining, it was possible for hospitals to identify highly probable patients to be readmitted and measures were put in place to mitigate the risk and there were better patient results. Even so, one must be mindful of the difficulties as well as the moral issues that come with applying data mining techniques in the healthcare industry.



QUESTIONS

1. In what way does data mining aid hospitals in preventing readmissions?

(Hint: Data mining enables the hospitals to stratify the patients according to risks, and then targeted strategies are employed to lower readmission rates and improve outcomes.)

2. What challenges are related to data mining in health care?

(Hint: Challenges concern ensuring high and effective quality of data to make clinicians understand how the model works for them to trust it and fairness and injustice issues about using such data towards the patients without bias.)

6

C H A P T E R

PREDICTIVE MODELLING IN BUSINESS ANALYTICS

CONTENTS

- 6.1 Introduction
- 6.2 Understanding the Predictive Modelling Concept
 - 6.2.1 Importance of Predictive Modelling
 - 6.2.2 Advanced Models in Business Analytics
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 - Activity
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- 6.4 Types of Predictive Models
 - 6.4.1 Trends in Predictive Analytics
 - Self Assessment Questions
 - Activity
- 6.5 Best Practices and Common Pitfalls When Implementing Models
 - Self Assessment Questions
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- 6.6 Summary
- 6.7 Multiple Choice Questions
- 6.8 Descriptive Questions
- 6.9 Higher Order Thinking Skills (HOTS) Questions
- 6.10 Answers and Hints
- 6.11 Suggested Readings & References

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**Case Objective**

The objective of this caselet is to apply predictive fitness analytics techniques for promoting a plan to increase health and wellness.

INTRODUCTORY CASELET**APPLICATION OF THE PREDICTIVE MODEL TO INCREASE HEALTH AND WELLNESS**

Due to increasingly advanced and connected medical devices, we now have a large dataset of fitness data gathered from people. This includes both regulated and unregulated health and wellness markets. However, all of this data is meaningless unless used to promote change. Consumers and the doctors are still trying to determine how to use this data in an actionable and insightful way. One answer from the data science perspective is to apply predictive fitness analytics techniques, analysing current and historical data and then promoting a plan to increase health and wellness.

Predictive analysis is a vital tool for measuring, benchmarking and improving health, fitness and wellness. It is universally agreed that predictive analysis can help increase the quality of healthcare, prevent adverse events, improve overall health and ideally decrease the cost of treatment. Employers are also using health and wellness programs to increase employees' engagement and productivity. The latest studies are trying to explore the connection between vitals collected from wearable health and fitness devices and predict approaching illnesses. Predicting potential and approaching illnesses can help us to take corrective measures or seek suitable treatment at the right time.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Understand the predictive modelling
- Explain the process of forming a predictive model
- Discuss the future trend of predictive modelling

6.1 INTRODUCTION

In the previous chapter, you learned about data mining, supervised learning and unsupervised learning.

The process of creating mathematical or computer representations of real-world business problems to make predictions, make choices, or solve issues is known as business analytics modelling. It allows businesses to turn massive amounts of data into useful information by applying a variety of statistical, mathematical and machine learning techniques.

These predictive models allow businesses to visualise different scenarios and their potential outcomes. They also aid in improving business procedures and addressing challenging business issues. Depending on the issue to be resolved and the available input data, the models might be either straightforward linear regressions or intricate machine learning models.

In this chapter, you will understand the concept of predictive modelling along with its importance and application. Then you will also learn about the process of framing a predictive model and understand the different types and trends associated with predictive modelling.

6.2 UNDERSTANDING THE PREDICTIVE MODELLING CONCEPT

Predictive modelling is a process used in predictive analytics to create a statistical model of future behaviour. Predictive analytics is the area of data mining involved with forecasting probabilities and trends. A predictive model consists of several predictors, which are variable factors that influence future behaviour or results. For example, in marketing a customer's gender, age, and purchase history might predict the likelihood of a future sale.

In predictive modeling, data for the relevant predictors is collected, followed by the formulation of a statistical model. Predictions are then generated, and the model is subsequently validated using additional data. The model may employ a simple linear equation or complex decision trees, planned out by sophisticated software.

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6.2.1 IMPORTANCE OF PREDICTIVE MODELLING

Predictive modelling is an important factor in business analytics, providing a methodical development for forecasting future outcomes based on historical data. Its applications range in multiple domains, offering practical tools for addressing specific business challenges. The importance of predictive modelling is as follows:

- ❑ **Decision support:** Predictive models are used to analyse historical trends and patterns to guide business decision-making processes.
- ❑ **Risk assessment:** These models are employed to identify potential risks and estimate their likelihood, enabling organisations to develop pre-emptive strategies.
- ❑ **Customer analysis:** Businesses utilise predictive analytics to analyse customer behaviour and preferences, informing targeted marketing strategies and customer relationship management.
- ❑ **Resource allocation:** Predictive modelling assists in forecasting resource needs, facilitating efficient allocation and cost management.
- ❑ **Anomaly detection:** These models are implemented in fraud detection systems to identify deviations from expected patterns in financial or operational data.
- ❑ **Market forecasting:** Predictive techniques are applied to anticipate market trends, aiding in strategic planning and product development.

These applications demonstrate the role of predictive modelling in enhancing operational efficiency and supporting evidence-based practices in business contexts. Business analytics uses data, statistical techniques, and technology to generate insights that support decision-making. Business Analytics consists of the following four types, as already discussed in Chapter 1: Introduction to Business Analytics.

- ❑ **Descriptive analytics:** This analysis answers “What happened?” by summarising historical data.

Example: A retail company analyses sales data from the past year to identify which products performed well and during which seasons.

- ❑ **Diagnostic analytics:** This analysis explains “Why did it happen?” by identifying the causes of past outcomes.

Example: A manufacturing firm investigates a decline in production efficiency by analysing machine performance data and workforce schedules.

- ❑ **Predictive analytics:** This analysis focuses on “What might happen?” by using historical data and statistical models to forecast future trends.

Example: An airline predicts passenger demand for different routes and seasons to optimise ticket pricing.

- **Prescriptive analytics:** This analysis addresses “What should we do?” by recommending optimal actions based on predictions and scenarios.

Example: A logistics company uses prescriptive analytics to determine the most efficient delivery routes considering traffic conditions, fuel costs and delivery deadlines.

For understanding predictive models, it is essential to start with basic modelling to build a foundational understanding of the problem, data and relationships. Basic modelling lays the groundwork for more advanced analysis by simplifying the initial steps of the process. The following are the steps of basic modelling:

1. **Understanding the problem:** The first step in any modelling approach is to define the business problem or objective clearly. This involves identifying the goals and understanding how they align with organisational priorities. Additionally, it is crucial to determine the variables involved and hypothesise their relationships. The key steps in understanding the problem are:
 - a. Define the business objective (e.g., forecasting sales, improving customer retention).
 - b. Identify key variables and their potential influence on the outcome (e.g., seasonality, pricing, customer demographics).
2. **Collecting data:** Data collection is central to modelling. The quality and relevance of the data determine the reliability of the model’s output. Data sources can be internal or external, or a mix of both and often provide the most comprehensive dataset. The data sources include:
 - ◆ Internal systems like sales data, inventory logs and operational metrics.
 - ◆ External sources like market research, competitor analysis, industry benchmarks and economic indicators.
3. **Simple models:** Simple models explore data relationships and generate initial insights. These models are easy to interpret, providing catalyst to more complex approaches. Some of the types of simple models are:
 - ◆ **Descriptive models:** Summarise historical data to identify trends and patterns (e.g., monthly sales trends over the past year).
 - ◆ **Rules-based models:** Use predefined rules or conditions, often represented as “if-then” logic (e.g., “if inventory falls below a threshold, reorder automatically”).

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- ◆ **Linear relationships:** Represent relationships using simple mathematical equations, such as $y = mx + b$, where y is the dependent variable, x is the independent variable, m and b represent the slope and intercept, respectively.
- 4. **Visualisation:** Visualisation tools help in understanding data patterns and relationships. Graphs, charts and scatter plots help to identify trends, outliers and correlations, guiding the development of more refined models. Some of the common visualisations include:
 - ◆ Line charts for trend analysis (e.g., revenue over time)
 - ◆ Scatter plots to observe correlations (e.g., sales vs. advertising spend)
 - ◆ Bar graphs to compare categorical data (e.g., sales by product category)

Basic modelling ensures a solid start to the analytics process, enabling businesses to explore data effectively and build a foundation for developing sophisticated predictive models.

6.2.2 ADVANCED MODELS IN BUSINESS ANALYTICS

Advanced models in business analytics provide specialised methods to address business challenges. They offer deeper insights by exploring relationships, categorising data, identifying patterns and optimising decision-making processes. Some of the most used models have been described with examples:

- **Regression analysis:** Regression analysis evaluates the relationship between dependent and independent variables to predict outcomes and uncover trends. This model can handle linear, polynomial and multivariate relationships. Beyond prediction, it is also used for hypothesis testing and determining which variables significantly affect outcomes.

Example: A marketing department applies regression analysis to measure the impact of digital ad spend, pricing strategies, and competitor activities on monthly revenue. By identifying which factors drive the most sales, the company can allocate resources more effectively.

- **Classification models:** Classification models utilise algorithms to assign data points to predefined categories or classes. They are foundational in machine learning and employ techniques such as decision trees, logistic regression and support vector machines. These models often work with labelled datasets for supervised learning tasks.

Example: A healthcare organisation uses classification models to predict patient health outcomes, such as classifying individuals

as “at-risk” or “not at risk” for diabetes based on attributes like age, weight and lifestyle habits.

- ❑ **Clustering models:** Clustering models group data points that share similar features into clusters without predefined labels, making them an unsupervised learning method. These models are valuable for exploring unstructured data and identifying latent patterns.

Example: A telecommunication company leverages clustering models to segment customers into groups based on usage patterns, such as high-data users, voice-call-centric customers and occasional users, enabling tailored service plans.

- ❑ **Optimisation models:** Optimisation models identify the best possible solution to a problem within a set of constraints. These constraints could include budget limits, resource availability, or time constraints. Common optimisation techniques include linear programming, nonlinear programming and integer programming.

Example: A supply chain manager uses optimisation models to determine the most cost-effective shipping routes and schedules, considering factors such as fuel costs, delivery times and warehouse capacities.

6.2.3 INTEGRATION OF TECHNOLOGY

Integrating advanced technologies with business analytics models enhances their functionality and scalability. These technologies empower businesses to process large datasets, uncover insights and implement solutions efficiently. The advanced technologies that are integrated for forming better models are explained as follows:

- ❑ **Data warehousing:** Data warehousing combines and arranges data from multiple sources into a centralised repository. This structure supports advanced querying, reporting, and analytics by ensuring data consistency and accessibility. Data warehouses often employ extract, transform, and load processes to prepare data for analysis.

Example: A retail company uses a data warehouse to integrate data from in-store sales, online transactions and customer loyalty programs. This allows them to generate unified reports, such as total sales per product category across all channels.

- ❑ **Machine learning:** Machine learning enables systems to improve predictions and performance by learning from data. It uses advanced algorithms to identify patterns and relationships. Machine learning models can be supervised, unsupervised, or reinforced, depending on the application.

Example: A financial institution applies machine learning to detect fraudulent transactions by analysing anomalies in customer

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spending patterns. These systems become increasingly accurate over time as they learn from new data.

- ❑ **Simulation:** Simulation involves creating digital models of real-world processes or systems to test various scenarios and predict outcomes without the associated risks or costs. It often uses tools such as Monte Carlo simulations and discrete-event simulations.

Example: A car manufacturer uses simulation to predict the impact of varying production rates on supply chain efficiency and costs. The company can identify the most efficient production schedule by testing different scenarios.

By combining these advanced models and technologies, organisations can address challenges ranging from operational inefficiencies to strategic planning, fostering innovation and competitiveness in dynamic business environments. You will study the process for framing a predictive model in the next section.



SELF ASSESSMENT QUESTIONS

1. Which of the following is not an application of predictive modelling?
 - a. Decision support
 - b. Risk assessment
 - c. Random guessing
 - d. Customer analysis
2. Descriptive analytics focuses on explaining “Why did it happen?” by identifying the causes of past outcomes. (True/False)
3. What type of analytics focuses on determining “What might happen?” using historical data and statistical models?
 - a. Descriptive analytics
 - b. Diagnostic analytics
 - c. Predictive analytics
 - d. Prescriptive analytics
4. Data warehousing integrates data from multiple sources into a central repository to support advanced querying and analytics. (True/False)



ACTIVITY

Research a company of your choice and identify how predictive modelling could be applied in its business processes. Present a report discussing potential benefits, challenges, and real-world applications of predictive analytics for the company.

6.3 FRAMING PREDICTIVE MODELS

The process of creating predictive models generally follows a structured approach regardless of the methodology used. Each step contributes to building an accurate and effective model that addresses the business problem and supports decision-making. A step-by-step process for framing predictive models is shown in Figure 6.1:

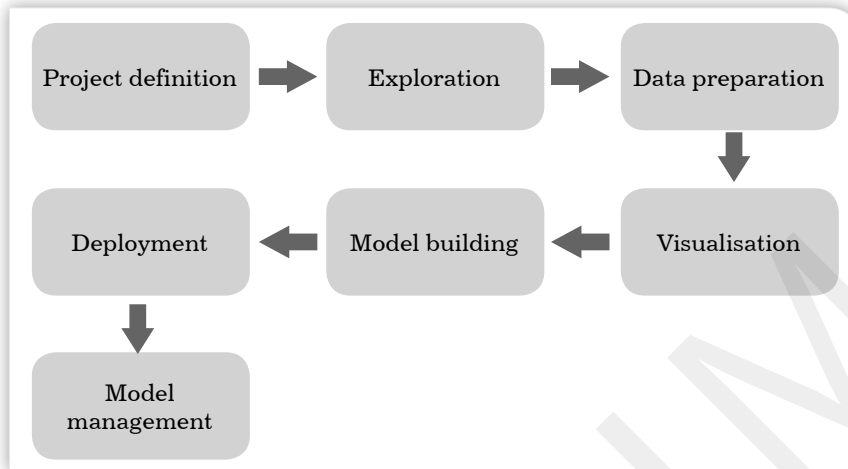


Figure 6.1: Step-by-step Process for Framing the Predictive Model

The following is the description of the steps for the process of creating predictive models:

1. **Project definition:** In this initial stage, it is crucial to clearly define the business objectives and desired outcomes of the project. These objectives are translated into predictive analytic goals and tasks, ensuring the model addresses the core business needs. The project definition helps identify the scope of the problem, the expected impact and how success will be measured. Clear objectives provide direction for all subsequent steps.
2. **Exploration:** During the exploration phase, the source data is analysed to understand its quality, structure and relevance to the business problem. This step is vital for identifying the most appropriate data and the best approach for building the model. Exploration helps determine the types of data that will be useful, the relationships between variables, and any issues in the data that need to be addressed.
3. **Data preparation:** Data preparation is one of the most time-consuming yet essential steps in predictive modelling. This stage involves selecting the relevant data, extracting it from various sources, and transforming it into a usable format. This process may include cleaning the data, handling missing values, and normalising or scaling the data to ensure compatibility with the modelling technique.

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4. **Visualisation:** Visualisation plays a crucial role in analysing data relationships and helps interpret the results of the model-building process. Visual tools like graphs, charts and plots detect patterns, trends and correlations between variables. Visualisation can guide the model development process by providing insights into how features interact and influence the target variable. It also helps stakeholders better understand the data and model results.
5. **Model building:** At this stage, predictive models are developed, tested and validated using the prepared data. Model building involves choosing the appropriate algorithm, training the model on the data and evaluating its performance. Testing includes checking how well the model generalises to unseen data. Validation ensures that the model meets the predefined metrics and objectives, such as accuracy or predictive power.
6. **Deployment:** Once the model has been validated, the next step is deployment. This phase involves applying the model's results to business decisions or processes. Deployment can range from providing insights to business users to integrating models into operational applications for automating decisions and processes. Monitoring the model's performance after deployment is essential to ensure it continues to meet business needs.
7. **Model management:** Model management focuses on maintaining and optimising predictive models after deployment. This step involves monitoring the model's performance over time to ensure it remains accurate and relevant. It also includes managing access, promoting the reuse of models and minimising redundant activities. Model management ensures that the model can be adapted or improved to keep pace with changes in business requirements or data.



SELF ASSESSMENT QUESTIONS

5. Which of the following is the primary focus of the “Exploration” phase in the predictive modelling process?
 - a. Cleaning the data
 - b. Analysing the quality, structure and relevance of the data
 - c. Developing predictive models
 - d. Deploying the model into business processes
6. The “Deployment” phase of the predictive modelling process includes the development of new predictive models. (True/False)



ACTIVITY

Define the business objectives for a predictive modelling project. Identify key data sources, describe the data exploration process, and outline how you would prepare the data for modelling. Discuss the importance of model validation and deployment in decision-making.

6.4 TYPES OF PREDICTIVE MODELS

Predictive analytics employs various techniques to uncover patterns and make informed predictions based on data. Each model type has unique strengths, making them suitable for different scenarios. Some of the key types of predictive analytical models are as follows:

- ❑ **Decision trees:** Decision trees are a straightforward and interpretable predictive model that organises data into hierarchical branches based on specific variables. Each branch represents a choice or condition, and the leaves at the end represent outcomes or decisions. A typical model of the decision tree which shows the internal node, branches and leaves is shown in Figure 6.2:

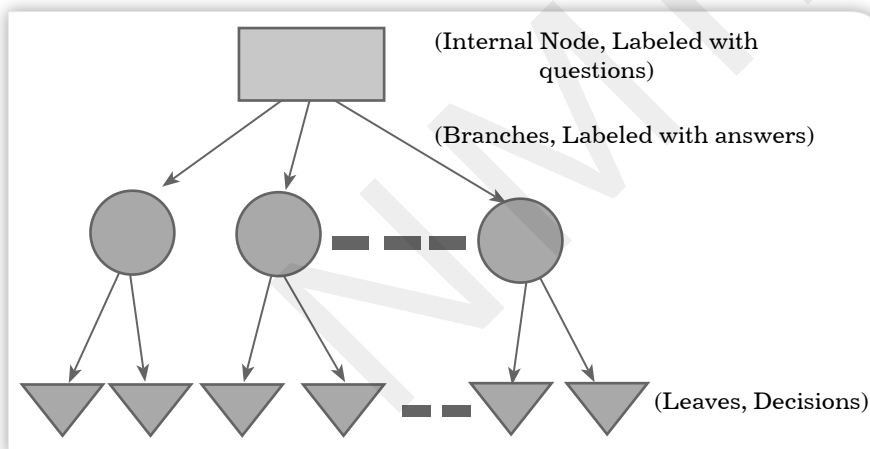


Figure 6.2: Decision Tree Model

Use case: A retail store uses a decision tree to identify the factors influencing customer purchases. Variables such as product price, promotions, and customer demographics define the branches, helping managers determine which promotions drive the most sales.

- ❑ **Regression models:** Regression models are widely used in statistical analysis to identify relationships between input variables (independent variables) and the target outcome (dependent variable). The model determines a mathematical equation that explains or predicts the outcome based on the

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input data. A regression model with a regression line, data points, dependent and independent variables is shown in Figure 6.3:

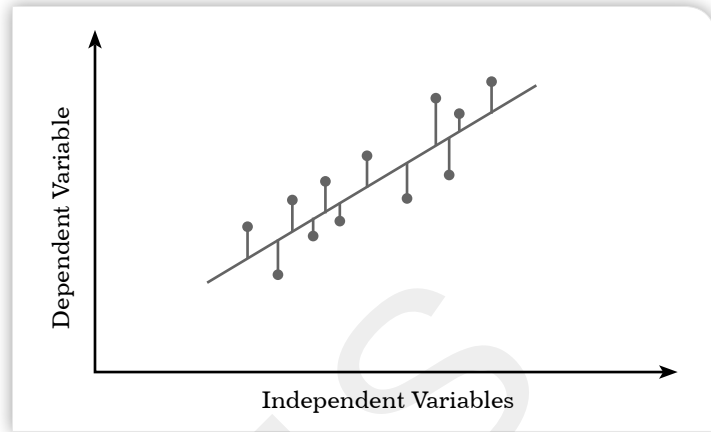


Figure 6.3: Regression Model

Use case: A financial analyst uses regression to predict stock performance based on factors such as price-to-earnings ratio, market trends and economic indicators. The model helps understand how changes in these variables affect stock returns.

- ❑ **Neural networks:** Neural networks are advanced models that use layers of artificial neurons to process data. They excel in handling complex, non-linear relationships and large datasets, making them ideal for intricate prediction tasks. The architecture of an artificial neural network is shown in Figure 6.4:

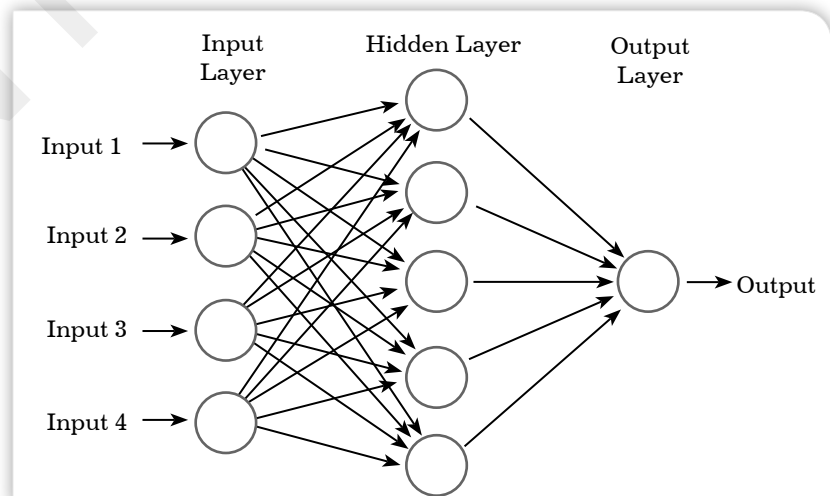


Figure 6.4: Architecture of an Artificial Neural Network

Use case: An e-commerce company uses neural networks to recommend products to customers. The model analyses purchase history, browsing behaviour, and preferences to provide personalised suggestions, boosting customer engagement and sales.

- ❑ **Clustering models:** Cluster models group similar data points based on shared attributes without predefined labels. This technique is useful for segmenting datasets to uncover hidden patterns or relationships. Figure 6.5 shows the visualisation of the cluster model which has two clusters:

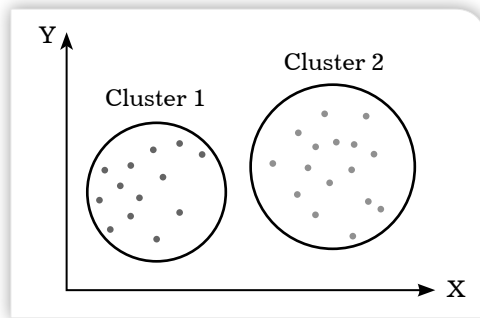


Figure 6.5: Cluster Model

Use case: Amazon.com uses cluster models to segment customers based on purchase frequency, average order value and account age. These clusters help design targeted marketing campaigns and identify growth opportunities.

- ❑ **Time-series models:** Time-series models focus on data points that are collected over time, analysing trends, seasonality and periodic patterns. These models predict future events based on past time-related data. A time-series model showcasing the Price vs. Year graph is shown in Figure 6.6:

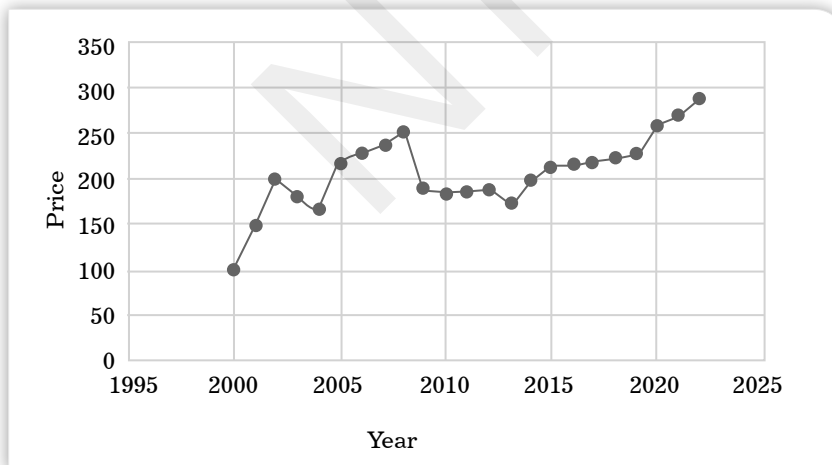


Figure 6.6: Time-series Model Visualisation

Use case: A call centre uses time-series models to predict peak customer service hours based on historical call volume data. This helps allocate staff efficiently to meet demand during busy periods.

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These models provide a toolkit for tackling a wide range of predictive analytics tasks, enabling organisations to derive actionable insights and improve decision-making across various domains.

6.4.1 TRENDS IN PREDICTIVE ANALYTICS

Predictive analytics is advancing quickly due to the rapidly evolving technology and its impact is being felt across industries. As businesses seek to gain a competitive edge by turning to predictive analytics to uncover valuable insights from their data, here are five predictions that will likely shape the future:

- ❑ **The demand for real-time data will rise:** Real-time data is becoming crucial for accuracy and relevancy as it updates predictive models as frequently as possible, keeping insights aligned with the changing reality at hand. As the need for such data keeps increasing, businesses will be able to make decisions that are not only timely but also accurate.
- ❑ **The rise of prescriptive analytics:** As we have read in Chapter 1, prescriptive analytics comes after predictive analytics, where the focus is no longer on predicting but on taking actions that will influence those predictions. By acting upon predicted insights before they come into place, organisations can steer the outcomes toward better pathways. Prescriptive analytics provides the power to optimise processes and outcomes to help one gain an edge in fast-changing landscapes.
- ❑ **Organisations will adopt synthetic data:** There is a phrase by Lisa Schwarz, Senior Director of Global Product Marketing at Oracle NetSuite, “Artificial Intelligence (AI) is only as good as the data that it is based on.” Equally, the strength of predictive analytics depends on the presence of quality data. However, the problem is that often there is a lack of necessary data in many organisations for building strong predictive models. The gap between these two can be bridged with the help of emerging tools to create “synthetic” data, or artificially generated data, that closely resembles real-world use cases. An organisation can thus start its predictive analytics journey even in the absence of sufficient data.
- ❑ **Adoption of machine learning and AI will continue to go up:** Next-generation companies are already using the power of machine learning and AI to predict with remarkable accuracy. As such, these techniques proved their worth in competitions as the organisations will continue to gain popularity in real-world issue solving.
- ❑ **Predictive analytics tools will democratise:** Today, it is not easy to implement and use predictive analytics effectively without specialised skills, which limits its adoption to technical experts.

That will change as industries shift toward user-friendly predictive analytics tools. The focus will be on intuitive usability, allowing non-technical users to leverage domain knowledge and generate predictions from their data. This will make predictive analytics more accessible and affordable, accelerating widespread adoption and greater business value.

The constantly changing field of predictive analytics reveals how organisations use data to their advantage. Organisations unlock new insights and opportunities, and they will keep evolving as these trends keep developing. Predictive analytics is going to play a big role in driving these trends through success, innovation and growth. Next, we will learn about the best practices and common pitfalls when implementing the predictive models.



SELF ASSESSMENT QUESTIONS

7. Which predictive model is best suited for analysing complex, non-linear relationships and large datasets?
 - a. Decision trees
 - b. Regression models
 - c. Neural networks
 - d. Cluster models
8. Time-series models are used to group similar data points based on shared attributes without predefined labels. (True/False)



ACTIVITY

Analyse a dataset using any two predictive models discussed above. Compare their outputs to determine which model better suits a specific business scenario, such as sales forecasting or customer segmentation.

6.5

BEST PRACTICES AND COMMON PITFALLS WHEN IMPLEMENTING MODELS

To develop effective and applicable predictive models for business decision-making, it is essential for analysts to follow ethical practices and maintain a disciplined approach. Often, the most successful models are built by avoiding common pitfalls such as bias, lack of clarity, or failure to consider the full scope of business needs. Analysts should prioritise transparency, ensure data integrity, and engage in clear communication with stakeholders to build models that are both reliable and relevant to business goals.

Additionally, it is important to regularly review and update models to keep them aligned with evolving business conditions. As industries and markets change, models must adapt to remain effective.

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Periodic adjustments are necessary to account for new data, shifts in strategy, or external factors that might influence business decisions. By maintaining a flexible approach and continuously refining model management practices, businesses can ensure the long-term sustainability and effectiveness of their predictive models.

Some of the best practices to consider when implementing models are:

- ❑ **Understand business objectives:** It is important to identify the business issue at hand and ensure the objectives of the model fit within the standards of the business. Be proactive in constant communication with the users of the model so that they are served as per the purpose of the model.
- ❑ **Data quality and preprocessing:** Spend resources on removing data errors, filling in the gaps, and other aspects of data quality control. Data preprocessing and feature engineering cannot be performed without a clear understanding of the industry to which the data applies. This knowledge allows for proper customisation of data preparation processes to suit the business and aims in question.
- ❑ **Feature selection:** It is important to include only those features in the model that are appropriate to prevent overfitting and enhance model understanding. In addition, employ domain expertise to influence the selection of features and prioritise those that are relevant to the business issue, as illustrated in Figure 6.7:

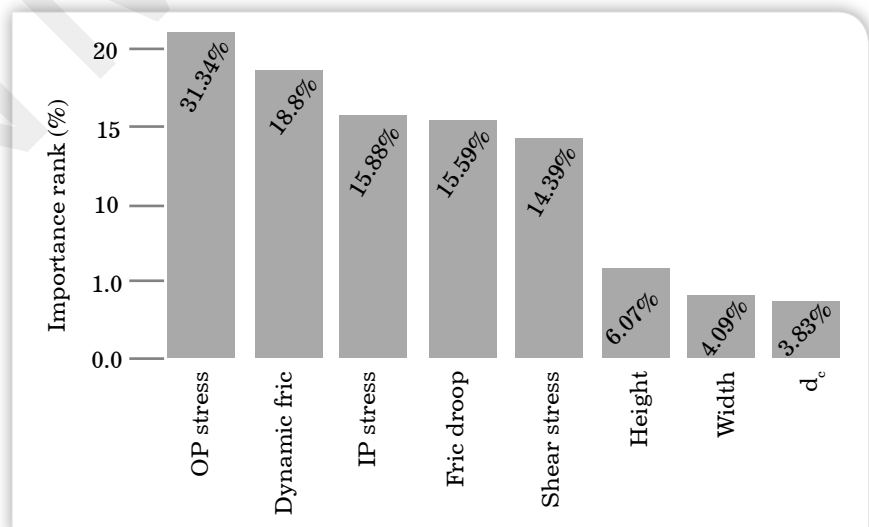


Figure 6.7: Influence of Feature Selection

- ❑ **Model selection:** The analysis and models should fit the nature of the business problem. When it comes to choosing analysis and models, issues of interpretation, scalability and complexity of implementation should be considered.

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- ❑ **Validation and evaluation:** It is necessary to adopt effective validation techniques to maintain the generalisability of the model. Moreover, appropriate metrics that support the business and are easy to understand are used.
- ❑ **Explainability and interpretability:** It is important to use any model that can easily be understood, especially when presenting to different stakeholders, and try as much as possible to explain the outputs of the model to gain confidence from the stakeholders.
- ❑ **Integration with business processes:** It is necessary to ensure the complete incorporation of the model into the running business operations by automating the deployment and monitoring aspects of the model to ensure that the model remains current about new data.
- ❑ **Collaboration and communication:** It is important to encourage analysis of complex issues across various data scientists, business analysts, and subject matter experts by making the technical insights interpretable in terms of business solutions.

Some common pitfalls encountered while implementing models are:

- ❑ **Lack of business understanding:** Not completely comprehending the business situation may result in models that do not meet the expectations of the stakeholders.
- ❑ **Ignoring data quality:** Failure to clean and preprocess the data can lead to unclean or prejudiced models.
- ❑ **Overfitting and underfitting:** Complicated models may overfit the training data but will not generalise to any new data; on the other hand, simple models have their disadvantages.
- ❑ **Inappropriate metrics:** Inaccurate model evaluation may arise from the use of metrics that are out of step with company goals.
- ❑ **Model black box:** Choosing extremely complex models without considering interpretability may have an impact on stakeholder support.
- ❑ **Poor model integration:** If the model is not incorporated into the existing business procedures, it may lose all of its value.
- ❑ **Lack of monitoring:** Failure to watch model performance over time will lead to loss of its predictive power.
- ❑ **Insufficient communication:** Successful deployment and acceptance of a model is achieved when there is proper interaction between the two groups.

Some of the effective methods to avoid overfitting in business analytics:

- ❑ **Cross-validation:** Employ methods such as cross-validation, where the given data is extended into k total divisions; k-1 of

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those are used to fit the model being tracked, and the remaining one is used for testing it. This last aspect consists of the rotation of various data subsets to assess the performance of the model's ability to generalise on different datasets.

- ❑ **Train with more data:** Increasing the size of the training dataset allows the model to learn the actual concepts of the data rather than the noise, which can result in a universal model. This is not always feasible but on almost all occasions, increasing data helps in enhancing the strength of the model.
- ❑ **Feature selection and engineering:** Eliminating or merging relevant features can minimise the model's complexity, hence minimising its chances of overfitting. Measures such as Principal Component Analysis (PCA) or regularisation techniques, including L1 and L2 regularisation may assist in such situations.
- ❑ **Simplify the model:** Employ simplistic models such as linear regression or decision trees with fewer depths instead of over-the-top complex models. This is because complex models tend to overfit the data, especially with small datasets.
- ❑ **Data augmentation:** If necessary, enhance the dataset through modified data points transformation (e.g., noise addition, resizing) for dataset vertical expansion and overfitting of the model to the training set patterns minimisation.



SELF ASSESSMENT QUESTIONS

9. What is the common pitfall in model implementation related to business understanding?
 - a. Train with more data
 - b. Lack of business understanding
 - c. Effective communication
 - d. Regular monitoring
10. Data preprocessing requires domain knowledge to ensure proper customisation for the business needs. (True/False)



ACTIVITY

Explore model implementation via a case study: align models with business needs, analyse pitfalls, and discuss best practices for effective deployment in groups.

6.6 SUMMARY

- ❑ Predictive modelling is a process used in predictive analytics to create a statistical model for predicting future behaviour.

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- ❑ Predictive analytics is the area of data mining involved with forecasting probabilities and trends.
- ❑ Predictive modelling is an important factor in business analytics, providing a methodical development for forecasting future outcomes based on historical data.
- ❑ Advanced models in business analytics provide specialised methods to address the business challenges. They offer deeper insights by exploring relationships, categorising data, identifying patterns and optimising decision-making processes.
- ❑ Integrating advanced technologies with business analytics models enhances their functionality and scalability.
- ❑ Predictive analytics employs various techniques to uncover patterns and make informed predictions based on data such as decision trees, regression models, cluster models and time-series models.
- ❑ Prescriptive analytics provides the power to optimise processes and outcomes to help one gain an edge in fast-changing landscapes.
- ❑ The most successful models are built by avoiding common pitfalls such as bias, lack of clarity, or failure to consider the full scope of business needs.
- ❑ It is important to regularly review and update models to keep them aligned with evolving business conditions. As industries and markets change, models must adapt to remain effective.



KEY WORDS

- ❑ **Business analytics:** Business analytics involves using data, statistical methods, and technology to support decision-making in areas such as marketing, risk management and resource planning.
- ❑ **Machine learning:** Machine learning is a technology that helps systems learn patterns from data, improving predictions and detecting unusual activities over time.
- ❑ **Prescriptive analytics:** Prescriptive analytics focuses on recommending the best actions to take, helping organisations improve efficiency and achieve better outcomes.
- ❑ **Predictive modelling:** Predictive modelling is the process of using statistical methods to forecast future outcomes based on data factors such as customer demographics and past purchases.
- ❑ **Regression analysis:** Regression analysis is a statistical method used to predict one variable based on the values of other related variables, such as sales trends influenced by pricing.

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6.7 MULTIPLE CHOICE QUESTIONS

1. Which of the following is not a key step in the basic modelling process for predictive analytics?
 - a. Data preparation
 - b. Exploration
 - c. Project definition
 - d. Implementation of AI
2. What type of analytics answers the question, “What might happen?”
 - a. Descriptive analytics
 - b. Diagnostic analytics
 - c. Predictive analytics
 - d. Prescriptive analytics
3. In predictive modelling, which stage involves transforming raw data into a usable format?
 - a. Model building
 - b. Exploration
 - c. Data preparation
 - d. Visualisation
4. Which predictive model is most suitable for categorising patients as “at-risk” or “not at risk” for a disease?
 - a. Clustering model
 - b. Classification model
 - c. Time-series model
 - d. Optimisation model
5. What is the primary purpose of regression analysis in predictive modelling?
 - a. Grouping similar data points into clusters
 - b. Uncovering relationships between variables to predict outcomes
 - c. Detecting anomalies in datasets
 - d. Assigning labels to unlabelled data points
6. Which visualisation tool is best suited for observing correlations between two variables?
 - a. Line chart
 - b. Bar graph
 - c. Scatter plot
 - d. Pie chart
7. What role does simulation play in predictive modelling?
 - a. It integrates machine learning algorithms into models.
 - b. It identifies the best solutions within constraints.
 - c. It predicts outcomes by testing various scenarios in a digital environment.
 - d. It groups data points into predefined clusters.
8. Which trend is expected to make predictive analytics more accessible to non-technical users?
 - a. Integration of machine learning

- b. Rise of prescriptive analytics
 - c. Use of real-time data
 - d. Development of user-friendly tools
9. What is a key benefit of using synthetic data in predictive modelling?
- a. Eliminating the need for data visualisation
 - b. Enhancing model deployment efficiency
 - c. Bridging the gap when real-world data is insufficient
 - d. Reducing data preparation time
10. Which of the following statements about clustering models is true?
- a. They predict outcomes based on labelled datasets.
 - b. They group data points with similar features without predefined labels.
 - c. They utilise decision trees to categorise data points.
 - d. They optimise resource allocation within constraints.

6.8 DESCRIPTIVE QUESTIONS

1. Describe the process of predictive modelling and explain its significance in forecasting business trends.
2. What are the primary steps involved in basic modelling for predictive analytics, and how do they aid in understanding the problem?
3. Outline the role of visualisation in predictive modelling and its importance in identifying data patterns.
4. Compare and contrast simple models and advanced models in predictive analytics, providing examples for each.
5. What are the key applications of predictive modelling in business, and how do they contribute to operational efficiency?

6.9 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. A telecommunication company wants to segment customers based on their usage patterns to develop tailored marketing strategies. Which predictive modelling technique should the company use?

a. Regression	b. Clustering
c. Classification	d. Neural Networks

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2. A financial institution uses machine learning for fraud detection in transactions. What additional step can enhance the accuracy of its predictive model?
 - a. Using unsupervised learning to label fraudulent transactions
 - b. Incorporating external data sources like transaction timestamps and IP addresses
 - c. Increasing the size of the labelled training dataset
 - d. Both b and c
3. A retailer uses a decision tree model to predict customer churn based on demographic and transaction data. If the model overfits the training data, what corrective action should be taken?
 - a. Increase the depth of the decision tree
 - b. Prune the decision tree by limiting its branches
 - c. Use a clustering algorithm instead of a decision tree
 - d. Add more features to the model
4. A financial firm aims to forecast stock market trends using regression models. Discuss the potential challenges the firm might face and strategies to overcome them.
5. A retail company is transitioning from descriptive analytics to prescriptive analytics for inventory management. What steps should they take to ensure the effective adoption of prescriptive techniques?

6.10 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No	Answer
Understanding the Predictive Modelling Concept	1.	c. Random guessing
	2.	False
	3.	c. Predictive analytics
	4.	True
Framing Predictive Models	5.	b. Analysing the quality, structure and relevance of the data
	6.	False
Types of Predictive Models	7.	c. Neural networks
	8.	False
Best Practices and Common Pitfalls When Implementing Models	9.	b. Lack of business understanding
	10.	True

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No	Answer
1.	d. Implementation of AI
2.	c. Predictive analytics
3.	c. Data preparation
4.	b. Classification model
5.	b. Uncovering relationships between variables to predict outcomes
6.	c. Scatter plot
7.	c. It predicts outcomes by testing various scenarios in a digital environment.
8.	d. Development of user-friendly tools
9.	c. Bridging the gap when real-world data is insufficient
10.	b. They group data points with similar features without predefined labels.

HINTS FOR DESCRIPTIVE QUESTIONS

- Predictive modelling uses data and statistical techniques to forecast future outcomes. It helps businesses predict trends such as customer behaviour, market conditions and risks. For example, in marketing, it predicts customer purchases, while in finance, it forecasts stock prices. Refer to Section 6.2
- The process includes understanding the problem, data collection, model building, validation, deployment and management. These steps ensure that the model aligns with business objectives and provides reliable predictions for decision-making. Refer to Section 6.3
- Predictive modelling enhances decision-making across industries. In retail, it predicts sales and customer preferences; in healthcare, it forecasts patient outcomes; in finance, it assesses risk; and in manufacturing, it anticipates equipment failures. Refer to Section 6.2
- Unsupervised learning identifies patterns in data without labelled outcomes. It's useful for customer segmentation, fraud detection and market analysis, helping uncover hidden structures in the data. Refer to Section 6.4
- Clustering groups data based on similarities and helps businesses segment customers identify trends and manage risks. For example, clustering can help target specific customer groups with tailored marketing strategies. Refer to Section 6.4

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ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No	Answer
1.	b. Clustering
2.	d. Both b and c
3.	b. Prune the decision tree by limiting its branches
4.	Challenges in forecasting stock market trends include data noise and market volatility. The firm can address these by using ensemble methods, combining multiple models to reduce bias and improve predictions. They should also incorporate risk-adjusted metrics to account for market fluctuations and avoid overfitting to historical data.
5.	To successfully adopt prescriptive analytics, the retail company must first integrate real-time inventory data and forecast demand patterns. Next, it should implement optimisation algorithms to recommend inventory levels based on demand forecasts. Continuous monitoring and adjusting the model will ensure optimal stock levels while minimising costs and stockouts.

6.11 SUGGESTED READINGS & REFERENCES**SUGGESTED READINGS**

- ❑ Pandey, A., Rajput, D. S., & Mohbey, K. K. (Eds.). (2020). *Predictive analytics using statistics and big data: Concepts and modeling*. Bentham Science Publishers.
- ❑ Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modelling*. Springer.

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- ❑ Forbes Technology Council. *Five key trends shaping the future of predictive analytics*. Forbes. Retrieved from <https://www.forbes.com/councils/forbestechcouncil/2023/10/05/five-key-trends-shaping-the-future-of-predictive-analytics/>

CASE STUDY

NOTES

PREDICTING AND PREVENTING TURNOVER AT HEWLETT-PACKARD (HP)

SCENARIO

HP, a global organisation with over 300,000 employees, faced significant challenges with employee turnover. In some sales divisions, turnover rates reached 20 per cent, resulting in an average tenure of only 4-5 years. High turnover imposed substantial costs, including recruitment expenses, low productivity, onboarding time and the loss of valuable knowledge and networks. Replacing mid-level employees costs HP approximately 150 per cent of their annual salaries, amounting to millions of dollars annually.

To address this issue, two scientists at HP leveraged two years of employee data to create a predictive model known as the Flight Risk score. This score quantified the likelihood of employees leaving, enabling managers to intervene proactively.

PROCESS

The following steps outline how HP used predictive analytics to address employee turnover:

1. **Data analysis:** The scientists leveraged data analysis by employing:
 - ◆ Historical employee data was collected, including pay, promotions, performance ratings and tenure.
 - ◆ Predictive models were developed to identify patterns and factors contributing to employee attrition.
2. **Model development:** The scientists created a predictive model using the following steps:
 - ◆ The Flight Risk score was created to predict attrition likelihood for each employee.
 - ◆ Intricate relationships among variables were uncovered, such as the increased likelihood of leaving when employees were promoted without receiving a substantial raise.
3. **Managerial access and training:** The scientists prevented data tampering by using the following steps:
 - ◆ Access to Flight Risk scores was restricted to select high-level managers.
 - ◆ Managers were trained to interpret the scores and understand the associated confidentiality and ethical considerations.
4. **Intervention strategies:** The scientists helped the managers by creating intervention strategies such as:



Case Objective

The objective of this case study is to use predictive analytics to identify employees at risk of leaving the organisation and implement targeted interventions to reduce turnover, minimise costs and maintain business continuity.

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CASE STUDY

- ◆ Managers were informed about key risk factors contributing to turnover and encouraged to develop retention strategies.
 - ◆ Proactive measures, such as addressing pay gaps or recognising high-performing employees, were implemented.
5. **Outcome monitoring:** The scientists monitored the outcome by:
- ◆ Using the system as an early warning tool, enabling prompt interventions, when necessary.
 - ◆ Tracking employee retention rates over time, providing feedback on the effectiveness of intervention strategies and allowing for continuous improvement in attrition management.

RESULTS

- ❑ HP identified actionable insights into why employees left the company, enabling targeted retention strategies.
- ❑ Proactive measures helped reduce turnover costs and maintain organisational stability.
- ❑ By applying predictive analytics, HP saved an estimated US \$ 300 million in costs related to employee attrition (Siegel, 2013).

CONCLUSION

HP's innovative use of predictive analytics provided managers with actionable insights to address employee attrition effectively. The Flight Risk score acted as an early warning system, fostering proactive decision-making and significant cost savings. This case highlights the value of integrating predictive analytics into HR practices to drive data-driven strategies for workforce management.

**QUESTIONS**

1. What factors were identified as contributing to employee turnover at HP?
(**Hint:** Analyse relationships between pay, promotions, performance ratings and turnover trends to determine why employees were leaving the organisation.)
2. How did the Flight Risk score benefit managers in addressing employee attrition?
(**Hint:** Provided predictive insights, enabled proactive interventions and helped managers create retention strategies based on data-driven risk factors.)

7

C H A P T E R

SIMPLE LINEAR REGRESSION

CONTENTS

- 7.1 Introduction
- 7.2 Concept of Simple Linear Regression
 - 7.2.1 Simple Linear Regression Model
 - 7.2.2 Estimated Regression Equation
 - 7.2.3 Least Squares Method
 - Self Assessment Questions
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- 7.3 Assessing the Fit of Simple Linear Regression Model
 - 7.3.1 Sum of Squares Method
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 - Self Assessment Questions
 - Activity
- 7.4 Summary
- 7.5 Multiple Choice Questions
- 7.6 Descriptive Questions
- 7.7 Higher Order Thinking Skills (HOTS) Questions
- 7.8 Answers and Hints
- 7.9 Suggested Readings & References

N O T E S

**Case Objective**

The objective of this caselet is to harness the power of simple linear regression at AdAnalytics Solutions, unravelling the relationship between advertising spending and sales revenue to optimise campaigns and elevate return on investment.

INTRODUCTORY CASELET**THE ANALYTICAL QUEST FOR ADVERTISING EFFICIENCY**

AdAnalytics Solutions undertook a comprehensive study to analyse how advertising expenditure relates to sales revenue. In particular, the analytics focused on developing a predictive model that would help estimate sales performance based on advertising spend using simple linear regression. While studying extensive historical data from multiple ad campaigns, the team identified certain trends and patterns that indicate the existence of correlation.

In accordance with the techniques of simple linear regression, the analytics unit analysed figures with the aim of calculating a regression equation which would express the relationship between advertising costs and sales figures. This equation served not only as an indicator for previous campaigns and their results but also became the basis for easier forecasting in the next marketing activities.

The results of the linear regression model allowed AdAnalytics Solutions to recommend using advertising expenditures to its customers. The expected resources of the model allowed for a strategic use of resources to ensure that every campaign maximised sales.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Define the concept of simple linear regression
- Explain the concept of assessing the fit of simple linear regression

7.1 INTRODUCTION

In the previous chapter, you learned about the basic concept of modelling used in business analytics. Then you learned about the linear models, binary models, best practices and common pitfalls when implementing models.

In business analysis, regression acts as a detective tool, systematically uncovering the intricate relations between various factors and their impact on outcomes. Specifically, linear regression proves invaluable for dissecting linear associations between variables. It assumes a linear relationship between an independent variable (e.g., advertising spending) and a dependent variable (e.g., sales), offering a practical framework for businesses to model and predict changes. This approach, grounded in data, becomes a strategic guide for decision-making in a complex market landscape.

In this chapter, we will learn about simple linear regression models and study how to assess their fit.

7.2 CONCEPT OF SIMPLE LINEAR REGRESSION

Simple linear regression is a powerful statistical tool used for data analysis and predictive modelling. Simple linear regression is a statistical method about the relationship between the dependent variable and one or more independent variables. The term “linear” signifies that we are dealing with a straight-line relationship. It is a versatile tool used across various disciplines, from finance to biology, to uncover relationships between variables and make informed predictions.

The primary objective of Simple Linear Regression (SLR) is to identify the optimal straight line that explains the relationship, and this line can also be used for prediction. It is a simple method of explaining and interpreting the relationship that exists between the two variables, hence it is useful for both analysis and prediction.

Some of the applications of SLR are as follows:

- ❑ **Sales forecasting:** Companies could apply SLR to previous data like advertising expenditures and seasonality, among others to estimate the future sales potential. For instance, to establish the

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sales potential, a company may look at sales income and correlates with planned advertising expenditures.

- ❑ **Cost analysis:** Businesses can determine how adjustments to one variable, such as production volume, impact expenses. For example, to improve budgeting and cost control, a manufacturer could employ SLR to assess how production levels affect overall costs.
- ❑ **Pricing strategy:** SLR can enable a company to understand the demand elasticity of its products. By external factors and the sales performance, a business organisation can make effective pricing decisions to optimise its revenues.
- ❑ **Customer satisfaction analysis:** The services may explore the influence of several factors, such as the speed of service versus the quality of the product, as far as the consumer satisfaction index is concerned. That helps in the perfection of the services provided to customers.
- ❑ **Marketing effectiveness:** SLR can be used to evaluate the success of marketing efforts by examining the relationship between marketing expenditure and customer satisfaction. Future marketing plans can benefit from this understanding.
- ❑ **Risk management:** SLR is frequently used in the finance industry to examine the link between risk factors (such as economic indicators) and asset returns to help firms make well-informed investment decisions.
- ❑ **Employee performance evaluation:** Employers can use SLR to guide training investments and performance management strategies by understanding the relationship between employee training hours and performance measures.
- ❑ **Inventory management:** Companies can optimise stock levels, save holding costs, and ensure product availability by examining the relationship between inventory levels and sales.

7.2.1 SIMPLE LINEAR REGRESSION MODEL

A SLR model in business analytics is a statistical technique used to examine the relationship between two variables- the dependent variable (the one being predicted) and the independent variable (the predictor). The goal of this model is to find a linear equation that best describes how the independent variable influences the dependent variable.

Some of the key components of a SLR model are as follows:

- ❑ **Dependent variable (Y):** This is the variable, whose value is changed in relation to the values of the independent variable. For example, sales revenue could be the dependent variable.

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- ❑ **Independent variable (X):** This is the variable, which is used to find or predict the value of dependent variable (for example, advertisement expenditure).
- ❑ **Linear relationship:** Simple regression assumes that the relationship between dependent and independent variables is a straight line.

The mathematical representation of a SLR model can be expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

X is the independent variable (the predictor),

Y is the dependent variable (the outcome to be predicted),

β_0 is the y-intercept of the regression line,

β_1 is the slope of the line (indicating the change in Y for a one-unit change in X),

ϵ is the error term, capturing the variability in Y not explained by X .

Figure 7.1 shows the visualisation of a SLR model correctly indicating the regression line, independent and dependent variables and the datapoints:

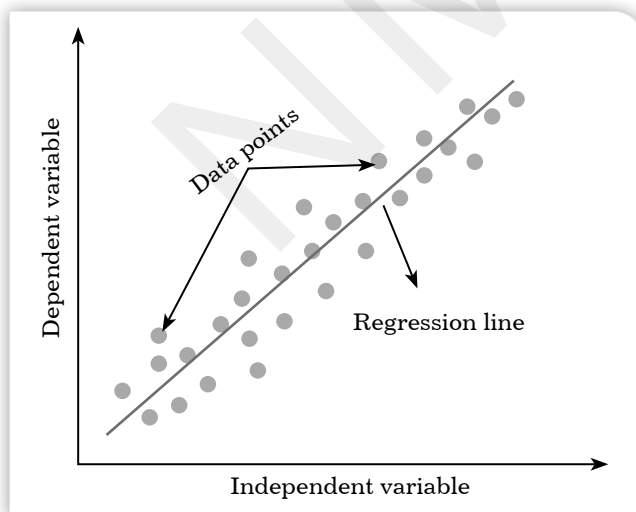


Figure 7.1: An SLR Model

7.2.2 ESTIMATED REGRESSION EQUATION

The estimated regression equation is one of the most significant aspects for understanding the relationship between independent variables and an outcome variable in business analytics.

NOTES

Regression analysis permits the application of numerical-data modelling and analysis in businesses, thereby making it possible to base decisions on empirical evidence.

The estimated regression equation is typically expressed in the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Where:

Y : It represents the dependent variable (the outcome we are trying to predict).

β_0 : The value β_0 describes a point at which the regression line crosses the y-axis. It commonly represents the value of Y that can be expected when the values of independent variables are all zeros.

$\beta_1, \beta_2, \dots, \beta_n$: The values $\beta_1, \beta_2, \dots, \beta_n$ are the values of associated independent variables coefficients $X_1, X_2, \dots, X_n$. These coefficients show the degree of change in the dependent variable due to alteration of each given predictor.

ϵ : It is the error term, accounting for the variability in Y that the models cannot explain.

The steps for deriving the estimated regression equation using Excel are as follows:

1. Prepare your data by arranging the data in the correct columns in Excel as shown in Figure 7.2:

	A	B
1	TV	Sales
2	230.1	22.1
3	44.5	10.4
4	17.2	9.3
5	151.5	18.5
6	180.8	12.9
7	8.7	7.2
8	57.5	11.8
9	120.2	13.2
10	8.6	4.8
11	199.8	10.6
12	66.1	8.6
13	214.7	17.4
14	23.8	9.2
15	97.5	9.7
16	204.1	19
17	195.4	22.4
18	67.8	12.5
19	281.4	24.4
20	69.2	11.3

Figure 7.2: Data of TV Quantity and Sales

NOTES

2. Select the columns in the Excel worksheet. From the Insert tab, select the Scatter Plot option. The visualisation will appear on the worksheet automatically. Figure 7.3 shows the visualisation of the scatter plot:

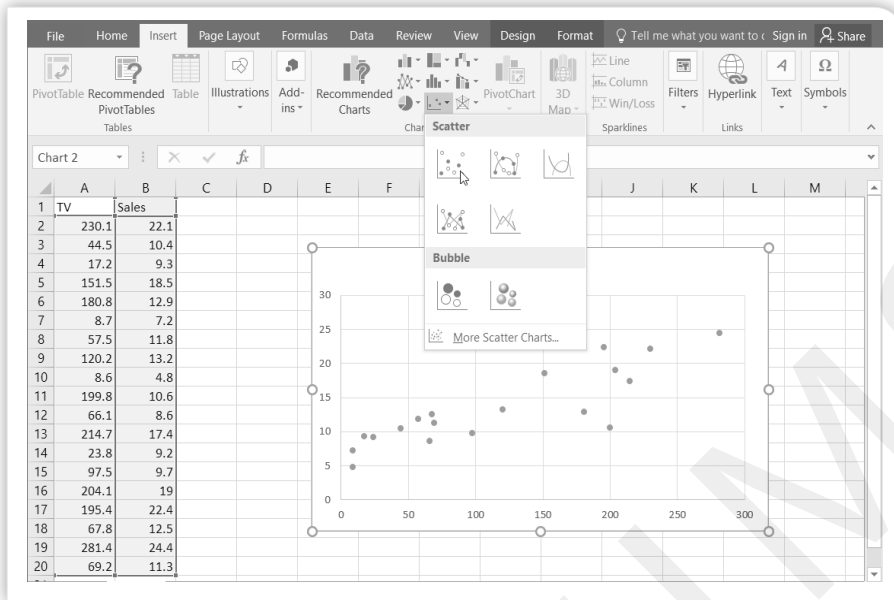


Figure 7.3: Visualisation of the Scatter Plot

3. Select More Options from Trendline dropdown from the Chart Elements option. Figure 7.4 shows the process of element addition in the chart:

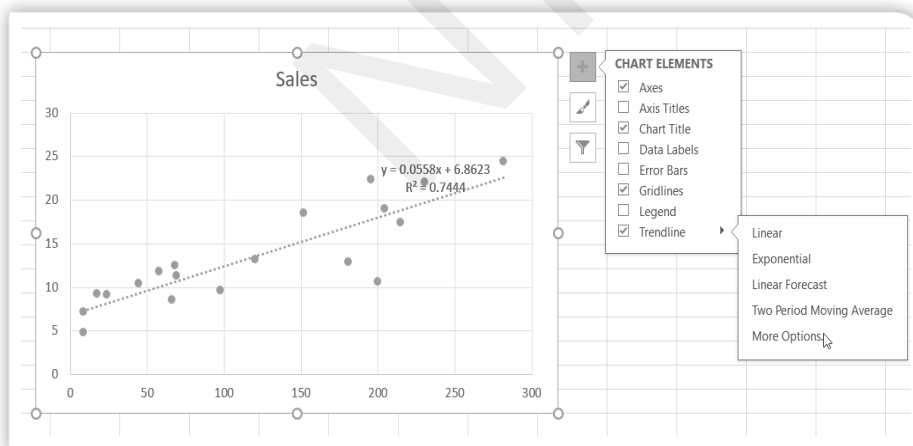


Figure 7.4: Element Addition in the Chart

4. Format the visualisation according to your need by selecting the Linear option from Trendline Options and checking the Display Equation on chart and Display R-squared value on chart in the Format Trendline dialogbox which opens on the right side of the Excel window.

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Figure 7.5 shows the formatting options in the scatter plot:

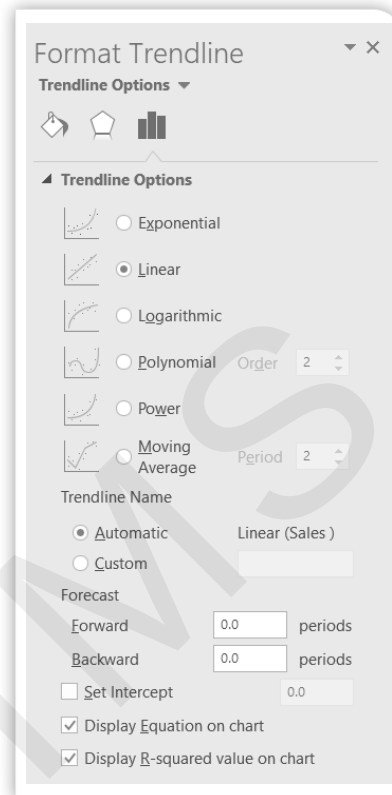


Figure 7.5: Formatting Options in Scatter Plot

5. View the plot in Excel after selecting OK. Figure 7.6 shows the view of the scatter plot with trendline, estimated regression equation and R^2 value:

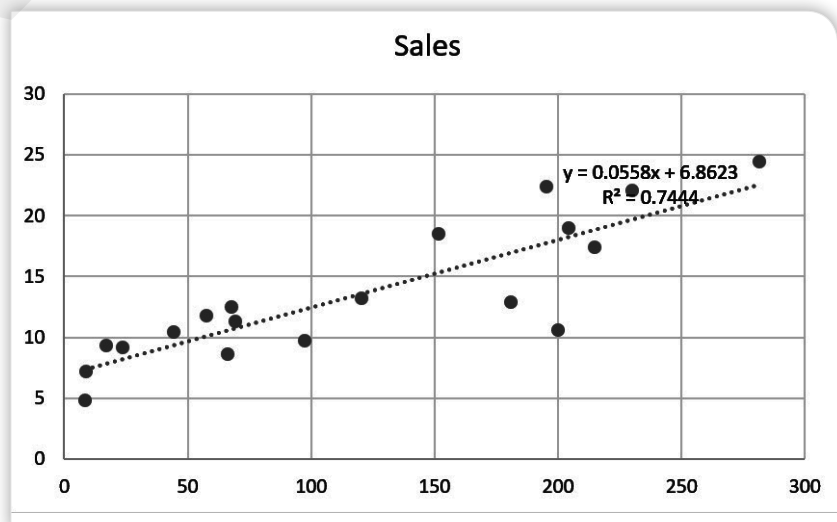


Figure 7.6: Scatter Plot with Trendline, Estimated Regression Equation and R^2 Value

7.2.3 LEAST SQUARES METHOD

The least squares method is a widely used statistical approach in business analytics for determining the link between two or more variables. Regression analysis is an important example of this technique, which involves estimation of one or more variables based on other variables. In this method, the best-fitting line or curve for the data available is the one where the sum of the squares of the vertical distance of each data point from the curve is the least or minimised.

In business analytics, it is perhaps most often used for forecasting, analysing trends and optimisation procedures. For example, many companies can readily do least squares regression modelling to forecast sales owing to historical data, and that helps them in the proper management of stock levels and in planning the market campaigns. Figure 7.7 shows the visual of the least square in linear regression:

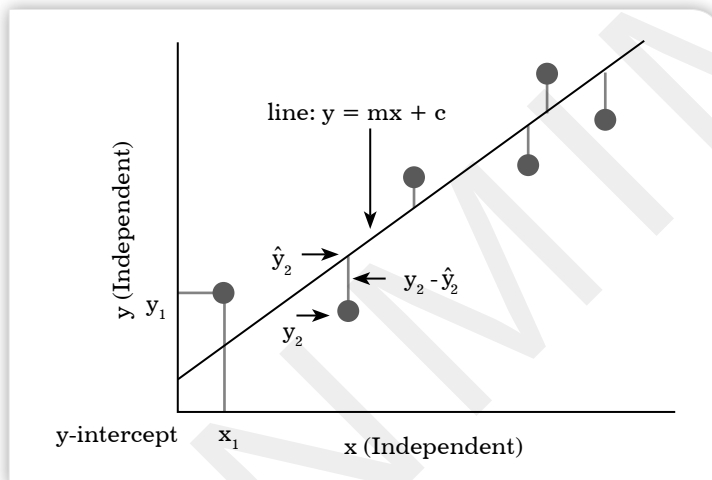


Figure 7.7: Least Square in Linear Regression

The classification of the least squares method as shown in Figure 7.7 are as follows:

- ❑ **Model representation:** This is a linear model represented as $y=mx+c$, where in this case, y is the dependent variable, x is the independent variable, m is the slope and c is the y-intercept.
- ❑ **Objective function:** The goal is to reduce the residual sum of squares, also expressed as $\sum(y_i - \hat{y}_i)^2$, where y_i is the measured data point and $\hat{y}_i$ is the estimated data point. The main purpose of objective function is to make the difference between the actual data points (y_i) and predicted points ($\hat{y}_i$) as small as possible by minimising the sum of the squared differences. This process in turn helps in improving the accuracy of the model.
- ❑ **Parameter estimation:** It entails determining the parameters m and b that help in minimising the objective function. This is done

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usually by taking partial derivatives of the objective function taken with respect to m and b and equating them to zero.

- ❑ **Solving equations:** The least squares method provides the optimal values of m and b that fit the linear equation to the observed data by minimising the overall difference between the actual and the predicted values.
- ❑ **Best-fitting line:** The best-fitting line equation is expressed when the appropriate parameters are calculated that minimise the sums of squares of differences between the points and the line fit, fulfilling the best linear representation of the data.

The least square estimated of x and y are:

$$m = \frac{\sum (X_i - \bar{X}) (Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

Where:

$\bar{X}$, $\bar{Y}$ are the means of X and Y , respectively.

X_i and Y_i are the individual datapoints of X and Y , respectively.

Example: Find the line of best fit using the least squares method for the given data on study hours and exam scores:

Study hours (in hours): [2, 3, 4, 5, 6]

Exam scores (out of 100): [65, 70, 75, 80, 85]

Solution: Let's denote study hours as x (independent variable) and exam scores as y (dependent variable). First, calculate the means of x and y values, denoted by $\bar{X}$ and $\bar{Y}$, respectively.

$$\bar{X} = \frac{(2+3+4+5+6)}{5} = 4$$

$$\bar{Y} = \frac{(65+70+75+80+85)}{5} = 75$$

Now, calculate the necessary values for the least squares method:

X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X}) \times (Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
2	65	-2	-10	20	4
3	70	-1	-5	5	1
4	75	0	0	0	0
5	80	1	5	5	1
6	85	2	10	20	4
Σ		0	0	50	10

Now, calculate the slope (m) using the formula:

$$m = \frac{\sum (X_i - \bar{X}) (Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{50}{10} = 5$$

Next, calculate the intercept (c) using the formula:

$$c = Y - mX = 75 - (5 \times 4) = 55$$

Thus, the equation of the line of best fit is $y = 5x + 55$. The equation $y = 5x + 55$ represents a straight line used to model the relationship between two variables, x and y. Here, x is the independent variable (input), and y is the dependent variable (output). The number 5 is the slope of the line, indicating that for every unit increase in x, y increases by 5. The number 55 is the y-intercept, showing the value of y when x equals zero. In practical terms, this line predicts the value of y based on x, following a consistent pattern of change. For example, if x increases by 2, y will increase by 10 (since $5 \times 2 = 10$).

After learning about the concept of SLR along with its model and equation, we will learn about the concept of assessing the fit of a SLR model.



SELF ASSESSMENT QUESTIONS

- Which variable in a simple regression model is used to make predictions?
 - Dependent variable (Y)
 - Error (ϵ)
 - Independent variable (X)
 - Slope (β_1)
- Which of the following options is an application of SLR in business analytics?
 - Website design optimisation
 - Cost analysis
 - Customer survey analysis
 - Product testing
- In the regression equation, the error term accounts for the variability in the dependent variable not explained by the predictor. (True/False)



ACTIVITY

Apply SLR to analyse business data, predict outcomes like sales from advertising spend, and derive strategic insights for students' business analytics comprehension.

N O T E S

7.3

ASSESSING THE FIT OF SIMPLE LINEAR REGRESSION MODEL

In business analytics, SLR is typically employed to comprehend and estimate the relationship between two variables—one being the dependent variable (outcome), while the other is the independent variable (predictor). Once a model for linear regression has been developed, the stage that follows consists of the examination of the model's fit, which refers to the degree to which the dependent variable's variation is accounted for by the regression model. Fitting a SLR model gives its users an understanding of the power and usefulness of the model and its predictive capabilities in forecasting and planning strategies, which is crucial to analysts and decision makers.

Some of the key metrics for assessing a model fit are as follows:

- ❑ **R-squared (coefficient of determination):** R-squared is the ratio of variance in the dependent variable, which is due to the independent variable. Its values are within the range of 0 and 1, with higher values signifying better fits.
- ❑ **p-value for the slope coefficient:** The p-value that is attached to the slope of the independent variable tests the efficacy of the relationship between dependent and independent variables. A p-value below a certain threshold (typically 0.05) indicates that the relationship is likely not due to chance, strengthening the model's validity.

The p-value in Figure 7.8 which shows $p = 0.16425$ exceeds the significance level (0.05), so we fail to reject the null hypothesis. This indicates insufficient evidence to conclude that the population mean salary differs from 80,000. The mean observed wage (216,500) may result from random variation, supporting the null hypothesis at the 5% significance level. For better understanding of the steps to find p-value in Excel, refer to Chapter 4.

	A	B	C	D	E	F
1	Position	Level	Salary			
2	Business Analyst	1	45000		Mean ($\bar{X}$)	216500
3	Junior Consultant	2	50000		Standard Deviation (s)	285054
4	Senior Consultant	3	60000		Count (n)	10
5	Manager	4	80000			
6	Country Manager	5	110000		Standard Error of Mean (SEM)	90142
7	Region Manager	6	150000		Degrees of Freedom (df)	9
8	Partner	7	200000		Hypothesised Mean (μ)	80000
9	Senior Partner	8	220000			
10	C.level	9	250000			
11	CEO	10	1000000		t-statistic	1.5143
12					p-value	0.16425
13						

Figure 7.8: p-value Output in Excel

- ❑ **Residual Standard Error (RSE):** The average deviation of the observed values from the regression line is computed through RSE, which in turn describes how accurate the model would be in predicting real life cases. As the RSE decreases, it implies that the estimated values of the model are closely approximating the true data values, hence the model is appropriate.

**NOTE**

The p-value help us to reject or fail to reject the null hypothesis in hypothesis testing.

Example: The study of a small-scale bookstore aims to investigate the possible connection between the number of hours spent by the staff members in arranging the book displays (X) and the total number of books sold on a daily basis (Y). The investigation takes a visual approach, employing a scatter plot to discern the degree to which a linear regression model could encapsulate the observed trend. A coherent linear pattern in the scatter plot would suggest a noteworthy correlation between the time invested in display organisation and the overall daily book sales.

A deeper dive involves scrutinising the vertical distances between data points and an imaginary regression line, shedding light on the consistency or variance in this relationship. This real-world analysis aims to elucidate whether the commitment of time to book display organisation translates into consistent and predictable changes in total daily book sales, offering valuable insights for strategic decision-making in the bookshop's daily operations.

7.3.1 SUM OF SQUARES METHOD

The sum of squares method is a statistical technique in business analytics that is used to measure the total variation within a dataset. By examining the differences between each data point and the mean, the method helps assess the degree of dispersion or spread in data. This method is critical for evaluating patterns and making informed decisions. The sum of squares is commonly applied in variance analysis, regression analysis and clustering, playing a fundamental role in identifying trends, optimising processes and minimising errors.

NOTES

Figure 7.9 shows the key components of the sum of square method in linear regression:

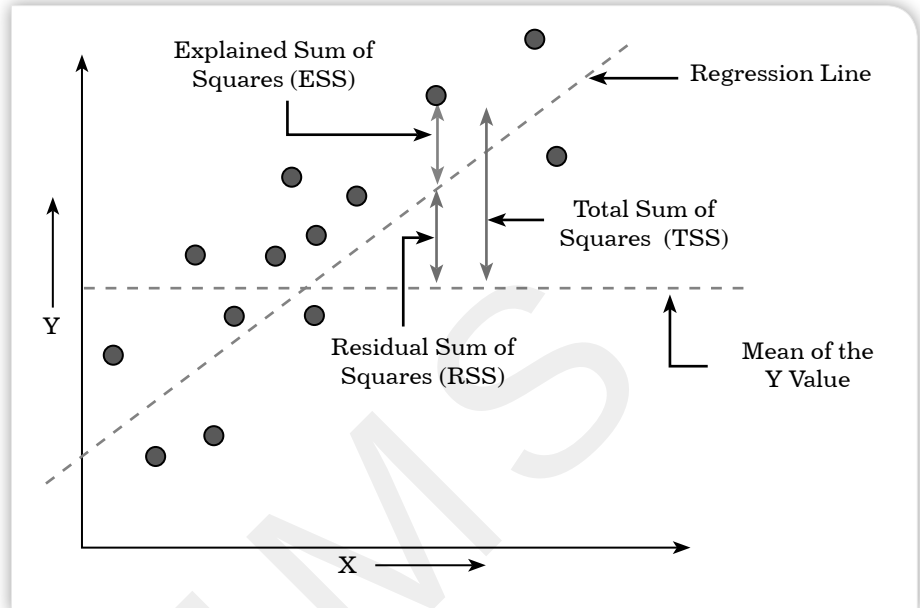


Figure 7.9: Sum of Squares in Linear Regression

The description of the components shown in Figure 7.7 is as follows:

- ❑ **TSS:** It is the sum of all squared deviations of each point from the mean. It is used to quantify the total variance present in the data.
- ❑ **ESS:** In regression, this component reflects the portion of total variation explained by the predictive model.
- ❑ **RSS:** It is commonly referred to as the error sum of squares. This metric quantifies the amount of variation or error in the model that remains unexplained by calculating the total of the squared differences between the actual values and the values predicted by the model.

7.3.2 THE CO-EFFICIENT OF DETERMINATION

The co-efficient of determination, or R squared (R^2), is one of the essential metrics because businesses use it in analysing the variation of the independent variables on the dependent variable and thus determining the performance of a given statistical model. It also helps carry out model accuracy assessment in most cases of regression, where the models handle the dependent variables and explain the percentage of the variance that can be accounted for by the independent variables.

In simple terms, R^2 is a measure of the extent to which the data points fall on the regression line or fit the regression model. It takes a value between 0 and 1:

- ❑ **0:** The model does not account for any of the variation in the dependent variable.
- ❑ **1:** The model accounts for all the variation in the dependent variable perfectly.

An R^2 value approaching 1 implies a higher dependence or relationship between the variables in question, but values nearer to 0 show an insignificant relationship. In business, this measurement enables the analysts to view the accuracy of the prediction or regression models and help the organisations make better decisions in relying on forecasts based on data.

The co-efficient of determination is commonly used to:

- ❑ **Evaluate model performance:** This aids analysts in understanding the effectiveness of models utilised in financial risk analysis, client segmentation and sales forecasting.
- ❑ **Compare models:** Companies frequently evaluate several models and choose the one with a higher R^2 since it suggests better predictive capabilities.

7.3.3 MODEL ASSUMPTIONS

There are some important requirements that must be satisfied for the analysis and interpretation to be valid. In SLR, these assumptions offer foundations for making correct inferences and reliable forecasts.

The major assumptions of the linear regression model are:

- ❑ **Linearity:** The assumption of linearity states that the relationship between the Independent Variable (IV) and the Dependent Variable (DV) must be linear. This means that changes in the IV should result in proportional changes in the DV across all IV levels. A linear relationship ensures that the regression model correctly captures the underlying pattern in the data. If this assumption is violated, predictions may be inaccurate, and the coefficients may not reflect the true effect of the IV on the DV. Scatterplots of the IV versus the DV or residual plots can be used to check for linearity as shown in Figure 7.10. For example, in a study examining the effect of study hours on exam scores, a linear relationship implies that increasing study hours consistently improves scores without unexpected fluctuations.

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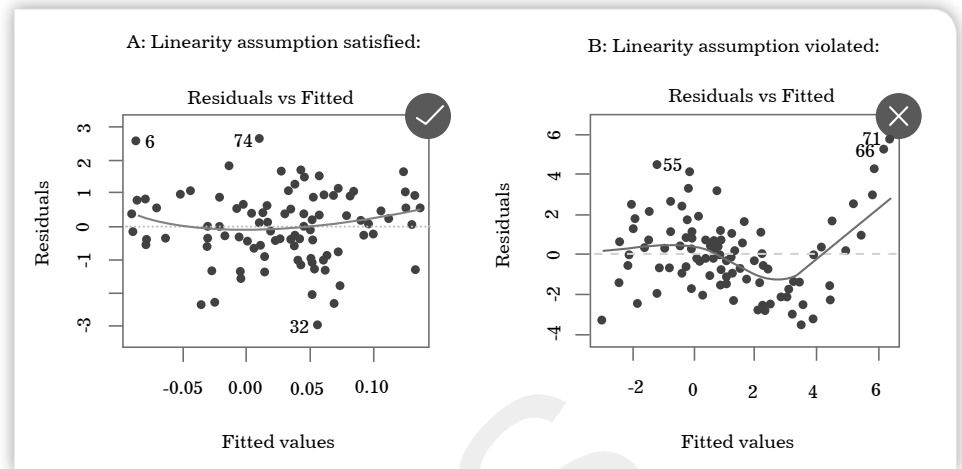


Figure 7.10: Satisfied and Violated Assumptions of Linear Regression

- ❑ **Independence of observations:** The independence of observations assumes that the value of one observation does not influence or depend on the value of another. This assumption is critical in ensuring unbiased and accurate parameter estimates in the regression model. Violations of independence often occur in clustered or time-series data, where observations may be naturally dependent. For instance, in a classroom setting, one student's exam score should not be affected by another student's performance. The Durbin-Watson statistic is commonly used to test independence in time-series data. Ensuring independence prevents systematic bias and ensures that the results of the regression are valid.
- ❑ **Homoscedasticity:** Homoscedasticity refers to the assumption that the variance of the residuals, or errors, remains constant across all levels of the independent variable. This means that the spread of the residuals should be roughly the same regardless of the value of the IV. If this assumption is violated, resulting in heteroscedasticity, the regression model may produce inefficient estimates, leading to biased hypothesis tests. A residual plot, where residuals are plotted against predicted values, can be used to check for homoscedasticity. For example, in a regression model examining the effect of study hours on exam scores, the prediction errors should not suddenly increase for students who studied more.
- ❑ **Normality of residuals:** The normality assumption requires that the residuals (differences between observed and predicted values) are normally distributed. This assumption is particularly important for hypothesis testing and constructing confidence intervals in smaller sample sizes. If residuals deviate significantly from normality, p-values and confidence intervals may be unreliable. Normality can be assessed using visual tools like histograms or Q-Q plots and statistical tests such as the Shapiro-

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Wilk test. For instance, in a regression model predicting exam scores based on study hours, the residuals should form a bell-shaped curve, with most errors being small and only a few being large.

- ❑ **No multicollinearity:** For SLR, the assumption of no multicollinearity ensures that independent variables are not highly correlated with one another. High multicollinearity makes it difficult to isolate the individual effects of each IV on the DV, leading to inflated standard errors and unreliable coefficient estimates. Variance Inflation Factor (VIF) is commonly used to detect multicollinearity, with values exceeding 5 or 10 indicating a potential problem. For example, in a model predicting exam scores, if both study hours and library time are highly correlated, it becomes challenging to determine whether studying or spending time in the library has a stronger impact on the scores.
- ❑ **No autocorrelation:** The assumption of no autocorrelation states that the residuals should not be correlated with one another. This is particularly relevant for time-series data, where observations may depend on previous values. Autocorrelation can lead to underestimated standard errors and biased statistical tests. The Durbin-Watson test is commonly used to detect autocorrelation, with values near 2 indicating no autocorrelation. For example, in a time-series model predicting monthly sales, if the residuals for one month are positively correlated with the residuals for the previous month, the model's predictions and inference may be unreliable.

Disregarding these assumptions can produce skewed estimate results, non-credible conclusions, as well as incorrect forecasts. Evaluation of such assumptions is important and should be done using various diagnostic tools such as residual plots, normal probability plots and statistical tests. Should the assumptions fail, some corrections or transformations may be needed to carry out a more effective analysis.



SELF ASSESSMENT QUESTIONS

4. Which metric measures the amount of variance in the dependent variable explained by the model?
 - a. p-value
 - b. Residual standard error
 - c. R-squared
 - d. Mean squared error
5. Which of the following represents the total variation in the data?
 - a. Residual Sum of Squares (RSS)
 - b. Explained Sum of Squares (ESS)
 - c. Total Sum of Squares (TSS)
 - d. Adjusted R-squared

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6. A high p-value for the slope coefficient indicates a strong relationship between the dependent and independent variables. (True/False)



ACTIVITY

Analyse business data, build regression models and evaluate fit using R-squared, sum of squares and assumptions. Discuss implications for decision-making in group presentations.

7.4 SUMMARY

- ❑ SLR is a powerful statistical tool used for data analysis and predictive modelling.
- ❑ A SLR model is a statistical technique used to examine the relationship between two variables- the dependent variable (the one being predicted) and the independent variable (the predictor).
- ❑ The Least Squares Method is a mathematical approach used to estimate the coefficients of the regression line by minimizing the Residual Sum of Squares (RSS).
- ❑ The sum of squares measures the variability in the dependent variable Y and is used to assess how well the model fits the data.
- ❑ The co-efficient of determination, or R squared R^2 , is one of the essential metrics because businesses use it in analysing the variation of the variation in dependent variable explained by independent variable and thus determining the performance of a given statistical model.
- ❑ Model assumptions in SLR are very important requirements that must be satisfied for the analysis and interpretation to be valid.



KEY WORDS

- ❑ **Estimate:** It refers to the value that has not been directly observed but is known based on certain statistical procedures or machine learning algorithms and calculated from the data given.
- ❑ **Homoscedasticity:** It is the assumption in statistics that the variability of errors in a regression model remains constant across all levels of the independent variable, indicating consistent variance.
- ❑ **Optimisation:** It is the process of adjusting model parameters or features to achieve the best possible performance or outcome, often involving minimising or maximising an objective function.

- ❑ **Random error:** This is an unpredictable variation in data caused by uncontrolled factors, influencing measurement imprecision but averaging out over time; inherent in statistical models or measurement processes.
- ❑ **Regression:** In machine learning, regression is a statistical technique that models the relationships between independent and dependent variables in order to draw conclusions or make predictions.

7.5 MULTIPLE CHOICE QUESTIONS

1. What does the error term (ϵ) in a regression model represent?
 - a. Predicted outcome
 - b. Variability in Y not explained by X
 - c. Total sum of squared errors
 - d. Average of all predictors
2. Which of the following domain uses linear regression the most?
 - a. Architecture
 - b. Astronomy
 - c. Finance
 - d. Linguistics
3. Which of these describes a straight-line relationship between the dependent and independent variables?
 - a. Linear relationship
 - b. Curvilinear relationship
 - c. Logarithmic relationship
 - d. Quadratic relationship
4. In a regression equation, if all independent variables are zero, the dependent variable value can be predicted by:
 - a. β_1
 - b. ϵ
 - c. X
 - d. β_0
5. Which equation form represents an estimated regression equation with multiple independent variables?
 - a. $Y = \beta_0 + \beta_1 X + \epsilon$
 - b. $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$
 - c. $Y = \beta_0 + \epsilon$
 - d. $Y = \beta_0 + \beta_1 + \beta_2$
6. Which assumption in linear regression states that residuals should not be correlated with each other?
 - a. Linearity
 - b. Homoscedasticity
 - c. Independence of residuals
 - d. Normality of residuals

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7. Which p-value indicates that the relationship between variables is statistically significant in regression?
 - a. 0.01
 - b. 0.05
 - c. 0.10
 - d. 0.50
8. Which of the following values of R-squared (R^2) indicates a perfect fit of the model?
 - a. 0
 - b. 0.5
 - c. 1
 - d. 0.25
9. Which of the following assumptions of SLR involves the relationship being represented by a straight line?
 - a. Homoscedasticity
 - b. Linearity
 - c. Independence of residuals
 - d. Normality of residuals
10. What does an Adjusted R-squared value account for?
 - a. The number of predictors in the model
 - b. The relationship between the variables
 - c. The variance of the residuals
 - d. The number of data points

7.6 DESCRIPTIVE QUESTIONS

1. What is the concept of SLR?
2. What is the least squares method in SLR?
3. How do you assess the fit of a SLR model?
4. What is the sum of squares method used for in regression analysis?
5. What is the co-efficient of determination (R^2) in SLR?

7.7 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. Alice applies the least squares method to find the best-fitting line for the data. The goal is to minimise the sum of the squared differences between the observed sales and the predicted sales. How does the least squares method help Alice in determining the best-fit line for her SLR model?
2. At Innovate Inc., Maria is analysing the relationship between employee training hours and productivity. She develops a

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simple regression model to predict productivity (in units) based on the number of training hours. After analysing the data, she finds that the regression equation is:

Productivity = $5 + 2.5 * \text{Training Hours}$. What can Maria conclude about the relationship between training hours and productivity?

3. At RetailPro, Sara has built a SLR model to predict the monthly sales of a product based on advertising spend. After fitting the model, she checks the SS values to assess the model's fit. If the TSS is 1000 and the RSS is 700, what can Sara infer about the model's fit?
4. Tom, a business analyst at FinanceSolutions, is using the SS method to assess the performance of a regression model that predicts loan default based on income and credit score. He notices that the RSS is much smaller than the TSS. What does this indicate about his model?
 - a. The model has a high level of unexplained variation.
 - b. The model explains most of the variation in the loan default data.
 - c. The model is overfitting the data.
 - d. The model has large residual errors.
5. George, a senior data scientist at GreenEnergy Solutions, is reviewing the assumptions of a SLR model he developed to predict energy consumption based on weather data. He notices that the residuals are randomly scattered without any discernible pattern. What assumption of the linear regression model is George verifying with this check?
 - a. Linearity of the relationship between variables
 - b. Normality of the residuals
 - c. Homoscedasticity (constant variance of residuals)
 - d. Independence of the residuals

7.8 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No	Answer
Concept of Single Linear Regression	1.	c. Independent variable (X)
	2.	b. Cost analysis
	3.	True

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Topic	Q. No	Answer
Assessing the Fit of Simple Linear Regression Model	4.	c. R-squared
	5.	c. Total Sum of Squares (TSS)
	6.	False

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No	Answer
1.	b. Variability in Y not explained by X
2.	c. Finance
3.	a. Linear relationship
4.	d. β_0
5.	b. $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$
6.	c. Independence of residuals
7.	b. 0.05
8.	c. 1
9.	b. Linearity
10.	a. The number of predictors in the model.

HINTS FOR DESCRIPTIVE QUESTIONS

1. SLR refers to a statistical technique to determine the association of two or more variables by fitting a linear line to the observed data. The dependent variable (Y) is determined by the independent variable (X) using the equation $Y = \beta_0 + \beta_1 X + \epsilon$, where β_0 is the intercept, β_1 is the slope, and ϵ is the error term. Refer to Section 7.2
2. The method of least squares is a statistical method that estimates the slope and intercept of a straight-line regression analysis by minimising the squared differences between the observed and estimated values of the dependent variable. This method makes sure that the most suitable line is found by minimising errors or residuals between the data points and regression lines. Refer to Section 7.2
3. In a SLR model, the fit of the data can also be evaluated through the use of the sum of squares method, co-efficient of determination (R^2), and checking the assumptions of the model. These are useful to evaluate how well the model fits the data in terms of explaining the changes in the dependent variable and predicting its value. Refer to Section 7.3
4. In this method, the total variation of the dependent variable is divided into parts: variation attributed to the regression (sum

of squares due to regression) and that which is not attributable to the model (sum of squares due to residuals). This makes it possible to evaluate the goodness of fit of the model. Refer to Section 7.3

5. R-square, or the co-efficient of determination, quantifies the extent to which changes in the independent variable of the regression explain the variation of the dependent variable. The R percentage values are between zero and one, where one implies perfect correlation. Refer to Section 7.3

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No	Answer
1.	The least squares method helps Alice by minimising the sum of squared residuals, which are the differences between the observed sales (actual values) and the predicted sales (fitted values). This results in the optimal line that best represents the linear relationship between advertising expenses and sales, ensuring the model provides the closest possible predictions.
2.	Maria's regression equation suggests that for every additional hour of training, productivity increases by 2.5 units. The intercept of 5 indicates that when no training is provided (0 hours), the expected productivity is 5 units. This implies a positive relationship between training hours and productivity at Innovate Inc.
3.	Sara can infer that a significant portion of the variation in sales is explained by the model. The SSR value of 700 indicates that 700 out of 1000 units of variation in sales are accounted for by the regression model. This suggests that the model has a good fit, as a large portion of the total variation is explained by advertising spend.
4.	b. The model explains most of the variation in the loan default data
5.	c. Homoscedasticity (constant variance of residuals)

7.9 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

- ❑ Essentials of Business Analytics, 2e, Camm, Cochran, Fry, Ohlmann, Anderson, Sweeney, Williams
- ❑ Martin, P. (2022) Linear Regression: An introduction to statistical models. Los Angeles: SAGE.
- ❑ Gujarati, D.N. (2019) Linear regression: A mathematical introduction. Los Angeles: SAGE.

N O T E S**WEB-REFERENCES**

- ❑ Wikipedia contributors. (n.d.). *Sum of squares*. *Wikipedia, The Free Encyclopedia*. Retrieved from https://en.wikipedia.org/wiki/Sum_of_squares
- ❑ Department of Statistics, Penn State University. (n.d.). 2.3 - *The simple linear regression model*. *STAT 462: Applied Regression Analysis*. Retrieved from <https://online.stat.psu.edu/stat462/node/79/>

CASE STUDY

NOTES

UNRAVELLING STUDENT ABSENTEEISM AND ACADEMIC PERFORMANCE WITH LINEAR REGRESSION ANALYSIS

SCENARIO

In a secondary school setting in the UK, the aim is to investigate the intricate link between student absenteeism and academic performance. The study spans multiple academic terms, utilising a robust dataset that includes detailed attendance records and corresponding academic outcomes.

DATA COLLECTION

A meticulous data collection process captures various aspects of student absenteeism, encompassing types (authorised, unauthorised) and durations. Simultaneously, comprehensive academic scores are gathered for a representative cohort, ensuring precision and comprehensiveness.

CHALLENGES

- ❑ **Recording system enhancement:** Creating a valid structure for noting down and storing instances of assumptions made, taking into account the difficulty in telling the difference between excused absence and non-excused absence.
- ❑ **Bias mitigation:** Ensuring equal treatment of data collected for all teachers despite possible differences in the way reported.
- ❑ **External factors consideration:** Investigating external factors such as health issues or family situations that might influence both absenteeism and academic performance.

ANALYSIS AND RESULTS

Leveraging simple linear regression analysis, the study quantifies the intricate relationship between absenteeism and academic scores. The evaluation of predictive analytics informs actionable insights, potentially shaping targeted interventions to enhance student engagement and performance.

SOLUTION

Simple linear regression is employed to model the correlation relationship between absenteeism and academic performance. Rigorous data validation protocols, categorisation of diverse types of absenteeism, and exploration of external factors contribute to a comprehensive analysis.



Case Objective

The objective of this case study is to discern the correlation relationship between student absenteeism and academic scores using simple linear regression.

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CASE STUDY



QUESTIONS

1. How can you incorporate external factors into an analysis to provide a more holistic understanding of the absenteeism academic performance relationship?

(Hint: Explore the integration of additional variables, such as health records or socio-economic indicators.)

2. What actionable insights could be derived from the linear regression analysis, and how might these inform tailored strategies to address absenteeism?

(Hint: Consider interventions like mentorship programs or targeted support services.)

8

C H A P T E R

MULTIPLE LINEAR REGRESSION

CONTENTS

- 8.1 Introduction
- 8.2 Understanding the Concept of Multiple Regression
 - 8.2.1 Application of Multiple Regression in Various Industries
 - Self Assessment Questions
 - Activity
- 8.3 Forward, Backward and Stepwise Regression
 - 8.3.1 Forward Regression
 - 8.3.2 Backward Regression
 - 8.3.3 Stepwise Regression
 - Self Assessment Questions
 - Activity
- 8.4 Assessing the Fit of Multiple Regression Model
 - Self Assessment Questions
 - Activity
- 8.5 Validation of Multiple Regression Models
 - 8.5.1 Coefficient of Multiple Determination (R-Square)
 - 8.5.2 Residual Analysis in Multiple Regression
 - Self Assessment Questions
 - Activity
- 8.6 Challenges in Multiple Regression
 - Self Assessment Questions
 - Activity
- 8.7 Summary
- 8.8 Multiple Choice Questions
- 8.9 Descriptive Questions
- 8.10 Higher Order Thinking Skills (HOTS) Questions
- 8.11 Answers and Hints
- 8.12 Suggested Readings & References

N O T E S

**Case Objective**

The objective of this caselet is to demonstrate the practical application and effectiveness of multiple regression analysis in optimising advertising spending for revenue growth in e-commerce.

INTRODUCTORY CASELET**OPTIMISING ADVERTISING SPENDING FOR REVENUE GROWTH**

Within a competitive e-commerce environment, an established online marketplace aimed to increase its revenue generated through sales by improving its advertising efficiency. The business analytics team applied multiple regression analysis to understand how different variables affecting sales were interrelated. Key independent variables included advertising spend across different channels, website traffic and social media engagement. The dependent variable was total monthly sales.

The team identified the most impactful variables through forward regression and constructed a multiple regression model. The analysis revealed that advertising spending on social media and website traffic significantly influenced sales. The model also considered interaction effects, providing nuanced insights. With this information, the retailer strategically allocated resources, focusing on channels exhibiting a higher impact on sales. Hence, the organisation noted considerable income growth, proving the impact of multiple regression in guiding the business strategies for its expansion.

The regression model enabled the company to integrate its advertising plan with revenue targets, demonstrating how efficient statistical methods in managing resources can result in favourable business results.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Describe the significance of the interpretation of multiple regression
- Evaluate the importance of forward, backward and stepwise regression
- Describe assessing the fit of the multiple regression model
- Describe the concept of validation of multiple regression models
- Employ various challenges in multiple regression

8.1 INTRODUCTION

In the previous chapter, you learnt about the simple linear regression model and assessed the fit of a simple regression model.

In business analytics, multiple regression serves as a powerful tool for predictive modelling, guiding strategic planning and resource allocation. Moreover, the integration of multiple regression in business analytics underscores the essence of leveraging data as a strategic asset. Multiple regression serves as the compass guiding businesses through dynamic landscapes, enabling adaptability and informed pivots based on empirical evidence, thereby enhancing competitiveness and fostering sustainable growth.

Multiple linear regression uses statistical techniques for forecasting the impact of multiple independent variables on a dependent variable, thus benefiting managers in predicting performance outcomes. This helps managers know which factors contribute the most and which ones the least to the affected variable. In this chapter, we will learn about the interpretation of multiple regression. In the next sections, we examine forward, backward and stepwise regression, understand the fit and validation of multiple regression models. At the end of the chapter, we will cover the challenges of multiple regression.

8.2 UNDERSTANDING THE CONCEPT OF MULTIPLE REGRESSION

Multiple regression is a statistical technique involving the analysis of relationships between a dependent variable and multiple independent variables. It evaluates each variable's contribution, helping to identify helper variables within the data. The coefficients and significance levels provide information on the influence of different variables on the subjects under investigation. This allows a more nuanced understanding of the studied phenomena.

NOTES

The mathematical expression about multiple regression between the dependent variable and multiple independent variables can be written as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_p x_p + \varepsilon$$

Where:

y is the dependent variable

β_0 is the intercept

$\beta_1, \beta_2, \dots, \beta_p$ are the coefficients representing the impact of each independent variable $x_1, x_2, \dots, x_p$ on the dependent variable

ε is the error term, representing unobserved factors affecting y

Example: Imagine you are studying the factors affecting house prices. The dependent variable (β) is the price of the house and the independent variables include:

x_1 : Size of the house (in square feet)

x_2 : Number of bedrooms

x_3 : Distance from the nearest metro stations (in kilometers)

The regression model in this example can be written as:

$$\text{Price} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

Where:

β_0 : The predicted price of a house when size, number of bedrooms, and distance are all zero (which is not realistic, but it helps set the baseline)

β_1 : The increase in house price for every additional square foot of size, keeping other variables constant

β_2 : The increase in house price for each additional bedroom, keeping other variables constant

β_3 : The decrease in house price for each additional mile away from the nearest metro station, keeping other variables constant

ε : Consists of things like neighbourhood quality, market conditions, or other unmeasured factors

The steps for performing multiple regression in MS Excel are as follows:

1. Prepare your data by arranging the data in columns in MS Excel as shown in Figure 8.1:

NOTES

	A	B	C	D	E
1	Month	Advertising Budget (x_1)	Store Location Quality (x_2)	Product Variety (x_3)	Monthly Sales (Y)
2	1	10	7	5	1000
3	2	12	6	6	1050
4	3	14	8	6	1200
5	4	13	5	7	1150
6	5	16	7	6	1250
7	6	18	8	7	1300
8	7	19	7	9	1350
9	8	20	9	8	1450
10	9	21	7	8	1400
11	10	22	7	7	1550
12	11	23	9	6	1600
13	12	25	6	5	1650
14	13	30	8	7	1700
15	14	28	7	6	1650
16	15	27	6	5	1550
17	16	24	5	8	1500
18	17	22	8	7	1400
19	18	26	7	6	1600
20	19	29	7	7	1700
21	20	33	9	5	1750
22	21	35	7	8	1800
23	22	37	8	6	1850
24	23	40	7	7	1900
25	24	45	9	5	2000

Figure 8.1: Data in Excel For Multiple Regression

- Go to the File tab, click Options, and then click Add-ins. In the Manage box, select Excel Add-ins and click Go. In the Add-ins box, check Analysis ToolPak and click OK.
- Select the Data Analysis ToolPak in the Data tab to start the process as shown in Figure 8.2:

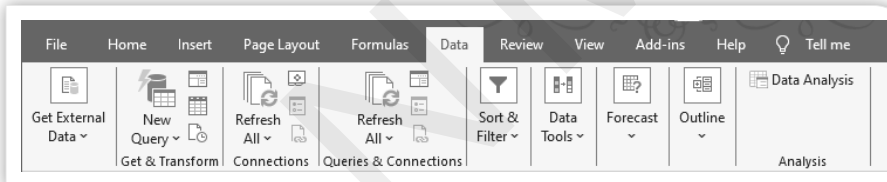


Figure 8.2: Selecting Data Analysis ToolPak

- Select Regression and click OK in the Data Analysis dialog box as shown in Figure 8.3:

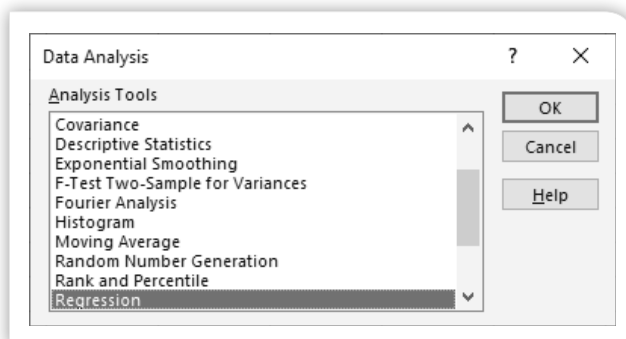


Figure 8.3: Data Analysis Dialog Box

NOTES

- Input regression parameters by setting the Y range with the dependent variable column, and the X range with independent variables, check Labels, Residuals, Standardized Residuals, Residual Plots and Line Fit Plots, and choose the output location for results in the New Workbook as shown in Figure 8.4:

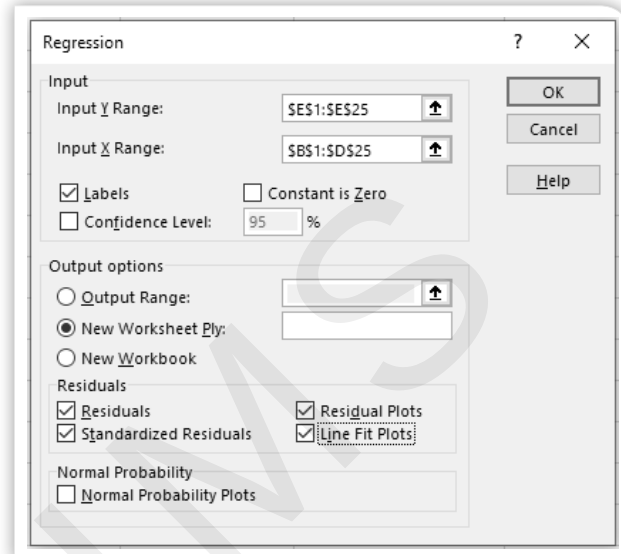


Figure 8.4: Selecting Range for Input Y and Input X Range

- After clicking OK, Excel will perform the multiple regression analysis and provide output, including coefficients, standard errors, t Stat, p-values, R-square, etc., as shown in Figure 8.5:

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.97076598							
R Square	0.94238659							
Adjusted R Square	0.93374458							
Standard Error	68.7837296							
Observations	24							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	1547771.804	515923.935	109.047129	1.45134E-12			
Residual	20	94624.02911	4731.20146					
Total	23	1642395.833						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	743.599164	161.7106223	4.59833222	0.00017402	406.2767169	1080.92161	406.2767169	1080.921611
Advertising Budget (X1)	28.4906638	1.691180059	16.8466177	2.7778E-13	24.96292404	32.0184036	24.96292404	32.01840361
Store Location Quality (X2)	6.08047833	13.85146335	0.43897733	0.6653838	-22.81316791	34.9741246	-22.8131679	34.97412457
Product Variety (X3)	4.48196554	15.88210266	0.28220228	0.78068806	-28.64752008	37.6114512	-28.6475201	37.61145115

Figure 8.5: Multiple Regression Output

According to the output shown in Figure 8.5:

- ◆ **Coefficients:** The coefficients help form the regression equation. From the output of the regression data, the for multiple regression equation is: $Y = 743.60 + 28.49x_1 + 6.08x_2 + 4.48x_3$. This allows the prediction of monthly sales based on the values of the Advertising Budget, Store Location Quality, and Product Variety.

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- ♦ **F-significance:** The Significance F value (1.45134E-12) is well below 0.05, indicating the model is statistically significant. This in turn shows that the regression equation is valid, and the independent variables effectively explain the variation in sales.
- ♦ **Multiple R:** Multiple R (0.9708) measures the strength of the linear relationship between observed and predicted sales values. A value close to 1 indicates a strong positive correlation, and the absence of the negative sign shows a positive direction of the relationship between the variables.
- ♦ **R-square (R^2):** The R^2 value (0.9424) shows that 94.24% of the variation in sales is explained by the independent variables. A high R^2 value suggests a good fit between the model and the data.
- ♦ **Adjusted R-square:** Adjusted R^2 (0.9337) adjusts R^2 for the number of variables, penalising for unnecessary variables. It is preferred over R^2 when comparing models with different numbers of variables to identify the better fit.
- ♦ **Standard error:** The standard error (68.78) measures the average distance between the observed and predicted sales values. A lower standard error indicates more accurate predictions and a tighter confidence interval.
- ♦ **p-values:** p-values determine the significance of independent variables. Variables with $p < 0.05$ (e.g., Advertising Budget) significantly contribute to the model, while those with $p > 0.05$ (e.g., Store Location Quality, Product Variety) can be excluded.

7. Inspect the residual result which is shown in Figure 8.6:

RESIDUAL OUTPUT				
Observation	Predicted Monthly Sales (Y)	Residuals	Standard Residuals	
1	1093.478978	-93.4789783	-1.457394603	
2	1148.861793	-98.86179316	-1.541315988	
3	1218.004077	-18.00407748	-0.28069461	
4	1175.753944	-25.75394419	-0.401519785	
5	1268.904927	-18.9049268	-0.294739403	
6	1336.448698	-36.44869833	-0.568257561	
7	1354.376918	-4.376918286	-0.06823884	
8	1390.546573	59.45342676	0.926915386	
9	1414.241699	-14.24169868	-0.222036817	
10	1444.330875	105.6691247	1.647446461	
11	1480.50053	119.4994697	1.863070023	
12	1514.758457	135.2415426	2.108498593	
13	1678.336664	21.66333574	0.33774469	
14	1610.792893	39.20710727	0.611262848	
15	1571.739785	-21.73978504	-0.338936582	
16	1493.633212	6.366788169	0.099262132	
17	1450.411354	-50.41135364	-0.785943921	
18	1553.811565	46.18843492	0.720106028	
19	1637.685044	62.31495623	0.971528386	
20	1760.925203	-10.925203	-0.170330615	

Figure 8.6: Residual Output

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Interpret the outcome as shown in Figure 8.6:

- ◆ **Predicted Monthly Sales (Y):** The predicted sales values are calculated using the regression equation derived from the coefficients. These values represent the expected monthly sales based on the inputs given for the advertising budget, store location quality and product variety.
 - ◆ **Residuals:** Residuals are the differences between the observed (actual) sales values and the predicted sales values. They show how far off the predictions are from the actual data. For instance, in Observation 1, the residual is -93.48 , indicating the model misjudged the actual sales by 93.48 units. Positive residuals suggest the prediction was lower than the actual value, while negative residuals indicate the prediction was higher.
 - ◆ **Standard Residuals:** Standard residuals are normalised residuals, calculated to make comparisons easier. These values are scaled based on the overall standard error of the residuals. Standardised residuals outside the range of -2 to 2 indicate potential outliers. For example, Observation 12 has a standard residual of 2.11, suggesting it deviates significantly from the model's prediction.
8. Compare the Residual Plots and Line Fit Plots of the different variables as provided below:
- ◆ Scatter chart for Advertising Budget (x_1) is shown in Figure 8.7:

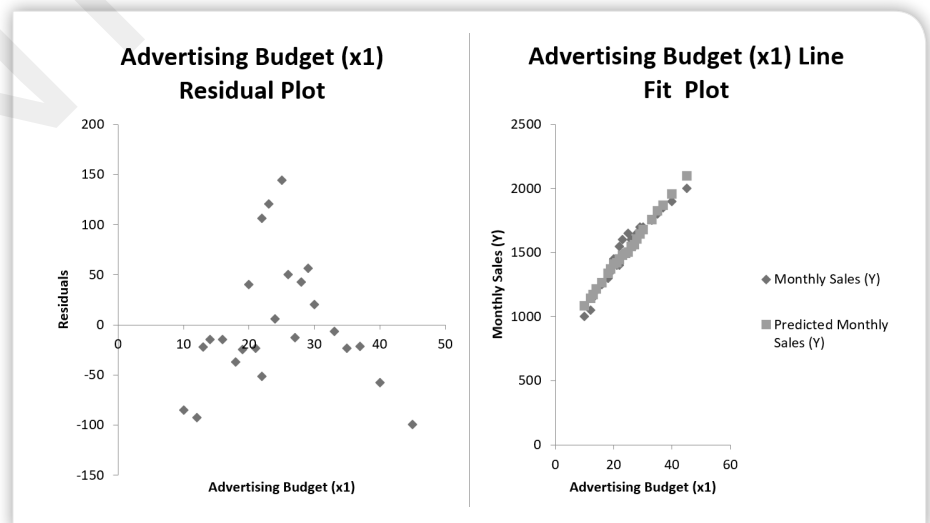


Figure 8.7: Residual Plot and Line Fit Plot of Advertising Budget

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- ◆ Scatter chart for Store Location Quality (x_2) is shown in Figure 8.8:



Figure 8.8: Residual Plot and Line Fit Plot of Store Location Quality

- ◆ Scatter chart for Product Variety (x_3) is shown in Figure 8.9:

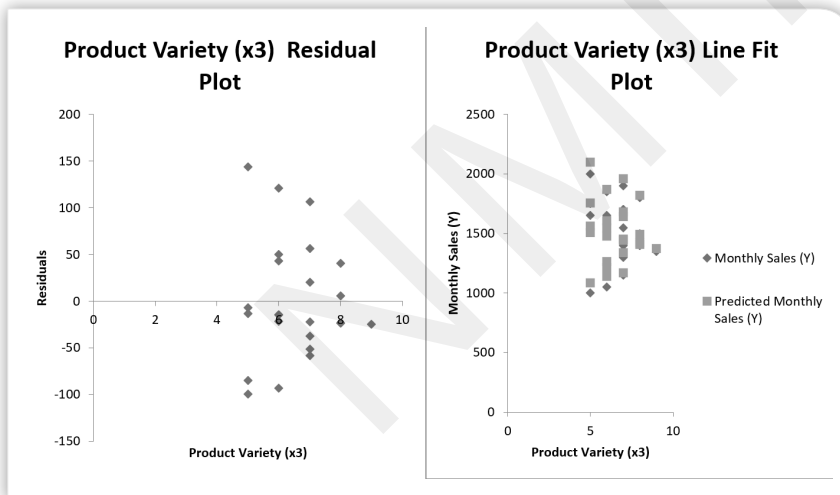


Figure 8.9: Residual Plot and Line Fit Plot of Product Variety

8.2.1 APPLICATION OF MULTIPLE REGRESSION IN VARIOUS INDUSTRIES

Some applications of multiple linear regression in different industries are:

- ❑ **Retail sector:** In the retail industry domain, multiple regression finds application in:
 - ◆ **Demand forecasting:** Retailers use multiple regression to predict future sales by looking at past sales, seasonal trends,

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promotions, holidays and the economy. This helps them manage stock, avoid running out of items and plan sales events.

- ◆ **Pricing strategy:** By analysing customer demographics, competition and customer sensitivity to price changes, retailers can predict how price changes or discounts will affect sales. This helps them set prices that maximise profits.

Example: A retail chain uses multiple regression to forecast sales based on advertising spend, store location, product pricing, and seasonal trends. This helps them decide which stores need more stock, how to price items, and when to run marketing campaigns.

- **Healthcare industry:** In the healthcare industry domain, multiple regression finds application in:

- ◆ **Patient outcomes prediction:** Multiple regression models help predict patient outcomes based on age, health history, treatment type, and lifestyle. This helps healthcare providers identify at-risk patients and create targeted care plans.

- ◆ **Resource allocation:** Hospitals use regression analysis to predict the need for resources like beds, staff and supplies based on patient data and disease trends.

Example: A health institute uses multiple regression to predict how long patients will recover after surgery, considering factors like age, type of surgery, other health issues and hospital stay duration. This helps improve patient care and resource management.

- **Banking and financial services:** In the banking and financial services domain, multiple regression finds application in:

- ◆ **Credit risk assessment:** Banks use multiple regression to assess creditworthiness by analysing income, credit score, employment status and debt-to-income ratio. This helps them set loan interest rates and reduce default risks.

- ◆ **Fraud detection:** Multiple regression helps identify suspicious transactions by analysing patterns in transaction frequency, amount, location and customer profile, aiding in fraud detection.

Example: Banks use multiple regression to assess the risk of loan default by looking at applicants' income, credit score, employment status, debts and savings. This helps them make better lending decisions.

- **Marketing and advertising:** In the marketing and advertising domain, multiple regression finds application in:

- ◆ **Customer segmentation and targeting:** Marketing teams use regression modelling to segment customers based on

behaviour, demographics and purchase history. This helps create targeted campaigns with higher engagement and conversion rates.

- ◆ **Campaign performance analysis:** Regression analysis helps measure the impact of different campaign efforts (like ad spend and exposure) on revenue and brand awareness, improving future strategies.

Example: A marketing team uses multiple regression to evaluate the effectiveness of various marketing channels (social media, email, TV commercials, influencer marketing) on sales. This helps them allocate funds more effectively.

- **Manufacturing industry:** In the manufacturing industry domain, multiple regression finds application in:

- ◆ **Quality control:** Manufacturers use multiple regression to analyse factors like materials, machine settings, temperatures, and operator skills to improve product quality and control quality-affecting factors.
- ◆ **Predictive maintenance:** Regression models help predict machinery failure based on operating hours, usage intensity, environmental conditions and maintenance history, reducing costs and downtime.

Example: A car manufacturer uses multiple regression to estimate production costs based on raw material prices, labour costs, production volume and machine maintenance. This helps them reduce costs without compromising quality.

Next, we will learn about forward, backward and stepwise regressions in the next section.



SELF ASSESSMENT QUESTIONS

1. Which of the following is true about interpreting the R-squared (R^2) value in multiple regression?
 - a. An R^2 value of 0 indicates a perfect fit between the model and the data.
 - b. R^2 measures the percentage of variation in the dependent variable explained by the independent variables.
 - c. R^2 always increases when more independent variables are added, regardless of their relevance.
 - d. An R^2 value of 1 means the regression model has no error.
2. In multiple regression, the p-value of an independent variable less than 0.05 indicates that the variable has no significant contribution to the model. (True/False)

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ACTIVITY

Use a real-world dataset to build and interpret a multiple regression model, emphasising coefficients using Excel.

8.3

FORWARD, BACKWARD AND STEPWISE REGRESSION

When building a regression model, it is often challenging to determine which variables are the most important. Forward, backward, and stepwise regression methods address this problem by helping to identify the most significant variables. Forward regression adds variables one at a time, backward regression removes the least significant variables and stepwise regression combines both approaches. These methods streamline the model-building process, making it more efficient and accurate. We discuss each method in the next sections.

8.3.1 FORWARD REGRESSION

In forward regression, the model is built up gradually, one variable at a time. After the addition of the new variable, the improvement in the fit of the model is assessed, and a decision is made whether to keep the variable in the model or not. Figure 8.10 shows the process of forward regression graphically:

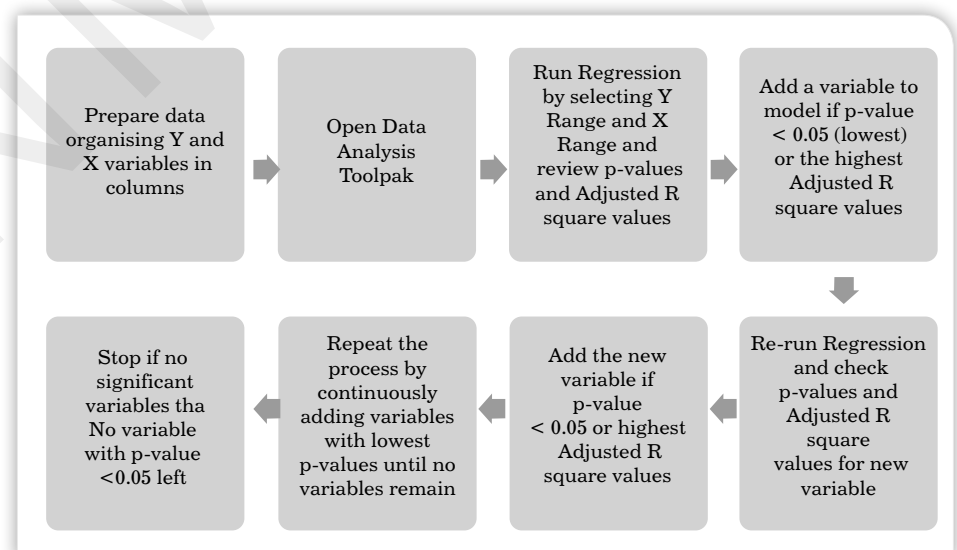


Figure 8.10: Process of Forward Regression in Excel

A step-by-step explanation of forward regression in the context of business analytics in Excel is provided as follows:

1. **Define the problem:** Imagine you are a business analyst in an e-commerce company, and your task is to forecast the sales of

a product using multiple variables such as advertising spend, website traffic and social media engagement.

2. **Data collection:** Gather data for the dependent variable (e.g., sales) along with potential independent variables (e.g., advertising spend, website traffic, social media interaction).
3. **Preliminary analysis:** Perform a basic analysis to explore the relationships between variables using correlation analysis, scatter plots or other visualisations to understand potential variables.
4. **Forward regression:** The steps for applying forward regression are:
 - i. **Build an empty model:** Begin with a model containing only the intercept (no independent variables).
 - ii. **Select the best variable:** Identify the independent variable most strongly correlated with the dependent variable. The selection is based on statistical measures like the lowest p-value or highest R-squared.
 - iii. **Update the model:** Add the selected variable to the model and assess its impact on the model's performance (e.g., check p-values, Adjusted R²).
 - iv. **Repeat:** Continue adding one variable at a time that provides the most significant improvement to the model, either based on p-values, R-squared or other criteria. Stop when no further improvements are possible or when you reach a predetermined number of variables or level of significance.

The practical applications of forward regression are:

- ❑ **Predictive modelling:** Forward regression is widely used in fields like economics, marketing and finance to predict outcomes based on multiple variables. For instance, predicting sales performance based on advertising budget, customer demographics and product features.
- ❑ **Feature selection in machine learning:** In machine learning and data science, forward regression helps in selecting the most relevant features (variables) for predictive models, improving model accuracy by removing irrelevant or highly collinear features.
- ❑ **Marketing optimisation:** Businesses can use forward regression to optimise their marketing strategies. By identifying which marketing efforts (e.g., online ads, social media campaigns, email marketing) most impact sales, companies can allocate resources more effectively.

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- ❑ **Risk management:** In finance or insurance, forward regression can identify which factors (e.g., market trends, economic indicators or customer behaviour) most significantly contribute to risks, allowing businesses to take preventive actions.
- ❑ **Customer behaviour analysis:** For customer retention or churn prediction, forward regression can determine which factors (e.g., usage patterns, customer satisfaction, support interactions) are most predictive of customer behaviour, enabling targeted strategies to improve customer loyalty.
- ❑ **Healthcare predictions:** In healthcare, forward regression can be used to predict patient outcomes based on various health indicators, like blood pressure, age and medical history, assisting doctors in identifying risk factors and potential treatment plans.
- ❑ **Employee performance analysis:** Forward regression can be applied in human resources to determine which employee characteristics (e.g., experience, education, work hours) are most correlated with job performance or turnover, helping businesses make data-driven hiring and training decisions.
- ❑ **Energy consumption forecasting:** For utilities and energy companies, forward regression can help forecast energy demand based on variables such as weather patterns, historical consumption and time of day, enabling better resource management and efficiency.
- ❑ **Agricultural yield prediction:** In agriculture, forward regression can be used to predict crop yields based on variables like weather conditions, soil quality and irrigation, helping farmers optimise resource allocation and planning.

These applications highlight how forward regression can be a versatile tool in various industries for improving decision-making and resource allocation.

8.3.2 BACKWARD REGRESSION

Backward regression is also a stepwise technique used in multiple regression analysis for the selection of variables. However, when one talks about backward regression, it starts with the end goal of building a model that contains all independent variables and continues to remove the variables that are least related to the dependent variable as illustrated in Figure 8.11:

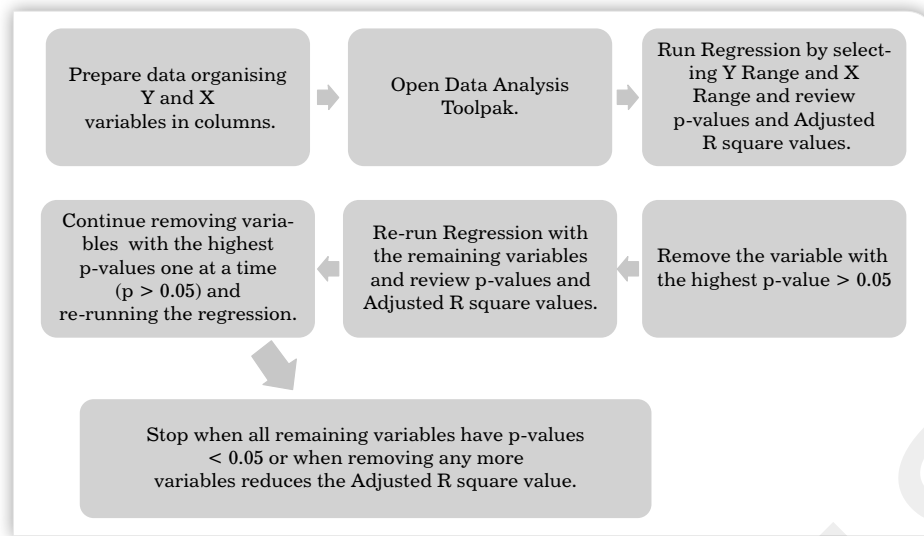


Figure 8.11: Process of Backward Regression

A step-by-step explanation of backward regression in the context of business analytics using Excel is provided as follows:

1. **Define the problem:** Consider the scenario where you are a business analyst at an e-commerce company, tasked with predicting product sales using various factors such as advertising spend, website traffic, and social media engagement.
2. **Data collection:** Collect data for the dependent variable (e.g., product sales) along with the potential independent variables (e.g., advertising spend, website traffic, social media engagement).
3. **Preliminary analysis:** Perform a basic analysis to explore relationships between the variables using correlation analysis, scatter plots, and other visualisations to understand potential variables.
4. **Backward regression:** The steps for applying backward regression are:
 - i. **Start with a full model:** Begin with a model that includes all potential explanatory variables (independent variables).
 - ii. **Eliminate the least significant variable:** Based on statistical measures such as extreme p-values, high multicollinearity, or low R-squared values, remove the variable that contributes the least to the model.
 - iii. **Update the model:** Refine by excluding the least significant variable and assess its impact on model performance.
 - iv. **Repeat:** Continue the process iteratively, removing one variable at a time, until you reach a stopping point where all remaining variables are statistically significant or the model is simplified to a desired level.

NOTES

This structured approach ensures the model is refined by eliminating unnecessary variables while retaining the dependent variable.

Some of the practical applications of backward regression are:

- ❑ **Medical research:** In healthcare and medical research, backward regression can be applied to identify which risk factors or medical variables (e.g., lifestyle factors, biomarkers) are most significant in predicting patient outcomes or disease progression, while eliminating less relevant factors.
- ❑ **Financial risk assessment:** Financial analysts use backward regression to evaluate which financial indicators (e.g., stock prices, interest rates, inflation) are most predictive of market behaviour or investment performance, simplifying models to focus on the most significant variables.
- ❑ **Econometric analysis:** Economists use backward regression to refine economic models by starting with a broad set of variables (e.g., GDP, unemployment, inflation rates) and removing less significant ones. This process helps to identify the key drivers of economic trends or policy outcomes.
- ❑ **Customer segmentation:** In customer analytics, backward regression can be used to identify the least important variables in segmentation models, helping businesses focus on the core characteristics that define high-value customer segments.
- ❑ **Supply chain optimisation:** Backward regression is applied in supply chain management to refine models predicting demand, inventory levels, or lead times, removing less relevant variables like minor variables of supply chain disruptions; thereby, improving operational efficiency.

These applications show how backward regression is valuable for refining models, simplifying analysis, and ensuring that only the most meaningful variables are included in decision-making processes.

8.3.3 STEPWISE REGRESSION

Stepwise regression is another method which makes an automatic selection of key variables in a model. This saves the time and effort of a data scientist or analyst developing the model since he/she will not need to manually consider the selection of variables for a model.

A step-by-step explanation of stepwise regression in the context of business analytics is as follows:

1. **Define the problem:** Imagine you are a business analyst in an e-commerce company, and you need to forecast product sales using multiple variables, such as advertising spend, website traffic, and social media engagement. The goal is to identify the best subset of predictors for the model.

2. **Data collection:** Gather data for the dependent variable (e.g., sales) and potential independent variables (e.g., advertising spend, website traffic, social media interaction).
3. **Preliminary analysis:** Conduct basic analysis to explore relationships between variables using correlation analysis, scatter plots or other visual tools to identify trends and multicollinearity.
4. **Stepwise regression:** The steps for applying stepwise regression are:
 - i. **Build an empty model:** Start with no variables in the model (only the intercept).
 - ii. **Add variables forward:** Test all candidate variables that are not yet in the model at each step. Add the variable that provides the largest improvement to the model (e.g., lowest p-value or maximum R-squared increase). The inclusion of variables is subject to a significance threshold (e.g., p-value < 0.05).
 - iii. **Remove insignificant variables:** After adding a new variable, re-evaluate the significance of all variables currently in the model. Remove any variables that no longer meet the significance threshold (e.g., p-value > 0.05).
5. **Repeat:** Continue adding and removing variables iteratively, one at a time, until no further variables can be added or removed based on the significance criteria.
6. **Stopping rule:** Stop the process when no variables can be added or removed that would significantly improve the model's performance.

Stepwise regression offers several advantages such as:

- ❑ **Automatic variable selection:** Stepwise regression automates the process of selecting significant variables, reducing the manual effort required in model building.
- ❑ **Efficient model building:** It provides an efficient method to handle many potential variables, identifying those that contribute most to the model.
- ❑ **Model simplicity:** By eliminating irrelevant variables, stepwise regression can produce a simpler, more interpretable model.
- ❑ **Improved model performance:** The method can improve model performance by excluding redundant or collinear variables, which may otherwise distort results.
- ❑ **Flexibility:** It allows both forward selection and backward elimination, offering flexibility in constructing the model.

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- ❑ **Handles large datasets:** Stepwise regression is particularly useful when dealing with datasets containing a high number of variables, helping to streamline analysis.
- ❑ **Quick iterations:** The method allows for rapid iteration and refinement of models, which can be advantageous in exploratory data analysis.
- ❑ **Adaptability:** Stepwise regression can adapt to changes in data, making it suitable for scenarios where variables or their relationships may evolve over time.

In general, the use of stepwise regression is to create an accurate and regression model that can deal with complex relationships among data. It is a widely employed and effective strategy in model selection for a wide range of domains. Stepwise regression presents several drawbacks such as:

- ❑ **Biased result:** Stepwise regression can lead to biased estimates, as well as increased Type I error rates and false positives.
- ❑ **Non-reproducible results:** The outcomes of stepwise regression may lack reproducibility, as they can change depending on factors such as sample size, variable order and significance thresholds.
- ❑ **Violates assumptions:** Stepwise regression methods can violate essential assumptions of regression analysis, including assumptions of independence, normality and linearity.
- ❑ **Ignores interactions:** The process of evaluating interactions and higher-order terms can be especially complex and difficult within a stepwise regression framework.
- ❑ **Might not return an optimal final model:** The final model produced by stepwise regression may not represent the most optimal solution for the given data.
- ❑ **Over-simplifications:** Stepwise regression often results in over-simplified models, which may not accurately reflect the true complexity of the data.
- ❑ **Depends on the computer program:** Stepwise regression heavily depends on the computer program used to select variables, often without regard to their actual relevance or meaning within the study.
- ❑ **Choosing appropriate α -values:** Stepwise regression heavily depends on the computer program used to select variables, often without regard to their actual relevance or meaning within the study.

The use of stepwise regression should be done carefully because of its limitations and the errors (termed as biases) as a result of the variable selection process. The modelled variable should be validated to make sure that it can be relied upon in prediction.



NOTE

The significance of mentioning forward and backward in stepwise regression highlights its iterative nature: forward selection adds variables progressively based on significance, while backward elimination removes insignificant ones, ensuring the model includes only variables that meaningfully improve its performance.

The differences between forward, backward, and stepwise regression techniques are shown in Table 8.1:

TABLE 8.1: DIFFERENCES BETWEEN FORWARD, BACKWARD AND STEPWISE REGRESSION

Aspect	Forward Regression	Backward Regression	Stepwise Regression
Approach	Begins with zero variables and adds each variable one by one.	Starts with all variables and removes them one by one.	Combines Forward and Backward methods. Adds or removes variables iteratively.
Selection Criteria	Adds variable based on criteria (e.g., p-value, Adjusted R square).	Removes variables based on criteria (e.g., p-value, Adjusted R square).	Uses criteria to decide whether to add or remove variables at each step.
Use Cases	Useful when there is no prior knowledge of important variables.	Useful when starting with a large set of variables.	Ideal when there is uncertainty about the importance of variables and both addition/removal need to be evaluated.
Computational Complexity	Low to Moderate, especially when there are few relevant variables.	Low to Moderate if there are many irrelevant variables.	High due to the iterative process of adding/removing variables.
Risk of Multicollinearity	May face multicollinearity issues if variables are added without checking.	May reduce multicollinearity if non-significant variables are removed.	Balances risk by iteratively checking for multicollinearity while adding/removing variables.

After learning about the types of multiple regression, we will study about assessing the fit of a multiple regression model.

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SELF ASSESSMENT QUESTIONS

3. What method involves gradually building a model by adding one variable at a time?
 - a. Backward regression
 - b. Forward regression
 - c. Stepwise regression
 - d. Logistic regression
4. Stepwise regression heavily depends on the computer program used to select variables, often without regard to their actual relevance or meaning within the study. (True/False)



ACTIVITY

Use different regression methods (forward, backward and stepwise) on a dataset, compare model outcomes for fit, predictive power and simplicity, creating an understanding of model selection techniques.

8.4

ASSESSING THE FIT OF MULTIPLE REGRESSION MODEL

Evaluating the fit of a multiple regression model is an essential process when one is interested in the relationship of a dependent variable with multiple independent variables. This entails an assessment of the degree and the extent to which the model can account for changes in the dependent variable based on given independent variables. The goal in general terms is to evaluate whether this particular model is appropriate for the data and can therefore be used for this study, or highlight certain aspects that may need further research. Several statistical methods are used in this stage of 'model fit' to evaluate the ability of the model to explain data and its fit relative to the assumptions made and suggest modifiable aspects of the model. Some of the diagnostic tools for model fit are as follows:

- ❑ **R-squared (R^2):** R-squared is a statistical measure that calculates how well the independent variables in a regression model can explain the variation in the dependent variable. The value of R^2 ranges from 0 to 1, with values closer to 1 indicating a better fit of the model. However, while a higher R^2 suggests a better fit, it does not always imply that the model is the best one. This is because R^2 can increase with the addition of more variables, even if those variables do not genuinely contribute to explaining the variance in the dependent variable. In Excel, the R^2 value is included in the regression output under the label "Multiple R and R square" as shown in Figure 8.5.
- ❑ **Adjusted R-squared:** Adjusted R-squared is a modified version of R-squared that adjusts for the number of variables included

in the model. Unlike R^2 , which can artificially increase with the addition of more variables, adjusted R-squared corrects the inclusion of unnecessary or irrelevant variables. This makes it a more reliable measure when comparing models with different numbers of variables. A higher adjusted R-squared indicates a better model fit, as it accounts for both the number of variables and the proportion of variance explained. In Excel, adjusted R-squared can be found in the regression output under the label “Adjusted R Square” as shown in Figure 8.5.

- ❑ **Residuals analysis:** Residuals are the differences between the observed values and the predicted values generated by the regression model. Analysing residuals helps assess whether the model adequately fits the data or if there are underlying issues. Ideally, residuals should be randomly scattered around zero, with no discernible patterns. If patterns, such as non-linearity or heteroscedasticity (unequal variance of residuals), appear, it suggests that the model may not fully capture the relationships between variables. Residual analysis can be performed in Excel by examining the residuals and creating scatter plots to visually inspect for patterns that may indicate model misspecification or outliers as shown in Figure 8.6.
- ❑ **F-statistic:** The F-statistic tests the overall significance of the regression model by comparing the model with and without the independent variables. It assesses whether at least one of the independent variables has a significant effect on the dependent variable. A higher F-statistic indicates a stronger relationship between the dependent variable and the independent variables, suggesting that the model is a good fit. Additionally, the F-statistic is accompanied by a p-value, which tests the null hypothesis that all regression coefficients are equal to zero. If the p-value is less than 0.05, it suggests that the model is statistically significant and that the independent variables collectively explain the variation in the dependent variable. In Excel, the F-statistic and its corresponding p-value are found in the “ANOVA” table of the regression output named as “Significance F” as shown in Figure 8.5.
- ❑ **Sum of Squared Errors (SSE):** The SSE is a measure of the total difference between the observed values and the values predicted by the regression model. It also shows the average distance of data points from the regression line by calculating the sum of the squared residuals. A lower SSE indicates a better model fit, as it suggests that the predicted values are closer to the actual observed values. However, SSE alone does not provide sufficient information about the model’s overall performance, as it does not account for the number of variables or data points. In Excel, SSE can be found in the “ANOVA” table as “Residual SS,” or it can be calculated manually by summing the squared differences

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between the observed and predicted values as shown in Figure 8.5.

Each of these diagnostic tools plays a crucial role in evaluating the performance of a multiple regression model. R-squared and adjusted R-squared provide insight into how well the independent variables explain the dependent variable's variation. Residual analysis helps detect potential issues such as non-linearity or heteroscedasticity, while the F-statistic tests the overall significance of the model. Finally, SSE provides an absolute measure of model error, helping to assess the fit of the regression model.

After learning about the model fit, we will learn about the validation of multiple regression models in the next section.



SELF ASSESSMENT QUESTIONS

5. Which metric indicates the average distance of data points from the regression line?
 - a. R-squared
 - b. F-statistic
 - c. SSE
 - d. Adjusted R-squared
6. The value of R^2 ranges from 0 to 1, with values closer to 1 indicating a worse fit of the model. (True/False)



ACTIVITY

Evaluate multiple regression model fit using metrics like R-squared and residual analysis. Analyse a dataset, interpret the metrics and understand the model's predictive accuracy and appropriateness for the data.

8.5 VALIDATION OF MULTIPLE REGRESSION MODELS

Multiple regression models based on several independent variables are important in business analytics in forecasting outcomes. Nevertheless, it is very important to validate these models to help in making business decisions with utmost precision and reliability. Validation consists of evaluating the model fit and confirming how well the model can perform with new data. Proper validation helps businesses to build trust in the model for use in real-world situations.

Some important aspects of model validation are as follows:

- ❑ **Overfitting and underfitting:** Overfitting occurs when a model not only captures the relationship between variables but

also memorises the training data, including irrelevant noise, resulting in poor performance in new or unseen data. On the contrary, underfitting happens when a model does not capture the underlying trend of a given dataset. Validation is useful in identifying and mitigating such concerns.

- ❑ **Generalisability:** A properly evaluated regression model should allow for predictions to be made on other datasets effectively. This is referred as generalisability and ensures that the model could be relied upon out of the first data sample.
- ❑ **Bias and variance tradeoff:** There is a tradeoff between bias and variance when it comes to the performance of the model. If the bias is too high, underfitting will occur, but if the variance is high, the data will result in overfitting. Validation techniques help to address this trade-off, ensuring a balance between the two requirements.

Several methods are commonly used to validate multiple regression models:

- ❑ **Training and testing data approach:** In this simple method of model evaluation, the data sample is partitioned into two sets, the training data and the testing data. The modelling is done (fitted) using the training data and then evaluated on the test data. This approach gives an idea of the expected performance of the model on the new, unseen data.
- ❑ **Cross-validation:** Cross-validation is a valuable technique, especially when working with limited data. One popular method is k-fold cross-validation. In this approach, the data is divided into k subsets, known as folds. The model is then trained k times, each time using a different fold as the validation set and the remaining k-1 folds as the training set.

This method helps in making the most out of the available data by ensuring that every data point gets to be in the validation set exactly once and in the training set k-1 times. One of the key advantages of k-fold cross-validation is that it reduces the variance in model assessment, providing a more reliable estimate of the model's predictive performance. By averaging the results across different folds, it helps in giving a more trustworthy estimate of the model's ability to predict correctly.

- ❑ **Bootstrapping:** This method refers to taking multiple random samples from the given dataset and fitting the model on all drawn samples one after another. By taking the average of all these results, the analyst can infer how stable the performance of the model is.
- ❑ **Holdout validation:** This approach involves setting aside a portion of the data that is not used during the model-building process. Instead, this reserved data is used only once, after all

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model adjustments are complete, to perform a final evaluation. Holdout validation is often combined with methods like cross-validation or train-test splitting to ensure the model's performance is accurately assessed.

- ❑ **Residual examination:** As defined, residuals are the differences between the model-predicted values and the observed values. Analysis of residuals serves to reveal erroneous behaviour and identify heteroscedasticity and validity of the regression model with respect to the linear model assumptions.

8.5.1 COEFFICIENT OF MULTIPLE DETERMINATION (R-SQUARE)

The coefficient of multiple determination, denoted as R^2 , is one of the most important measures in regression analysis. It indicates how well the model can predict observed outcomes. More specifically, R^2 represents the percentage of the variance in the dependent variable that is explained by the independent variables in the regression model. This coefficient is crucial in multiple regression as it helps assess the performance of the model and the effectiveness of the variables.

In a multiple linear regression model, the coefficient of determination, R^2 , ranges between 0 and 1. An R^2 value of 1 means that the model has perfectly captured all the variability in the outcome, while an R^2 value of 0 indicates that the model cannot explain any of the variability. For instance, if ($R^2 = 0.75$), it means that the independent variables explain 75 per cent of the variance in the dependent variable. This also suggests that there could be other factors not included in the model that account for the remaining 25 per cent of the variance.

The formula for R^2 is as follows:

$$R^2 = 1 - \frac{SS_{tot}}{SS_{res}}$$

Where:

SS_{res} (Residual Sum of Squares) measures the variation in observed data points that the model does not explain.

SS_{tot} (Total Sum of Squares) represents the total variation in the dependent variable.

8.5.2 RESIDUAL ANALYSIS IN MULTIPLE REGRESSION

Residual analysis focuses on analysing variations by considering the differences between model-predicted and observed data points.

Residuals are defined mathematically as:

$$\text{Residual} = \text{Observed Value} - \text{Predicted Value}$$

If the dependent variable is perfectly predicted by the model, all the residuals would be equal to zero, meaning that the model perfectly fits the data. To ensure reliable results, multiple regression analysis relies on a few key assumptions about the residuals:

- ❑ **Normality of residuals:** It is expected that the residuals will be normally distributed, hence they should exhibit a normal distribution centred at zero. This suggests that any variation from the model is random and does not tend to favour certain outcomes.
- ❑ **Homoscedasticity:** This implies that residuals must have the same variance, that is, the distribution of residuals should be almost the same at every level of the independent variables. Any change in residual variance (heteroscedasticity) may indicate that the predictive power of the model differs over the range of the data, and as a result, predictions may be weaker.
- ❑ **Independence:** The residuals must not be correlated, as that would imply the existence of some order between them. If residuals are found to be autocorrelated, it may suggest that some factor is missing or the model in use is outdated.
- ❑ **Linearity:** The independent variables and the dependent variable should be related linearly.

There are several ways through which business analysts analyse residuals to evaluate the model and improvements:

- ❑ **Residual plots:** This is among the most frequently used technique where the value of the residuals is placed on the vertical axis and a dependent variable or the predicted parameter values is placed on the horizontal axis. For a good fitting model, the residuals should be randomly scattered around the horizontal line with zero coordinates. Any behaviour such as a tendency to form a curve or even areas of excess concentration may be indicative of some assumption violations such as non-linearity or heteroscedasticity.
- ❑ **Q-Q plots:** A quantile-quantile plot (Q-Q plot) is a graphical tool used to evaluate the normality of residuals. In this form of plotting, the residuals are plotted in a normal distribution curve. If the residuals are not in line with the diagonal line, they may be indicating abnormal behaviour, suggesting problems with the model.
- ❑ **Leverage and influence measures:** These are used to check whether the model is overly affected by some points in the data. For instance, Cook's Distance and leverage plots can help depict the locations of influential points likely affecting the outcome of the model.

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- ❑ **Autocorrelation check:** Check for autocorrelation which is the correlation of residuals with their past values using the Durbin-Watson measure and other tests. The presence of autocorrelation in the residuals indicates that the model has violated the assumption of the independence of the data.

Although multiple linear regression is implemented very easily, there are certain challenges that occur when it is implemented. These challenges are provided in the next section.

**SELF ASSESSMENT QUESTIONS**

7. Homoscedasticity means that residuals have constant variance across levels of independent variables. (True/False)
8. A high R^2 value always indicates a model with low bias. (True/False)

**ACTIVITY**

Assess regression model accuracy by comparing the difference between observed and predicted values for normality and patterns, ensuring model adequacy and potential refinements.

8.6 CHALLENGES IN MULTIPLE REGRESSION

Multiple linear regression is a useful technique in research and analysis. However, it is not without challenges and limitations.

To overcome these challenges, it involves the use of various statistical techniques, understanding the underlying domain and an appreciation of the specific scenario within which the regression is going to be used. Periodic revisions of the model can also serve to protect the precision and accuracy of the model. Some common issues and the approaches to dealing with the problems are as follows:

- ❑ **Multicollinearity:** When a model contains two or more independent variables that have a very high correlation, that phenomenon is known as multicollinearity. This may distort the regression coefficients making it impossible to determine the effect of the individual variables separately. Techniques such as the Variance Inflation Factor (VIF) and Principal Component Analysis (PCA) can help in the identification and correction of the multicollinearity problem that arises in regression analysis from correlated or highly related independent variables.
- ❑ **Overfitting:** Overfitting occurs when the regression model is overly complex and starts picking the noise in the data instead of the actual associations. While these lead in high accuracy on the training dataset, the model fails to generalise on new data that

it has not been trained on. Employing regularisation techniques, such as Lasso or Ridge regression, can help address overfitting issues.

- ❑ **Heteroscedasticity:** This indicates a scenario in which the errors are generated with a non-constant variance across all levels of the independent variables. This assumption is essential in multiple regression analysis which, when violated, produces incorrect estimates and wrongful interpretations. Employing log or square root transformations can help reduce the variance.
- ❑ **Multidimensional:** As we increase the number of independent variables in the models, the complexity of the models increases making interpretation of such models difficult. Techniques, such as PCA and feature selection, are used to minimise the number of data features.
- ❑ **Data quality and missing data:** Business datasets often have missing values, erroneous entries or incomplete information; such inadequacies could impact the performance of regression models. Such issues can be addressed with imputation methods such as mean/mode imputation or predictive modelling for missing data issues.



SELF ASSESSMENT QUESTIONS

9. Heteroscedasticity implies that errors have a constant variance across levels of independent variables. (True/False)
10. Data quality issues can negatively affect regression model performance. (True/False)



ACTIVITY

Perform a sensitivity analysis on a multiple regression model, identifying and addressing outliers through variable removal or combination for improved model accuracy.

8.7 SUMMARY

- ❑ Multiple regression analyses the relationship between a dependent variable and multiple independent variables, assessing each variable's contribution to explain the phenomena under study.
- ❑ Forward regression adds variables sequentially based on their significance, useful for feature selection, predictive modelling and optimisation across industries.
- ❑ Backward regression starts with all variables, removing the least significant ones to simplify models for healthcare, finance and marketing.

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- ❑ Stepwise regression automates variable selection to improve model efficiency but risks biases, over-simplification, and non-reproducible results.
- ❑ Key tools for evaluating model fit include R^2 , adjusted R^2 , residual analysis, F-statistic and SSE. These measures ensure the model explains variability in the dependent variable while detecting specification issues.
- ❑ Model validation ensures reliability through training-test splits, cross-validation, bootstrapping and residual examination, addressing overfitting, underfitting and generalisability.
- ❑ R^2 measures how well independent variables explain the variance in the dependent variable, ranging between 0 (no explanation) and 1 (perfect explanation), aiding in model performance evaluation.
- ❑ Residual analysis checks assumptions like normality, homoscedasticity, independence, and linearity, using residual plots to identify patterns that may signal model issues or improvements.



KEY WORDS

- ❑ **Coefficients:** Coefficients within a regression equation represent the influence of independent variables on the dependent variable, providing a quantified measure of their relationships.
- ❑ **Model assumptions:** These fundamental assumptions need to be satisfied for a regression model to be considered valid. These include linearity, independence of errors, homoscedasticity, normality of residuals and no multicollinearity.
- ❑ **R-squared:** This metric represents the explanatory power of the regression model by showing the percentage of variability in the dependent variable that it can account for.
- ❑ **Multicollinearity:** It refers to the presence of high correlations among variable variables. It can lead to issues in the model, such as unstable coefficients or inflated standard errors.

8.8 MULTIPLE CHOICE QUESTIONS

1. Which of the following is a key factor in fraud detection using multiple regression in banking?
 - a. Loan interest rate
 - b. Customer transaction patterns
 - c. Employee background checks
 - d. Market trends

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2. What is the primary purpose of multiple regression analysis?
 - a. To predict a dependent variable using a single independent variable
 - b. To analyse relationships between multiple independent variables and a dependent variable
 - c. To simplify data visualisation
 - d. To ensure linearity in all data relationships
3. What statistical criterion is used in stepwise regression to select variables?
 - a. Maximum likelihood
 - b. Mean squared error
 - c. p-value
 - d. Standard deviation
4. Which regression method starts with all variables in the model and removes the least significant variables step by step?
 - a. Forward regression
 - b. Backward regression
 - c. Stepwise regression
 - d. Simple regression
5. Which of the following measures in multiple regression adjusts for the number of independent variables in the model?
 - a. R-squared
 - b. Adjusted R-squared
 - c. F-statistic
 - d. Residual Sum of Squares
6. What does heteroscedasticity in residual analysis indicate?
 - a. Unequal variance of residuals
 - b. Non-linear relationships between variables
 - c. Perfectly normal distribution of residuals
 - d. A high R-squared value
7. Which method divides data into training and testing sets for validation?
 - a. Bootstrapping
 - b. Cross-validation
 - c. Holdout validation
 - d. Train-test split

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8. Which assumption suggests that residuals should be normally distributed?
 - a. Independence
 - b. Linearity
 - c. Normality
 - d. Homoscedasticity
9. What is a significant issue in MLR when two or more independent variables are highly correlated?
 - a. Overfitting
 - b. Multicollinearity
 - c. Heteroscedasticity
 - d. Multidimensionality
10. What is a common outcome of having non-constant error variance in regression analysis?
 - a. Reliable estimates
 - b. Inefficient estimates
 - c. Stable interpretations
 - d. Increased accuracy

8.9 DESCRIPTIVE QUESTIONS

1. What is the purpose of interpreting coefficients in a multiple regression model?
2. How is multiple regression used in the healthcare industry?
3. What is backward regression in multiple regression analysis?
4. What is a common method used to assess the fit of a multiple regression model?
5. How does overfitting impact the performance of a multilinear regression model?

8.10 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. Sarah, a marketing manager, is analysing how customer age, income and education level influence purchasing behaviour using multiple regression. After interpreting the coefficients, she finds that income has the strongest positive effect on purchases. What should Sarah conclude based on this result?
2. A healthcare organisation wants to predict patient satisfaction based on factors like wait time, doctor's communication skills

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and hospital amenities. They use multiple regression analysis. Which industry-specific insights can they gain from this model?

3. Jack is using a stepwise regression method, where he as well as adds removes variables during the model-building process. What is the key advantage of stepwise regression?
4. Maria is building a multiple regression model with several variable variables, including age, income and years of experience. She observes high correlations between income and years of experience. What challenge does this indicate?
 - a. Heteroscedasticity
 - b. Multicollinearity
 - c. Homoscedasticity
 - d. Endogeneity
5. Alex is developing a regression model to predict employee satisfaction. After adding many variables, the model shows a high R-Square, but the predictions do not generalise well to new data. Which challenge is Alex most likely facing?
 - a. Heteroscedasticity
 - b. Underfitting
 - c. Overfitting
 - d. Linearity

8.11 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
Understanding the Concept of Multiple Regression	1.	b. R^2 measures the percentage of variation in the dependent variable explained by the independent variables.
	2.	False
Forward, Backward and Stepwise Regression	3.	b. Forward regression
	4.	True
Assessing the Fit of Multiple Regression Model	5.	c. SSE
	6.	False
Validation of Multiple Regression Models	7.	True

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Topic	Q. No.	Answer
Challenges in Multiple Regression	8.	False
	9.	False
	10.	True

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. Customer transaction patterns
2.	b. To analyse relationships between multiple independent variables and a dependent variable
3.	c. p-value
4.	b. Backward regression
5.	b. Adjusted R-squared
6.	a. Unequal variance of residuals
7.	d. Train-test split
8.	c. Normality
9.	b. Multicollinearity
10.	b. Inefficient estimates

HINTS FOR DESCRIPTIVE QUESTIONS

1. The coefficients of multiple regression analysis are helpful since they show the impact of individual independent variables on the dependent variable. Each of the coefficients represents the expected change in the dependent variable for a one unit increase in the respective independent variable on the other variables being constant. Refer to Section 8.2
2. In the field of healthcare, multiple regression analysis is employed for evaluating the determinants of patient outcomes—one of which is the estimation of the duration of recovery in relation to age, treatment administered and other health complications. It helps identify key variables influencing health metrics for better treatment planning. Refer to Section 8.2
3. Backward regression begins with all candidate variables in the model and removes the least significant ones step-by-step, continuing until only significant variables remain. It helps simplify models by retaining only influential variables. Refer to Section 8.3

NOTES

4. The most common method to assess the fit of a multiple regression model is by examining the R-squared value, which indicates how well the independent variables explain the variation in the dependent variable. Additionally, Adjusted R-squared, F-statistic, and p-values for individual coefficients are also used to evaluate the model's effectiveness. Refer to Section 8.4
5. Overfitting is a common phenomenon that occurs when the model complexity is high, leading to the fitting of noise rather than the actual relationship in the data. Such a model tends to give good performance on its training data, while it is usually very poor with any additional external data and therefore not very useful, as its generalisation performance is greatly reduced. Refer to Section 8.6

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	Sarah should conclude that as income increases, customers are likely to make more purchases. The positive coefficient for income suggests a direct relationship between higher income and higher purchasing behaviour.
2.	The healthcare organisation can gain insights into the factors that most significantly influence patient satisfaction, such as how much wait time or communication skills impact overall satisfaction. This can guide improvements in services.
3.	The main benefit of stepwise regression is that it helps to build efficient models by selecting significant variables for the model through inclusion and exclusion of variables method based on statistical criteria.
4.	b. Multicollinearity
5.	c. Overfitting

8.12 SUGGESTED READING & REFERENCES

SUGGESTED READINGS

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CASE STUDY

NOTES

PREDICTING EMPLOYEE PERFORMANCE

SCENARIO

For the last two decades, VideoStori Private Limited has been expanding its business. The company's HR department needed to comprehend the factors influencing employee performance ratings. Utilising insights from some strategies, the HR department devised a few talent development initiatives to enhance overall workforce performance.

These strategies include targeted training programmes and recruitment tactics based on education levels. Because of this, they have a hypothesis that suggests that the performance of employees may be affected by a variety of factors, including the number of training hours, the number of years of experience, and the level of education.

DATA

The data was collected from more than one hundred employees by the HR manager, and it included the following variables as shown in following table:

Table: Employee Data of VideoStori				
Employee	Education level	Years of experience	Training hours	Performance rating
1	2	5	20	75
2	3	8	15	82
3	1	3	25	65
4	4	10	30	90
5	2	6	18	78
6	3	9	22	85
7	2	4	12	70
8	4	12	35	95
9	3	7	28	80
10	1	2	10	60

According to the table:

- ❑ **Dependent variable (Y):** Employee Performance rating (on a scale of 1 to 100)
- ❑ **Independent variable:** The independent variables include:
 - ◆ X_1 : Education level (measured as the highest degree obtained) coded as 1 for high school, 2 for Bachelor's, 3 for Master's and 4 for PhD.



Case Objective

The objective of this case study is to understand factors impacting employee performance (education, experience, and training hours) to enhance workforce performance via tailored HR initiatives at VideoStori Private Limited.

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- ◆ X_2 : Years of experience
- ◆ X_3 : Number of training hours

PROCESS

1. **Data preparation:** Clean and organise data, ensuring proper coding for categorical variables.
2. **Variable selection:** Choose relevant independent variables (education, experience and training hours).
3. **Model specification:** Formulate the multiple regression equation based on chosen variables.
4. **Regression analysis:** Use statistical software to run the regression analysis on the collected data.
5. **Interpret coefficients:** Examine coefficients to understand the impact of each variable on the dependent variable (employee performance).
6. **Evaluate significance:** Assess p-values to determine the significance of each variable.
7. **Model validation:** Check assumptions like normality, homoscedasticity and multicollinearity.
8. **Equation interpretation:** Derive insights from the equation, understanding the baseline performance and the impact of each variable.
9. **Recommendations:** Based on findings, suggest HR initiatives to enhance employee performance, considering the significant variables.



QUESTIONS

1. Obtain the equation for multiple regression to estimate the employee performance based on all three factors, which are the education level, training hours and the years of experience. What does the intercept of β_0 represent?

(Hint: Use the multiple regression equation format: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$.)

2. Each additional level of education (X_1), year of experience (X_2) and training hour (X_3) is associated with an increase of how many points in employee performance rating?

(Hint: Examine the coefficients in the multiple regression equation to determine the points increase associated with each additional level of education, year of experience and training hour.)

9

C H A P T E R

LOGISTIC REGRESSION

CONTENTS

- 9.1 Introduction
- 9.2 Concept of Logistic Regression Model
 - 9.2.1 Estimation of Parameter in Logistic Regression
 - 9.2.2 Interpretation of Logistic Regression Parameters
 - Self Assessment Questions
 - Activity
- 9.3 Accuracy of Logistics Model
 - 9.3.1 Accuracy Paradox
 - 9.3.2 Sensitivity, Specificity and Precision
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 - 9.3.4 Area Under the Curve (AUC) and Lorenz Curve
 - Self Assessment Questions
 - Activity
- 9.4 Summary
- 9.5 Multiple Choice Questions
- 9.6 Descriptive Questions
- 9.7 Higher Order Thinking Skills (HOTS) Questions
- 9.8 Answers and Hints
- 9.9 Suggested Readings & References

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**Case Objective**

The objective of the caselet is to formulate a predictive model based on logistic regression to identify variables that mostly cause employee attrition.

INTRODUCTORY CASELET**EMPLOYEE ATTRITION FORECASTING**

In the rapidly changing world of technology, AlphaSoft is seeing an increasing rate of turnover in its workforce. Using logistic regression, they investigated a large dataset of demographics, job, satisfaction, tenure and attrition, with careful pre-treatment and application of forward selection to arrive at key predictors: job satisfaction and tenure, bringing much greater complexity into the dynamics of attrition.

Validated through precise accuracy metrics, the predictive model generates actionable insights. Targeted approaches that focus on upgrading job satisfaction and tenure policies are found to be the critical interventions. These interventions were fine-tuned for departments that have experienced higher rates of attrition within the organisation so that the environment they work in encourages employees to remain there.

Continual model refinement is still of paramount importance in order to stay on top of the constantly shifting fine print of workforce behaviours. The process helps AlphaSoft act ahead of attrition and cultivate a happier and more engaged workforce.

AlphaSoft's predictive model is a directional compass giving a delicate understanding of the complexities of attrition. The insights helped in strategic interventions so that a workplace culture sets up a system that promotes employee satisfaction and retention in the workplace.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Discuss the concept of a logistic regression model
- Evaluate logistic regression model performance using accuracy and sensitivity

9.1 INTRODUCTION

In the previous chapter, you read about multiple linear regression, its interpretation and its applications. Further, you have learnt forward, backward and stepwise regression. You have also learnt how to assess the fit of the multiple regression model.

Logistic regression is considered an advancement of traditional linear regression. It is a statistical method to model the relationship of a set of independent variables (predictors) with a categorical dependent variable. For example, it may be “Yes/No” or it may be coded as “1” or “0”. Such scenarios are common in real-world situations, for instance, whether an adoption of a search engine occurs or not, whether a job is done, or if a response to a marketing offer is provided.

The objective of binary logistic regression aligns with earlier chapters: constructing an optimal and straightforward model to comprehend the correlation between the variables (X's) and the categorical dependent variable (Y). Its aim is to assess how well the variables predict the categorical outcome and evaluate the model's “goodness-of-fit,” similar to regular linear regression.

In this chapter, you will explore the concept of logistic regression along with its essentials like the estimation of parameter in logistic regression and the interpretation of various parameters. You will also learn about the accuracy paradox, sensitivity, specificity, ROC curves and AUC and Lorenz Curve.

9.2 CONCEPT OF LOGISTIC REGRESSION MODEL

A logistic regression model is a statistical model that helps to model and understand a categorical dependent variable in terms of one or more independent variables. A logistic regression model is primarily

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used when the dependent variable is binary or categorical and relates to “Yes/No,” “Success/Failure,” or “1/0” as shown in Figure 9.1:

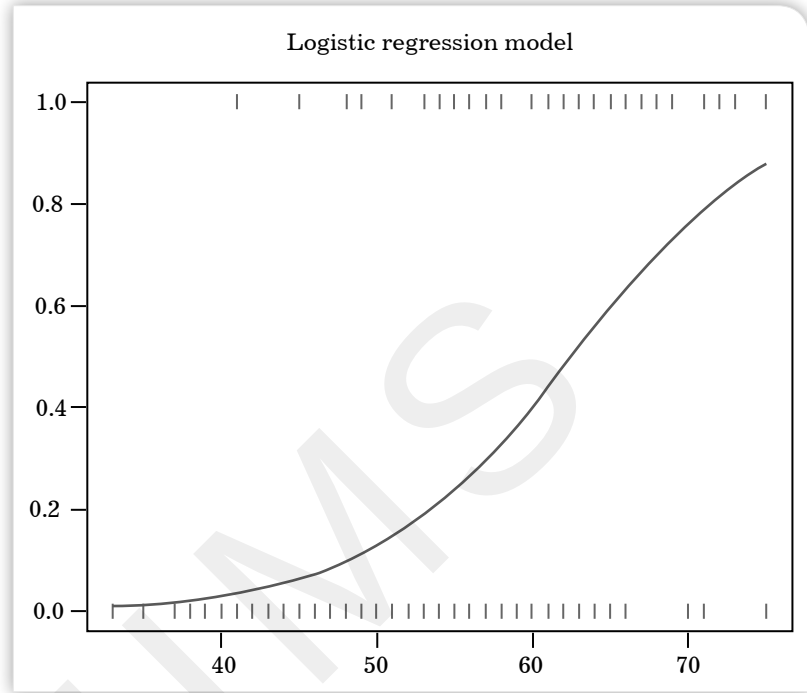


Figure 9.1: Logistic Regression Visual Representation

The logistic regression model utilises the logistic function (also known as the sigmoid function) to predict the probability of an event or outcome occurring. Logistic regression predicts the likelihood of an observation belonging to a class or category, unlike linear regression, which predicts a continuous variable. The logistic regression model converts the output of a linear regression model into a number between 0 and 1 via the logistic function. This output represents the probability of the event occurring, based on the given levels of the independent variables.

The logistic regression model estimates coefficients for every independent variable, showing their effect on the log-odds of the event occurring. This helps a researcher understand the direction and strength of the relationship between independent variables and the odds of a categorical outcome.

Logistic regression is used to predict the probability of an event occurring (e.g., pass or fail). The output is a probability between 0 and 1, calculated using the logistic (sigmoid) function. The logistic function for a given set of input features x and parameters β (weights) is:

$$(P = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

Where:

$X_1, X_2, \dots, X_n$ are the values of the independent variables

$P(Y = 1 | X)$ represents the probability that the outcome Y equals 1 given the values of the independent variables $X_1, X_2, \dots, X_n$

e is the base of the natural logarithm, which is equal to 2.71828183

β_0 is the intercept term

$\beta_1, \beta_2, \dots, \beta_n$ are the coefficients associated with each independent variables $X_1, X_2, \dots, X_n$

Some examples that prove the usage of logistic regression are as follows:

- ❑ **Disease risk prediction:** In conducting a study about heart disease prediction, the logistic regression model can be applied in terms of analysing data regarding the patients by considering variables that are age, cholesterol levels, and blood pressure, by which the chance of having heart disease of any patient can be predicted.
- ❑ **Assessing creditworthiness:** Logistic regression is applied in creditworthiness assessment. For example, credit card companies use this method to assess the creditworthiness of applicants. Some of the variables that help predict the possibility of defaulting on payment by the customer include income, credit history and debt-to-income ratio.
- ❑ **Customer churn prediction:** Logistic regression is used in the telecommunication industry to predict customer churn. The factors used to estimate the probability of losing a customer are customer satisfaction, contract length and usage of the service.
- ❑ **Employee turnover prediction:** The logistic regression model can easily be applied by the HR departments towards the probable turnover of the employee. Job satisfaction, work-life balance, and job tenure are some of the variables that have been found and used in regression analysis to explain and mitigate these risks of turnover.
- ❑ **Economic event forecasting:** In economics, logistic regression is applied for binary event forecasting, for example, to predict whether the economy is likely to go into recession or not. Predictors may be used in the form of indicators for unemployment, inflation and GDP growth.
- ❑ **Species habitat modelling:** The logistic regression is one method applied in ecology to explore the habitat preference of species. Given environmental variables such as temperature and precipitation, alongside vegetation cover, the scientist would be able to predict whether such a species could likely appear in a given location or not.

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- ❑ **Defective product prediction:** Logistic regression is used in manufacturing to predict the probability of a defective product concerning other quality controls. This can be useful as the problems identified are done sooner in the production process.
- ❑ **Dropout prediction:** In the education sector, logistic regression can predict the chances of a dropout from school. Predictors include attendance, academic performance and socio-economic background.
- ❑ **Environmental science:** In environmental science, logistic regression is used for predicting the habitat of a species based on environmental conditions and variables.

9.2.1 ESTIMATION OF PARAMETER IN LOGISTIC REGRESSION

To use logistic regression for predicting binary outcomes, it is essential to estimate the model's parameters, for example, the coefficients. This process is done using the Maximum Likelihood Estimation (MLE) function, which intends to find values of the coefficients to fit the data that maximises the likelihood of observing the given data.

In simple terms, MLE tries to find the coefficients that measure how well your model explains the data. It answers the question: "How likely is it to observe this data, given these parameter values?" MLE finds the "best-fit" parameters for your data by maximising the likelihood of the observed data under your chosen model.

The likelihood function calculates how likely it is to observe the given data for different values of the parameters β . For logistic regression, the likelihood of observing the data given the parameters is the product of probabilities for all data points:

$$L(\beta) = P(y_1 | x_1, \beta) \times P(y_2 | x_2, \beta) \times \dots \times P(y_n | x_n, \beta)$$

Where:

y_i is the observed outcome (0 or 1) for each data point

$P(y | x, \beta)$ is the probability predicted by the logistic function for each data point

For each data point, if the outcome y_i is 1, the likelihood is the predicted probability $P(y_i=1 | x_i, \beta)$. If the outcome is 0, the likelihood is $1 - P(y_i=1 | x_i, \beta)$.

Since working with products can be difficult, especially when the numbers are small, to simplify the math, we usually work with the log-likelihood, which is the natural logarithm of the likelihood:

$$\ell(\beta) = \sum_{i=1}^m [y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))]$$

This simplification is done as the function simplifies the multiplication of probabilities into addition. It also avoids the issue of underflow with small numbers (products close to zero).

To find the best parameters β , we need to maximise the log-likelihood function $l(\beta)$. This is done through optimisation methods, such as Gradient Descent. The method starts with an initial guess for β and iteratively updates it. The algorithm calculates the gradient (the slope or direction of steepest increase) of the log-likelihood and incrementally increases or decreases β in the direction that increases $l(\beta)$.

9.2.2 INTERPRETATION OF LOGISTIC REGRESSION PARAMETERS

In logistic regression, the parameters (β) are the weights assigned to the input features $x_1, x_2, \dots, x_n$. These parameters control how each feature influences the probability of the outcome and hence need to be interpreted concerning the effect changes in the independent variables make on the log-odds of the outcome variable. The log-odds is the natural logarithm of the odds of the outcome, and the odds are the ratio of the probability of the event happening to the probability of it not happening. Log odds are a way to express probabilities in a format that is particularly useful in statistical models like logistic regression.

To understand them, start with odds, which compare the likelihood of an event happening to the likelihood of it not happening. For example, if there is a 75% chance of rain, the odds are 3:1 (calculated as $0.75 / 0.25$). Log odds are simply the natural logarithm ($\ln$) of these odds, transforming the ratio into a single number. In this case, the odds of 3 become a log odds value of $\ln(3) = 1.0986$. This transformation is helpful because probabilities are limited to values between 0 and 1, while log odds can range from negative infinity to positive infinity, making them more flexible for mathematical models. In essence, log odds are a way to represent probabilities in a way that simplifies calculations and modeling in statistics.

Mathematically:

$$\square \text{ Odds} = \frac{P(y=1|x)}{P(y=0|x)}$$

$$\square \text{ Log-odds} = \log\left(\frac{P(y=1|x)}{P(y=0|x)}\right)$$

If β_i is the coefficient for a feature x_i , it tells us how much the log-odds of the outcome will increase or decrease with a one-unit increase in x_i .

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We can interpret the sign of a coefficient in two ways, namely positive β_i and negative β_i . The theory behind positive and negative coefficients, along with examples, is as follows:

- ❑ **Positive β_i :** A positive coefficient indicates that as the feature x_i increases, the log-odds of the outcome occurring (e.g., success, an event happening) increase, meaning the probability of the outcome happening also increases.

Example: If $\beta_1 = 0.5$ for the feature “hours studied,” it means that for each additional hour of study, the log-odds of passing the exam increase by 0.5.

- ❑ **Negative β_i :** A negative coefficient indicates that as the feature x_i increases, the log-odds of the outcome decreases, meaning the probability of the outcome happening decreases.

Example: If $\beta_2 = -0.3$ for the feature “number of absences,” it means that for each additional absence, the log-odds of passing the exam decrease by 0.3.

To make interpretation more intuitive, we often exponentiate the coefficients *exp* β_i , which gives the Odds Ratio (OR). The odds ratio simplifies understanding of how a one-unit change in a feature x_i affects the odds of the outcome:

- ❑ **OR > 1:** A one-unit increase in x_i increases the odds of the outcome
- ❑ **OR < 1:** A one-unit increase in x_i decreases the odds of the outcome
- ❑ **OR = 1:** A one-unit increase in x_i does not affect the odds

Example: Consider a logistic regression model predicting whether a student passes (1) or fails (0) based on the number of hours studied and absences. The equation is:

$$P(\text{pass}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \times \text{hours_studied} + \beta_2 \times \text{absences})}}$$

Suppose the coefficients are:

- ❑ $\beta_0 = -1.5$ (intercept)
- ❑ $\beta_1 = 0.4$ (coefficient for hours studied)
- ❑ $\beta_2 = -0.2$ (coefficient for absences)

The interpretations are:

- ❑ $\beta_1 = 0.4$: Each additional hour studied increases the log-odds of passing by 0.4. The odds ratio is $e^{0.4} \approx 1.49$, meaning the odds of passing increase by 49% per extra hour of study.
- ❑ $\beta_2 = -0.2$: Each additional absence decreases the log-odds of passing by 0.2. The odds ratio is $e^{-0.2} \approx 0.82$, meaning the odds of passing decrease by 18% per additional absence.

- β_0 : The intercept represents the log-odds of passing when all features are zero (e.g., no hours studied with no absences). It serves as a baseline probability.

By exponentiating the coefficients, they become odds ratios, that are easier to interpret practically. After learning about the concept of logistic regression model, we will learn about the accuracy of a logistic model.



SELF ASSESSMENT QUESTIONS

1. Which statistical method is commonly used to estimate parameters in logistic regression?
 - a. Mean Squared Error (MSE)
 - b. MLE
 - c. Ordinary Least Squares (OLS)
 - d. Root Mean Square Error (RMSE)
2. In logistic regression, the interpretation of coefficients involves:
 - a. Direct impact on the outcome variable
 - b. Calculating the mean of independent variables
 - c. Understanding odds ratios
 - d. Linear relationship with the dependent variable
3. MLE tries to find the coefficients that make the predictions align as closely as possible with the actual outcomes in the data. (True/False)
4. An odds ratio greater than 1 indicates that as the predictor decreases, the odds of the outcome increase. (True/False)



ACTIVITY

Create a simulated dataset and apply logistic regression to predict a categorical outcome. You can generate synthetic data for independent variables and then use logistic regression to predict the outcome variable.

9.3 ACCURACY OF LOGISTICS MODEL

Logistic regression is an essential tool for solving classification problems, especially binary classification as mentioned above. It computes the probability values to determine how likely it is that an observation belongs to a particular class. Its core objective is to classify an outcome as either positive (e.g., class 1) or negative (e.g., class 0),

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represented as the probability $P(Y=1)$. This enables decision-makers to categorise observations into two groups: class 1 (positive) or class 0 (negative). The core aspect of classification involves determining a cut-off probability (P_c) to guide decision-making.

The decision-making process in classification involves comparing the predicted probability of an event, $P(Y=1)$, to a cutoff threshold, P_c . If the predicted probability is less than P_c , the observation is classified as the negative class ($Y=0$). The observation is classified as the positive class ($Y=1$) if it is equal to or greater than P_c . Based on the threshold probability, this step determines which class each observation belongs to. Mathematically, this can be expressed as:

- $Y = 0$, if $P(Y=1) < P_c$
- $Y = 1$, if $P(Y=1) \geq P_c$

For example, consider using logistic regression to predict customer churn in a subscription service. Here, churn is a binary outcome: $Y=1$ for churn (customer leaves) and $Y=0$ for no churn (customer stays). The logistic regression model predicts churn probability, $P(Y=1)$, based on usage, subscription duration, and engagement metrics.

9.3.1 ACCURACY PARADOX

The accuracy paradox in logistic regression happens when a model has a generally high accuracy but cannot predict the minority class accurately and therefore performs poorly in practical or meaningful terms. This paradox often arises in situations involving imbalanced datasets, where one class significantly outnumbers the other(s). High accuracy can be achieved for a model by getting a majority of the cases correctly predicted while ignoring the minority class, which leads to false positive results. The effect of imbalanced dataset in comparison to a balanced dataset can be seen in Figure 9.2:

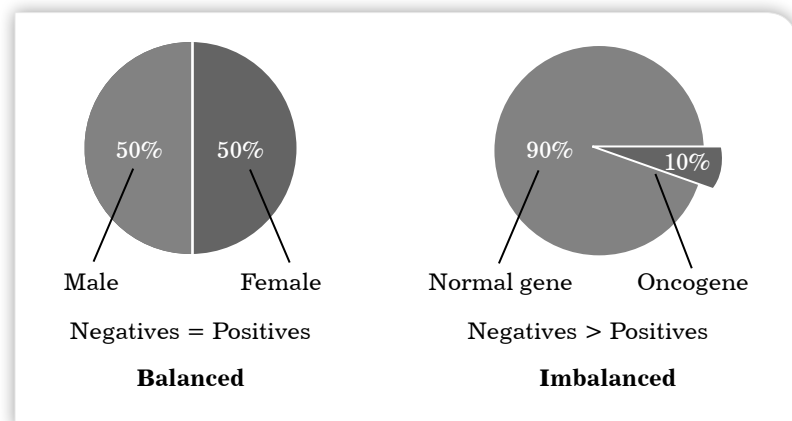


Figure 9.2: Effect of Imbalanced Dataset

Some of the implications of the accuracy issues in this context are as follows:

- ❑ **Misleading metrics:** When relying solely on accuracy as a metric, the model's performance can be misrepresented, as accuracy does not account for class imbalances or the importance of certain outcomes.
- ❑ **Overlooking critical insights:** Models may fail to prioritise important classes or events, such as detecting fraud or diagnosing cancer, which could lead to critical insights being ignored.
- ❑ **Poor decision-making:** When optimising models purely for accuracy may result in decisions that do not align with the intended business or research objectives.

The above issues emphasise the importance of considering additional metrics, such as sensitivity, specificity and precision, to provide a more comprehensive evaluation of the model's performance, especially in situations where class distribution is imbalanced. Some of the model evaluation and tuning related to accuracy assessment in logistic regression are as follows:

- ❑ **Evaluation metrics:** Evaluation metrics such as precision, recall, and F1-score can offer deeper insight into the model performance on both majority and minority classes compared to high-accuracy figures alone.
- ❑ **Resampling techniques:** Resampling techniques such as oversampling the minority class and under-sampling the majority class so that the learning can happen for both classes could enhance accuracy.
- ❑ **Class weight adjustments:** Adjustments of weights to classes in the logistic regression algorithm to get the instances of the minority class right to reduce the imbalance.
- ❑ **Ensemble methods:** Application of techniques such as bagging or boosting, to enhance the model's ability to capture patterns in the minority class by combining predictions from multiple models.
- ❑ **Threshold adjustment:** Adjust the classification threshold to prioritise sensitivity over specificity, ensuring the model is more sensitive to the minority class and reducing the impact of the imbalance on predictions.

9.3.2 SENSITIVITY, SPECIFICITY AND PRECISION

Some of the key performance metrics used in logistic regression for evaluating the accuracy of classification models are sensitivity, specificity and precision. To understand these metrics, it is important

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to first understand about True Positives (TP) and True Negatives (TN) as shown in Figure 9.3:

		Actual Value (as conformed by experiment)	
		Positives	Negatives
Predicted Value (predicted by the test)	Positives	TP True Positives	FP False Positives
	Negatives	FN False Negatives	TN True Negatives

Figure 9.3: Confusion Matrix of True Positives and True Negatives



NOTE

A confusion matrix is a table that compares predicted values to actual values to evaluate the performance of a classification model. In predictive analytics, a confusion matrix (sometimes also called a table of confusion) is a table with two rows and two columns that reports the number of TP, False Negatives (FN), False Positives (FP), and TN.

TP refer to instances where the model correctly identifies the presence of a condition or class while TN refer to instances where the model correctly identifies the absence of a condition or class. In this context, a confusion matrix is a table used to evaluate the performance of a classification model, displaying the counts of TP, TN, FP and FN to assess how well the model classifies data.

- ❑ **Sensitivity (true positive rate or recall):** This metric evaluates the ability of the model to correctly identify instances of the positive class. In the context of customer churn, sensitivity measures the proportion of actual churn cases that the model correctly predicts. Sensitivity can be calculated using the following formula:

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

- ❑ **Specificity (true negative rate):** Specificity measures a model's ability to correctly identify cases of the negative class. For example, while creating a customer churn model using logistic regression, specificity measures the proportion of non-churn cases that the model correctly predicts. Specificity can be calculated using the following formula:

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}}$$

- ❑ **Precision (positive predictive value):** Precision quantifies the accuracy of the model when it predicts positive instances. In the context of customer churn, precision measures the proportion of predicted positive cases that are actually positive. Precision can be calculated using the following formula:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Example: Suppose we have the following confusion matrix as shown in Table 9.1:

TABLE 9.1: CONFUSION MATRIX		
	Actual No Churn	Actual Churn
Predicted No Churn	800	50
Predicted Churn	30	120

Find the sensitivity, specificity and precision based the above confusion matrix table.

Solution: Here,

$$TP = 120$$

$$FP = 30$$

$$TN = 800$$

$$FN = 50$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{120}{120 + 50} = \frac{120}{170} = 0.7059$$

In this example, sensitivity measures the ability of the model to correctly identify customers who actually churned. Approximately, 70.59% of the actual churn cases are correctly predicted by the model.

$$\text{Specificity} = \frac{TN}{TN + FN} = \frac{800}{800 + 30} = \frac{800}{830} = 0.9639$$

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Specificity assesses the model's ability to correctly identify customers who did not churn. In this case, approximately 96.39% of the actual non-churn cases are correctly predicted by the model.

$$\text{Precision} = \frac{TP}{TP + FN} = \frac{120}{120 + 30} = \frac{120}{150} = 0.8$$

Precision quantifies the accuracy of the model when it predicts true positives correctly. In this example, the model's predictions of customer churn have a precision of 80%, meaning that 80% of the predicted positive cases are positive.

To summarise, these metrics provide a comprehensive view and understanding of the model's performance from various perspectives.

9.3.3 RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

The ROC curve is a graphical representation that explains the trade-off between sensitivity and specificity. This curve is an effective visual tool for evaluating the biases of the classification models. The ROC curve offers insights into how well a model can differentiate between positive and negative cases by plotting sensitivity against specificity at different threshold values. The Area Under the ROC Curve (AUC) serves as a concise metric, quantifying the overall performance of the model. A higher AUC indicates superior discrimination, ranging from 0 (no discrimination) to 1 (perfect discrimination), with 0.5 representing performance equivalent to random chance.

The utility of the ROC curve spreads beyond binary classification, making them applicable for comparing models across different domains and treating imbalanced datasets. This comprehensive assessment helps in evaluating a model's discriminative power and offers nuanced insights into its overall effectiveness in making accurate predictions.

Let us understand this with an example: A medical research team is investigating whether Body Mass Index(BMI) can be used as a predictor of a human heart disease called atherosclerosis. The team collected data from 30 subjects, each with a recorded BMI and an autopsy result indicating whether they were "Diseased" or "Healthy." The goal of the research team is to determine whether BMI can effectively distinguish between "Diseased" and "Healthy" individuals by creating an ROC Curve.

The medical team performed the following steps with the help of Excel using the sensitivity and specificity values from each threshold, and plot the ROC curve to visually evaluate how well BMI distinguishes between diseased and healthy individuals.

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1. Open the `bmi_autopsy_results.csv` file in Excel. The data should look like as shown in Figure 9.4:

	A	B
1	BMI	Autopsy Result
2	30.1	Healthy
3	33.7	Diseased
4	31.3	Healthy
5	30	Diseased
6	27.3	Healthy
7	32.2	Diseased
8	27.6	Diseased
9	37.6	Diseased
10	39.2	Diseased
11	26.4	Diseased
12	35.4	Healthy
13	29.6	Healthy
14	30.5	Diseased
15	38.4	Diseased
16	21.2	Diseased
17	19.9	Healthy
18	18.4	Healthy
19	36.3	Diseased
20	30	Healthy
21	37.1	Diseased
22	39.5	Diseased
23	35.6	Healthy
24	28.2	Diseased
25	35.2	Healthy
26	20.6	Healthy
27	30	Healthy
28	21.2	Healthy
29	38.8	Healthy
30	29.5	Healthy
31	27.1	Diseased

Figure 9.4: `bmi_autopsy_results.csv` File

2. Select the Pivot Table option in the Insert tab in the Menu Bar as shown in Figure 9.5. The dialog box of the pivot table opens in Excel.

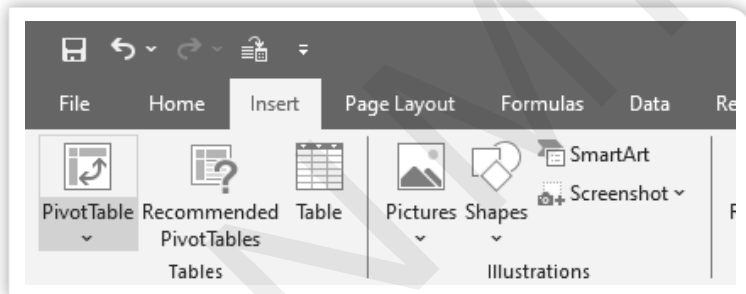


Figure 9.5: Selecting Pivot Table

3. Inspect the Pivot Table dialog box. The data in the worksheet has been automatically chosen by Excel. Select Existing Worksheet for placing the Pivot Table. Select the D2 tab in the worksheet for the location. The Pivot Table should look like as given in Figure 9.6:

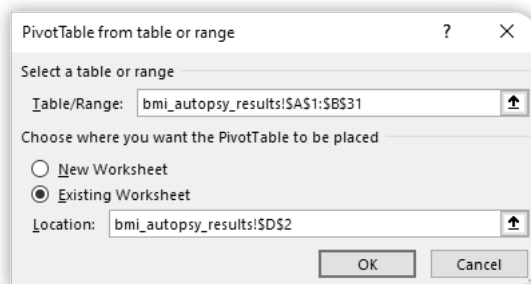


Figure 9.6: Pivot Table Dialog Box

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4. Select the OK button in the dialog box. Excel gives the following outputs. In the cell D2, it gives a box with the name PivotTable1. Excel also opens a PivotTable field on the right side of the window as shown in Figure 9.7:

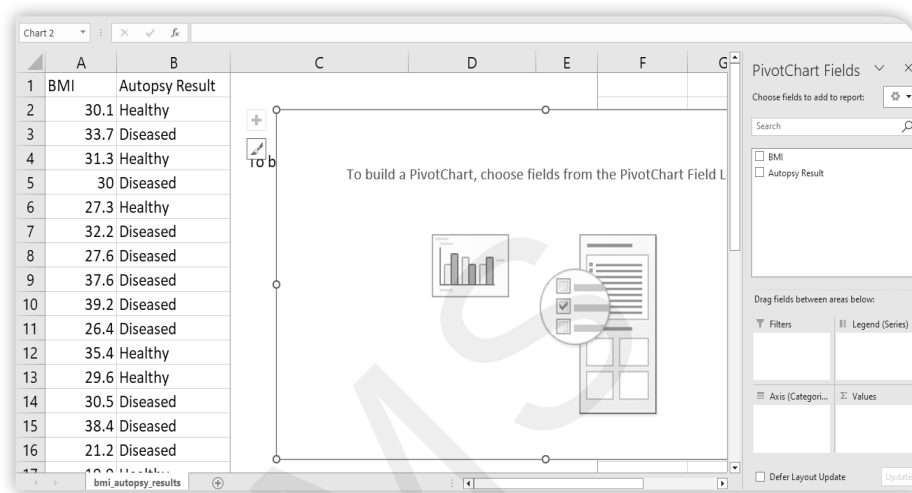


Figure 9.7: Pivot Table Output

5. Drag and drop BMI to the Rows area and Autopsy Result to the Columns and Values area respectively. The output of this step is shown in Figure 9.8:

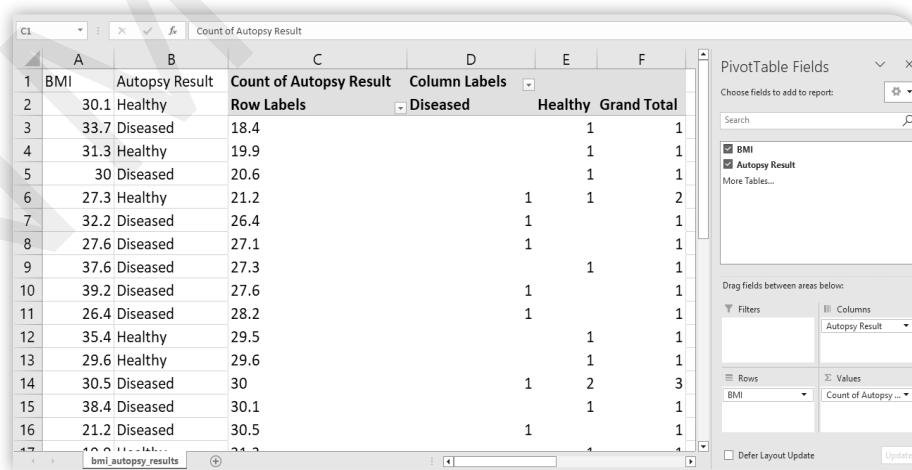


Figure 9.8: Auto Counting Done by Excel

6. Enter Sensitivity and False Positive in cells H2 and I2 respectively. Let us assume 18.4 to be the cut-off in the case that, BMIs over 18.4 has disease. Enter the following formula =SUM(E4:E\$33)/E\$34 in cell H4. Drag the formula by clicking on the small square on the box till cell I33.

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The output is shown in Figure 9.9:

	C	D	E	F	G	H	I
4	18.4			1	1	1	1
5	19.6				1	1	1
6	19.9		1	1	1	0.93333333	0.93333333
7	20.6			1	1	0.93333333	0.93333333
8	21.2			1	1	0.93333333	0.93333333
9	26.4		1		1	0.93333333	0.93333333
10	27.1		1		1	0.86666667	0.86666667
11	27.3			1	1	0.8	0.8
12	27.6		1		1	0.8	0.8
13	28.2		1		1	0.73333333	0.73333333
14	29.5			1	1	0.66666667	0.66666667
15	29.6			1	1	0.66666667	0.66666667
16	30		1		1	0.66666667	0.66666667
17	30.1			1	1	0.6	0.6
18	30.5		1		1	0.6	0.6
19	31.3			1	1	0.53333333	0.53333333
20	32.1			1	1	0.53333333	0.53333333
21	32.2		1		1	0.53333333	0.53333333
22	33.7		1		1	0.46666667	0.46666667
23	35.1			1	1	0.4	0.4
24	35.2			1	1	0.4	0.4
25	35.4			1	1	0.4	0.4
26	35.6			1	1	0.4	0.4
27	36.3		1		1	0.4	0.4
28	37.1		1		1	0.33333333	0.33333333
29	37.6		1		1	0.26666667	0.26666667
30	38.4		1		1	0.2	0.2
31	38.8			1	1	0.13333333	0.13333333
32	39.2		1		1	0.13333333	0.13333333
33	39.5		1		1	0.06666667	0.06666667
34	Grand Total		15	15	30		

Figure 9.9: Output of Sensitivity

7. Enter the formula $\text{=SUM(F4:F\$33)/F\$34}$ in cell I4 and implement step 6 again to find the values of False Positive. The output is shown in Figure 9.10:

	H	I
	Sensitivity	False Positive
	1	1
	1	0.93333333
	0.93333333	0.93333333
	0.93333333	0.86666667
	0.93333333	0.8
	0.93333333	0.73333333
	0.86666667	0.73333333
	0.8	0.73333333
	0.8	0.66666667
	0.73333333	0.66666667
	0.66666667	0.66666667
	0.66666667	0.6
	0.66666667	0.53333333
	0.6	0.53333333
	0.6	0.46666667
	0.53333333	0.46666667
	0.53333333	0.4
	0.53333333	0.33333333
	0.46666667	0.33333333
	0.4	0.33333333
	0.4	0.26666667
	0.4	0.2
	0.4	0.13333333
	0.4	0.06666667
	0.33333333	0.06666667
	0.26666667	0.06666667
	0.2	0.06666667
	0.13333333	0.06666667
	0.13333333	0
	0.06666667	0

Figure 9.10: Output of False Positives

NOTES

8. Select the Sensitivity values. Select Scatter with Smooth Lines and Markers chart in the Insert option From the Menu bar. Drag the chart to an Empty cell. In this case, the chart has been dragged to cell K2 as shown in Figure 9.11:

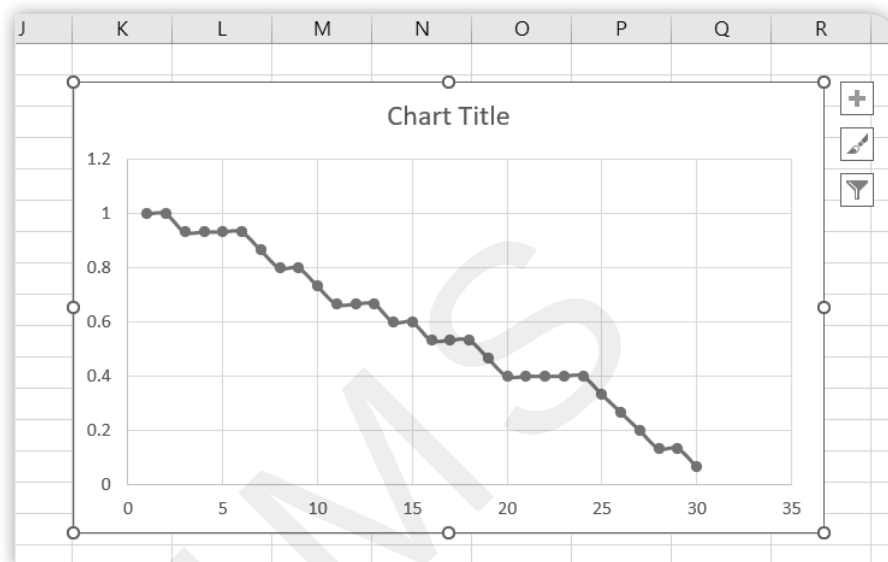


Figure 9.11: Scatter Chart

9. Select Layout 1 from the Quick Layout option as shown in Figure 9.12:

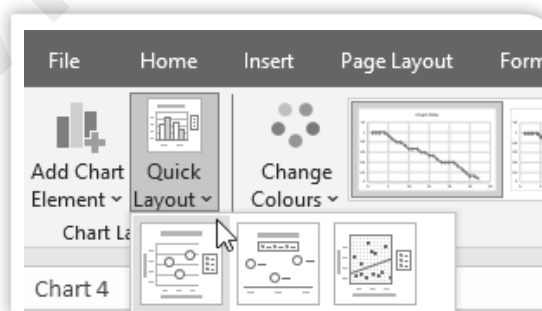


Figure 9.12: Selecting Scatter Chart Layout

10. Select any data point in the chart. Click on Select Data from Chart Design as shown in Figure 9.13:

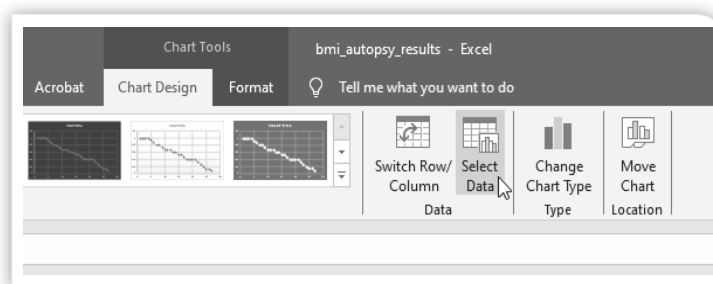


Figure 9.13: Clicking Select Data from Chart Design

N O T E S

11. Select the Edit button in the Select Data Source dialog box as shown in Figure 9.14:

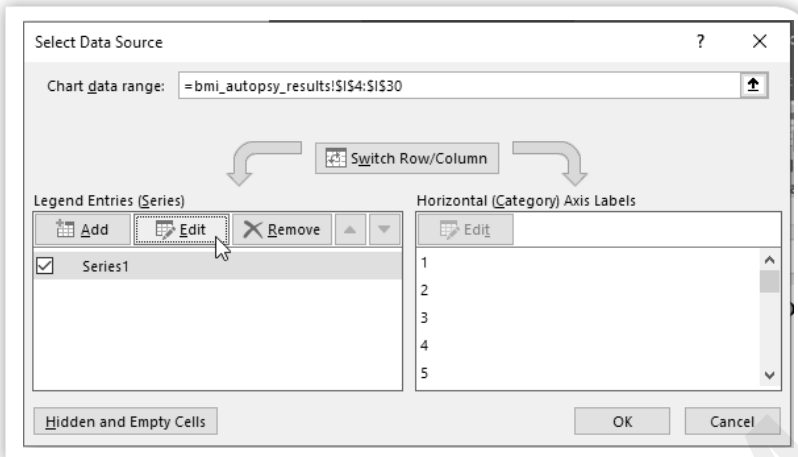


Figure 9.14: Selecting the Edit Button

12. Select the False Positive column in Series X values as shown in Figure 9.15. Click OK button two times.

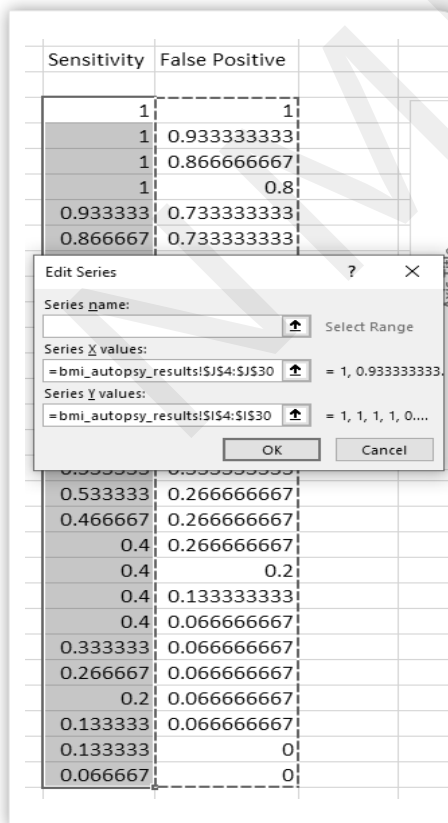


Figure 9.15: Selecting X-axis

NOTES

13. Change the axis titles and the chart title in the Scatter chart as shown in Figure 9.16:

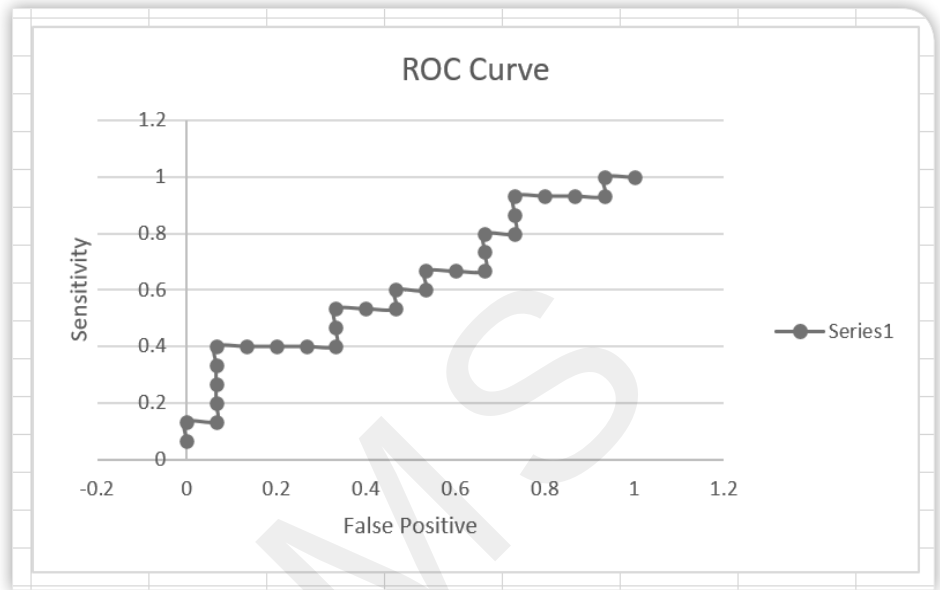


Figure 9.16: Renaming the Axes and Chart Title

14. Double-click on any of the axes and change the minimum bound to 0 and maximum bound to 1 as shown in Figure 9.17:

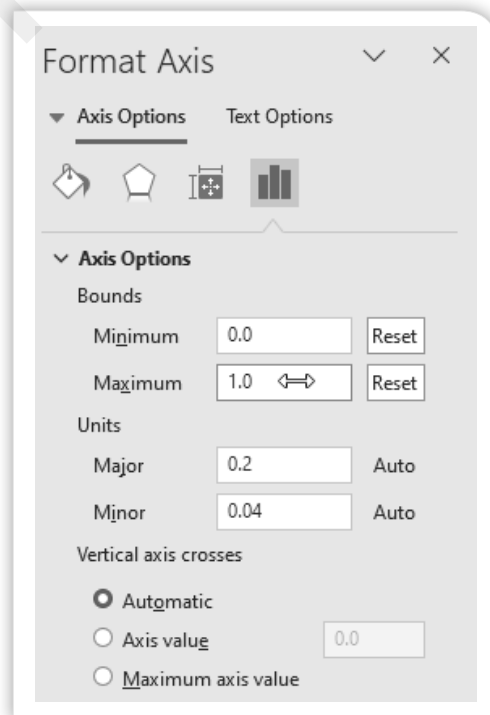


Figure 9.17: Changing the Axis Bounds

NOTES

The final output of the ROC Curve of the dataset after increasing the chart height is shown in Figure 9.18:

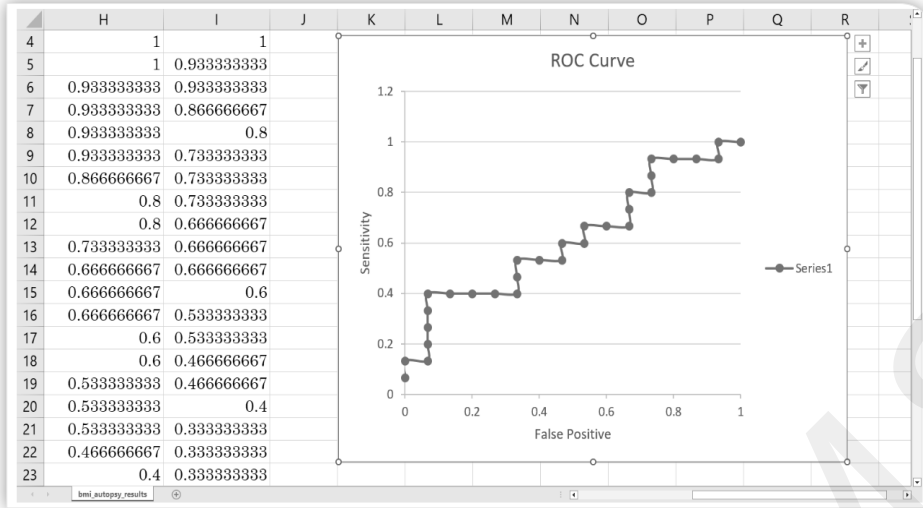


Figure 9.18: ROC Curve in Excel

Since the Scatter chart shows that data points after the specific data points become 1 (that means, the data points after the specific point become true for false positives) the specific point becomes the cut-off for the diagnostic test for the range of BMI above that will risk in having atherosclerosis occurring in the hearts of the individuals. The cut-off which is 26.4 is shown in Figure 9.19:

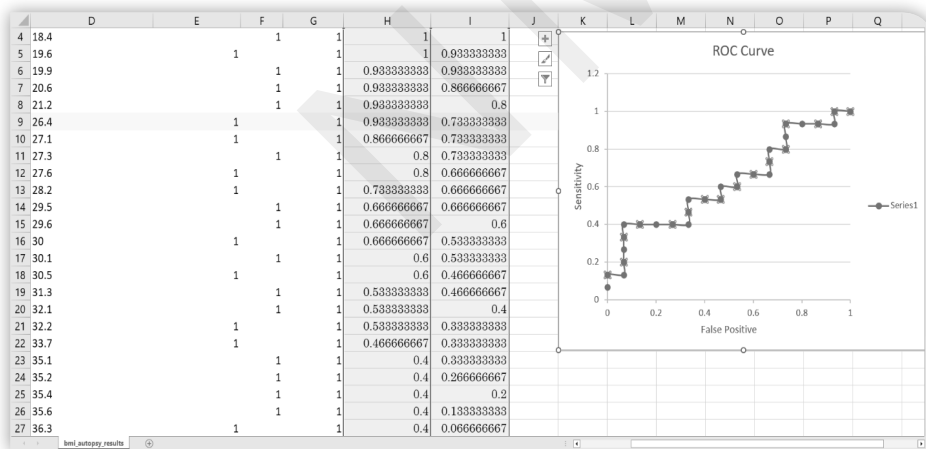


Figure 9.19: Cut-off Point of the BMI

9.3.4 AREA UNDER THE CURVE (AUC) AND LORENZ CURVE

The AUC is a metric used to quantify the performance of a classification model, particularly in the context of a ROC curve. The ROC curve visually represents a model's ability to separate the positive and negative examples at different classification thresholds. AUC calculates the area under this curve, summarising the model's

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overall discriminatory power. It ranges from 0 to 1, with 1 indicating perfect discrimination and 0.5 representing performance equivalent to random chance.

The applications of AUC Curve in logistic regression are:

- ❑ **Model comparison:** AUC is commonly used to compare the performance of different classification models. A model with a higher AUC is generally considered to have better discriminatory power.
- ❑ **Threshold selection:** ROC curves help visualise the trade-off between sensitivity and specificity in different classification thresholds. AUC aids in selecting an optimal threshold based on the specific requirements of the application. For example, in a medical test, one might prioritise minimising false positives or false negatives.
- ❑ **Imbalanced datasets:** AUC is robust when dealing with imbalanced datasets, where one class significantly outnumbers the other. It provides a comprehensive assessment of a model's ability to discriminate between classes without being overly sensitive to class imbalances.
- ❑ **Cross-domain model evaluation:** AUC facilitates the comparison of models across different domains and datasets. This makes it a valuable metric for assessing the generalisability of a classification model.

The AUC of a ROC curve quantifies how well a model distinguishes between positive and negative classes. Here, we compare three models with AUC values of 0.7, 0.85 and 0.9 to understand their practical implications. Table 9.2 shows the model discrimination and its practical implementation:

TABLE 9.2: PRACTICAL IMPLEMENTATION OF AUC VALUES

AUC Value	Model Discrimination	Practical Implications
0.7	Fair	The model is better than random guessing but may struggle in cases requiring high precision or recall. Ideal for exploratory analyses or low-stakes decisions.
0.85	Good	Indicates strong discrimination. Suitable for applications where both sensitivity and specificity are critical, such as customer churn prediction or credit scoring.
0.9	Excellent	Excellent ability to distinguish classes. Best for high-stakes scenarios like medical diagnosis or fraud detection where minimising errors is essential.

The Lorenz Curve shows inequality or concentration in distribution and can be used in classification problems to evaluate model performance. In the context of logistic regression, the Lorenz curve is sometimes adapted to analyze ranking performance or model calibration, though this is less common than AUC.

The Lorenz curve can represent the cumulative distribution of predicted probabilities or scores.

It shows how well the logistic regression model ranks positive instances compared to negative instances. For example, if sorted by predicted probabilities, you can plot the cumulative share of true positives against the cumulative share of the population.

For binary classification, it plots the cumulative percentage of true positives against the cumulative percentage of the population or predicted positives, helping to visualise how prediction probabilities are distributed.

When comparing classification models, the Lorenz Curve can provide insights into how well a model distinguishes positive instances across the entire dataset, especially in imbalanced datasets where class representation is unequal.

A Lorenz curve closer to the diagonal indicates poorer ranking performance. A highly curved Lorenz curve (bowed outward) indicates better separation between positive and negative instances.

The applications of the Lorenz Curve in logistic regression include:

- ❑ **Model evaluation in imbalanced settings:** The Lorenz Curve can highlight how well the model identifies positive cases across the spectrum of predictions, offering a complement to the AUC.
- ❑ **Visualising concentration of positive instances:** The curve visually represents how positive predictions are distributed, which can help refine model tuning by adjusting thresholds to capture more of the positive instances earlier.



SELF ASSESSMENT QUESTIONS

5. Sensitivity measures the ability of a model to correctly identify instances of the negative class. (True/False)
6. The accuracy paradox primarily affects models dealing with balanced datasets. (True/False)
7. Which metric evaluates the proportion of predicted positive cases that are actually positive in the context of customer churn prediction?
 - a. Sensitivity
 - b. Specificity

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- c. Precision
- d. Recall
- 8. What does the AUC signify in binary classification?
 - a. The accuracy of the model at a specific threshold
 - b. The trade-off between sensitivity and specificity
 - c. The probability of ranking a randomly chosen positive instance higher than a randomly chosen negative instance
 - d. The measure of how well a model ranks instances within different percentiles



ACTIVITY

Using a dataset with class imbalance, implement oversampling and class weight adjustments. Compare model performance using accuracy, sensitivity and precision before and after balancing.

9.4 SUMMARY

- ❑ Logistic regression is an extension of linear regression designed for categorical dependent variables, commonly used in binary outcomes.
- ❑ The logistic regression model uses the sigmoid function to predict the probability of binary outcomes. Unlike linear regression, it transforms the output to probabilities between 0 and 1, making it suitable for classification tasks.
- ❑ Logistic regression coefficients represent log-odds ratios, requiring consideration of odds ratios for interpretation. Positive and negative coefficients indicate the direction of the relationship with the log-odds.
- ❑ The accuracy paradox highlights challenges in using accuracy as a metric in imbalanced datasets, where a model predicting the majority class might yield high accuracy but overlook the minority class.
- ❑ Sensitivity, specificity and precision are commonly used metrics offering detailed insights into a model's performance in identifying positive and negative instances, providing a balanced evaluation.
- ❑ The ROC curve illustrates the trade-off between sensitivity and specificity at different thresholds. The AUC quantifies overall model performance.
- ❑ AUC measures the probability of ranking positive instances higher than negative ones.



KEY WORDS

- ❑ **Accuracy paradox:** It is a phenomenon that occurs when a classification model achieves high accuracy but fails to capture instances of the minority class. Accuracy paradox is especially common in imbalanced datasets, where one class dominates the data.
- ❑ **AUC:** This is a metric derived from the ROC curve, representing the likelihood that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance by the model.
- ❑ **Classification cut-off probability:** It is the threshold probability used to classify observations into different classes.
- ❑ **Log-likelihood function:** This is a function that represents the probability of observing the given set of outcomes given a particular set of parameter values.
- ❑ **Logistic model:** This is a statistical method used for analysing the relationship between a categorical dependent variable and one or more independent variables.
- ❑ **Lorenz Curve:** It is a curve used in binary classification to evaluate how well a model ranks positive instances compared to negative instances.
- ❑ **MLE:** This is a statistical method used to calculate the parameters of a model by identifying the values that maximise the likelihood of the observed data.
- ❑ **Precision, recall (sensitivity), specificity:** Metrics used to evaluate the performance of a classification model. Precision measures the proportion of predicted positive cases that are actually positive, recall measures the proportion of actual positive cases that the model correctly predicts, and specificity measures the proportion of actual negative cases that the model correctly predicts.

9.5 MULTIPLE CHOICE QUESTIONS

1. What type of dependent variable is primarily used in a logistic regression model?
 - a. Continuous
 - b. Categorical or binary
 - c. Ordinal
 - d. Ratio
2. In logistic regression, what is the purpose of the logistic function?
 - a. To estimate the continuous outcome of a dependent variable
 - b. To constrain predicted probabilities between 0 and 1

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- c. To classify outcomes into multiple classes
- d. To minimise errors in categorical predictions
- 3. Which method is used to estimate parameters in logistic regression?
 - a. Least Squares Method
 - b. MLE
 - c. K-Means Clustering
 - d. Bayesian Estimation
- 4. In a logistic regression model, a positive coefficient for an independent variable indicates:
 - a. A decrease in log-odds of the outcome
 - b. No effect on the outcome
 - c. An increase in log-odds of the outcome
 - d. A reduction in the probability of the outcome occurring
- 5. Which metric measures a model's ability to correctly identify true negatives in a logistic regression model?
 - a. Sensitivity
 - b. Specificity
 - c. Precision
 - d. Recall
- 6. What is the AP in logistic regression?
 - a. A model performs better on a minority class than the majority class
 - b. High accuracy achieved by ignoring the minority class
 - c. The model perfectly predicts both classes
 - d. Low accuracy due to a balanced dataset
- 7. What is the purpose of the AUC in the context of logistic regression?
 - a. To measure the accuracy of the model
 - b. To summarise the model's overall discriminatory power
 - c. To calculate the model's sensitivity and specificity
 - d. To determine the coefficients for independent variables
- 8. Which of the following best describes the role of the Lorenz Curve in logistic regression?
 - a. It estimates the probability of a positive instance.
 - b. It represents the model's overall accuracy.
 - c. It visualises the concentration of positive instances in imbalanced datasets.
 - d. It helps calculate log-odds for independent variables.

9. When interpreting logistic regression coefficients, the intercept represents:
 - a. The odds of the outcome when all predictors are at their mean.
 - b. The probability of the outcome occurring given all predictors.
 - c. The log-odds of the outcome when all predictors are zero.
 - d. The mean probability of the outcome across all observations.
10. Which technique is not typically used to address class imbalance in logistic regression?
 - a. Resampling techniques b. Adjusting class weights
 - c. Threshold adjustment d. Cross-validation

9.6 DESCRIPTIVE QUESTIONS

1. Discuss the process of estimating parameters in logistic regression, emphasising the role of MLE.
2. Explain the interpretation of logistic regression parameters, focusing on odds ratios and the direction of relationships.
3. Elaborate on the classification table, sensitivity, specificity and precision in the context of logistic regression. Provide examples for better understanding.
4. Explain the concept of AUC in the context of ROC curves. How does the AUC value relate to the performance of a binary classification model, and what does it signify about the model's ability to discriminate between classes?
5. Imagine you have two ROC curves for two different classifiers, and their corresponding AUC values are 0.85 and 0.60. Compare the performance of these two classifiers based on their ROC curves and AUC values. What insights can you draw about the discrimination ability, sensitivity and specificity of each model?

9.7 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. Suppose you have constructed a logistic regression model to predict customer churn in a telecommunications company. The model's sensitivity is 80 per cent, but its specificity is only 60 per cent. Which of the following statements is most likely true, given this information?
 - a. The model is accurately identifying most customers who are likely to churn.
 - b. The model is incorrectly classifying many non-churning customers as churners.

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- c. The model is suffering from overfitting and needs to be simplified.
 - d. The model's cut-off probability needs to be adjusted to increase specificity.
2. You are comparing two logistic regression models for predicting credit card default. Model A has a higher AUC than Model B, but Model B has a higher specificity. Which model would you generally prefer to use in practice?
 - a. Model A, because it has a higher AUC, indicating better overall discrimination.
 - b. Model B, because it has higher specificity, suggesting a better balance of sensitivity and specificity.
 - c. It depends on the specific costs associated with false positives and false negatives in the credit card default scenario.
 - d. Neither model is ideal, as both have shortcomings in their performance metrics.
3. A researcher employs forward selection for ROC curve analysis to choose variables for a predictive model of hospital readmission. The AUC improvement for a particular variable is 0.03, but the change is not statistically significant ($p\text{-value} = 0.07$). Should the variable be included in the model?
 - a. Yes, because the AUC improvement is relatively high, suggesting a significant contribution.
 - b. No, because the change in AUC is not statistically significant at the 0.05 level.
 - c. It depends on the researcher's judgement and the importance of the variable in a clinical context.
 - d. It is impossible to determine about the variable and the model without further information.
4. Imagine you are building a logistic regression model to predict student dropout on an online learning platform. You have data on course engagement, learning styles, and socio-economic factors. Discuss two potential challenges you might face when interpreting the model's results due to multicollinearity and confounding variables.
5. A marketing team uses a logistic regression model to predict customer response to a new loyalty programme offer. The model achieves a high accuracy of 85%. However, upon closer inspection, the team discovers that the majority of predicted "positive" responders are actually false positives. Explain two possible reasons for this phenomenon and suggest potential strategies for overcoming them.

9.8 ANSWERS AND HINTS**ANSWERS FOR SELF ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Concept of Logistic Regression Model	1.	b. MLE
	2.	c. Understanding odds ratios
	3.	True
	4.	False
Accuracy of Logistics Model	5.	False
	6.	False
	7.	c. Precision
	8.	c. The probability of ranking a randomly chosen positive instance higher than a randomly chosen negative instance.

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. Categorical or binary
2.	b. To constrain predicted probabilities between 0 and 1
3.	b. MLE
4.	c. An increase in log-odds of the outcome
5.	b. Specificity
6.	b. High accuracy achieved by ignoring the minority class
7.	b. To summarise the model's overall discriminatory power
8.	c. It visualises the concentration of positive instances in imbalanced datasets.
9.	c. The log-odds of the outcome when all predictors are zero.
10.	d. Cross-validation

HINTS FOR DESCRIPTIVE QUESTIONS

1. Estimating parameters in logistic regression involves finding coefficients that maximise the likelihood of observed outcomes. This process utilises MLE, and iteratively optimises parameter values to fit the logistic regression model to the data. Refer to Section 9.2
2. Logistic regression parameters are interpreted through odds ratios, indicating the impact of independent variables on the log-odds of the event. Understanding the direction and strength

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of these relationships involves considering the odds ratios and the sign (positive/negative) of the coefficients. Refer to Section 9.2

3. Sensitivity, specificity and precision metrics provide insights into different aspects of a logistic regression model's performance. Sensitivity measures true identification, specificity measures true negative identification, and precision evaluates the accuracy of positive predictions. Refer to Section 9.3
4. AUC represents the area under the ROC curve, assessing a model's ability to distinguish between classes. Higher AUC suggests better discrimination. It captures the trade-off between sensitivity and specificity, providing an overall performance measure. Refer to Section 9.3
5. A higher AUC indicates better discrimination. Compare the AUC values; a higher value (0.85) suggests superior performance compared to a lower one (0.60). Analyse the ROC curves to understand each model's trade-off between sensitivity and specificity. Refer to Section 9.3

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	b. The model is incorrectly classifying many non-churning customers as churners.
2.	c. It depends on the specific costs associated with false positives and false negatives in the credit card default scenario.
3.	c. It depends on the researcher's judgement and the importance of the variable in a clinical context.
4.	Challenges in student dropout prediction are: • Multicollinearity: If various learning styles or socio-economic factors are highly correlated, they might mask the individual effects in the model, making it difficult to disentangle their true impact on dropout. • Confounding variables: Factors like unexpected life events or changes in study goals might indirectly influence dropout but wouldn't be included in the model, potentially distorting the estimates of other variables.
5.	The reasons for the phenomenon (false positive) are: • Selection bias: If the training data used to build the model over-represents a specific customer segment, the model might be biased towards predicting their behaviour, leading to false positives for other segments.

Q. No.	Answer
	• Class imbalance: If “positive responders” are a much smaller proportion of the customer base, the model might prioritise overall accuracy by simply predicting the majority class (non responders), leading to high false positives for the underrepresented minority class. The strategies to overcome these issues are: • Data preprocessing: Techniques like data normalisation can reduce multicollinearity. Gathering more diverse training data and addressing class imbalance can mitigate bias. • Model validation and optimisation: Using separate validation datasets and employing techniques like cross-validation can help identify and address bias and false positives. • Domain expertise: Incorporating knowledge of the specific context and customer segments can guide interpretation and model refinement.

9.9 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

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- ❑ *Statistics 101: Logistic regression, logit and regression equation (2015)* YouTube. Available at: https://www.youtube.com/watch?v=NmjT1_nClzg
- ❑ *Logistic regression in R, clearly explained!!!! (2018)* YouTube. Available at: https://www.youtube.com/watch?v=C4N3_XJJ-jU

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**Case Objective**

The objective of this case study is to leverage logistic regression to detect fraudulent transactions and protect customers in real-time.

CASE STUDY**FRAUD DETECTION IN FINANCIAL TRANSACTIONS AT AMERICAN EXPRESS****SCENARIO**

American Express implemented a fraud detection system based on a logistic regression model. This model analysed various factors to identify patterns and anomalies indicative of fraudulent activity.

KEY FEATURES

Key features of the model are:

- ❑ **Spending patterns:** The model analysed spending amounts, frequency and category. Unusual spending amounts, sudden changes in spending habits, and purchases from unfamiliar categories could raise suspicion.
- ❑ **Location data:** Transactions occurring far from the cardholder's usual location could be indicative of fraud. The model analysed GPS data or transaction locations to identify geographical inconsistencies.
- ❑ **Transaction times:** Late-night or early morning transactions, especially for high-value purchases, could be a red flag. The model analysed the time of day and transaction amount to identify unusual patterns.

MODEL TRAINING AND IMPROVEMENT

The logistic regression model was trained on a large dataset of historical transactions, including both fraudulent and legitimate ones. This dataset helped the model learn the patterns and characteristics of both types of transactions.

American Express continuously monitors and updates the model with new data and evolving fraud trends. This ensures the model's accuracy and effectiveness in detecting even the most sophisticated fraudulent activities.

OUTCOME

The implemented model achieved a remarkable 99 per cent accuracy rate in identifying fraudulent transactions. This significantly reduced the number of fraudulent charges hitting accounts, saving American Express millions of dollars.

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The system also allowed for real-time flagging of suspicious transactions. This enabled American Express to block fraudulent charges before they could impact customers' accounts, protecting them from financial losses.

Additionally, internal investigation costs were reduced as the model streamlined the process of identifying and investigating suspicious activity.

BENEFITS

- ❑ **Reduced financial losses:** By preventing fraudulent charges, American Express saved millions of dollars annually.
- ❑ **Enhanced customer trust:** Protecting customers from fraud strengthened their trust and loyalty towards the company.
- ❑ **Improved efficiency:** Real-time detection and flagging of suspicious transactions optimised the investigation process and saved time.
- ❑ **Proactive approach:** Continuous monitoring and model updates ensured the system stayed ahead of evolving fraud trends.

CHALLENGES AND FUTURE CONSIDERATIONS

- ❑ Fraudsters constantly adapt their tactics, requiring continuous monitoring and model updates to stay ahead.
- ❑ Balancing accuracy with false positives is crucial to avoid inconveniencing legitimate customers.
- ❑ Integrating the fraud detection system with other security measures and data sources can further enhance its effectiveness.



QUESTIONS

1. While the American Express fraud detection system achieves impressive accuracy, some argue it could lead to false positives impacting legitimate customers. Can you discuss the potential ethical and logistical challenges of balancing accuracy with minimising inconvenience to customers?

(Hint: Consider exploring themes like user experience, trust, and potential solutions for reducing false positives while maintaining security.)

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CASE STUDY

2. Logistic regression remains a powerful tool, but the landscape of fraud is constantly evolving. Can you discuss some potential limitations of this approach and speculate on how American Express or other financial institutions might need to adapt their fraud detection strategies in the future?

(Hint: Think about emerging technologies like AI and machine learning, the increasing sophistication of fraudsters, and potential data privacy concerns associated with advanced fraud detection methods.)

10

C H A P T E R

FORECASTING AND TIME-SERIES

CONTENTS

- 10.1 Introduction
- 10.2 Concept of Forecasting
 - 10.2.1 Techniques for Forecasting
 - 10.2.2 Understanding the Theory of Forecasting Techniques
 - Self Assessment Questions
 - Activity
- 10.3 Concept of Time-Series Analysis
 - 10.3.1 Time-Series Pattern
 - 10.3.2 Understanding the Techniques Used in Time-Series Analysis
 - 10.3.3 Application of Time-Series Principles
 - Self Assessment Questions
 - Activity
- 10.4 Summary
- 10.5 Multiple Choice Questions
- 10.6 Descriptive Questions
- 10.7 Higher Order Thinking Skills (HOTS) Questions
- 10.8 Answers and Hints
- 10.9 Suggested Readings & References

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**Case Objective**

The objective of this caselet is to improve inventory management in the retail industry and optimise supply chain operations by accurately forecasting future sales.

INTRODUCTORY CASELET**RETAIL SALES FORECASTING**

Republic Mart, a major retail industry, faced challenges due to the dynamic market conditions, seasonality, promotional events, economic trends, and shifts in consumer behaviour, all in an attempt to overcome the problems of forecasting. Raghav, the Chief Technical Officer (CTO), applied an approach that involved the collection and cleaning of daily sales data, including details regarding promotions, holidays and other related events.

Raghav improved the dataset with features such as exponential smoothing, moving averages, and indicator variables for special events, along with external economic indicators. He carefully tested time-series forecasting models and machine learning algorithms. The dataset was split into training and validation sets to develop an ensemble model that drew together many forms of forecasting methods. This approach effectively captured short-term and long-term trends, seasonality and event impacts.

The introduced ensemble model not only resulted in drastic stockouts and hence improved customer satisfaction but also reduced excess inventories and optimised the working capital for the retail chain. Besides this competitive advantage, embracing a more strategic data-driven decision-making process and flexibility through continuing monitoring as well as periodical model retraining to align with changes in market conditions allowed them to achieve such competitive advantages.

**LEARNING OBJECTIVES**

After studying this chapter, you will be able to:

- Define the concept of forecasting and its techniques
- Demonstrate the techniques used in time-series analysis

10.1 INTRODUCTION

In the previous chapter, we discussed the logistic regression model's concept and application. We also discussed the methods to assess the model's fitness.

Forecasting is a specialised field within the predictive analytics domain that focuses on predicting future values based on past observations. Forecasting can be applied in many industries such as finance, economics, meteorology and manufacturing. Time-series analysis is a forecasting method based on historical time-stamped data for some event of interest. Time-series analysis involves developing models using past data to make informed decisions.

This chapter begins by introducing the concept of forecasting and its techniques. Further, the chapter delves into the core concepts and techniques for time-series modelling.

10.2 CONCEPT OF FORECASTING

Forecasting is a systematic process of making predictions or estimates about future events or trends based on historical data, observation, and analysis. It involves the information use of past available data to get proper future predictions so that individuals, organisations or even systems can prepare themselves and make decisions accordingly while allocating resources. Forecasting is a critical management function and an important exercise in various fields such as business, economics, meteorology and finance.

In that aspect, business forecasting is in itself a decision-making mechanism that enables businesses to tackle any future uncertainty. Businesses employ forecasting to allocate budgets strategically.

Methods for making predictions based on historical data in forecasting may range from simple statistical models to complex machine learning algorithms with the overall aim of reducing uncertainty. Some of the typical features of forecasting are:

- ❑ It focuses on future events.
- ❑ It indicates the likelihood of future events.
- ❑ It evaluates historical data.
- ❑ It employs statistical tools and techniques.
- ❑ It incorporates personal observations.

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A forecasting process is shown in Figure 10.1:

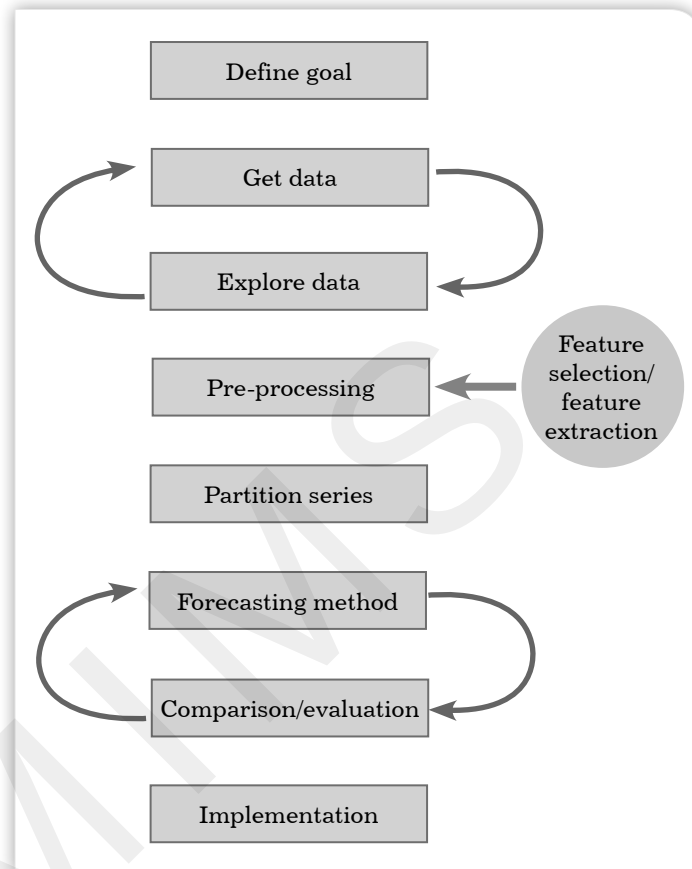


Figure 10.1: Forecasting Process

The following are the steps taken in forecasting:

1. **Define goal:** Clearly state the forecasting goals.
2. **Collect data:** Collect necessary data to for model building and forecasting.
3. **Understand data:** Study the data characteristics for patterns, trends, as well as underlying structures. This can include visual and simple descriptive statistics to get an overall view of the data.
4. **Preprocessing:** Enhance the data quality by handling missing values, removing outliers, and normalising or transforming variables.
5. **Feature selection/feature extraction:** Determine and/or derive the most relevant features of the set that contribute toward the accuracy of the forecast. This enhances model performance by focusing on meaningful predictors.
6. **Partition series:** Split the data up into training, testing, and other segments needed to validate the forecasting model's accuracy.

7. **Forecasting approach:** Choose an appropriate forecasting approach, such as Auto-Regressive Integrated Moving Average (ARIMA) exponential smoothing, and machine learning models to generate a forecast.
8. **Comparison/evaluation:** Compare the forecast results with the actual data to measure model accuracy. This could be done while computing metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE).
9. **Implementation:** Use the model to make forecasts on new or future data.

10.2.1 TECHNIQUES FOR FORECASTING

As you have studied above, forecasting involves predicting future events based on historical data or expert judgment. Considering this, forecasting techniques can be broadly categorised into qualitative and quantitative methods as shown in Figure 10.2:

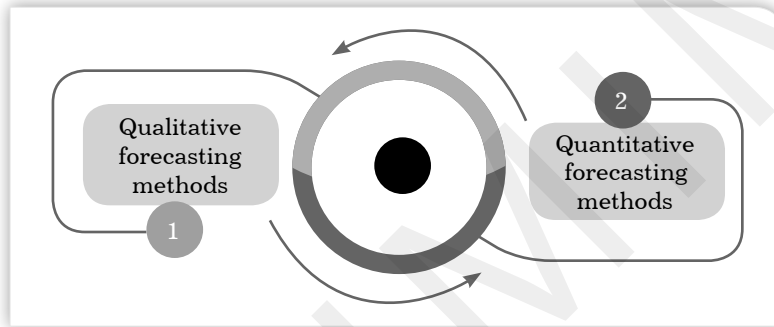


Figure 10.2: Techniques of Forecasting

Following is a detailed explanation of both categories and some of their types:

- ❑ **Qualitative forecasting techniques:** Qualitative techniques of forecasting depend on subjective judgments, experts' opinions, and qualitative data for predicting future events. These are applied when little or no historical data is available or the reliability of such historical data is questionable. This approach is also useful when forecasting requires outcomes that cannot be objectively expressed or assessed. Some of the most prominent qualitative techniques of forecasting are:
 - ◆ **Expert judgement:** This technique relies on the opinions, experience, and views of an expert in a particular field.
 - ◆ **Market research:** It is a systematic process of searching, analysing, and interpreting information about a market, including statistics about potential customers, competitors and the industry. It helps in understanding the preferences of customers or market trends.

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- ◆ **Delphi method:** The Delphi method is a structured, iterated forecasting technique where the opinions of the experts are sought through controlled surveys or questionnaires. It is applied when there is a need to synthesise diverse expert opinions.



NOTE

Qualitative forecasting techniques are often used in industries and sectors where external factors, such as changes in government policies, technological advancements, or shifts in consumer preferences, play a significant role in shaping future outcomes.

- **Quantitative forecasting techniques:** These techniques are used to make predictions on future events or values by using numerical data and statistical models. This method makes inferences based on patterns from historical data, mathematical models and statistical techniques. It is very widely applied today in fields such as finance, economics, business, and operations. These methods base their forecasting on past data patterns and relationships, hence allowing organisations to predict trends and, in turn, make appropriate decisions. Examples of the most common quantitative techniques for forecasting are:
 - ◆ **Time-series analysis:** A statistical method of analysing and modelling time-series data to identify, forecast patterns, trends and behaviours over a period of time.
 - ◆ **Box-Jenkins (ARIMA):** Box-Jenkins method or ARIMA, developed by George Box and Gwilym Jenkins, combines autoregression, integration, and moving average components in modelling and forecasting future values of a time-series. It is also considered a common tool for time-series forecasting using statistical modes.
 - ◆ **Regression analysis:** It is a statistical technique applied to determine the relationship between independent variables (predictors) and at least one dependent variable(s) of a response. The basic application of regression analysis describes the nature and strength of the relationship, building predictions, and inferences about causal relationships between variables.
 - ◆ **Time-series decomposition:** It is another technique that decomposes time-series data into the specific components that comprise the original data. These are largely trend, seasonality, and the residual (or error) components. Such decomposition of a time-series data helps to understand the underlying structural relationships and different patterns.
 - ◆ **Forecasting with neural networks:** Neural networks are effective models in identifying complex relationships and

pattern capturing in data. Some of the common examples of time-series forecasting using neural networks include Recursive Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs).

- ◆ **Monte Carlo simulation:** This is another technique, that uses randomness to analyse data. The method is useful in problems involving large numbers of variables that interact in complex ways or are inherently random.

10.2.2 UNDERSTANDING THE THEORY OF FORECASTING TECHNIQUES

To fully grasp forecasting, one must understand the unique assumptions that support different forecasting techniques. It also involves choosing the most suitable models by analysing the characteristics of the data and the specific goals of the task, whether it's short-term demand planning, financial projections, or other forecasting applications. These considerations are described below:

- ❑ **Model assumptions and limitations:** Each model has its own requirement for specific use-case and hence it is critical to understand the specific model assumptions before applying the model. For instance, linear regression assumes that a linear relationship is present between variables whereas ARIMA assumes stationarity. Stationarity refers to the condition when a time-series data's mean and variance remain constant over an extended time.
- ❑ **Model selection criteria:** These criteria can include considerations such as simplicity, interpretability, computational efficiency and predictive performance.
- ❑ **Forecasting time horizon:** A key consideration for model selection is to distinguish between short-term and long-term forecasting. Some methods are better suited for shorter time horizons compared to others.
- ❑ **Uncertainty and risk analysis:** Recognition of uncertainty in the forecast is another important consideration. Methods, such as Monte Carlo simulations or prediction intervals, are used to give a possible range of outcomes that may be expected, which then allows decision-makers to consider the extent of risk for informed decision-making.
- ❑ **Data pre-processing:** Understanding and extent of data pre-processing requirements for cleaning, normalisation, and missing values for better modelling and application.

Having learned about forecasting and its techniques, we will study about time-series and its techniques in the next section.

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**SELF ASSESSMENT QUESTIONS**

1. Which of the following is a qualitative forecasting technique that involves gathering opinions from experts in a structured manner?
 - a. Market research
 - b. Expert judgment
 - c. Moving average
 - d. Exponential smoothing
2. What is the primary goal of forecasting?
 - a. Analysing historical data
 - b. Predicting future events
 - c. Understanding present circumstances
 - d. Identifying outliers in the data
3. Forecasting is primarily concerned with predicting past events accurately. (True/False)

**ACTIVITY**

You are part of a team responsible for predicting the weather for a local community. Your goal is to provide accurate forecasts to help residents plan their activities avoiding sunny and rainy weather conditions.

10.3 CONCEPT OF TIME-SERIES ANALYSIS

Time-series analysis is a method of analysing sequential data collected over a period of time. Time-series data is different from other types of data as the analysis can suggest how variables of interest change over time. In other words, in time-series analysis, time is a crucial variable because it shows how the data alters over time.

Time-series analysis requires a large number of data points to ensure its uniformity and reliability. It ensures that any trends or patterns uncovered are not outliers and can decipher seasonal variance. Additionally, time-series data can be used for forecasting data based on historical data.

Using data visualisations, business users can identify seasonal trends and understand their underlying causes. When organisations analyse data over an interval of time, they can employ time-series forecasting to predict the likelihood of future events. Such analysis can show predictable changes in the data, like seasonality or cyclic behaviour, which provides a better understanding of data variables and helps in better forecasting.

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The factors that influence the values of an observation in a time-series are referred to as the components of the time-series. The detailed description of the four components of a time-series analysis is as follows:

- ❑ **Trend:** Trends are the components of a time-series analysis that display the general trend of data values to increase, decrease or remain stationary during a period of time. In other words, trends specify the long-term changes in the data and can uncover growth or decline with time.

Population, agricultural production, items manufactured, number of births and deaths in a certain geographical location, number of industries in a city, and number of schools or colleges in a state are examples of trends. Figure 10.3 shows the trend of a time-series data of a rental service for the dates January 11th to January 13th, which shows an upper trend indicating an increase in the usage of the service:

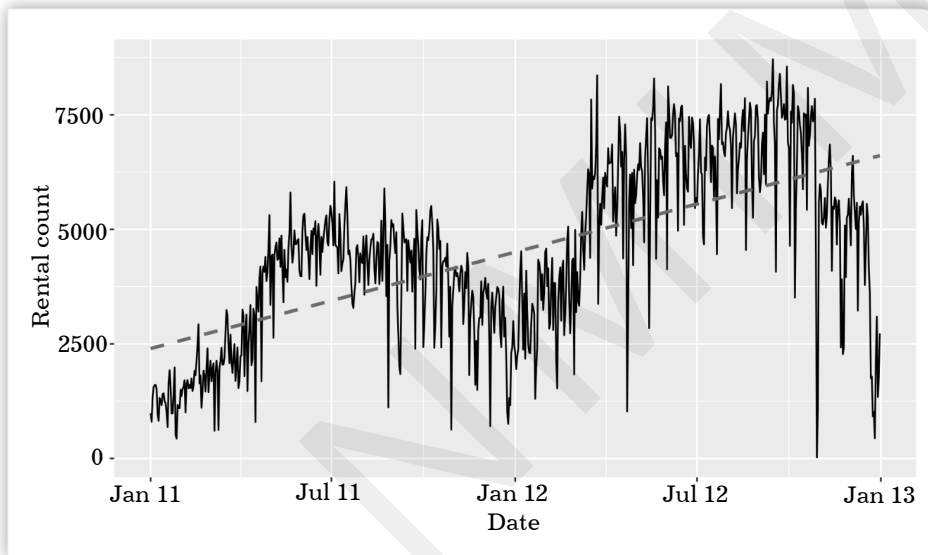


Figure 10.3: Trend of a Time-Series Data

The following are the types of trends of a time-series data:

- ◆ **Upward trend:** This type of trend suggests a general increase of data values over time.
- ◆ **Downward trend:** This type of trend suggests a general decrease of data values over time.
- ◆ **Horizontal trend:** This type of trend suggests that the data values remain constant with time.
- ◆ **Non-linear trend:** This is the type of trend that displays a more complex pattern of data changes over time, including upward or downward trends that may change direction or magnitude.

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◆ **Damped trend:** This type of trend suggests a gradual decline in the degree of change over time, where the rate of change slows down with time.

- **Seasonality:** Seasonality in time-series refers to predictable patterns that repeat over a regular time period such as day, week, month, or year. These patterns happen due to repeating events, such as holidays, weekends, or the changing of seasons, and appear in various types of time-series data, such as sales, weather, or stock prices. A visualisation of seasonality by taking a hypothetical example of Airline passengers from the year 1950 till 2020 is shown in Figure 10.4:

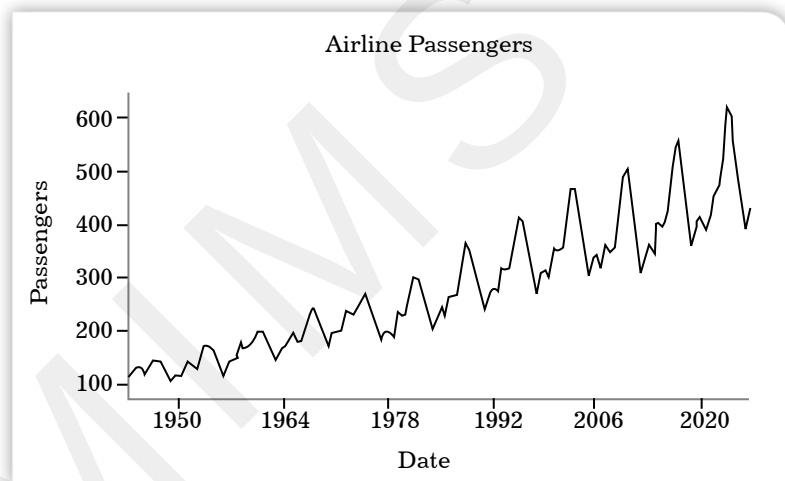


Figure 10.4: Seasonality in Time-Series Data

There are several types of seasonality in time-series data, including:

- ◆ **Weekly seasonality:** This is a type of seasonality in time-series data that repeats over a 7-day period and is usually seen in data such as sales, energy usage, or transportation patterns.
 - ◆ **Monthly seasonality:** This is a type of seasonality repeats over a 30- or 31-day period and is usually seen in data such as sales or weather patterns.
 - ◆ **Annual seasonality:** This type of seasonality repeats over a 365- or 366-day period and is usually seen in data such as sales, agriculture, or tourism patterns.
 - ◆ **Holiday seasonality:** This is a type of seasonality is caused by special events such as holidays, festivals, or sporting events and is usually seen in data such as sales, traffic, or entertainment patterns.
- **Irregular or random fluctuations (Noise):** Noise represents the random, unpredictable factors in data that are not part of any visible pattern. It can be caused by various factors such as measurement errors, external shocks, or inherent randomness in

the system. To develop an accurate model, it is critical to manage and mitigate noise, allowing the model to focus on genuine patterns rather than random variability. Noise vs. noise-free data is shown in Figure 10.5, where the straight line represents the noise-free data and the zig-zag line shows the observed data with noise:

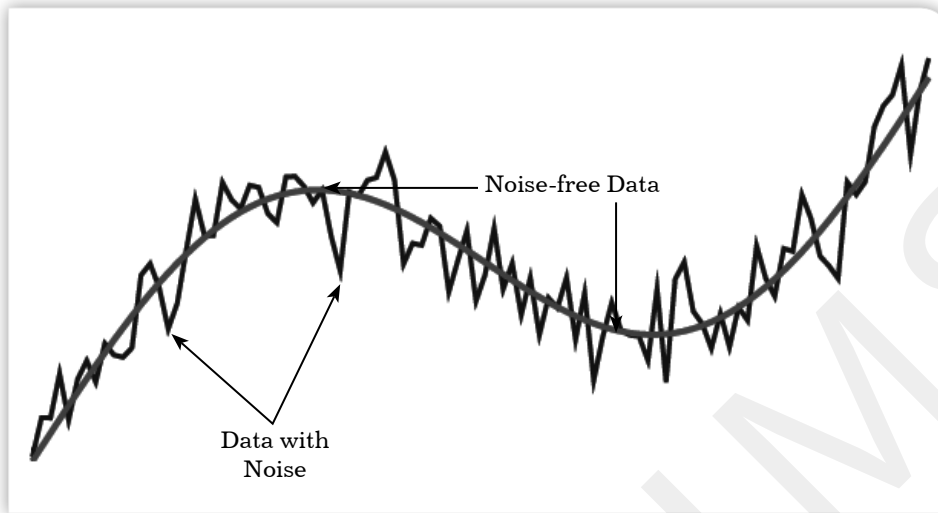


Figure 10.5: Noise vs Noise-free Data Visualisation

Some types of noise that can be found in the time-series analysis are:

- ◆ **White noise:** A type of time-series with no trend or seasonality, and where the data is a sequence of random values. It's practically impossible to predict white noise.
- ◆ **Random noise:** A form of random variation that's almost always present in a time-series. Reducing noise in the data is often a significant part of the analysis process.

10.3.1 TIME-SERIES PATTERN

Time-series patterns are repeating structures or behaviours observed in sequential data points over a regular interval of time. These time-series patterns provide the foundation for the underlying dynamics of a system, enabling precise predictions and extracting valuable insights. A thorough knowledge of these time-series patterns enables analysts to select adequate modelling techniques, validate assumptions and make precise predictions. Some common time-series patterns can be given as:

- **Horizontal pattern:** Horizontal patterns refer to the absence of a distinguishable trend, indicating a stable or constant value. This type of time-series pattern is applicable for recognising periods of stability or equilibrium in the dataset. As the horizontal patterns

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suggest a lack of long-term movement either increase or decrease in fluctuations in data, and the data values remain relatively constant over time identifying these patterns are essential. The importance of identifying horizontal patterns is:

- ◆ **Stable processes:** Horizontal patterns indicate stable systems, such as consistent production levels, demand, or performance metrics, essential in operational planning and monitoring.
 - ◆ **Baseline detection:** Horizontal patterns often serve as baselines for comparing periods with significant deviations or disruptions.
 - ◆ **Simplified forecasting:** Horizontal patterns in time-series are simpler to forecast as they rely primarily on the mean value and random variations.
- **Trend pattern:** Trend patterns are a long-term systematic movement in a particular direction. It reflects the overall direction in which the time-series is headed. Trends can be linear, indicating a constant rate of change, or non-linear, demonstrating more complex patterns. Identifying and understanding trends is essential for making informed decisions, as they provide insights into the underlying growth or decline in the data.
 - **Seasonality pattern:** Seasonality involves regular and predictable variations that occur at specific intervals. These intervals can be daily, weekly, monthly, or even yearly, often tied to external factors such as weather, holidays, or events. Detecting seasonality is crucial for adjusting strategies or predictions based on recurring patterns, ensuring that the influence of these regular fluctuations is appropriately considered.
 - **Cyclic patterns:** Cyclic patterns are the patterns that represent fluctuations happening at intervals at a longer rate than seasonality. Unlike seasonality, the cyclic pattern does not have any fixed period and can expand over several periods. Recognising cyclic behaviour is important for understanding wider economic or business cycles, which in turn helps in long-term planning and decision-making. The difference between the seasonal pattern and the cyclic pattern is shown in Table 10.1:

TABLE 10.1: DIFFERENCE BETWEEN SEASONALITY AND CYCLICAL PATTERN

Aspect	Seasonality	Cyclical Pattern
Definition	Regular and predictable fluctuations attached to specific time intervals	Irregular fluctuations occurring over long periods
Duration	Short-term, typically within a year (e.g., monthly, quarterly)	Long-term, spanning several years or decades

Aspect	Seasonality	Cyclical Pattern
Cause	Influenced by calendar events, weather, or cultural factors	Driven by economic, political, or industry-related factors
Predictability	Highly predictable and recurring	Less predictable and irregular in timing and duration
Example	Higher ice cream sales in summer	Economic recessions and expansions (business cycles)

- **Level shifts:** Level shifts are abrupt changes in the baseline of the time-series. These shifts can be caused by structural changes in the system, interventions, or sudden external events. Detecting and understanding level shifts is crucial for assessing the impact of significant changes on the overall data-generating process. Figure 10.6 shows us a visualisation of a level shift in a time-series data:

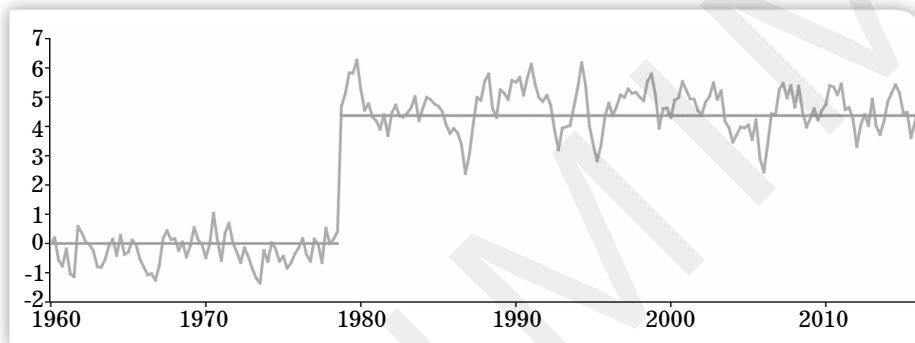


Figure 10.6: Time-Series Data with Level Shift

10.3.2 UNDERSTANDING THE TECHNIQUES USED IN TIME-SERIES ANALYSIS

Understanding the techniques in the time-series analysis is essential for extracting valuable insights from sequential data. Moving averages, exponential smoothing, and machine learning models are used for the detection of trends, seasonality, and cyclical patterns in time-series. These techniques are important to ensure accurate prediction, anomaly detection and decision-making. Through the analysis of time-series data, analysts can look out for hidden patterns, predict future events and optimise processes.

For industries heavily reliant upon predictive insights, to navigate operational and strategic goals, there is a high dependency on time-series analysis. For example, in finance, time-series is applied for stock price prediction, portfolio management and economic forecasting. In health care, time-series analysis is used to track patients' vital signs, predict outbreaks of diseases, and optimise resource distribution. In

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supply chain management, it helps in demand forecasting, inventory management, and production scheduling. With the importance that data-driven decisions are gaining, expertise in time-series analysis allows individuals a competitive edge.

In this section, we discuss the various models used for time-series analysis:

- **Auto-Regressive (AR) model:** An AR model is a type of time-series model that represents the relationship between a variable and its past values. In other words, the variable value at a given time point is assumed to be a linear combination of its previous values. The “Auto” in Auto-Regressive signifies that the variable is regressed on its past values.

The general form of an AR model of order (p), often denoted as AR(p), is given by: $Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$

Where:

Y_t is the value of the time-series at time t

c is a constant term

$\phi_1, \phi_2, \dots, \phi_p$ are the autoregressive coefficients

$Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$ are the past values of the time-series

ε_t is the white noise error term at time t .

Figure 10.7 shows the visualisation of AR models with different parameters:

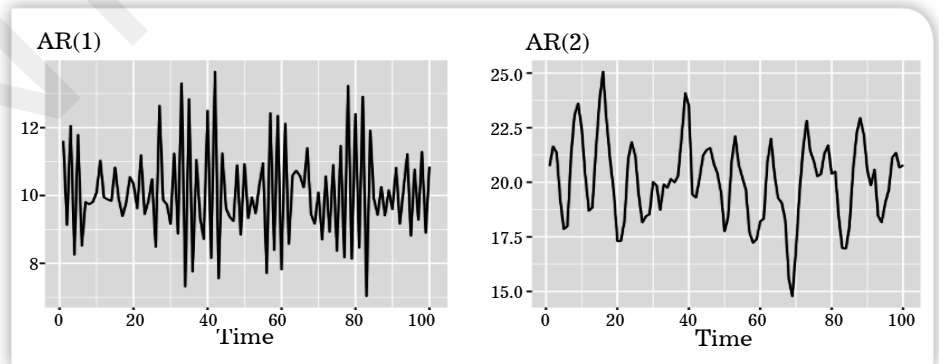


Figure 10.7: Two examples of data from AR models with different parameters.

Left: AR (1) with $y_t = 18 - 0.8y_{t-1} + \varepsilon_t$.

Right: AR (2) with $y_t = 8 + 1.3y_{t-1} - 0.7y_{t-2} + \varepsilon_t$.

In both cases, ε_t is normally distributed white noise with mean zero and variance one.

For an AR (1) model:

- ◆ When $\phi_1 = 0$, y_t is comparable to white noise

- ◆ When $\phi_1 = 1$ and $c = 0$, y_t is comparable to a random walk
- ◆ When $\phi_1 = 1$ and $c \neq 0$, y_t is comparable to a random walk with drift
- ◆ When $\phi_1 < 0$, y_t tends to oscillate around the mean.

**NOTE**

A random walk in time-series data happens when the value at each step is equal to the previous value plus a random white noise. Random walk with drift adds a constant (drift) to each step, creating a systematic upward or downward trend over time, revealing some persistent changes.

AR models are normally restricted to stationary data, in which case some constraints on the values of the parameters are required. For example,

- ◆ For an AR (1) model: $-1 < \phi_1 < 1$
- ◆ For an AR (2) model: $-1 < \phi_2 < 1, \phi_1 + \phi_2 < 1, \phi_2 - \phi_1 < 1$

□ **Moving Averages (MA) model:** MA models are basic techniques of time-series analysis that are used for removing fluctuations, showing trends in time-series data, and identifying patterns in the sequence of data. In moving averages, the average of consecutive data points is computed over specified periods. There exist many types of moving averages, such as:

- ◆ **Simple Moving Average (SMA):** This method calculates the average of a fixed number of past observations, and it often uses techniques to smooth short-term fluctuations.
- ◆ **Weighted Moving Average (WMA):** WMA is a variation of the SMA, in which the weight given to different observations is different to reflect greater significance on more recent data points.
- ◆ **Exponential Moving Average (EMA):** EMA is another variation of the WMA that gives exponentially greater weights to recent observations and is therefore more sensitive to change in the recent observations. Each type of moving average has different strengths and is more suited to different demands for forecasting and analysis.
- ◆ **ARIMA Model:** The ARIMA model is a powerful and widely used time-series forecasting technique that combines three key components:
 - ✓ **AR:** This component represents the relationship between an observation and a specified number of its lagged (previous) observations. For instance, AR terms capture

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how today's value depends on past values, such as yesterday's value or that of the day before.

- ✓ **Integrated (I):** This step involves differencing the data to make the time-series stationary. A stationary time-series has a constant mean and variance over an extended period essential for ARIMA to perform efficiently.
- ✓ **MA:** This component models the relationship between an observation and the residual errors from past observations. MA terms smooth the fluctuations by considering past forecast errors.

The ARIMA model is defined by three parameters:

- ✓ **p:** The order of the AR component (number of lag observations to include).
- ✓ **d:** The degree of differencing required to make the series stationary.
- ✓ **q:** The order of the MA component (number of lagged forecast errors included).

A step-by-step application of moving average technique using Excel is provided as follows:

1. Open an Excel spreadsheet with the properties data. Dataset can be downloaded from <https://www.kaggle.com/datasets/htagholdings/property-sales>. Select the file named as “raw_sales.csv” from the two csv files in the zip file. The data in Excel can be seen in Figure 10.8:

	A	B	C	D	E	F	G
1	datesold	postcode	price	propertyType	bedrooms		
2	2/7/2007 0:00	2607	525000	house	4		
3	2/27/2007 0:00	2906	290000	house	3		
4	3/7/2007 0:00	2905	328000	house	3		
5	3/9/2007 0:00	2905	380000	house	4		
6	3/21/2007 0:00	2906	310000	house	3		
7	4/4/2007 0:00	2905	465000	house	4		
8	4/24/2007 0:00	2607	399000	house	3		
9	4/30/2007 0:00	2606	1530000	house	4		
10	5/24/2007 0:00	2902	359000	house	3		
11	5/25/2007 0:00	2906	320000	house	3		
12	6/26/2007 0:00	2902	385000	house	3		
13	6/27/2007 0:00	2906	305000	house	3		
14	6/27/2007 0:00	2612	850000	house	4		
15	6/28/2007 0:00	2904	765000	house	4		
16	6/30/2007 0:00	2615	517000	house	4		
17	7/2/2007 0:00	2914	800000	house	5		
18	7/3/2007 0:00	2906	336000	house	3		
19	7/6/2007 0:00	2615	535000	house	5		
20	7/7/2007 0:00	2602	900000	house	4		
21	7/8/2007 0:00	2600	327000	house	1		

Figure 10.8: Sales Data

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2. Select the Data Analysis ToolPack in the Data tab for starting the process as shown in the Figure 10.9:

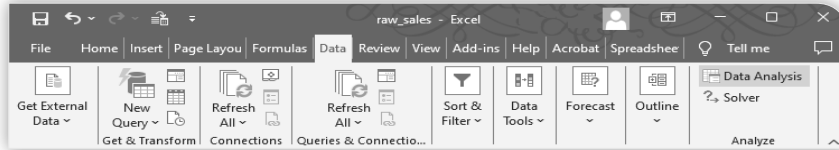


Figure 10.9: Select Data Analysis from Data

3. Select Moving Average and click the OK button in the Data Analysis dialog box as shown in Figure 10.10:

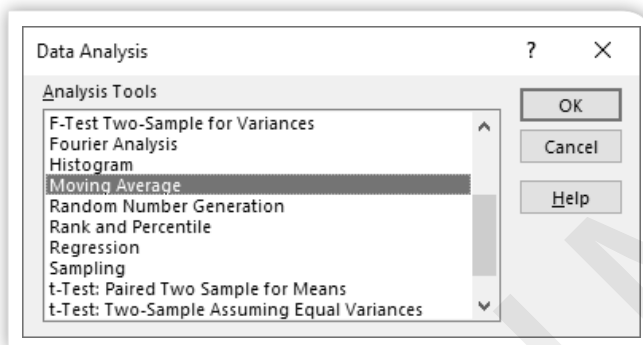


Figure 10.10: Selecting Moving Average Option

4. Set Input range with the dependent variable column which in this case is the “price” column in the csv file. Check Labels if headers exist. In this process, the header has not been selected as shown in Figure 10.11. Keep the interval for the moving average output as 2.
5. Select the output location for results in the Output Range shelf as shown in Figure 10.11. For this process, the F2 cell has been selected.

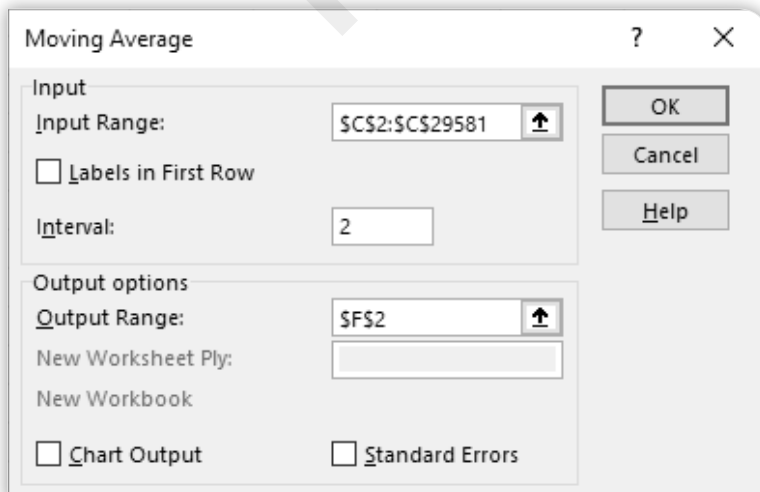


Figure 10.11: Selecting Parameters for Moving Average

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6. Click the OK button in the Moving Average dialog box and the moving averages of the “price” column has been formed as shown in Figure 10.12:

	A	B	C	D	E	F	G	H
1	datesold	postcode	price	propertyType	bedrooms			
2	2/7/2007 0:00	2607	525000 house		4	#N/A		
3	2/27/2007 0:00	2906	290000 house		3	407500		
4	3/7/2007 0:00	2905	328000 house		3	309000		
5	3/9/2007 0:00	2905	380000 house		4	354000		
6	3/21/2007 0:00	2906	310000 house		3	345000		
7	4/4/2007 0:00	2905	465000 house		4	387500		
8	4/24/2007 0:00	2607	399000 house		3	432000		
9	4/30/2007 0:00	2606	1530000 house		4	964500		
10	5/24/2007 0:00	2902	359000 house		3	944500		
11	5/25/2007 0:00	2906	320000 house		3	339500		
12	6/26/2007 0:00	2902	385000 house		3	352500		
13	6/27/2007 0:00	2906	305000 house		3	345000		
14	6/27/2007 0:00	2612	850000 house		4	577500		
15	6/28/2007 0:00	2904	765000 house		4	807500		
16	6/30/2007 0:00	2615	517000 house		4	641000		

Figure 10.12: Moving Average Output

The moving average column provides a smoothed view of the property price fluctuations over time, reducing the impact of individual price volatility. This helps identify broader price trends, such as whether property prices in the region are generally increasing, decreasing, or remain stable.

By comparing the actual property prices (“price” column) with the moving average values, we can identify short-term deviations or anomalies, such as unusually high or low sales prices compared to the average trend.

- ❑ **Exponential smoothing:** Exponential smoothing is a method of forecasting which provides exponentially decreasing weights to historical data. This technique is useful when the time-series includes any sort of trend or seasonality. The main types of exponential smoothing are:
 - ◆ **Simple exponential smoothing:** It is mainly applicable to time-series data without any trend or seasonality, thus providing the smoothest possible forecast from the past alone.
 - ◆ **Double exponential smoothing:** This type of smoothing includes both trend and level, and is therefore suited for data that depicts a linear trend.
 - ◆ **Triple exponential smoothing (Holt-Winters method):** This considers both trend and seasonality and especially proves to be very effective for seasonal time-series prediction. Exponential smoothing models assume a balanced approach for conducting time-series forecasting, especially where future observations are likely to follow the past pattern.

The characteristics of exponential smoothing techniques are:

- ◆ **Adaptability:** Exponential smoothing methods are adaptive, giving more weight to recent observations. This adaptability allows them to quickly adjust to changes in the underlying patterns of the time-series.
- ◆ **Simplicity:** The simplicity of these methods makes them easy to implement and interpret. They require minimal parameter tuning, making them accessible to users with varying levels of expertise.
- ◆ **Flexibility:** Exponential smoothing techniques can handle time-series data with different characteristics. From simple patterns without trend or seasonality to more complex data with multiple components, these methods can be tailored accordingly.
- ◆ **Weighted averaging:** The weighted averaging approach allows for a smooth and gradual transition between observations. The weights assigned to data points diminish exponentially, ensuring a balance between capturing recent trends and maintaining stability.
- ◆ **Forecasting:** Exponential smoothing is particularly effective for short-term forecasting. Its ability to adapt to recent changes in the data makes it well-suited for predicting values in the near future.
- ◆ **Seasonality incorporation:** The Holt-Winters method, with its inclusion of a seasonality component, is specifically designed to capture and incorporate repeating patterns in the data. This feature makes it robust for time-series with pronounced seasonal variations.

10.3.3 APPLICATION OF TIME-SERIES PRINCIPLES

The application of time-series principles is widespread across various fields, and it plays a crucial role in understanding, analysing, and predicting temporal patterns in data. Some of the key applications of time-series principles are as follows:

- ❑ **Stock market analysis:** Time-series principles are used to analyse stock prices, identify trends, and make predictions for investment decisions.
- ❑ **Economic indicators:** Time-series analysis helps economists' study economic indicators such as GDP, inflation rates and unemployment over time.
- ❑ **Sales forecasting:** Businesses use time-series forecasting to predict future sales based on historical sales data, enabling better inventory management and resource planning.

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- ❑ **Demand forecasting:** Time-series principles assist in forecasting the demand for products and services, optimising production and supply chain processes.
- ❑ **Weather forecasting:** Meteorologists use time-series analysis to study historical weather data and make predictions about future weather patterns, temperature changes and precipitation.
- ❑ **Disease surveillance:** Time-series principles are applied in tracking and predicting the spread of diseases, analysing patient health data and planning healthcare resources.
- ❑ **Patient monitoring:** Monitoring and analysing patients' vital signs over time using time-series methods helps in early detection and intervention.
- ❑ **Energy consumption forecasting:** Time-series analysis assists in forecasting energy consumption patterns, optimising energy production and managing resources efficiently.
- ❑ **Market research:** Time-series principles are used to analyse market trends, consumer behaviour, and sales data to make informed marketing strategies.
- ❑ **Advertising effectiveness:** Analysing the impact of advertising campaigns over time helps marketers optimise strategies and allocate resources effectively.
- ❑ **Traffic flow prediction:** Time-series principles are applied to predict traffic patterns, congestion and optimise transportation systems.
- ❑ **Logistics planning:** Forecasting shipment volumes and delivery times using time-series analysis aids in efficient logistics planning.
- ❑ **Population studies:** Time-series methods are used in demographic studies to analyse population growth, migration patterns, and other social phenomena over time.
- ❑ **Network traffic prediction:** Time-series principles help in forecasting network traffic, optimising bandwidth, and ensuring the efficient functioning of telecommunication networks.
- ❑ **Climate change studies:** Time-series analysis is employed in studying long-term climate trends, analysing environmental data, and making predictions about climate change impacts.
- ❑ **Inventory management:** Time-series forecasting assists retailers in managing inventory levels, anticipating demand fluctuations, and reducing stockouts or overstock situations.

**SELF ASSESSMENT QUESTIONS**

4. What does seasonality in time-series data refer to?
 - a. Random fluctuations at fixed intervals
 - b. Predictable variations at regular intervals
 - c. Unexplained variability over time
 - d. Long-term systematic movements
5. Which of the following is a characteristic of white noise in time-series data?
 - a. Uncorrelated elements
 - b. Zero mean
 - c. Consistent variance
 - d. No discernible pattern
6. Why is identifying and understanding trends essential in time-series analysis?
 - a. Trends make the data more complex
 - b. Trends facilitate anomaly detection
 - c. Trends have no impact on predictions
 - d. Trends provide insights for forecasting and decision-making
7. Simple exponential smoothing is suitable for time-series data with both trend and seasonality. (True/False)
8. Horizontal patterns in time-series suggest a lack of long-term movement, indicating a constant level in the data. (True/False)

**ACTIVITY**

Analyse historical stock prices using time-series principles, identify trends, and make predictions for investment decisions using Excel.

10.4 SUMMARY

- ❑ Forecasting is a systematic process of making predictions or estimates about future events or trends based on historical data, observations and analysis.
- ❑ Qualitative forecasting techniques involve subjective judgements, expert opinions, and qualitative data to make predictions about future events.

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- ❑ Time-series analysis is a statistical technique used to analyse and interpret temporal data, where observations are recorded over regular intervals of time.
- ❑ Time-series patterns are recurring structures or behaviours observed in sequential data points over regular intervals of time.
- ❑ Moving averages are fundamental techniques in time-series analysis that help smooth out fluctuations, reveal trends, and identify patterns in sequential data.
- ❑ Exponential smoothing is based on the idea that recent observations have a greater influence on the forecast than older observations.



KEY WORDS

- ❑ **ARIMA:** This is a forecasting model integrating AR and MA components to analyse and predict stationary time-series data.
- ❑ **Cyclic patterns:** The type of irregular patterns or long-term fluctuations in time-series data caused by economic, political, or industry-related factors, without fixed timing or predictability.
- ❑ **Exponential smoothing:** This is a forecasting method that applies exponentially decreasing weights to past data points to generate smoothed values, and reduce noise in time-series data.
- ❑ **Monte Carlo simulation:** It is a statistical method that uses random sampling to simulate and analyse complex systems or processes, assess risks and predict outcomes in uncertain scenarios.
- ❑ **Neural networks for forecasting:** This is a machine learning approach using artificial neural networks, like RNNs and LSTMs, to identify patterns and predict future values in time-series data effectively.
- ❑ **Time-series decomposition:** This is a technique that breaks down time-series data into distinct components—trend, seasonality, and residual—to understand and analyse underlying patterns for improved forecasting.
- ❑ **White noise:** White noise is random, unpredictable data fluctuations with no discernible pattern. In modelling, it represents errors or variations that cannot be explained by the model.

10.5 MULTIPLE CHOICE QUESTIONS

1. Which of the following is a key characteristic of forecasting?
 - a. Focuses only on past events
 - b. Does not evaluate data from the present

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- c. Provides useful insights to reduce uncertainty
 - d. Does not incorporate personal observations
2. The Delphi method in forecasting is best suited for situations where:
- a. Historical data is abundant and reliable
 - b. Experts' opinions need to be synthesised through structured iteration
 - c. Market research data needs statistical analysis
 - d. Only numerical data is available
3. Which of the following time-series components involves fluctuations over irregular periods without a fixed duration?
- a. Seasonality
 - b. Trend
 - c. Cyclic pattern
 - d. Level shifts
4. In an ARIMA model, what does the "I" component represent?
- a. Auto-Regressive relationship
 - b. Moving average errors
 - c. Integration through differencing
 - d. Incorporation of exponential smoothing
5. What distinguishes exponential smoothing from simple moving averages?
- a. It assigns equal weight to all observations.
 - b. It assumes no trend or seasonality.
 - c. It gives more weight to recent observations.
 - d. It is only suitable for non-seasonal data.
6. What type of time-series pattern involves abrupt changes in the baseline of the data?
- a. Horizontal pattern
 - b. Level shifts
 - c. Trend pattern
 - d. Seasonal pattern
7. Which forecasting technique is most suitable for modelling data with complex, non-linear relationships?
- a. Box-Jenkins (ARIMA)
 - b. Neural networks

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- c. Time-series decomposition
- d. Moving averages
- 8. What is a fundamental difference between seasonality and cyclic patterns in time-series analysis?
 - a. Seasonality is less predictable, while cyclic patterns are highly predictable.
 - b. Cyclic patterns are driven by calendar events, while seasonality is influenced by external factors.
 - c. Seasonality has fixed durations, while cyclic patterns have irregular durations.
 - d. Cyclic patterns are short-term, while seasonality spans several years.
- 9. When the AR parameter of an AR (1) model is equal to 1, what type of behaviour does the model exhibit?
 - a. White noise
 - b. Random walk
 - c. Stationary oscillation
 - d. Exponential decay
- 10. Which technique uses randomness to simulate and analyse complex systems or processes?
 - a. Time-series decomposition
 - b. Monte Carlo simulation
 - c. Regression analysis
 - d. Exponential smoothing

10.6 DESCRIPTIVE QUESTIONS

- 1. How does exponential smoothing work in time-series?
- 2. What are the applications of time-series principles?
- 3. Compare and contrast qualitative and quantitative forecasting methods.
- 4. Explain how the identification and understanding of seasonality in time-series analysis can significantly impact decision-making in a specific real-world scenario.
- 5. Compare and contrast the characteristics and applications of moving averages and exponential smoothing in the time-series analysis.

10.7 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. A company is expanding its operations globally. Design a comprehensive time-series forecasting strategy that incorporates both global economic trends and local market dynamics.
2. An e-commerce platform utilises forecasting models to predict product demand and manage inventory. Assess the application of exponential smoothing in this context.
3. You are analysing a time-series dataset and notice a significant trend and seasonality. Which forecasting method is specifically designed to handle both trend and seasonality components?
 - a. Moving average
 - b. Exponential smoothing
 - c. None of the above
 - d. Both a & b
4. You are working with a dataset representing monthly stock prices which exhibit a strong upward trend with no clear seasonality. Which method would you choose for forecasting, considering the characteristics of the data?
 - a. Time-series analysis
 - b. Delphi's method
 - c. Exponential smoothing method
 - d. Moving average method
5. A company experiences sudden changes in demand due to external factors such as promotions or economic events. How might the choice between double and triple exponential smoothing impact the model's ability to adapt to such abrupt changes?
 - a. Moving average is more adaptable
 - b. Time-series analysis is more adaptable
 - c. Both methods are equally adaptable
 - d. Either method is suitable for abrupt changes

10.8 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
Concept of Forecasting	1.	b. Expert judgment
	2.	b. Predicting future events
	3.	False

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Topic	Q. No.	Answer
Concept of Time-Series Analysis	4.	b. Predictable variations at regular intervals
	5.	d. No discernible pattern
	6.	d. Trends provide insights for forecasting and decision-making
	7.	False
	8.	True

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	c. Provides useful insights to reduce uncertainty
2.	b. Experts' opinions need to be synthesised through structured iterations
3.	c. Cyclic pattern
4.	c. Integration through differencing
5.	c. It gives more weight to recent observations.
6.	b. Level shifts
7.	b. Neural networks
8.	c. Seasonality has fixed durations, while cyclic patterns have irregular durations.
9.	b. Random walk
10.	b. Monte Carlo simulation

HINTS FOR DESCRIPTIVE QUESTIONS

1. Exponential smoothing is a popular and effective technique used in time-series to capture and predict trends in data. The fundamental idea behind exponential smoothing is to assign different weights to historical data, with recent data points receiving higher weights. Refer to Section 10.3
2. Traders and investors use these forecasts to make informed decisions. In economics, time-series models help in understanding and predicting economic trends, inflation rates, and GDP growth. Epidemiologists utilise time-series analysis to track and predict the spread of diseases. Refer to Section 10.2
3. Qualitative methods are subjective and rely on non-numeric information, while quantitative methods are objective and rely on numerical data. Qualitative methods are more flexible and adaptable to changing circumstances, making them suitable for scenarios with high uncertainty. Quantitative methods, on the

other hand, provide a more systematic and rigorous approach, offering precise forecasts based on historical patterns. Refer to Section 10.2

4. Seasonality, seen in industries like retail, affects sales patterns. For instance, holiday seasons drive peaks. Understanding this allows businesses to optimise inventory, plan promotions and enhance overall performance. Refer to Section 10.3
5. Moving averages smooth trends with lag, while exponential smoothing adapts quickly. One should choose the appropriate one based on data volatility and forecasting needs. Moving averages for stability, exponential smoothing, for adaptability and recent trends. Refer to Section 10.3

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	To design a comprehensive time-series forecasting strategy for a company expanding its operations globally, it is crucial to integrate both global economic trends and local market dynamics. Firstly, establish a robust data collection system that aggregates relevant global economic indicators such as GDP growth, inflation rates and exchange rates. Next, employ time-series forecasting models that can accommodate both global and local factors. Regularly update the forecasting models to account for shifts in global economic conditions and changes in local market dynamics.
2.	Exponential smoothing is well-suited for an e-commerce platform's demand forecasting. Its adaptability and simplicity enable accurate predictions, aiding in optimal inventory management and resource planning for efficient business operations.
3.	d. Both a & b
4.	c. Exponential smoothing method
5.	d. Either method is suitable for abrupt changes

10.9 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

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CASE STUDY

NOTES

DEMAND FORECASTING IN PHARMACEUTICAL SUPPLY CHAINS

SCENARIO

The Pharma_Log acts as a wholesaler and being a part of a supply chain for pharmaceutical products provides a link between manufacturing companies in EU and CIS and distributors in Central Asia (see Figure A). The affiliated distributor Pharma_Dis is located in the emerging market and resells only in-market products. While the Pharma_Dis has direct access to market and sales data records, the Pharma_Log has access to data provided by manufacturers (historical sales data, product availability, future product changes, stockouts, planned marketing activities, etc.).

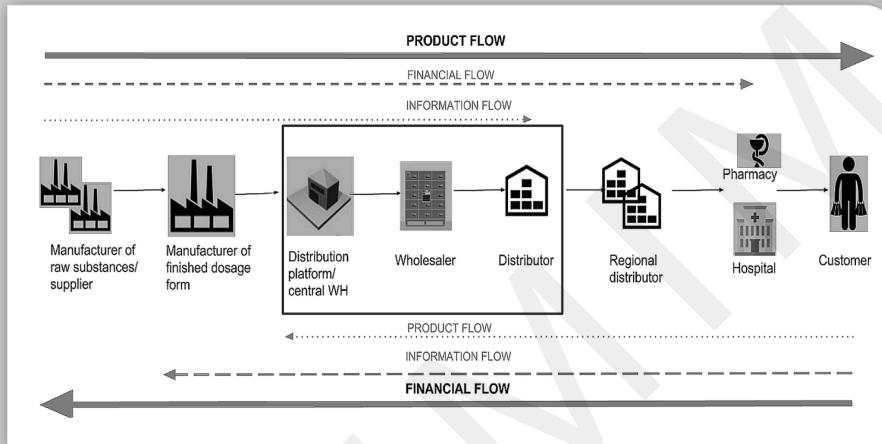


Figure A: Supply Chain of Pharmaceutical Products

(Source: <https://www.sciencedirect.com/science/article/pii/S1877050919301061>)

PROBLEM

The Pharma_Dis is a startup company, which faced increasing challenges in demand forecasting. On one hand, there is a lack of historical data sets as it is operating during the period of less than a year. On the other hand, volatility comes in with an emerging market environment. As a result, available population data for demand forecasting has relatively high fluctuations (see Figure B) which are difficult to model and hard to fit within such a short data range.



Case Objective

The objective of this case study is to discuss the state-of-art methods and challenges in demand forecasting for the pharmaceutical industry, presenting a case study and comparing SMA, multiple linear regressions and symbolic regression models.

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CASE STUDY

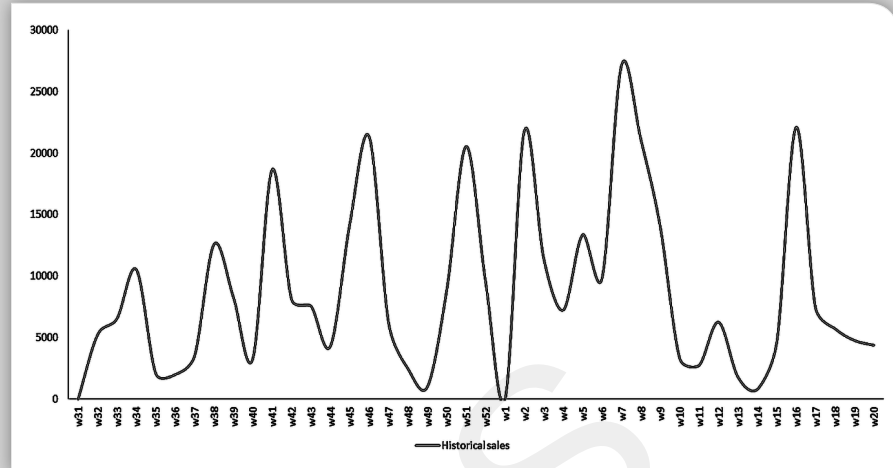


Figure B: Historical Sales for the Product Family

(Source: <https://www.sciencedirect.com/science/article/pii/S1877050919301061>)

The purchase orders were generated based on the product demand projected by regional distributors based on the sales data and stocks available at the distribution platform. The demand forecasting function is performed by the wholesaler using market sales data from the distributor. Weekly forecast is calculated based on the moving average of the last 13 weeks of sales. No additional data is considered while historical sales of the product provided by a manufacturer might be of a high value and can be potentially integrated into demand forecasting and purchase order processing.

The availability of market data and access to manufacturer's data provide sufficient ground for setting up collaborative forecasts as a joint project of two companies. In addition, collaborative demand forecasting will allow introducing mechanisms for sharing demand information and incorporating the jointly derived demand forecasts into replenishment decisions in a supply chain.

APPROACH USED

The general algorithm used for in-market product forecasting includes the following main steps:

- ❑ Trending historical data sets to examine market and product performance
- ❑ Applying the effects of ex-trend events which occur but are not reflected in the historical data

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- ❑ Converting trended data into the forecast outputs. In the simplest case, trending of historical data is performed by drawing a projected line (trend) into future

Methods for obtaining market and product baseline functions via forecast calculations are analysed in the next section. To identify ex-event and quantify their effects on the forecast appear to be the challenging tasks for forecasters. Converting trended data to forecasts outputs allows integration of supply and demand data.

The above-mentioned steps have been applied to create an integrated procedure for in-market pharmaceutical product demand forecasting and Purchase Orders (PO) generation.

Demand forecasting experiments have been performed for a specific pharmaceutical product ACT0002UZ01. The historical weekly sales product data contains 41 data points. Three experimental scenarios based on the application of different forecasting methods are investigated applying the simple moving average method, multiple linear regression, and symbolic regression with genetic programming. For regression scenarios, the following factors potentially affecting the product demand have been taken into consideration: a distributor price list; the discounted selling price of the product; a week number of sales in a month; and weekly average currency rate. For each scenario, forecasting output and forecast errors are analysed and applications feasibility and implications are provided.

CONCLUSION

The supply chain of pharmaceutical products is characterised by high complexity which is considered as one of the main barriers to improving performance of a pharmaceutical supply chain. Demand forecasts form basis for all strategic and planning in pharmaceutical logistics and supply chain management. Benchmarking studies show that forecasting is a relatively new task for the pharmaceutical industry that might explain the dominant position of simple methods performed mainly in Excel spreadsheets.



QUESTIONS

1. How does the volatility in the emerging market affect the demand forecasting process for Pharma_Dis?

(**Hint:** Volatility leads to fluctuations in market data, making it difficult to model and fit accurate demand forecasts due to the short data range.)

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2. Why might integrating manufacturer-provided data into demand forecasting improve the accuracy of the forecasts?

(Hint: Manufacturer data, including historical sales and product changes, offers additional insights that could enhance forecasting accuracy by supplementing market data from the distributor.)

(Source: <https://www.sciencedirect.com/science/article/pii/S1877050919301061>)

11

C H A P T E R

DECISION THEORY

CONTENTS

- 11.1 Introduction
- 11.2 Concept of Decision Theory
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**Case Objective**

The objective of this caselet is to understand how decision theory helped researchers in eradicating poliovirus.

INTRODUCTORY CASELET**U.S. CENTERS FOR DISEASE CONTROL AND PREVENTION**

Throughout history, polio has been one of the most terrifying diseases known to humans. Polio is caused by the poliovirus. Wild Polioviruses (WPVs) refer to polioviruses that exist in the wild, rather than viruses derived from vaccines. WPVs are typically transmitted when contaminated water or food is ingested by an uninfected person, which can be common in areas with poor hygiene or poor sanitation. In the 1950s, polio vaccines were developed separately by Jonas Salk and Albert Sabin. These vaccines led to the eradication of WPV transmission in the United States by 1979 and throughout the Western Hemisphere by 1991. However, polio remains endemic to several countries around the world, which puts all other populations at risk of infection.

The Global Polio Eradication Initiative (GPEI) seeks to eradicate polio worldwide. It is financed by a variety of public and private donors including the World Health Organization, the U.S. Centers for Disease Control and Prevention (CDC), and the Bill & Melinda Gates Foundation. The CDC collaborated with Kid Risk Inc. to use analytics tools to evaluate the policy options available for pursuing global polio eradication in terms of benefits, risks and costs. The researchers used a variety of decision analysis tools, including decision trees, to consider the various options available to potentially eradicate polio around the world. These policy decisions include options such as using Oral Poliovirus Vaccines (OPVs), Inactivated Poliovirus Vaccines (IPVs), and whether to continue routine vaccinations after WPV transmission has been eradicated. The decision trees allowed the researchers to capture much of the complexity inherent in such a large-scale decision problem and to evaluate the relative risks, benefits, and costs of different policy options. The uncertainties in costs were modelled using 47 different probability distributions for cost inputs, and cost data were collected through extensive data collection efforts led by the Global Polio Laboratory Network.

The findings from the decision analysis modelled researchers to understand that the speed with which medical professionals respond to a polio outbreak can be more effective than having extremely high rates of WPV immunity through vaccinations. The models also allowed the researchers to calculate the cost trade-offs of eradicating polio worldwide versus responding to greater polio outbreaks. The research provided extensive evidence of the value of the GPEI by showing that the net benefits of countries covered by the GPEI approach US\$ 40–50 billion. The work done by these researchers was recognised with the 2014 INFORMS Franz Edelman Award based on the enormous financial and global health impacts of these analytical models and the derived insights.

(Source: Business Analytics Descriptive, Predictive, Prescriptive. Cengage Learning)



LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Explain the concept of decision theory
- Understand decision-making without probabilities along with their approaches
- Interpret decision-making with probabilities

11.1 INTRODUCTION

In the previous chapter, we discussed forecasting and time-series analysis techniques and patterns.

The study of decision theory focuses on techniques for choosing the best path of action when several options are open and it is not possible to predict with confidence what will happen next. Decision theory is a systematic approach of making decisions under uncertainty, providing a framework for analysing and evaluating alternatives. Decision-making strategies can be categorised into decision-making without probabilities, such as the optimistic, conservative, and minimax approaches, and decision-making with probabilities, such as expected monetary value and decision trees.

This chapter explores the fundamentals of decision theory, the probabilistic and non-probabilistic approaches of decision theory, and the associated methods in each approach.

11.2 CONCEPT OF DECISION THEORY

Decision theory is a framework for making logical and informed choices under conditions of uncertainty or risk. It combines principles from mathematics, statistics, economics, and psychology to evaluate and select the best course of action among several alternatives. Decision theory provides a structured approach to making decisions under uncertain, complex conditions, often involving multiple goals. It bridges the gap between theoretical principles and practical decision-making assisting businesses to make the best possible decisions with due consideration for limitations and trade-offs. Decision theory is often applied in various fields to analyse alternatives and their potential consequences in a structured manner.

Decision theory analyses choices under various decision contexts such as certainty, uncertainty, or risk, and evaluates them using factors like payoffs and utility, and structures decisions through components like agents, models and techniques.

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Decision scenarios describe the situational frameworks or conditions under which decisions are made. They include:

- ❑ **Decisions under certainty:** When the outcomes of all choices are completely known

Example: Choosing between fixed interest rate plans

- ❑ **Decisions under uncertainty:** When the outcomes of choices are unknown, and probabilities cannot be calculated

Example: Launching a new product in a market that has not been tested

- ❑ **Decisions under risk:** When the probabilities of different outcomes are known or can be estimated

Example: Investing in stocks with historical data on returns

The quantitative or qualitative measures that are used to compare and evaluate choices are called decision factors. They include:

- ❑ **Payoffs:** These refer to the benefits or losses that result from making a specific choice. For example, the profit earned from an investment can be considered a payoff.
- ❑ **Utility:** A measure of the level of satisfaction or personal preference associated with a decision, particularly for outcomes that are not purely monetary. For example, choosing a career based on personal happiness rather than salary reflects the concept of utility.
- ❑ **Risk:** This represents the likelihood of loss or the variability in outcomes associated with a decision. An example of risk is the possibility of losing money in a business venture.
- ❑ **Expected value:** It is the weighted average of all possible outcomes, calculated by considering their probabilities. For example, it can be used to predict future returns on investments based on different scenarios.

11.2.1 COMPONENTS AND APPLICATION OF DECISION THEORY

The components of decision theory are the essential building blocks that guide the decision-making process. They provide a structured framework for analysing problems, evaluating options, and selecting the best course of action.

The components of decision theory is shown in Figure 11.1:

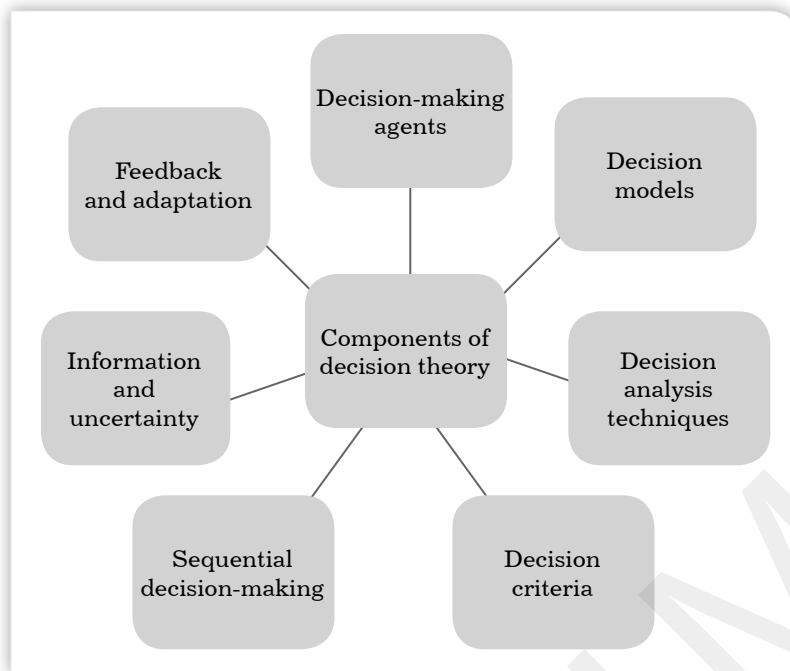


Figure 11.1: Components and Application of Decision Theory

The components shown in Figure 11.1 are explained in detail:

- ❑ **Decision-making agents:** These are the individuals, groups, organisations, or systems responsible for making decisions. These agents can operate independently or collaboratively, depending on the context of the decision. Their goals, preferences, and constraints significantly influence the choices they make. For example, a company's management team acts as a decision-making agents when deciding whether to launch a new product.
- ❑ **Decision models:** These are frameworks that use mathematical, statistical, or computational tools to represent and analyse decision problems. These models simplify complex problems, making them easier to evaluate systematically. They can include deterministic models, which apply to decisions made under certainty, and probabilistic models, which are used for decisions under risk or uncertainty. For instance, a cost-benefit analysis model can be used to assess the potential value of investment opportunities.
- ❑ **Decision analysis techniques:** These are the methods and tools used to solve decision problems by comparing alternatives and identifying the best option. Common techniques include decision trees, optimisation methods, game theory and utility theory. These techniques help decision-makers visualise possible outcomes, evaluate trade-offs and address uncertainties. For example,

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a decision tree can illustrate the potential outcomes of various business strategies.

- ❑ **Decision criteria:** These refer to the rules or guidelines that decision-makers use to evaluate and choose between alternatives. These criteria can be quantitative, such as maximising profit, or qualitative, such as improving customer satisfaction. They provide a clear basis for making consistent and rational decisions. For instance, a company may use sustainability as a key decision criterion when selecting suppliers, prioritising long-term environmental goals over short-term cost savings.
- ❑ **Sequential decision-making:** It involves making decisions in multiple steps over time, where each decision influences future choices and outcomes. This approach is particularly useful in dynamic or long-term scenarios. Sequential decision-making takes into account both immediate and long-term consequences of actions. For example, a pharmaceutical company may decide to invest in a new drug by following a step-by-step process, from initial research to market launch, adjusting its strategy based on the results at each stage.
- ❑ **Information and uncertainty:** Information is a critical factor in decision-making because it affects the accuracy of predictions and the evaluation of choices. Decision-makers use data and evidence to reduce uncertainty and improve the quality of their decisions. Information can be perfect, meaning it is complete and accurate, or imperfect, meaning it is partial or unreliable. For example, market research data can help a company predict customer demand for a new product, reducing uncertainty and guiding its strategy.
- ❑ **Feedback and adaptation:** Feedback mechanisms enable decision-makers to assess the outcomes of their actions and adjust their strategies accordingly. This iterative process is particularly important in dynamic environments where conditions change over time. For instance, a business might revise its marketing strategy based on customer feedback and sales performance, ensuring continuous improvement and alignment with market demands.

Some of the applications of decision theory in real-life scenarios are as follows:

- ❑ **Marketing and advertising:** Decision theory can help marketers use their resources wisely by looking at things like who they want to reach, where to advertise, and how much money they have. Marketers can use decision trees to see the possible results of different campaign strategies and pick the one that is likely to give the best outcome.

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- ❑ **Public policy and governance:** Decision theory aids policy makers in evaluating different options for public infrastructure projects. By considering factors like economic impact, environmental effects, and public opinion, decision-makers can use decision analysis to select the most beneficial and politically feasible course of action.
- ❑ **Supply chain management:** Companies use decision theory to optimise supplier selection based on factors like cost, reliability and delivery times. Decision analysis can help in modelling the uncertainties associated with each supplier's performance and assessing the overall risk in the supply chain.
- ❑ **Human resources:** Decision theory can be useful in hiring decisions as it assesses individuals considering multiple criteria, including skills, experience, and how well the individual 'fits' within the company culture. Hiring managers could use decision trees or utility theory to determine and calculate the value each candidate would bring to the organisation.
- ❑ **Healthcare management:** Use of decision theory applies resources such as the hospital bed, medical equipment and staff. This will show the total number of patients at a given time, the rise or fall in demand, and budgetary limits, among others, which best use resources for better management of patient care.
- ❑ **Environmental policy:** An analysis or selection of climate change mitigation policies is carried on by employing the theory of decision. The policymakers analyse the net fiscal, practicality, and long-term effectiveness implications of varying policies for reducing carbon emissions, using renewable sources of energy, and so forth.
- ❑ **Technology investment:** Within a technology firm, decision theory aids in measuring the expected payback from research and development that is currently underway. Based on the criteria including market demand, competitiveness, and technological feasibility, decisions connecting those initiatives that fall within a corporate strategy can then be made.
- ❑ **Educational planning:** Universities can use decision-making methods to improve how they admit students and give scholarships. By looking at grades, activities outside of class, and chances for success, schools can make smart choices that consider fairness and variety.

11.2.2 BRANCHES OF DECISION THEORY

Decision theory is an interesting field that explores how decisions should be made. It can be broadly divided into two main branches,

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(a) Normative decision theory and (b) Descriptive decision theory as shown in Figure 11.2:

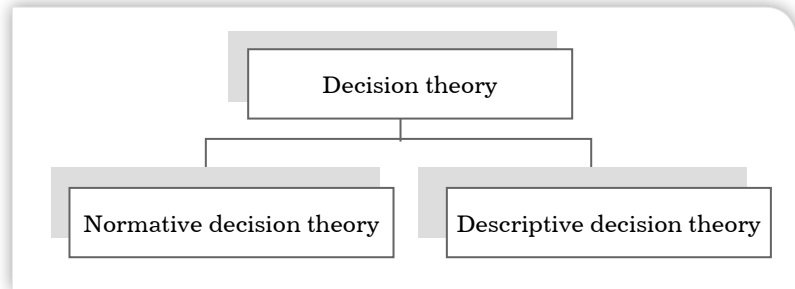


Figure 11.2: Branches of Decision Theory

The branches of decision theory according to Figure 11.2, are described as follows:

- ❑ **Normative decision theory:** This branch identifies the optimal decisions based on certain criteria and models. The term “normative” refers to establishing, relating to, or deriving from a standard or norm, often involving setting guidelines or standards for what is considered acceptable or ideal. Accordingly, normative decision theory prescribes how decisions should be made to achieve the best outcomes, assuming rational behaviour and perfect information.

Normative decision theory sets standards for rational decision-making under ideal conditions, encompassing expected utility theory and Bayesian decision theory. Decision analysis applies these principles to develop tools and methodologies, including decision support systems, aiding individuals in making more informed decisions.

Example: Consider a financial advisor helping a client in deciding investment options. There are two investments: Investment A and Investment B. The financial advisor needs to recommend an investment option that maximises return while considering the client’s risk tolerance. This involves looking for the decision that offers the highest expected return percentage with a risk level that aligns with the client’s preferences. The normative decision theory involves creating a decision table based on expected returns and risks. Table 11.1 shows a decision table for investment options based on expected returns and risks:

Table 11.1: Decision Table for Investment Options

Decision Factors	Investment A	Investment B
Expected return	10%	8%
Risk	5%	3%
Decision	Maximise return	Maximise return

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- ❑ **Descriptive decision theory:** Unlike normative theory, descriptive decision theory examines how decisions are actually made in real life. It takes into account human behavior, biases, and irrationalities, providing a more realistic view of decision-making processes.

Descriptive decision theory, is also known as optimal decision theory, and examines decisions made by individuals with irrational behaviour, utilising diverse frameworks. As this decision theory is focused on real-world scenarios, it acknowledges deviations from normative principles, notably through behavioural economics.

In operational research, companies use decision theory for its consideration of multiple outcomes and rational reasoning, emphasising on understanding how individuals think logically rather than idealistic thinking. It also analyses the reasons behind decisions, considering crucial elements like location and time in the decision-making process.

Example: Consider a scenario where an individual, Abhijeet, is deciding on a vacation destination. He has three options: beach resort, mountain retreat, and urban adventure. Table 11.2 shows the decision table for choosing a vacation destination:

Table 11.2: Decision Table for Choosing a Vacation			
Criteria	Beach Resort	Mountain Retreat	Urban Adventure
Scenery	High	Medium	Low
Activities	High	Medium	High
Weather preference	Warm	Cool	Any
Budget	Medium	High	Low
Accessibility	High	Medium	High

In table 11.2, preferences for each criterion are represented by the terms 'High', 'Medium' and 'Low', Abhijeet's decision-making process involves considering these criteria subjectively.

The following are the descriptive observations for Abhijeet's decision-making process:

- ❑ **Intuitive evaluation:** Abhijeet might intuitively assess each destination based on his personal preferences without assigning specific weights or scores to criteria.
- ❑ **Trade-offs:** He may make trade-offs, recognising that no destination is perfect in all aspects.
- ❑ **Influence of past experiences:** Abhijeet's past vacation experiences, positive or negative, might influence his current decision-making.

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- ❑ **Influence of external factors:** Abhijeet might be influenced by factors like recommendations from friends, recent news, or travel trends.
- ❑ **Dynamic decision process:** Abhijeet's preferences may change dynamically during the decision process as he considers and reconsiders different aspects.

As we have seen, each branch offers unique insights and applications, making decision theory a versatile and valuable field in various disciplines, including economics, psychology and management.

In decision theory, we use various tools and methods to make informed choices under uncertainty. Payoff tables help visualise the potential outcomes of different decisions. Expected value analysis calculates the average expected outcome, while techniques such as minimax evaluate decisions based on different risk preferences. Sensitivity and scenario analyses assess how changes in variables or predefined scenarios impact outcomes, and decision trees provide a visual representation of decision paths and outcomes.

11.2.3 PAYOFF TABLES IN DECISION-MAKING

A payoff table is a decision-making tool used in decision theory to classify and display the payoffs connected to several decision alternatives and possible conditions. It is mainly useful when a decision-maker must choose between different routes of action, and the outcomes rely on uncertain events or conditions. Payoff tables are commonly used in uncertain decision-making because they provide a straightforward method to assess potential gains and losses. They do this by displaying the payoff (profit or loss) for different outcomes based on two variables: different decision choices and different possible real-world scenarios.

The goal of decision analysis is to select between one or the other sequence of action (the decision), given historical knowledge of its outcome (called a payout) for many future scenarios (called states of nature or occurrences). The process of selecting different sequences of action is under the decision-maker's control, but not the short-term conditions of nature.

The initial step in the decision theory process is problem formulation. We start by forming a verbal statement of the problem. We then classify the decision alternatives; the uncertain future events, referred to as chance events; and the outcomes related to each combination of decision alternative and coincidental event outcomes.

Consider a construction project of the Public Work Department (PWD). PWD purchased land that will be the site of a new luxury condominium complex. The location provides a spectacular view

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of the lush surroundings, where the nearest railway station, metro stations, and schools are within a 10-minute walking distance. PWD plans to price the individual condominium units between ₹3 crore and ₹14 crore.

PWD commissioned preliminary architectural drawings for three different projects: one with 30 condominiums, one with 60 condominiums and one with 90 condominiums. The financial success of the project depends on the size of the condominium complex and the chance event concerning the demand for the condominiums. The statement of the PWD decision problem is to select the size of the new luxury condominium project that will lead to the largest profit given the uncertainty concerning the demand for the condominiums.

Given the statement of the problem, it is clear that the decision is to select the best size for the condominium complex. PWD has the following three decision alternatives:

- d_1 = a small complex with 30 condominiums
- d_2 = a medium complex with 60 condominiums
- d_3 = a large complex with 90 condominiums

A factor in selecting the best decision alternative is the uncertainty associated with the chance event concerning the demand for condominiums. When asked about the possible demand for the condominiums, PWD's president acknowledged a wide range of possibilities but decided that it would be adequate to consider two possible chance event outcomes: a strong demand and a weak demand.

In decision analysis, the possible outcomes for a chance event are referred to as the states of nature. The states of nature are defined so that they are mutually exclusive (no more than one can occur) and collectively exhaustive (at least one must occur); thus, one and the only one of the possible states of nature will occur. For the PWD problem, the chance event concerning the demand for condominiums has two states of nature:

- s_1 = strong demand for the condominiums
- s_2 = weak demand for the condominiums

Given the three decision alternatives and the two states of nature, a question arises about which complex size should PWD choose for the project. To answer this question, PWD needs to know the outcome associated with each possible combination of decision alternatives and state of nature. In decision analysis, we refer to the outcome resulting from a specific combination of a decision alternative and a state of nature as a payoff. As PWD wants to select the complex size that provides the largest profit, profit is used as the outcome.

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The payoff table with profits (in crores) is shown in Table 11.3:

Table 11.3: Payoff Table for the PWD Condominium Project (Crores)		
Decision Alternative	Strong Demand (s_1)	Weak Demand (s_2)
Small complex (d_1)	8	7
Medium complex (d_2)	14	5
Large complex (d_3)	20	-9

If a medium complex is built and demand turns out to be strong, a profit of ₹14 crore will be realised. We will use the notation V_{ij} to denote the payoff associated with decision alternative i and state of nature j . Using Table 11.3, $V_{31} = 20$ indicates that a payoff of ₹20 crores occurs if the decision is to build a large complex (d_3) and the strong demand state of nature (s_1) occurs. Similarly, $V_{32} = -9$ indicates a loss of ₹9 crores if the decision is to build a large complex (d_3) and the weak demand state of nature (s_2) occurs.

Although payoff tables help in evaluating trade-offs, but they are only a preliminary tool in the decision-making process. To finalise decisions, we will explore how to utilise payoff tables for decision-making without probability in the next section.



SELF ASSESSMENT QUESTIONS

- Which of the following is not a key concept of decision theory?
 - Decision-making agents
 - Risk, uncertainty and certainty
 - Budget allocation models
 - Decision analysis techniques
- Decision trees and payoff tables are examples of decision analysis techniques used in decision theory. (True/False)
- In decision theory, the branch that focuses on logically ideal decisions based on defined values is called:
 - Predictive decision theory
 - Optimal decision theory
 - Normative decision theory
 - Descriptive decision theory
- Sequential decision-making in decision theory focuses only on short-term outcomes without considering long-term implications. (True/False)

**ACTIVITY**

You are part of a consulting team advising a real estate company planning a new housing project. Using a payoff table, analyse three potential project sizes (small, medium, large) and two demand scenarios (strong, weak). Determine the optimal decision to maximise profit while considering risks. Create and analyse a payoff table based on the given scenario. Present your decision-making process and justify your recommendation using decision theory concepts.

11.3 DECISION-MAKING WITHOUT PROBABILITIES

Sometimes, the approaches to decision analysis do not need the knowledge of the probabilities of the states of nature. These methods are suitable in situations where the most promising outcome and the least promising outcome analysis are sufficient and in which the decision-maker has little trust in his/her ability to estimate the probabilities. Because different approaches sometimes lead to different decision recommendations, the decision-maker needs to learn and apply the approaches available and then select the specific approach that is the most suitable according to the judgement of the decision maker.

11.3.1 OPTIMISTIC APPROACH

The optimistic approach estimates each decision alternative in terms of the best payoff that can occur. The decision alternative that is recommended is the one that provides the best possible payoff. For a problem in which maximum profit is desired, as in the PWD problem, the optimistic approach would lead the decision maker to choose the alternative corresponding to the largest profit. For problems involving minimisation, this approach leads to choosing the alternative with the smallest payoff.

To illustrate the optimistic approach, we use it to develop a recommendation for the PWD problem. First, we need to establish the maximum payoff for each decision alternative. Then, we need to select the decision alternative that provides the overall maximum payoff. These steps systematically identify the decision alternative that provides the largest possible profit as shown in Table 11.4:

Table 11.4: Maximum Payoff for Each PWD Decision Alternative

Decision Alternative	Maximum Payoff
Small complex, d_1	8
Medium complex, d_2	14
Large complex, d_3	20 (maximum of the maximum payoff values)

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Because 20 which corresponds to d_3 is the largest payoff, the decision to construct the large condominium complex is the recommended decision alternative using the optimistic approach.

11.3.2 MINIMAX APPROACH

The minimax approach is also known as the minimax regret approach in decision analysis. Regret is the difference between the payoff linked with a unique decision substitute and the payoff associated with the decision that yields the most desirable payoff for a given state of nature. In other words, regret represents how much potential payoff one would miss by selecting a particular decision alternative, given that a specific state of nature will occur. This is why regret is often referred to as opportunity loss.

As its name implies, under the minimax approach in decision analysis, one would choose the decision alternative that minimises the maximum state of regret that would happen over all possible states of nature. This approach is neither purely optimistic nor purely conservative.

Suppose that PWD constructs a small condominium complex (d_1) and demand turns out to be strong (s_1). Table 11.4 shows that the resulting profit for PWD would be ₹8 crores. However, given that the strong demand state of nature (s_1) has occurred, we realise that the decision to construct a large condominium complex (d_3), yields a profit of ₹20 crore which would be the best outcome. The difference between the payoff for the best decision alternative (₹20 crore) and the payoff for the decision to construct a small condominium complex (₹8 crore) is the regret or opportunity loss related to the decision alternative d_1 when state of nature s_1 occurs. In this scenario, the opportunity loss or regret is ₹20 crore - ₹8 crore = ₹12 crore. Similarly, if PWD makes the decision to construct a medium condominium complex (d_2) and the strong demand state of nature (s_1) occurs, the opportunity loss, or regret, associated with d_2 would be ₹20 crore - ₹14 crore = ₹6 crore. Upon completion of the analysis, if PWD chooses to construct a large complex (d_3) and demand is strong, they would have no regret.

Generally, the following expression represents the opportunity loss or regret:

$$\text{Regret (Opportunity Loss); } R_{ij} = |V_j^* - V_{ij}|$$

Where:

R_{ij} is the regret associated with the decision alternative d_i and state of nature s_j

V_j^* is the payoff value corresponding to the best decision for the state of nature s_j

V_{ij} the payoff corresponding to the decision alternative d_i and state of nature s_j

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For minimisation problems, the best payoff, V_j^* , is the smallest entry in column j . Because this value is less than or equal to V_{ij} , the absolute value of the difference between V_j^* and V_{ij} ensures that the regret is always the magnitude of the difference.

Using the above equation and the payoffs in Table 11.1, we can compute the regret associated with each combination of decision alternative d_i and state of nature s_j . Because the PWD problem is a maximisation type of problem, V_j^* will be the largest entry in column j of the payoff table. So, to calculate the regret, we subtract each entry in a column from the largest entry in the column as shown in Table 11.5:

Table 11.5: Regret Table for the PWD Condominium Project (₹ Crores)		
	State of Nature	
Decision Alternative	Strong Demand (s_1)	Weak Demand (s_2)
Small complex (d_1)	12	0
Medium complex (d_2)	6	2
Large complex (d_3)	0	16

The next step in applying the minimax regret approach is to list the maximum regret for each decision alternative as shown in Table 11.6.

Table 11.6: Maximum Regret for Each PDC Decision Alternative	
Decision Alternative	Maximum Regret (₹ Crores)
Small complex (d_1)	12
Medium complex (d_2)	6 (Minimum of the maximum regret)
Large complex (d_3)	16

Selecting the decision alternative with the minimum of the maximum regret values results in the minimax regret decision. For the PWD problem as discussed above, the alternative to construct the medium condominium complex, with a corresponding maximum regret of ₹6 crores, is the recommended minimax regret decision.

The main criticism of the approaches discussed in this section is that they do not consider any information about the probabilities of the various states of nature limiting their effectiveness in decision-making under uncertainty. In the next section, we discuss the approach that utilises probability for selecting decision alternatives offering a more informed basis for selecting decision alternatives.

11.3.3 CONSERVATIVE APPROACH

The conservative approach determines each decision alternative based on the worst payoff that can happen. The decision alternative advised is the one that provides the best of the worst possible payoffs.

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For a problem in which the output measure is profit, like the PWD problem, the conservative approach would lead the decision maker to choose the alternative that maximises the minimum possible profit that could be obtained. For problems involving minimisation (e.g., when the output measure is cost instead of profit), this approach identifies the alternative that will minimise the maximum payoff.

To illustrate the conservative approach, we use it to develop a recommendation for the PWD problem. First, we identify the minimum payoff for each of the decision alternatives; then we select the decision alternative that maximises the minimum payoff as shown in Table 11.7:

Table 11.7: Minimum Payoff for Each PWD Decision Alternative	
Decision Alternative	Minimum Payoff (Crores)
Small complex (d_1)	7 (Maximum of the minimum payoff values)
Medium complex (d_2)	5
Large complex (d_3)	-9

As analysed in Table 11.7, 7 which corresponds to d_1 yields the maximum of the minimum payoffs, the decision alternative of a small condominium complex is recommended. This decision approach is considered conservative because it identifies the worst possible payoffs and then recommends the decision alternative that avoids the possibility of extremely 'bad' payoffs. In the conservative approach, PWD is guaranteed a profit of at least ₹7 crores. Although PWD may make more, it cannot accept a loss of ₹7 crores.

After exploring about decision-making without probabilities along with its approaches, we will learn about decision-making with probabilities in the next section.



SELF ASSESSMENT QUESTIONS

5. Which decision-making approach focuses on selecting the decision alternative with the maximum possible payoff?
 - a. Conservative approach b. Optimistic approach
 - c. Minimax regret approach d. Probabilistic approach
6. In the minimax regret approach, what does 'regret' represent?
 - a. The difference between the worst and best payoffs across all decision alternatives.
 - b. The difference between the payoff of a chosen alternative and the payoff of the best alternative for a specific state of nature.
 - c. The minimum payoff achievable by a decision alternative.
 - d. The probability of selecting an incorrect decision.

7. The optimistic approach guarantees the decision-maker will avoid extremely 'bad' payoffs. (True/False)
8. In the conservative approach, the decision alternative with the maximum of the minimum payoffs is recommended. (True/False)



ACTIVITY

Continuing from the previous activity, the real estate firm decided between small, medium, and large housing projects under strong and weak demand scenarios. Create a payoff table showing profits (₹ crores) for each combination. Apply the optimistic, conservative, and minimax regret approaches to recommend the best project size. Justify your choice.

11.4 DECISION-MAKING WITH PROBABILITIES

Decision-making with probabilities involves assessing and integrating some probability estimates into the decision-making process. In scenarios where uncertainty happens and different outcomes are achievable, decision-makers use probability to quantify the possibility of each outcome. This allows for a more systematic and quantitative analysis, allowing decision-makers to make informed choices based on the principles of probability theory.

11.4.1 EXPECTED MONETARY VALUE

In many decision-making situations, probability assessments for the states of nature can be obtained. When such probabilities are available, we can use the expected value approach to identify the best decision alternative. Expected Monetary Value (EMV) is a statistical technique used in risk management to quantify the potential financial impact of risks. It is calculated by multiplying the probability of a risk occurring by its monetary impact. EMV helps in decision-making by providing a clear picture of potential gains or losses from different scenarios. This method is particularly useful in project management for prioritising risks and planning contingencies. Let's first outline the expected value of a decision alternative and then apply it to the PWD problem.

Let, N is the number of states of nature and $P(s_j)$ is the probability of the state of nature. Because one and only one of the N states of nature can occur, the probabilities must satisfy two conditions, which are:

- ♦ $P(s_j) \geq 0$, for all states of nature
- ♦ $\sum_{j=1}^N P(s_j) = P(s_1) + P(s_2) + \dots + P(s_N) = 1$

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NOTE

In decision theory, states of nature refer to the different possible scenarios or outcomes that could occur, each of which has a probability associated with it. These scenarios are uncertain, meaning the decision-maker does not know which one will occur, but they can assign probabilities based on available information or assumptions.

s_j in the above condition represents a specific state of nature. For example, in the context of the PWD problem, s_1 could represent “strong demand” and s_2 could represent “weak demand.”

The Expected Value (EV) of decision alternative d_i is defined as follows:

$$EV(d_i) = \sum_{j=1}^N P(s_j) \times \text{Payoff}(d_i, s_j)$$

Where:

$P(s_j)$ is the probability of the state of nature s_j

$\text{Payoff}(d_i, s_j)$ is the payoff associated with decision alternative d_i and state of nature s_j

Now consider the PWD problem to see how the expected value approach can be applied.

PWD is optimistic about the potential for the luxury high-rise condominium complex. Suppose that this optimism leads to an initial subjective probability assessment of 0.8 that demand will be strong (s_1) and a corresponding probability of 0.2 that demand will be weak (s_2). Thus, $P(s_1) = 0.8$ and $P(s_2) = 0.2$. Using the payoff values in Table 11.1 and the above equation, we can compute the expected value for each of the three decision alternatives as follows:

$$EV(d_1) = 0.8(8) + 0.2(7) = 7.8$$

$$EV(d_2) = 0.8(14) + 0.2(5) = 12.2$$

$$EV(d_3) = 0.8(20) + 0.2(9) = 14.2$$

Thus, using the expected value approach, we find that the large condominium complex, with an expected value of ₹14.2 crores, is the recommended decision. The expected monetary value is visually demonstrated with the help of decision trees which is discussed in the next section.

11.4.2 DECISION TREE

A decision tree is a graphical demonstration of a decision-making process, outcomes and their probabilities. It is a powerful tool in decision analysis and can be used to visually represent complex

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decision problems, considering various alternatives, states of nature, and their associated probabilities and payoffs. Decision trees are particularly useful for making decisions under uncertainty and for systematically structuring the decision-making process.

The calculations performed in the previous section which was used to identify the decision alternative with the best-expected value can be conveniently carried out on a decision tree. Figure 11.3 shows the decision tree for the PWD problem with state-of-nature branch probabilities:

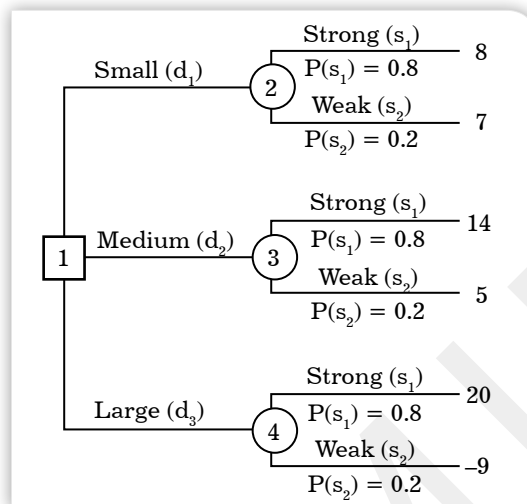


Figure 11.3: PWD Decision Tree with State-of-Nature Branch Probabilities

Working backward through the decision tree, we first compute the expected value at each chance node. In other words, at each chance node, we weight each possible payoff by its probability of occurrence. By doing so, we obtain the expected values for nodes 2, 3 and 4, as shown in Figure 11.4:

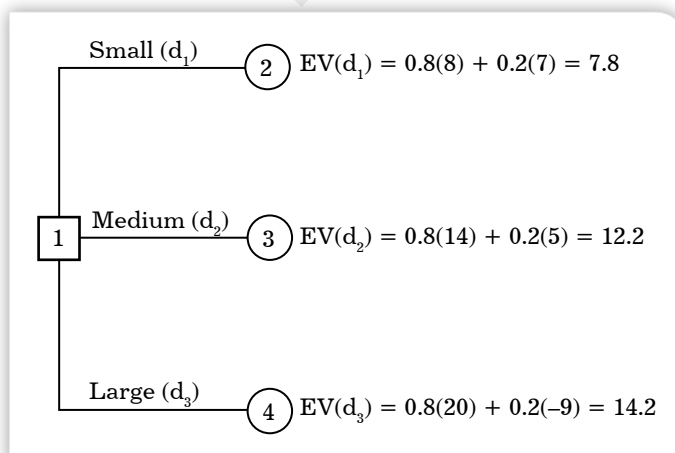


Figure 11.4: Applying the Expected Value Approach Using a Decision Tree for the PWD Condominium Project

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A decision tree begins with a root node, which has no incoming branches, as illustrated in Figure 11.4. The internal nodes, sometimes referred to as decision nodes, receive input from the outward branches of the root node. Both types of nodes perform assessments based on available attributes to create homogeneous subsets, which are represented by leaf nodes or terminal nodes. Every conceivable result in the dataset is represented by the leaf nodes.

**SELF ASSESSMENT QUESTIONS**

9. Which of the following conditions must be satisfied for the probabilities of the states of nature in decision-making probabilities?
 - a. The sum of probabilities must equal 1 and each probability must be between 0 and 1.
 - b. The probabilities must be independent of one another.
 - c. The sum of payoffs must equal 0.
 - d. The probabilities must be greater than 1 for at least one state of nature.
10. Decision trees are not useful for decision-making under uncertainty because they do not account for probabilities. (True/False)

**ACTIVITY**

You are assisting a construction firm in evaluating three project options (small, medium, and large). Demand probabilities are strong demand (0.6) and weak demand (0.4). Using provided payoffs (₹10 crores for small, ₹14 crores for medium, ₹18 crores for large in strong demand; and ₹5 crores, ₹6 crores and ₹8 crores in weak demand), calculate the expected value for each project. Construct a decision tree to visualise the options and recommend the best alternative based on the expected value approach.

11.5 SUMMARY

- ❑ Decision theory provides the basis to make decisions when things are uncertain, complex and consist of multiple goals.
- ❑ Decision theory analyses choices under various decision contexts such as certainty, uncertainty, or risk, and evaluates them using factors like payoffs and utility, and structures decisions through components like agents, models and techniques.
- ❑ Decision-making agents are the individuals, groups, organisations, or systems responsible for making decisions.

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- ❑ The decision theory is divided mainly into two parts such as normative decision theory and descriptive decision theory.
- ❑ Normative decision theory explores logically ideal decisions based on defined values, determining the most favourable course of action within set limits.
- ❑ The term “normative” refers to establishing, relating to, or deriving from a standard or norm, often involving setting guidelines or standards for what is considered acceptable or ideal.
- ❑ Descriptive decision theory, also known as optimal decision theory, examines decisions made by individuals with irrational behaviour, utilising diverse frameworks.
- ❑ A payoff table is a decision-making tool used in decision theory to classify and display the payoffs connected to several decision alternatives and possible conditions.
- ❑ The optimistic approach estimates each decision alternative in terms of the best payoff that can occur.
- ❑ The conservative approach determines each decision alternative based on the worst payoff that can happen.
- ❑ The minimax approach, also known as the minimax regret approach is the difference between the payoff linked with a unique decision substitute and the payoff associated with the decision that yields the most desirable payoff for a given state of nature.
- ❑ Decision-making with probabilities involves assessing and integrating some probability estimates into the decision-making process.
- ❑ In expected monetary value, the expected value of a decision alternative is the sum of weighted payoffs for each decision alternative.
- ❑ A decision tree is a graphical demonstration of a decision-making process, outcomes and their probabilities.



KEY WORDS

- ❑ **Chance node:** It is a part of a decision tree, where probabilities for different outcomes are applied to calculate expected values.
- ❑ **EMV:** This method is used to determine the best decision alternative by considering the probabilities of various outcomes.
- ❑ **Payoff table:** This is a table used to show the outcomes of different decision alternatives and states of nature.
- ❑ **Regret (Opportunity loss):** This is the difference between the payoff associated with a given decision and the best possible payoff for a specific state of nature. It is mainly used in the minimax approach for decision.

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- ❑ **Trade-offs:** Trade-offs refer to the process of balancing different factors or alternatives, where choosing one option typically means sacrificing another. In decision-making, trade-offs involve comparing the benefits and costs of different choices to make the best possible decision.

11.6 MULTIPLE CHOICE QUESTIONS

1. What is the primary goal of decision theory in business analytics?
 - a. To identify the most profitable investment opportunities
 - b. To transform data-informed insights into strategic actions
 - c. To predict future market trends
 - d. To minimise all business risks
2. Which of the following is not a key component of decision theory?
 - a. Decision-making agents
 - b. Optimisation and rational decision-making
 - c. Random chance events
 - d. Decision analysis techniques
3. Which of the following describes normative decision theory?
 - a. It examines decisions made by individuals with irrational behaviour.
 - b. It sets standards for rational decision-making under ideal conditions.
 - c. It models the actual decision-making process used by individuals.
 - d. It focuses on predicting the future outcomes of various decisions.
4. In the context of decision theory, what is a 'payoff table' used for?
 - a. To display the financial impact of each decision alternative
 - b. To show the cost of decision-making mistakes
 - c. To predict the future demand for a product
 - d. To track the probabilities of decision outcomes
5. Which of the following is a characteristic of descriptive decision theory (also known as optimal decision theory)?
 - a. It focuses on the ideal way a decision should be made under rational conditions.

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- b. It examines actual decisions made by individuals, acknowledging irrational behaviour.
 - c. It calculates the expected utility of each decision alternative.
 - d. It ignores human behaviour and focuses solely on the outcome of decisions.
6. Which approach in decision-making focuses on selecting the alternative with the best possible payoff?
- a. Conservative approach
 - b. Optimistic approach
 - c. Minimax approach
 - d. Expected value approach
7. In the conservative approach, which of the following is prioritised in decision-making?
- a. Maximum possible payoff
 - b. Worst possible payoff
 - c. Expected value of payoffs
 - d. Difference between the maximum and minimum payoffs
8. What does the minimax regret approach aim to minimise?
- a. Maximum payoff
 - b. Regret associated with a decision
 - c. Probability of a specific outcome
 - d. Difference between payoffs for the same state of nature
9. In decision-making with probabilities, the expected monetary value is calculated by:
- a. Summing the maximum payoffs for each alternative
 - b. Multiplying payoffs by their respective probabilities
 - c. Selecting the alternative with the minimum regret
 - d. Using a decision tree to visualise outcomes
10. What is the primary purpose of a decision tree in decision-making?
- a. To calculate the exact payoff for each decision alternative
 - b. To visually represent different decision alternatives, states of nature and associated probabilities
 - c. To find the decision alternative with the least regret
 - d. To identify the maximum payoff for each state of nature

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11.7 DESCRIPTIVE QUESTIONS

1. Explain the concept of decision theory and its applications in business analytics.
2. Describe payoff tables in decision-making.
3. Explain the optimistic approach.
4. Elaborate on the decision trees and how they help in expected monetary value.
5. Describe the minimax approach.

11.8 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. In a scenario where a company is considering three alternatives for expansion—building a small factory, medium factory, or large factory—based on past data, how would the optimistic approach influence the decision?
2. A business is evaluating two investment projects, Project A and Project B, based on their best and worst payoffs. If the conservative approach is applied, what would the final decision be?
3. A manager is choosing between three product designs with varying payoffs based on market conditions (strong demand, weak demand). How would the minimax regret approach guide the decision?
4. A company is selecting between three potential projects: A, B and C. Each project has different potential payoffs based on whether demand is strong or weak. If the company uses the optimistic approach, which project should it select?
 - a. Project A
 - b. Project B
 - c. Project C
 - d. It cannot be decided without more information
5. In a decision-making scenario involving multiple alternatives and possible outcomes, if the decision maker is highly risk-averse, which approach should they use to make their choice?
 - a. Optimistic approach
 - b. Conservative approach
 - c. Minimax regret approach
 - d. Expected monetary value approach

11.9 ANSWERS AND HINTS**ANSWERS FOR SELF ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Concept of Decision Theory	1.	c. Budget allocation models
	2.	True
	3.	c. Normative decision theory
	4.	False
Decision-Making without Probabilities	5.	b. Optimistic Approach
	6.	b. The difference between the payoff of a chosen alternative and the payoff of the best alternative for a specific state of nature.
	7.	False
	8.	True
Decision-Making with Probabilities	9.	a. The sum of probabilities must equal 1 and each probability must be between 0 and 1.
	10.	False

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. To transform data-informed insights into strategic actions
2.	c. Random chance events
3.	b. It sets standards for rational decision-making under ideal conditions.
4.	a. To display the financial impact of each decision alternative
5.	b. It examines actual decisions made by individuals, acknowledging irrational behaviour.
6.	b. Optimistic approach
7.	b. Worst possible payoff
8.	b. Regret associated with a decision
9.	b. Multiplying payoffs by their respective probabilities
10.	b. To visually represent different decision alternatives, states of nature and associated probabilities

HINTS FOR DESCRIPTIVE QUESTIONS

1. Decision theory is a framework that helps individuals and organisations make decisions under uncertainty, complexity

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and multiple objectives. It provides a structured methodology for analysing alternatives and their potential outcomes, bridging the gap between theoretical principles and practical decision-making. Refer to Section 11.2

2. A payoff table is a decision-making tool used in decision theory to classify and display the payoffs connected to several decision alternatives and possible conditions. It is mainly useful when a decision-maker must choose between different routes of action, and the outcomes rely on uncertain events or conditions. Refer to Section 11.2
3. The optimistic approach estimates each decision alternative regarding the best payoff. The decision alternative that is recommended is the one that provides the best possible payoff. Refer to Section 11.3
4. A decision tree is a graphical demonstration of a decision-making process, outcomes, and their probabilities. It is a powerful tool in decision analysis and can be used to visually represent complex decision problems, considering various alternatives, states of nature, and their associated probabilities and payoffs. Refer to Section 11.4
5. The minimax approach in decision analysis, one would choose the decision alternative that minimises the maximum state of regret that would happen over all possible states of nature. This approach is neither purely optimistic nor purely conservative. Refer to Section 11.3

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	The optimistic approach would focus on the maximum potential payoff for each alternative. The company would select the alternative with the highest possible profit. For instance, if the maximum payoff for the small factory is ₹10 crores, the medium factory ₹15 crores, and the large factory ₹25 crores, the decision would be to choose the large factory, as it offers the highest possible return.
2.	In the conservative approach, the decision-maker would focus on minimising risk by selecting the alternative with the best of the worst payoffs. For example, if Project A has the worst payoff of ₹5 crores and Project B ₹2 crores, the conservative approach would recommend Project A, as it offers a higher minimum payoff, ensuring less financial risk.

Q. No.	Answer
3.	The minimax regret approach aims to minimise the maximum regret. The manager would first calculate the opportunity loss for each decision alternative, comparing the best possible payoff for each state of nature. The decision alternative with the lowest maximum regret would be chosen. This approach helps avoid significant regret in any scenario by considering the worst-case opportunity loss.
4.	c. Project C
5.	b. Conservative approach

11.10 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

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N O T E S

CASE STUDY

LEADING AN ORGANISATION THROUGH TIMELY AND IMPACTFUL ACTION**Case Objective**

The objective of this case study is to provide a comprehensive understanding of the translation of decisions for a stronger team alignment and performance in New England insurance.

SCENARIO

Senior leaders at a top New England insurance provider need to develop the skills and behaviours for better and faster decision-making. This virtually delivered program spans four half-day sessions and includes individual assignments, facilitator-led presentations and simulation decision-making. Over the past two months, this program touched over 100 leaders, providing them with actionable models and frameworks to use back on the job.

PROBLEM

For one of New England's most iconic insurers, senior leaders are challenged to make timely, effective decisions. These leaders face decisions on three levels: those they translate to their teams, those they make themselves, and those that they influence. However, in a quickly changing highly regulated market, risk aversion can lead to slow and ineffective decisions. How can senior leaders practice in a safe environment the quick, yet informed, decision-making necessary for the job while simultaneously learning new models and techniques without the learning experience burdening their precious time?

SOLUTION

The effective decision-making program was artfully designed to immerse senior leaders in a 16 hours of hands-on experience, including reflection and feedback activities, applicable exercises, supporting content, and participation in a business simulation to practice the core content of the program utilising hypothetical pay-off tables. Participants worked together in small groups to complete these activities within a limited time frame, replicating the work environment in which these leaders must succeed. Continuous reflection and group discussion around results create real-time learning for leaders. Application exercises then facilitate the simulation experience and their work back on the job. The program employs a variety of learning methodologies, including:

- ❑ Individual assignments that incorporate content and frameworks designed to develop effective decision-making skills
- ❑ Guided reflection activities to encourage self-awareness and commitment to action

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- ❑ Large group conversations and live discussions focused on peer input around key learning points
- ❑ Small group activities, including virtual role plays designed to build critical interpersonal and leadership skills
- ❑ A dynamic business simulation in which participants are charged with translating, making and influencing difficult decisions
- ❑ Facilitator-led discussions and presentations

RESULT

Effective decision-making was developed in 2020 as an experience for senior-level leaders. After a successful pilot, the program was then rolled out to two more cohorts in 2021 and 2022. The senior-level leaders who participated in the program then requested we offer the same program to their direct reports. After some small adjustments to make the program more appropriate for director-level leaders, the program was launched in 2022 for approximately 100 directors.



QUESTIONS

1. How can senior leaders in New England's insurance company apply the effective decision-making program's models and frameworks to improve their decision-making skills in a fast-paced, regulated market?

(**Hint:** Focus on utilising guided reflection, business simulations, and peer discussions to translate decisions to their teams effectively.)

2. What role does the dynamic business simulation play in preparing senior leaders to translate, make, and influence decisions in a real-world context?

(**Hint:** Consider how the simulation replicates workplace challenges and integrates hypothetical payoff tables to practice decision-making within time constraints.)

(Source: <https://www.insight-experience.com/case-studies/effective-decision-making-leading-an-organization-through-timely-and-impactful-action>)

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C H A P T E R

DECISION TREES

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**Case Objective**

The objective of this caselet is to understand how LuxeTrends' enhances customer engagement using a decision tree model, optimising marketing strategies through precise customer segmentation based on demographics, behaviour and preferences.

INTRODUCTORY CASELET**ENHANCING CUSTOMER ENGAGEMENT AT LUXETRENDS - A DECISION TREE APPROACH**

LuxeTrends, a high-end fashion retailer, is keen on boosting customer engagement through personalised offerings and fraud detection strategies in its financial processes. To achieve this, LuxeTrends leverages decision tree models to understand customer preferences, detect anomalies in spending patterns and optimise engagement tactics. LuxeTrends analyses customer demographics, purchase history and browsing behaviour to create segmented marketing strategies. For instance, if a customer has repeatedly browsed luxury handbags but has yet to make a purchase, the decision tree can predict the best incentives—such as targeted discounts or exclusive previews—to convert interest into a sale.

Additionally, LuxeTrends integrates decision trees in its fraud detection mechanism. The model detects potentially fraudulent transactions by looking for trends in the historical transaction data, for example, finding high amounts of purchases in a distinct location that has not been used previously or abnormalities in the purchasing conduct of a customer. The decision tree approach enables the finance team to identify these high-risk transactions in real time thus enabling LuxeTrends to protect its clients while at the same time making sure that there is no interruption of service. This dual application of decision trees not only enhances customer engagement but also builds trust by ensuring financial security, ultimately fostering loyalty among LuxeTrends' clients.



LEARNING OBJECTIVES

After studying this chapter, you will be able to:

- Define key concepts of decision trees
- Develop decision trees using CHAID for analysis
- Decipher classification and regression decision tree
- Evaluate ensemble methods for improved model performance

12.1 INTRODUCTION

In the previous chapter, we learnt about the concept of decision theory. Next, we explored decision-making without probabilities. In the end, we explored decision-making with probabilities.

Decision trees are powerful tools used in data analysis for classification and regression tasks. This chapter covers key aspects of decision trees, including their elements, working mechanism and various types. It also explores the advantages and risks associated with decision trees. Additionally, topics like Chi-Square Automatic Interaction Detection (CHAID), Classification and Regression Trees (CART) and ensemble methods such as random forest are discussed, along with concepts like Gini impurity, entropy and Bonferroni correction. These methods play crucial roles in improving model accuracy and performance.

This chapter starts by discussing the concept of decision trees, their elements, how they work and the types of decision trees. Next, it discusses CHAID and CART. At the end, it discusses the ensemble methods.

12.2 CONCEPT OF DECISION TREES

Decision tree is a versatile and interpretable predictive modelling technique in machine learning and decision analysis. Decision trees are powerful tools for decision-making and predictive modelling in business analytics. They help break down complex decision processes into smaller, manageable parts and are particularly effective in classification and regression tasks. It works like a tree where each non-terminal node represents a decision or test on a feature. Each branch shows the result of that decision or test and each leaf node shows the final decision or outcome.

When building a decision tree, the data is split based on the most informative attributes to make the groups as similar as possible. This process continues until a stopping rule is met or the groups are pure

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enough. A typical decision tree which shows whether a person is fit based on Age > 30, junk food eating habits and exercise routine is shown in Figure 12.1:

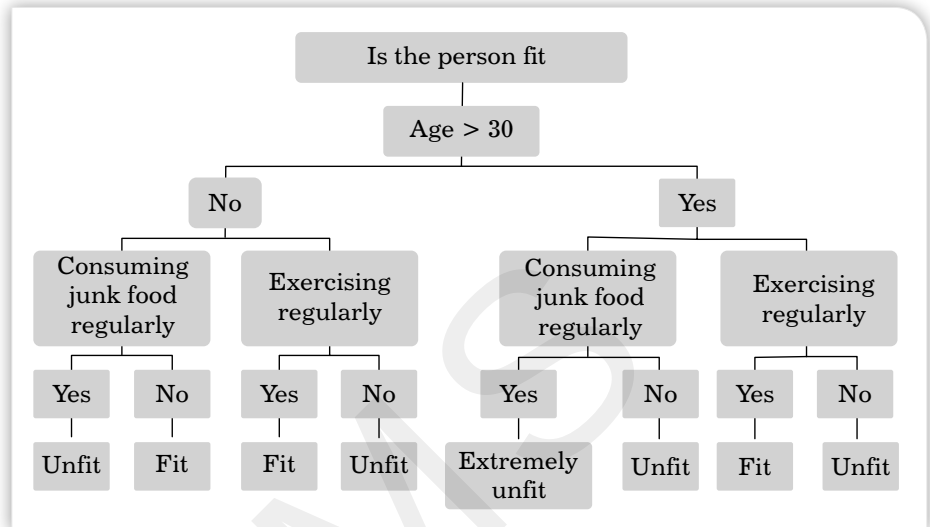


Figure 12.1: Typical Decision Tree Showing Decision Based on Three Different Factors

Decision trees can handle both qualitative and quantitative data. Their visual format makes complex decision processes easy to understand. They are often used for classification (labelling instances) and prediction (forecasting outcomes).

Decision trees, because of their simplicity and interpretability, have been used widely in so many real-life applications across domains. Some practical examples include:

- ❑ **Healthcare:** Decision trees help a doctor determine the most likely diagnosis based on symptoms and medical history, along with laboratory test results. For example, categorising patients according to the risk of certain diseases, such as diabetes or heart conditions and suggesting appropriate treatment protocols.
- ❑ **Finance:** Decision trees help in predicting the likelihood of loan applicants defaulting on credit given their level of income, employment history and credit score. They are also helpful in flagging suspicious transactions by identifying patterns not consistent with normal behaviour.
- ❑ **Retail and marketing:** Decision trees are also implemented in the segmentation of customers based on customer purchase behaviour to select appropriate marketing campaigns for them. Decision trees also predict what products a customer is most likely to purchase based on their preference and buying patterns.
- ❑ **Manufacturing:** Decision trees help in recognising the defective product by analysing the manufacturing data. They are also

helpful in finding the most efficient method of production to reduce costs and waste.

- ❑ **Education:** Decision trees are implemented for suggestions of courses to students based on their areas of interest and previous records.
- ❑ **Transportation and logistics:** Decision trees are used in the determination of the shortest or best route for delivering items to a place or travelling from one point to another.

Decision trees are especially suitable for problems where decisions need to be interpretable by possible non-technical stakeholders. This makes them a valid tool for operational and strategic problem-solving.

12.2.1 ELEMENTS OF DECISION TREES

A decision tree starts with a root node, which does not have any incoming branches. The outgoing branches from the root node feed into the internal nodes, also known as decision nodes. Based on the available features, both node types conduct evaluations to form homogenous subsets, denoted by leaf nodes, or terminal nodes. The leaf nodes represent all the possible outcomes within the dataset. A typical decision tree with its elements mentioned above is shown in Figure 12.2:

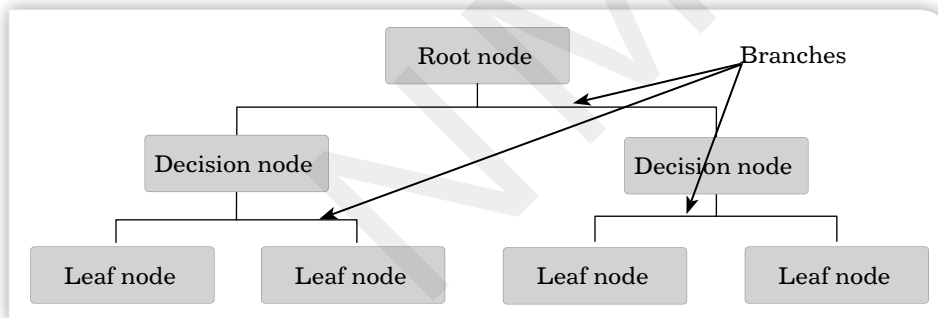


Figure 12.2: Elements of a Decision Tree

The elements of a decision tree, as shown in Figure 12.2, are discussed as follows:

- ❑ **Root node:** The root node is the highest-order node, which contains the entire data set. This is the initial node in the process of making a decision, whose purpose is to divide the corresponding data set utilising the most important attribute.
- ❑ **Branches:** The connections between the nodes on the tree are known as branches and depict the possible results of employing a decision rule at the given node. They further facilitate the movement from one decision node to another with the help of certain attribute values.

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- ❑ **Decision nodes (internal nodes):** These are the nodes in the tree where the data set splits based on particular attributes. The decision node has internal split conditions so that the overall tree can separate the particular data subsets into smaller data sections.
- ❑ **Leaf nodes (terminal nodes):** These are the terminal points in the tree structure and signify the conclusive classification (or decision) made after applying all the decision rules. Each leaf node indicates a certain resulting outcome or a prediction.

12.2.2 WORKING OF DECISION TREES

A decision tree consists of repeatedly dividing the data based on certain attributes, creating decision rules in each internal node and making predictions in the leaf nodes. The outcome of this process can be understood through a clear and easy-to-understand model that can be used for making those decisions in machine learning. The systematic working of a decision tree is shown in Figure 12.3:

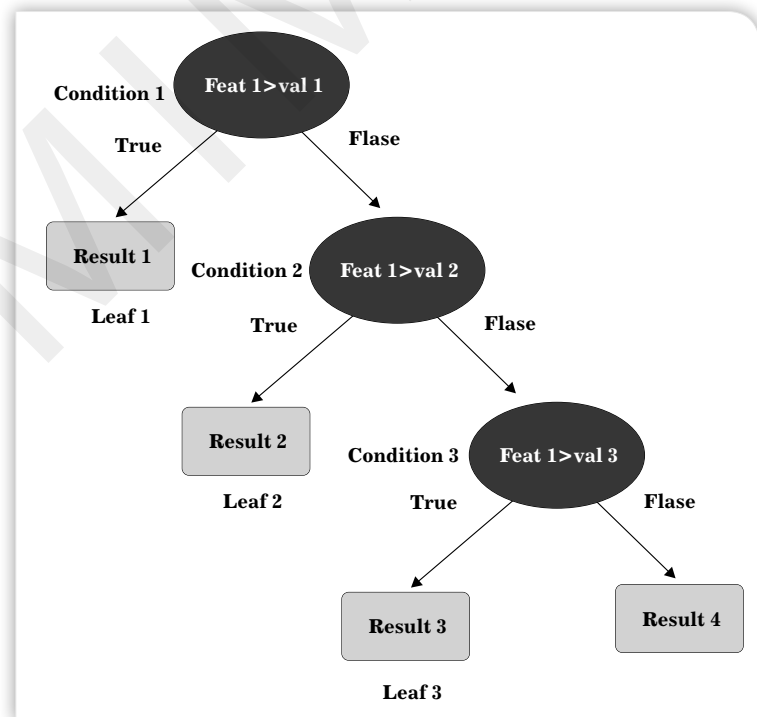


Figure 12.3: Working of a Decision Tree

The following steps are involved in the formation of a decision tree:

1. **Initialisation:** Start with the root node while including the whole dataset. The algorithm determines the most appropriate attribute for partitioning the data according to some parameter (for example, information gain or Gini impurity).

2. **Splitting:** Apply the designated attribute to divide the dataset into smaller parts. Each part reflects a value or a few values of the chosen attribute. This splitting process is continued at each internal node, giving rise to a tree-like structure.
3. **Recursive process:** Use the division process repeatedly on every subset within the internal nodes, thus forming a tree. This continues until some stopping rule is reached, either through a maximum depth, a minimum number of samples being present in a node or total segregation in the subsets.
4. **Decision rules:** Create a decision rule internal node according to the selection of an attribute and its threshold. These rules help in directing the data flow along the path in the tree. For instance, a rule about a numeric attribute will require assessing whether its value exceeds a given threshold.
5. **Predictions:** Move along the decision rules from the root node to a leaf to determine new examples. In classification, it may be the class that is the most common in that leaf, while in regression, it is usually the average value that corresponds to that leaf.

12.2.3 TYPES OF DECISION TREES

Different kinds of decision trees have their advantages and are appropriate for a particular task. The flexibility and ease of understanding of decision trees enable their application in different fields such as health care, finance and the like.

Some of the most common types of decision trees that are used today are as follows:

- ❑ **Classification trees:** These trees are employed for classification tasks where the goal is to categorise input instances into predefined labels or classes. The concept of a decision tree is based on the premise that the data can be classified by successively partitioning, or splitting, the data based on the values of the input features.
- ❑ **Regression trees:** These trees are designed for regression tasks, predicting continuous numerical values. Similar to classification trees, but instead of labels, the leaves of a regression tree contain predicted numerical values.
- ❑ **CART:** These trees can do both classification and regression. In CART, the splitting criteria can be modified according to the need of the task, for example using Gini impurity in the case of classification or Mean Squared Error in the case of regression.
- ❑ **ID3 (Iterative Dichotomiser 3):** ID3 is more oriented towards the classification of data compared to others. It utilises the information

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gain criterion to determine on which attribute to make the split at each node, and thus helps to reduce uncertainty, or entropy as much as possible.

- ❑ **C4.5 (Successor to ID3):** C4.5 extends ID3 to handle both classification and regression tasks. It introduces the information gain ratio to address the bias of ID3 towards attributes with more distinct values, promoting fairness in attribute selection.
- ❑ **CHAID:** CHAID is primarily used for statistical analysis to discover relationships between categorical variables. It employs Chi-square tests to detect significant associations between categorical variables, creating a tree structure that highlights interactions.

12.2.4 ADVANTAGES AND RISKS ASSOCIATED WITH DECISION TREES

Decision trees are flexible, and because of the flexibility of the technology, they may have applications across various domains. The following are some advantages of decision tree implementations:

- ❑ They are flexible and come in many forms to suit most business decision-making applications.
- ❑ The data they use is usually heterogeneous, such as numerical and textual data.
- ❑ They utilise dirty data with no need for much cleaning or standardisation.
- ❑ They make use of missing data values during their training process.
- ❑ They do not take much time to train or score new data, which is different from other varieties of ML models.
- ❑ They are considered explainable, or glass-box models, structured for direct interpretability. Explainable models are more easily verified and debugged by data scientists and more likely to be approved and adopted for business applications than some of their more complex ML cousins.
- ❑ Many free and commercial software packages offer different settings and types of decision tree algorithms.
- ❑ They provide information on feature selection and significance to help users understand which features bear the key impact on the prediction or label being assigned.

The risks associated with a decision tree are as follows:

- ❑ **Overfitting:** In decision trees, overfitting occurs when fluctuations employed by the model to make accurate predictions result in the

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learning of the training data to the detriment of the performance of other data samples, especially new ones.

- ❑ **Sensitivity to small changes:** Decision trees exhibit high sensitivity to even minor variations in data. For example, a small alteration in the training data can cause significant changes to the tree and hence affect the stability of the model.
- ❑ **Biased towards dominant classes:** Decision trees in classification tasks with imbalanced datasets may be biased towards the dominant class. The algorithm tends to favour the majority class, potentially leading to poor performance among minority classes.
- ❑ **Limited expressiveness:** Decision trees may have limited expressiveness for capturing complex relationships in the data, especially when compared to more advanced algorithms like ensemble methods or neural networks.
- ❑ **Unstable near decision boundaries:** Decision trees can be unstable near decision boundaries, making them sensitive to slight changes in input features. This sensitivity can result in different predictions for similar instances close to the decision boundary.

After learning about decision trees, we will learn about CHAID tree and its development in the next section.



SELF ASSESSMENT QUESTIONS

1. Which of the following is a common risk associated with decision trees?
 - a. Data privacy issues
 - b. Overfitting
 - c. Poor data collection
 - d. High computational cost
2. Decision trees take more time to train or score new data from other varieties of ML models. (True/False)
3. CART can be used for both classification and regression tasks. (True/False)



ACTIVITY

Imagine guiding a robot to decide if a student passes or fails using study hours and previous scores. Simplify rules for a clear pass/fail outcome by forming a hypothetical decision tree.

12.3

CHI-SQUARE AUTOMATIC INTERACTION DETECTION (CHAID)

The CHAID is a type of decision tree algorithm used to establish relationships between a dependent variable and a group of independent variables in business analytics. It is a qualitative, exploratory approach not constrained by assumptions of statistical distributions. CHAID is particularly effective in identifying associations between variables and assessing their importance for specific outcomes. This detailed orientation helps uncover insights not easily identified through conventional analyses.

CHAID is widely adopted for business predictions, especially when dealing with categorical variables. First introduced in the 1960s, the technique has become integral to fields like market research, healthcare and finance, where data-driven decisions are critical. Its proficiency lies in determining how a dependent variable relates to multiple independent variables, enabling segmentation, classification and prediction—all crucial functions in marketing and other domains.

The algorithm relies heavily on tree construction and statistical tests such as the Chi-square test to evaluate relationships. Through recursive splitting, CHAID creates a tree structure that visually represents significant patterns and interactions within the data. The resulting CHAID tree provides interpretable insights that support decision-making and predictive modelling.

The method partitions data into subgroups based on the values of one or more independent variables. This recursive process aims to maximise homogeneity in the dependent variable within each subgroup. Splitting continues until the subgroups are relatively homogeneous or a stopping criterion such as a minimum sample size at each node or failing to reject the null hypothesis in a Chi-square test is met.

Some of the key characteristics of CHAID are as follows:

- ❑ **Chi-square test:** The core of CHAID is the Chi-square test, which assesses the association between categorical predictors and the target variable. The algorithm selects the predictor with the strongest relationship to the target based on the highest chi-square value.
- ❑ **Splitting and merging:** CHAID splits nodes based on significant differences in the target variable distribution among the predictor categories. If two or more categories of a predictor do not significantly differ, they are merged. This merging process minimises overfitting and enhances interpretability.
- ❑ **Multi-level splits:** Unlike binary decision trees (e.g., CART), CHAID allows multi-level splits, meaning a node can be split

into more than two branches if it is statistically significant. This reduces the complexity of the tree and leads to simpler rules.

- ❑ **Handling missing values:** CHAID can effectively manage missing values by incorporating them into the analysis, either by treating them as a separate category or by using surrogate splits based on other available predictors.

12.3.1 CHAID TREE DEVELOPMENT

CHAID tree development is a type of decision tree algorithm, which assesses the quality of each split at every node of the tree using the chi-square statistic. It divides the observational data in such a way that they do not overlap and cover all possible segments, hence resulting in a clear set of decision rules. Many other decision tree methods are limited to handling only categorical predictors, but CHAID easily accommodates both categorical and continuous predictors making it applicable to a wider range of data types.

CHAID is based on a coherent set of procedures that facilitate the creation of decision trees. The first step involves dividing the dataset into smaller sets according to categorical characteristics, followed by an evaluation of the likelihood of each split occurring statistically. The following are the key steps involved in CHAID tree development:

1. **Selecting the best predictor:** CHAID begins by examining the relationship between each predictor variable and the target (dependent) variable using Chi-square tests. The predictor with the strongest association (highest Chi-square statistic and lowest p-value) is selected for the first split.
2. **Splitting the data:** The data is split into subsets based on the values or ranges of the selected predictor variable. Each subset corresponds to a distinct category or interval, ensuring maximum homogeneity of the target variable within each subset.
3. **Evaluating subsequent splits:** For each resulting subset, CHAID repeats the process of selecting the best predictor and splitting the data. This is performed recursively, creating a tree structure where each branch represents a decision path based on the relationships between variables.
4. **Stopping criteria:** The recursive splitting continues until a stopping criterion is met. These criteria may include:
 - a. A minimum node size (e.g., a subset contains fewer than a predefined number of observations)
 - b. A maximum tree depth
 - c. Non-significant Chi-square test results, indicating no further meaningful splits

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5. **Pruning the tree:** To prevent overfitting and enhance generalisability, CHAID prunes the tree by removing branches or nodes that do not significantly improve the model's performance. This process simplifies the tree and ensures it remains interpretable. Pruning is typically guided by statistical thresholds or cross-validation to strike a balance between model complexity and predictive accuracy.

12.3.2 BONFERRONI CORRECTION

The Bonferroni correction is a statistical method of hypothesis testing that adjusts the problem of multiple comparisons. In business analytics, for example, it is used often when one has to carry out many statistical tests at the same time to control the Type I error rate (the probability of making a false positive). The Bonferroni correction modifies the significance level in view of the fact that the chances of error increase when various tests are done.

The Bonferroni correction plays an important role in hypothesis testing in business analytics due to its robustness in managing the Type I error rate when several hypotheses are tested at the same time. The following points describe why it is important:

- ❑ **Multiple comparisons:** It is common for researchers to test multiple hypotheses (for example, test several marketing strategies or compare the performance of many departments within the organisation) simultaneously. Each hypothesis test has its fair share of chances of making a type I error, which is the error of not accepting the null hypothesis when it was supposed to be retained.
- ❑ **Inflated error rate:** With the increase of tests conducted, the possibility of finding at least one false positive increases too. In the absence of correction, conducting several tests raises the overall Family-Wise Error Rate (FWER).
- ❑ **Practical application:** While performing business analytics, we use Bonferroni's correction so that any inferences drawn from the tests are valid. For example, if one is testing several marketing strategies, the use of the correction means that any significant differences are not merely random variations, which decreases the chances of making a business decision based on a type.
- ❑ **Conservative approach:** The Bonferroni correction advocates a more cautious position aiming to achieve statistically significant results. Although it can be too stringent when many tests are performed, it reduces the risk of drawing excessively optimistic conclusions.

Some of the applications of Bonferroni correction are as follows:

- ❑ **Market research:** When assessing more than one element (e.g., price sensitivity, customer satisfaction or brand perception) in

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more than one customer segment, significant results can be made more accurate by applying the Bonferroni correction.

- ❑ **A/B testing:** The Bonferroni adjustment avoids errors across tests whenever a company runs multiple versions of the same website, product feature or marketing campaign.
- ❑ **Risk analysis:** It is also incorporated while conducting more than one risk assessment or assessing the impact of more than one risk factor in any financial assessed by way of a model, as a means of ensuring that the outcome of such decisions is statistically viable.

We will study about CART in the next section.



SELF ASSESSMENT QUESTIONS

4. CHAID is not bound to the assumptions of which statistical concept?
 - a. Normal distribution
 - b. Central limit theorem
 - c. Linear regression
 - d. Causation
5. Which type of variable is typically the dependent variable in CHAID analysis?
 - a. Numeric
 - b. Binary
 - c. Ordinal
 - d. Categorical
6. A higher Chi-square value in CHAID analysis indicates a stronger association between variables. (True/False)



ACTIVITY

Conduct a CHAID analysis on a dataset with customer demographics. Interpret the resulting tree to generate actionable insights for targeted marketing strategies. Identify significant factors influencing customer behaviour and tailor marketing efforts accordingly.

12.4 CLASSIFICATION AND REGRESSION TREE (CART)

CART is a powerful and widely used algorithms in business analytics that enables organisations to make data-driven decisions by modelling complex relationships between variables. It belongs to the class of algorithms known as decision trees which are models

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designed for classifying data or predicting a certain output value from a set of input variables. In a business environment, the models provided by CART are transparent and easy to understand, making them appropriate for analysing business problems such as market segmentation, predicting customers' purchases and risk assessment.

The process of building the CART model consists of using the whole dataset, and at each stage, it is subsequently divided into two sections. The algorithm determines the optimal feature and threshold that best classifies the data concerning the target variable at every decision node. The effectiveness of a split is usually assessed using criteria such as:

- ❑ **Gini impurity (for classification tasks):** This quantifies the amount of disorder in a given node. The Gini index should be reduced to the lowest level possible, which indicates that the classes are closely related in their respective nodes.
- ❑ **Mean Squared Error (MSE):** MSE refers to the mean of the square of the predicted and actual values in an experiment.

The following are the steps involved in the classification tree of CART:

1. **Tree construction:** The tree starts with the entire dataset as the root node. At each node, CART identifies the feature and corresponding split point that maximises the separation of classes (for classification tasks) using measures like Gini impurity or entropy.
2. **Recursive splitting:** The dataset is split into subsets based on the chosen feature and split point. This process continues recursively for each subset until a stopping criterion is met, such as a predefined tree depth or a minimum number of samples in a leaf node.
3. **Leaf nodes and predictions:** Terminal nodes represent the final subsets after recursive splitting. Each leaf node is assigned a class label, representing the predicted class for instances falling into that subset.

The steps involved in the regression tree of CART are:

1. **Tree construction:** Similar to classification, the tree begins with the entire dataset as the root node. Nodes are split based on features and split points, but in regression, the objective is to minimise the sum of squared differences between predicted and actual values within each subset.
2. **Recursive splitting:** The selected feature and split point are used to iteratively divide the dataset into subsets. Until a stopping criterion such as MSE, the minimum number of samples in a leaf node or a pre-determined tree depth is satisfied, the process keeps going.

3. **Leaf nodes and predictions:** Terminal nodes (leaf nodes) represent the final subsets after splitting. Each leaf node is assigned a predicted numerical value, often the mean or median of the actual values within that subset.

12.4.1 GINI IMPURITY INDEX

The Gini impurity index is a statistic that is commonly used in business analytics, especially in decision trees and machine learning models, to determine how pure a particular dataset is. It measures the probability of getting a randomly chosen element from the dataset incorrectly labelled if a random label were assigned according to the distribution of labels in the dataset. Simply put, Gini impurity refers to the extent of overlap of the data across the different classes.

The formula for the Gini impurity index is:

$$\text{Gini}(D) = 1 - \sum_{i=1}^k p_i^2$$

Where:

p_i is the probability of an element being classified into class i

k is the number of classes in the dataset.

Example: Imagine you have a dataset with 100 samples, and you want to classify them into two classes: A and B. Let us say that 40 samples belong to class A and 60 samples belong to class B. To calculate the Gini impurity, follow the given steps:

1. Calculate the proportion of each class:

$$\text{Proportion of class A } (P_A) = \frac{40}{100} = 0.4$$

$$\text{Proportion of class B } (P_B) = \frac{60}{100} = 0.6$$

2. Compute the Gini impurity using the formula:

$$\text{Gini impurity} = 1 - (P_A^2 + P_B^2)$$

Substituting the values: Gini impurity =

$$1 - (0.4^2 + 0.6^2) = 1 - (0.16 + 0.36) = 1 - 0.52 = 0.48$$

So, the Gini impurity for this dataset is 0.48. This value indicates the impurity level or diversity in the dataset, with 0 being completely pure (all samples in one class) and higher values indicating more impurity.

CART, specialising in both classification and regression trees, employs the Gini index as a classification metric. The purity of a node

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is assessed based on the degree of homogeneity in class labels within that node, with the degree ranging between 0 and 0.5. A value of 0 implies that all data belongs to a single class or variable, while 0.5 indicates diversity among classes or fields.

12.4.2 ENTROPY

Entropy, like the Gini index, is another impurity measure used in decision tree algorithms, particularly in algorithms like ID3 and C4.5. The concept of entropy is borrowed from information theory, where it calculates the amount of uncertainty in a dataset. In the context of decision trees, entropy is used to evaluate the impurity of a node.

A higher entropy value indicates more unpredictability, while a lower value indicates less randomness. In decision trees, entropy helps determine the best way to split data by measuring the disorder or impurity within subsets of data. The general goal is to reduce entropy, creating more homogeneous groups and improving the model's accuracy. A node with entropy 0 is pure, meaning all data points in the node belong to the same class. As entropy increases, the impurity or disorder within the node also increases.

The entropy for a node is calculated using the formula:

$$\text{Entropy } (S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Where:

p_i is the probability of data points in node t that belong to class i

Example: Imagine you have a fair six-sided die. The probability of each face (state) showing up is $\frac{1}{6}$.

To calculate the entropy:
$$S = \sum_{i=1}^6 -\frac{1}{6} \log_2 \frac{1}{6}$$

Since $\log_2 \frac{1}{6} = -\log_2 6$

Therefore,
$$S = -6 \times \frac{1}{6} \times (-\log_2 6)$$

$$S = \log_2 6 \approx 2.585$$

So, the entropy of rolling a fair six-sided die is approximately 2.585 bits. This value represents the uncertainty or randomness associated with the outcome of rolling the die. In other words, each face of the die has an equal chance of appearing, making the system highly uncertain. The entropy value (2.585 bits) quantifies this uncertainty.

During the construction of a decision tree, the goal is to find the feature that maximises information gain. Information gain is a measure of how much a particular feature reduces uncertainty in predicting the class labels of instances within a dataset. It is calculated for each feature by subtracting the weighted average of entropies after a split from the entropy before the split:

$$\text{Information Gain}(\text{Feature}) = \text{Entropy}(\text{Before Split}) - \sum_i \left(\frac{\text{Number of Data Points in subset } i}{\text{Total Number of Data Points}} \times \text{Entropy}(\text{Subset } i) \right)$$

Where:

Entropy (Before Split) is the entropy of the dataset before any split.

The sum is taken over each subset resulting from the split based on the chosen feature.

The Entropy (Subset i) is the entropy of the i^{th} subset.

The higher the information gain, the more effective the feature is in reducing uncertainty about the class labels. Therefore, during the construction of a decision tree, the feature that maximises information gain is chosen as the root node for the split.

We will learn about ensemble method in the next section.



SELF ASSESSMENT QUESTIONS

7. Which of the following criteria is commonly used for assessing the effectiveness of splits in CART classification tasks?
 - a. Root mean square error
 - b. Information gain
 - c. Gini impurity
 - d. Chi-square
8. What type of output does a regression tree predict?
 - a. Numerical value
 - b. Categorical value
 - c. Binary classification
 - d. Ordinal value
9. Information gain is used to evaluate the effectiveness of splits in decision trees. (True/False)

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ACTIVITY

Implement a CART on a real estate dataset, predicting property values based on location, size and market trends.

12.5 ENSEMBLE METHOD

The ensemble method is a powerful method used to enhance business analytics predictive models. This is done by integrating several predictive models, known as ‘weak learners,’ to develop an improved predictive strength model. The main premise stems from the fact that individual models may be prone to error but combining several would lessen the chances of making an overall error since there is a collective intelligence principle. Some patterns and relations that are more complicated than a simple regression can be understood through diverse models’ ensembles and cannot be comprehended through singular models.

In the case of decision trees, ensemble techniques are very effective. While decision trees are powerful, they can be sensitive to the noise in the data and tend to overfit the training data. Representation of such methods is the application of many decision trees together and taking advantage of them. In gaining stability in predictions and accuracy of the model, aggregation of predictions from different decision trees is done.

Types of ensemble methods are as follows:

- ❑ **Bagging (bootstrap aggregating):** Bagging helps make predictions more stable by training multiple models on different subsets of the data, created through random sampling with replacement. Each model makes its prediction, and the final result is obtained by averaging the predictions or using a majority vote. A common example of this approach is the random forest, which combines many decision trees for classification or regression tasks. An illustration of the bagging method is shown in Figure 12.4:

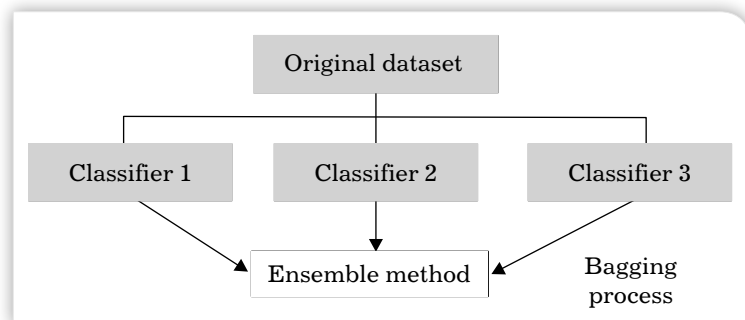


Figure 12.4: Bagging Method

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- ❑ **Boosting:** Boosting is an ensemble technique that combines the predictions of several weak predictors to create a strong, accurate prediction. The objective of boosting is to minimise the model bias by training a sequence of models, each model correcting the errors of the previous one. It focuses on the mistakes made and provides more importance to the wrongly classified cases over some time ensuring high accuracy. AdaBoost, Gradient Boosting and XGBoost are among a few models that are very effective for predictive performance and are frequently used in business analytics.

An illustration of the boosting method is shown in Figure 12.5:

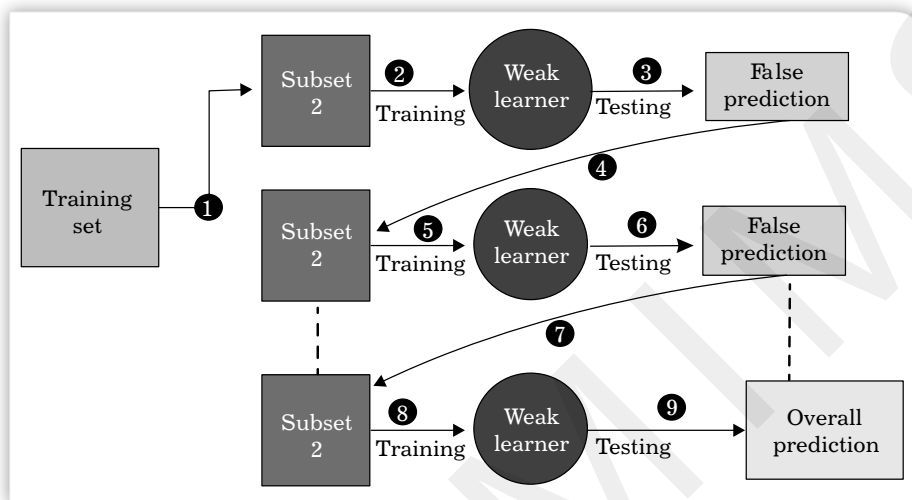


Figure 12.5: Boosting Method

- ❑ **Stacking:** Stacking is another technique in which multiple models or algorithms are put together to improve the accuracy of predictions. In this technique, a new model called a meta-model is used to aggregate different models' predictions by learning the most appropriate way to do so for different predictions. While bagging and boosting involves using the same kind of models, which can be referred to as homogeneous ensembles, stacking uses heterogeneous ensembles by incorporating different algorithms in the ensemble. This versatility in the approach is one of the reasons why stacking is very useful and works well when solving complicated business issues.

Some of the applications of ensemble method in business analytics are as follows:

- ❑ **Customer segmentation and targeting:** Random forests and gradient boosting models assist in dividing clients based on their interests, demographics, etc., which contributes to a deeper approach to marketing.

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- ❑ **Fraud detection:** Multiple models can be combined to increase the sensitivity and specificity of particular patterns related to fraud and its detection, thus improving the outcome.
- ❑ **Risk management and credit scoring:** Numerous models can be combined to produce ensemble methods, which allow for a more complete evaluation of credit risk, which is very important in lending and finance.
- ❑ **Sales forecast:** The application of more than one predictive model to sales forecasting promotes the accuracy of the forecasting, enabling a business to make rational decisions concerning stocking levels, staff and marketing strategy.

12.5.1 RANDOM FOREST

Random forest is a combination of many decision trees in which each of these trees makes individual predictions and the final output is calculated through some aggregation process. Random forests were invented to deal with the overfitting problem associated with the use of a single decision tree. It makes use of dual approaches, which are bagging and the use of random features during the construction of the individual trees. It is widely applied in both classification and regression tasks due to its versatility and ability to handle complex datasets. An example of a random forest is shown in Figure 12.6:

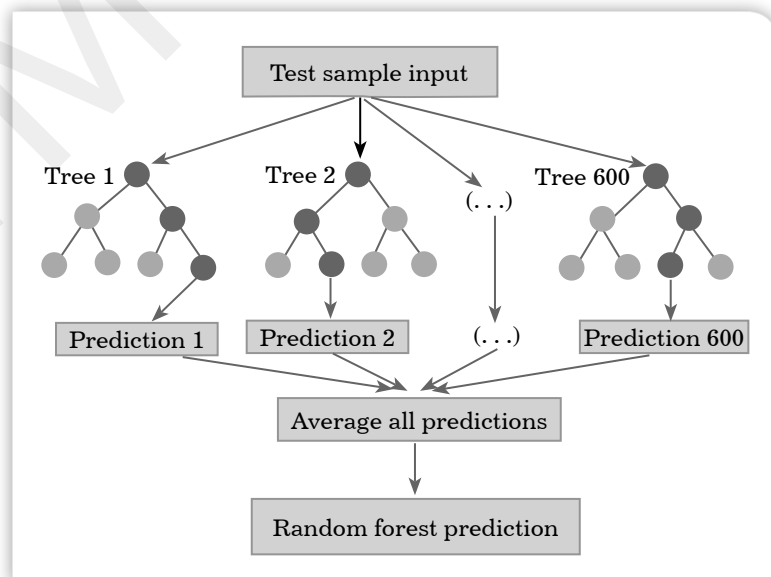


Figure 12.6: Random Forest

Random forests are versatile and robust, making them effective for both classification and regression tasks. The ensemble nature of random forests enhances predictive accuracy and reduces overfitting, making them a popular choice in machine learning. The random forest algorithm involves several steps, which are:

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1. **Data collection:** Gather a dataset with features and a target variable. Ensure the dataset is diverse and representative of the problem.
2. **Random sampling (bootstrap sampling):** Randomly select a subset of the dataset with replacement (bootstrap sampling). This creates multiple training datasets for building individual trees.
3. **Feature randomness:** At each node of the tree, select a random subset of features to consider for splitting. This helps in de-correlating the trees and improving overall performance.
4. **Building decision trees:** Construct multiple decision trees using the randomly sampled datasets and features. Each tree is grown to its maximum depth without pruning.
5. **Voting (ensemble):** Combine the predictions of all trees to make the final prediction. For classification, this involves a majority vote; for regression, it is often an average.
6. **Out-of-Bag (OOB) score:** Evaluate the performance of the random forest using the OOB samples, which are data points not included in the bootstrap sample for each tree.
7. **Prediction:** Apply the trained random forest model to new data for predictions. The model generalises well due to the diversity introduced by random sampling and feature selection.



SELF ASSESSMENT QUESTIONS

10. Random forests are not effective for both classification and regression tasks. (True/False)
11. Which ensemble method relies on training a series of models that correct previous errors?
 - a. Bagging
 - b. Random forest
 - c. Boosting
 - d. Clustering
12. Random forests use only one decision tree to make predictions, enhancing accuracy and reducing overfitting. (True/False)



ACTIVITY

Implement the ensemble method on healthcare data to enhance disease prediction models. Leverage the decision trees for improved diagnostic precision, benefiting healthcare decision-making.

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12.6 SUMMARY

- ❑ Decision tree is a versatile and interpretable predictive modelling tool employed in machine learning and decision analysis.
- ❑ The root node is the highest order node, which contains the entire dataset.
- ❑ Classification trees are employed for classification tasks where the goal is to categorise input instances into predefined labels or classes.
- ❑ The CHAID is a type of decision trees algorithms used to establish relationships between a dependent variable and a group of independent variables in business analytics.
- ❑ The Bonferroni correction is a statistical method of hypothesis testing that considers the problem of multiple comparison.
- ❑ CART are powerful, widely used algorithms in business analytics that enable organisations to make data-driven decisions by modelling complex relationships between variables.
- ❑ The Gini impurity index is a statistic that is commonly used in business analytics, especially in decision trees and machine learning models, to determine how pure a particular dataset is.
- ❑ Entropy is a notion which has been borrowed from the information theory to express a measure of disorder in a dataset in business analytics.
- ❑ The ensemble method is a more powerful method used for the enhancement of business analytics predictive models.
- ❑ Random forest is a combination of many decision trees in which each of these trees makes individual predictions and the final output is calculated through some aggregation process.



KEY WORDS

- ❑ **Ensemble methods:** It is a collection of techniques designed to improve predictive power through the “combination” of multiple models.
- ❑ **Type I error:** It occurs when a relationship that does not exist is mistakenly identified in a decision tree as the influence of a certain variable on the outcome thereby, leading to the wrong decision being made.
- ❑ **Noise:** It refers to irrelevant or random variations in data that do not represent meaningful patterns, potentially affecting the accuracy of decision tree models and other machine learning algorithms.

- ❑ **Pruning:** It involves selectively removing nodes from a decision tree to prevent overfitting, achieving a balance between model complexity and performance and improving generalisation to new data.

12.7 MULTIPLE CHOICE QUESTIONS

- Which type of decision tree is best suited for predicting numerical outcomes?
 - Classification tree
 - Regression tree
 - ID3
 - CHAID
- Which decision tree algorithm uses Chi-square tests for splitting?
 - ID3
 - C4.5
 - CART
 - CHAID
- Which impurity measure is commonly used in decision trees for classification tasks?
 - Mean squared error
 - Gini index
 - Pearson correlation
 - Cosine similarity
- What is the purpose of Bonferroni correction in hypothesis testing?
 - To decrease Type II error rate
 - To adjust for multiple comparisons
 - To increase the power of the test
 - To make tests simpler
- What is the criterion used by CHAID to evaluate the significance of splits?
 - p-value
 - Gini index
 - F-statistic
 - Cross-validation score
- What does the Bonferroni correction adjust for in statistical tests?
 - Sample size
 - Multiple comparisons
 - Regression coefficients
 - Data normalisation
- Which concept is borrowed from information theory to evaluate disorder in a dataset?
 - Decision boundaries
 - Gini impurity
 - Entropy
 - Cross-validation

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8. What is the range of Gini impurity values?
 - a. 0 to 1
 - b. 0 to 2
 - c. 0 to 0.5
 - d. 0.5 to 1
9. Which principle reduces errors when combining several predictive models in ensemble methods?
 - a. Maximum likelihood
 - b. Collective intelligence
 - c. Regression analysis
 - d. Dimensional reduction
10. Which method involves creating multiple decision trees to improve stability in predictions?
 - a. Regression analysis
 - b. Clustering
 - c. Decision tree pruning
 - d. Ensemble methods

12.8 DESCRIPTIVE QUESTIONS

1. What is a decision tree in business analytics? Explain its primary purpose.
2. Explain CHAID in the context of decision trees.
3. What are the main elements of a decision tree?
4. What is CART? Discuss its applications.
5. What is a random forest, and how does it improve the decision-making accuracy?

12.9 HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

1. A hospital needs to decide whether to admit patients based on symptoms like fever, cough and shortness of breath. They have a dataset of previous cases. How can a decision tree help them make this decision?
2. A financial analyst is using a decision tree to predict loan defaults. They need to understand what components of the tree are critical for this prediction. What are the key elements they should focus on?
3. A pharmaceutical company is testing multiple factors influencing drug efficacy using CHAID. They are concerned about inflated error rates. How does the Bonferroni correction help?
4. A retail company is trying to increase the precision of its sales forecasts. In order to achieve higher prediction performance than a single model, the data science team chooses to employ a technique that mixes numerous models. By combining the predictions of multiple base learners, they test a method that

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lowers the possibility of overfitting. Which method are they most likely using?

- a. Decision trees b. Linear regression
- c. Ensemble method d. K-Nearest neighbours

5. A healthcare company is using a machine learning model to predict whether patients have a particular disease based on their medical records. To avoid overfitting, they choose an algorithm that creates multiple decision trees using random samples of the data and features and then averages the results. Which algorithm are they using?

- a. K-Means clustering b. Random forest
- c. Decision tree d. Linear regression

12.10 ANSWERS AND HINTS

ANSWERS FOR SELF ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
Concept of Decision Trees	1.	b. Overfitting
	2.	False
	3.	True
Chi-Square Automatic Interaction Detection (CHAID)	4.	a. Normal distribution
	5.	d. Categorical
	6.	True
Classification and Regression Tree	7.	c. Gini impurity
	8.	a. Numerical value
	9.	True
Ensemble Method	10.	False
	11.	c. Boosting
	12.	False

ANSWERS FOR MULTIPLE CHOICE QUESTIONS

Q. No.	Answer
1.	b. Regression tree
2.	d. CHAID
3.	b. Gini index
4.	b. To adjust for multiple comparisons

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Q. No.	Answer
5.	a. p-value
6.	b. Multiple comparisons
7.	c. Entropy
8.	c. 0 to 0.5
9.	b. Collective intelligence
10.	d. Ensemble methods

HINTS FOR DESCRIPTIVE QUESTIONS

1. A decision tree is a term used for a tool in predictive modelling that incorporates a tree-like structure of decisions leading to various outcomes. Most often it is used to avoid complex decision-making processes and divides the given data into multiple small parts. Refer to Section 12.1
2. CHAID is a technique commonly used to create decision trees. It involves dividing the dataset using Chi-square tests for the best predictors of a given categorical outcome, which makes it helpful when it comes to exploratory data analysis. Refer to Section 12.3
3. The main elements of a decision tree are the root node (starting point), decision nodes (intermediate points with branches), leaf nodes (final outcomes) and branches (connections that represent decisions or rules). Refer to Section 12.1
4. CART is a decision tree approach that can be applied to both classification (where the outcomes are categorical) and regression (where the outcomes are continuous). It is also used for predictive modelling in applications such as customer profiling and predicting future sales. Refer to Section 12.4
5. A random forest is an ensemble method that utilises multiple decision trees built from random samples of the available data and their features. Its accuracy is enhanced since it combines the outputs of several individual trees, thus minimising the errors. Refer to Section 12.5

ANSWERS FOR HIGHER ORDER THINKING SKILLS (HOTS) QUESTIONS

Q. No.	Answer
1.	A decision tree can be developed using the past dataset, where nodes represent symptoms and branches represent their presence or absence. The tree will guide doctors in classifying patients' urgency for admission by identifying key symptoms that often lead to severe conditions, assisting in faster and more accurate decision-making.

Q. No.	Answer
2.	The analyst should focus on the root node, which represents the most significant feature (e.g., credit score), decision nodes indicating criteria (like income or debt-to-income ratio) and leaf nodes that show the prediction outcome (default or no default). These elements help interpret how each attribute influences the risk of default.
3.	By applying the Bonferroni correction, the company can control the Type I error rate across multiple hypotheses. It adjusts the p-value significance level, ensuring that only highly reliable splits remain in the CHAID model, minimising false positives in the drug efficacy analysis.
4.	c. Ensemble method
5.	b. Random forest

12.11 SUGGESTED READINGS & REFERENCES

SUGGESTED READINGS

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**Case Objective**

The objective of this case study is to demonstrate the practical application of decision trees in analysing customer data within the context of ElectroTech.

CASE STUDY**UNVEILING CUSTOMER INSIGHTS FOR IMPROVED LOYALTY AT ELECTROTECH****SCENARIO**

ElectroTech, a prominent online electronics retailer, has observed a concerning trend of diminishing customer loyalty over the past year. To address this challenge, ElectroTech's data science team is embarking on a mission to leverage decision tree analysis, aiming to unravel the factors influencing customer churn and devise strategies to bolster retention.

DATA COLLECTION

The data science team at ElectroTech compiles a robust dataset encompassing diverse customer-related variables, including but not limited to:

- ☐ Purchase history (frequency, average order value)
- ☐ Demographics (age, gender, location)
- ☐ Product preferences and categories
- ☐ Customer reviews and ratings

METHODOLOGY

In pursuit of a comprehensive understanding of customer behaviour, the team opts for a decision tree algorithm. This algorithm's ability to handle both categorical and numerical data aligns well with the varied nature of the dataset.

STEPS IN MODEL DEVELOPMENT

The steps involved in model development are:

1. Data preprocessing involves rigorously handling missing values, strategically encoding categorical variables and, if necessary, scaling features for balanced model performance.
2. Splitting the data is achieved through meticulous division of the dataset into training and testing sets, often using an 80-20 split.
3. Model training entails using the training set to construct the decision tree model, with careful selection of hyperparameters to optimise performance.
4. Model evaluation involves a thorough assessment of the model's performance on the testing set, utilising metrics such as accuracy, precision, recall and F1 score. Iterative fine-tuning of the model may be performed if required.

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PERFORMANCE METRICS

To evaluate the effectiveness of the decision tree model, ElectroTech's data science team employs the following performance metrics:

- ❑ **Accuracy:** This metric measures the proportion of correctly classified instances (correct churn predictions). A higher accuracy indicates that the model is reliably predicting customer churn and retention.
- ❑ **Precision:** This metric measures how well the model can detect positive churn cases. It is especially crucial in situations when false positive (incorrectly predicting churn) having a high cost.
- ❑ **Recall:** This metric measures the model's effectiveness to uncover each and every customer that is predicted to churn. A high recall is particularly important for ensuring that the efforts to retain at-risk customers are targeted for retention efforts.
- ❑ **F1 score:** This metric measures the harmonic mean of precision and recall. It is a metric that takes both extremes into account and finds the average computationally. The better the F1 score, the more improved the overall performance of the model, especially in churn vs. retention rate situations where they are not equal.

RESULTS

The decision tree model successfully uncovers key factors contributing to customer churn at ElectroTech. Prominent features include infrequent purchases, negative product reviews and a high number of customer service interactions. The model demonstrates a high accuracy rate on the testing set, indicating its effectiveness in predicting customer churn.

RECOMMENDATIONS

Informed by the decision tree insights, ElectroTech can implement targeted strategies to enhance customer retention. These strategies are:

- ❑ **Tailored promotions:** Launch personalised marketing campaigns targeting customers with infrequent purchases.
- ❑ **Product quality enhancement:** Address product issues raised in negative reviews to improve overall customer satisfaction.

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- ❑ **Streamlined customer support:** Enhance customer support processes to reduce the number of service interactions and improve customer experience.

CONCLUSION

This case study displays the application of decision trees in analysing customer data for predicting churn in the context of ElectroTech. By understanding the factors influencing customer retention, ElectroTech can make data-driven decisions to bolster customer satisfaction and fortify overall business performance.

**QUESTIONS**

1. Why did the data science team choose a decision tree algorithm for this analysis, and how does its ability to handle both categorical and numerical data align with the dataset's diverse nature?

(Hint: Consider the advantages of decision trees and how they suit the characteristics of the provided dataset.)

2. Why is it important to handle missing values and strategically encode categorical variables during the data pre-processing phase?

(Hint: Reflect on how missing values and categorical variables can affect the performance of a decision tree model.)